Data Communication & Computer Networks (CN)

The Complete Book — Theory + MCQs (Topic 1 to 7)

631 MCQs — har topic pehle (Theory), phir questions Easy → Medium → Hard

Reference Book: Data Communications and Networking — Behrouz A. Forouzan

Prepared for: RPSC Programmer • ACP • RSMSSB IA • SA • DD • BCI • SCI • REET • RAS • DSSSB • KVS • NVS • HTET • SSC CGL/CHSL • RRB • IBPS • UPSC & all Rajasthan + Central Govt. Exams (Computer Awareness)

Format: Har section: (full-page theory) → phir questions (Easy → Medium → Hard)

Question row: Left = question + options | Right = correct answer + Nukte (theory pages mein poori)

Language: Hindi second language; technical terms English mein (protocol, topology, jitter...)

Di culty: Har question par chip: · Easy / · Medium / · Hard — sections mein Easy se Hard tak

Coverage: 631 MCQs — 2011–2026 India-level exams (GATE, NET, KVS, EMRS, DSSSB, RPSC, SSC, RRB, IBPS...) + Forouzan

Syllabus Map — 7 Topics, 31 Sections

Topic 1. Introduction of Data Communication

- Section A - DC Components & Protocol Basics (Message, Sender, Receiver, Transmission Medium, Protocol) — 17 Q
- Section B - Characteristics (Delivery, Accuracy, Timeliness, Jitter) & Criteria (Performance, Reliability, Security) — 19 Q
- Section C - Transmission Mode (Direction of Data Flow: Simplex, Half-Duplex, Full-Duplex) — 9 Q
- Section D - Network Structure / Topology (Physical: Mesh, Star, Bus, Ring, Others; Logical: IEEE 802) — 21 Q
- Section E - Types of Networks (LAN, MAN, WAN, PAN, CAN, SAN; Internet, Intranet, Extranet; Point-to-Point, Multipoint; Client-Server, P2P) — 33 Q
- Section F - Network History (ARPANET to Today's Network) — 8 Q
- Section G - General Network Basics (WWW, E-mail, Protocols, Servers, Switching - from repo intro tests) — 17 Q

Topic 2. Network Models (OSI & TCP/IP)

- Section A — OSI Model: Basics, Layer Order & Support Layers — 12 Q
- Section B — OSI Layer Functions (Layer 1 → Layer 7) — 51 Q
- Section C — TCP/IP Model — 12 Q
- Section D — Comparison, Encapsulation & PDUs — 7 Q

Topic 3. Addressing (MAC, IPv4, IPv6, Port, DNS)

- Section A — Address Levels & MAC Address (Physical Address) — 10 Q
- Section B — IPv4 Addressing (Format, Classes & Special Addresses) — 33 Q
- Section C — Subnetting & CIDR — 27 Q
- Section D — IPv6 Addressing — 9 Q
- Section E — Port Numbers, DNS/URL & Address Resolution (ARP) — 14 Q

Topic 4. Networking Devices

- Section A — Physical-Layer Devices (Hub, Repeater, Modem & Simple Devices) — 9 Q
- Section B — Switch / NIC / Bridge — 10 Q
- Section C — Router / Brouter — 8 Q
- Section D — Gateway, Firewall & NAT Devices — 11 Q

Topic 5. Transmission Medium

- Section A — Signals & Performance (Encoding, Noise, Bandwidth, Data Rate) — 25 Q

Section B — Guided Media (UTP, Coaxial, Fiber Optic) — 16 Q

Section C — Unguided Media (Wireless) — 7 Q

Topic 6. OSI Layers in Detail

Section A — Data Link Layer: Flow & Error Control — 52 Q

Section B — Data Link Layer: Framing & MAC Sub-layer — 27 Q

Section C — Network Layer: IP, ICMP, ARP & Routing Protocols — 36 Q

Section D — Transport Layer: TCP/UDP, Ports & Sockets — 37 Q

Section E — Session, Presentation & Application Layer — 41 Q

Topic 7. Network Security

Section A — Security Goals & Attacks — 17 Q

Section B — Cryptography — 21 Q

Section C — Firewalls, IPSec & Security Protocols — 15 Q

Kaise padhein: Har section mein pehle padho (concepts, table, formulas) — phir questions karo. Question ka left side khud try karo, right side par answer aur Nukte (exam trick) check karo. Revision: last mein Nukte Rapid-Fire.

TOPIC 1 — Introduction of Data Communication

124 MCQs • Theory → Questions (Easy → Medium → Hard)

Ω Samjh — Section A - DC Components & Protocol Basics (Message, Sender, Receiver, Transmission Medium, Protocol)

Syllabus: DC Components - Message, Sender, Receiver, Transmission Medium, Protocol

Data Communication = do devices ke beech message ka exchange. Iske 5 components hote hain aur communication rules ko protocol kehte hain.

Data Communication ŋ Ȼ message -ų 5 components ʐ ȝ Ȼ protocol ʐ

- Message — information (data) jo bheji jaati hai: text, number, picture, audio, video.
  Message — (data) : text, number, picture, audio, video.
- Sender (source) — device jo message bhejta hai; Receiver (destination) — jise message milta hai.
  Sender (source) — message ; Receiver (destination) — ɢ message ɜ
- Transmission medium — rasta jisse message jaata hai (guided: wire/cable; unguided: wireless).
  Transmission medium — ⱬ ɢ message (guided: /cable; unguided: wireless)
- Protocol = communication ke rules (agreed set of rules). Har protocol mein 3 cheezein: Syntax (format/order), Semantics (har section ka meaning), Timing (speed + order).
  Protocol = ȝ Ȼ Ī protocol Ø ØSyntax (format/order), Semantics (Ȼ ŋ Timing (ȝ + t')
- Network = devices (nodes) + connecting links; data ko share karne ke liye banaya jata hai.
  Network = (nodes) + links; data ɠ
- Data flow direction ko transmission mode kehte hain (Simplex / Half-Duplex / Full-Duplex) — yeh Section C mein hai.
  Data ɬɛ transmission mode ʑ(Simplex / Half-Duplex / Full-Duplex) — Section C Ȼ

Component	Kaam (work)
Message	data jo jaana hai — the information
Sender	message ko bhejta hai — source device
Receiver	message ko paata hai — destination device
Transmission medium	message ka rasta — guided / unguided
Protocol	communication ke rules — syntax, semantics, timing

Q1. The physical path between the sender and receiver is called:
Sender receiver physical path _____ :

(A) Link / link (link) (B) Node / node (node)
(C) Segment / ермент (segment) (D) Socket / (socket)

[Source: Repo: BCI_SCI Subject Test-7 (UNIT TEST-2 BCI/SCI), Q20] · Easy

(A) Link

Nukta: Note: In Forouzan's 5 components, this same physical path is called the transmission medium. Both terms mean the physical path.

Note: Forouzan 5 components Ȼ physical path transmission medium Ȼ physical path

Q2. Identify the five components of a data communication system.

Data communication system components ɬ ɬG

(A) Message, Sender, Receiver, Transmission medium, Protocol (B) Message, Router, Switch, Cable, Software

(C) Data, Sender, Receiver, Hub, TCP/IP (D) Message, Sender, Receiver, Software, Hardware

[Source: Forouzan Ch1 (Q1-1) — Practice Set] · Medium

(A) Message, Sender, Receiver, Transmission medium, Protocol

Nukta: 5 components: Message → Sender → Medium → Receiver + Protocol (rules). favourite exam question.

5 components: Message → Sender → Medium → Receiver + Protocol (rules) Exam favourite question

Q3. The _____ is the physical path over which a message travels from sender to receiver.

_____ physical path ɢ message sender receiver

(A) Path / (B) Medium

(C) Protocol (D) Route / ę (path)

[Source: Repo: CN Topic Test-01 (UNIT TEST-1), Q35] · Medium

(B) Medium

Nukta: Medium = physical path between sender and receiver. Remember: Message -> Sender -> Medium -> Receiver.

Medium = sender receiver physical path. ØMessage -> Sender -> Medium -> Receiver.

Q4. The information to be communicated in a data communication system is called the _____.

Data communication system ØCommunicate ɬ information _____

(A) Medium (B) Protocol

(C) Message / message (message) (D) Transmission / (transmission)

[Source: Repo: CN Topic Test-01 (UNIT TEST-1), Q38] · Medium

(C) Message

Nukta: Message = the data/information to be communicated. 5 popular forms: text, numbers, images, audio, video.

Message = communicate ɬ data. 5 popular forms: text, numbers, images, audio, video.

Q5. Which of the following shares the data of two computers?

ɜ ĸ ɠ ɣ Ø computers data share ( ) ?

(A) Library / ľ (B) Network

(C) Grouping / (grouping) (D) Integrated system / ɬ ų

[Source: Repo: CN Topic Test-01 (UNIT TEST-1), Q13] · Medium

(B) Network

Nukta: Network = 2 or more devices connected to share data/resources.

Network = 2 ŏ devices Ø , data/ resources share ɠ

Q6. _____ refers to two characteristics: when data should be sent and how fast it can be sent.

_____ characteristics : data 3

(A) Semantics / (semantics) (B) Syntax / (syntax)

(C) Timing / (timing) (D) none of the above /

[Source: Repo: CN Topic Test-01 (UNIT TEST-1), Q44] · Medium

(C) Timing

Nukta: Protocol 3 elements = Syntax (format), Semantics (meaning), Timing (when & how fast).

Protocol 3 elements = Syntax (format), Semantics ( ) Timing (3).

Q7. What are the main elements of a protocol? 1. Syntax 2. Semantics 3. Timing

3 protocol (elements) 1. Syntax 2. Semantics 3. Timing

(A) 1, 2 only / 1 2 (B) 1, 3 only / 1 3

(C) 2, 3 only / 2 3 (D) 1, 2, 3 only / 1, 2 3

[Source: Repo: CN Topic Test-02 (UNIT TEST-2), Q3] · Medium

(D) 1, 2, 3 only

Nukta: Protocol = Syntax + Semantics + Timing.

Protocol = Syntax + Semantics + Timing.

Q8. Why are protocols needed in data communication?

Data communication protocols ?

(A) To enable communication between di erent types of systems by providing common rules / systems communication common rules (B) To increase the processing speed of the CPU / CPU processing speed

(C) To convert every message into a common language / message common language (D) Protocols are optional — devices can communicate without rules / Protocols optional — 3 rules devices communicate

[Source: Forouzan Ch1 (Q1-11) — Practice Set] · Medium

(A) To enable communication between di erent types of systems by providing common rules

Nukta: Protocol = agreement of rules (syntax + semantics + timing). Iske communication impossible.

Protocol = rules agreement (syntax + semantics + timing) 3 communication impossible

Q9. A set of rules that governs data communication is called a:

Data communication regulate (ȝ ȝŮ) rules set _____ :

(A) rule / ȝ (rule) (B) medium / medium (medium)

(C) Link / link (link) (D) Protocol / protocol (protocol)

[Source: Repo: CN Topic Test-03 (UNIT TEST-3), Q15] · Medium

(D) Protocol

Nukta: Protocol = rules set. ɲ without a protocol, two devices may be connected but cannot communicate.

Protocol = ȝ Ȼ set. Ȼȝ protocol devices to ɀ ȝ communicate ɨ

Q10. Consider: (1) Twisted pair cables (2) Microwaves and satellite signals (3) Repeaters (4) Analog transmissions (5) Fiber optics. Which of the above is considered as signal transmission medium in data communications?

ȝ Ȼ(1) Twisted pair cables (2) Microwaves satellite signals (3) Repeaters (4) Analog transmissions (5) Fiber optics ਸ Ȼ data communications Ȼsignal transmission medium ę Ȼਸ ?

(A) (1) and (5) / (1) (5) (B) (1) and (2) / (1) (2)

(C) (1), (2) and (5) / (1), (2) (5) (D) (1), (2), (3) and (5) / (1), (2), (3) (5)

[Source: Repo: CN Topic Test-01 (UNIT TEST-1), Q10] · Medium

(C) (1), (2) and (5)

Nukta: Guided media: twisted pair, coaxial, fiber. Unguided media: radio, microwave, infrared.

Guided media: twisted pair, coaxial, fiber. Unguided media: radio, microwave, infrared.

Q11. Data communication is the exchange of _____ between two devices via some form of transmission medium.

Data communication = do devices _____ exchange, ȝ transmission medium through

(A) only text messages / ǵ ɲtext messages (B) data (the message) - text, numbers, images, audio or video / data (message) - text, numbers, images, audio video

(C) only electrical signals / ǵ ɲelectrical signals (D) only hardware devices / ǵ ɲhardware devices

[Source: Extra (Forouzan) - fills syllabus gap] · Medium

(B) data (the message) - text, numbers, images, audio or video

Nukta: Data communication = exchange of DATA (message) between two devices over a medium.

Data communication = do devices DATA (message) exchange, medium through

Q12. The device that sends the data message in a data communication system is called the _____.

Data communication system ৎ device data message , use _____ ২

(A) Receiver / Receiver (ৃ ী) (B) Sender / Sender (ৃ)

(C) Protocol / Protocol (protocol) (D) Hub / Hub (hub)

[Source: Extra (Forouzan) - fills syllabus gap] · Medium

(B) Sender

Nukta: Sender = source device; Receiver = destination device.

Sender = source device; Receiver = destination device.

Q13. Which of the following best lists the popular forms of information (message) in data communication?

Data communication ৎnformation (message) popular forms achhe ?

(A) Only text and numbers / ু ুtext numbers (B) Only audio and video / ু ুaudio video

(C) Text, numbers, images, audio and video / text, numbers, images, audio video (D) Only binary digits / ু ুbinary digits

[Source: Extra (Forouzan) - fills syllabus gap] · Medium

(C) Text, numbers, images, audio and video

Nukta: 5 popular forms of message: text, numbers, images, audio, video.

Message 5 popular forms: text, numbers, images, audio, video.

Q14. ARPANET stands for:

ARPANET :

(A) Advanced Research Projects Agency Network (B) Advanced Recheck Projects Agency Internet

(C) American Research Projects Agency Network (D) Advanced Research Protocol Agency Network

[Source: IBPS/SBI computer-awareness banks (via tpointtech & meritnotes)] · Medium

(A) Advanced Research Projects Agency Network

Nukta: This is the network on which TCP/IP was later installed — the grandmother of the Internet.

network ে TCP/IP ৎ — Internet ি .

Q15. In a computer network, nodes are:

Computer network ৎodes ২

(A) the computer that originates the data / computer data originate (shurua) (B) the computer that routes the data / computer data route

(C) the computer that terminates the data / computer data terminate (ী) (D) all of the mentioned / ূ (all of the above)

[Source: Repo: CN Topic Test-01 (UNIT TEST-1), Q30] · Hard

(D) all of the mentioned

Nukta: Node = any connected device of the network (computer, printer, router, etc.).

Node = network ু connected device (computer, printer, router, ৎ).

Q16. Consider the statements: S1: The physical medium of connection is known as a link. S2: The computers connected to it are known as nodes.

Ø 3 ØS1: connection physical medium link S2: computers nodes z

(A) Both S1 and S2 are false / Ø1 S2 z
(B) S1 true but S2 false / S1 3 S2
(C) S1 false but S2 true / S1 3 S2
(D) Both S1 and S2 true / Ø1 S2 z

[Source: Repo: CN Topic Test-02 (UNIT TEST-2), Q2] · Hard

(D) Both S1 and S2 true

Nukta: Link = physical medium; Node = connected device (computer).

Link = physical medium; Node = õ device (computer).

Q17. Which one of the following is NOT one of the five basic components of a data communication system?

3 K g y Ø data communication system five basic components Ø ? ?

(A) Message / Message (message) (B) Protocol / Protocol (protocol)
(C) Router / Router (router) (D) Transmission medium / Transmission medium (medium)

[Source: Extra (Forouzan) - fills syllabus gap] · Hard

(C) Router

Nukta: ₦5 components: Message, Sender, Receiver, Transmission Medium, Protocol. Router = device, component y

Ø5 components: Message, Sender, Receiver, Transmission Medium, Protocol. Router = device , component ?

Ώ Samjh — Section B - Characteristics (Delivery, Accuracy, Timeliness, Jitter) & Criteria (Performance, Reliability, Security)

Syllabus: Characteristics - Delivery, Accuracy, Timeliness, Jitter | Properties/Criteria - Performance, Reliability, Security

Ek achhi communication service ki 4 characteristics hoti hain — Delivery, Accuracy, Timeliness, Jitter — aur service ko 3 criteria par taula jata hai: Performance, Reliability, Security.

Ğ r communication service ɬ 4 characteristics ʐ— Delivery, Accuracy, Timeliness, Jitter — service 3 criteria : Performance, Reliability, Security

- Delivery — message sahi destination par pahunche (galat jagah nahi).
  Delivery — message destination õ (ʁ)
- Accuracy — data bina errors ke pahunche; corrupted data deliver nahi hona chahiye.
  Accuracy — data ȝ Ůɬ Ȼ õ ; data deliver ʁ ɛ
- Timeliness — message time par pahunche, late nahi (real-time: audio/video).
  Timeliness — message õ (real-time: audio/video ʛ ę)
- Jitter — packets ke beech delay ka variation; audio/video mein uneven playback ki wajah.
  Jitter — packets delay Ȼ - ; audio/video Ȼ playback
- Performance ko measure karte hain Throughput (asli data rate) aur Delay/Latency se; dono opposite hain — jyada load matlab kam throughput, jyada delay.
  Performance Throughput (ľ ȝ data rate) Delay/Latency ʐ Ȼ ʐ
- Reliability = failure frequency + recovery speed (MTBF — mean time between failures).
  Reliability = ȝ ɬ + ɻ ɬ ȝ (MTBF)
- Security = data ko unauthorized access/modification/destruction se bachana (encryption, password, firewall).
  Security = data unauthorized access/modification/destruction (encryption, password, firewall)

Characteristic	Matlab (meaning)
Delivery	sahi jagah pahunchna
Accuracy	bina error ke
Timeliness	time par
Jitter	delay ka variation (uneven delay)

Q18. Which physical layer impairment mainly causes jitter in VoIP tra c?

VoIP tra c Ø itter ħ kaaran physical layer impairment ?

(A) Attenuation / attenuation (signal ɬ shakti) (B) Delay variation / delay (variation)

(C) Distortion / distortion (vikriti) (D) Crosstalk

[Source: Repo: BCI Subject Test-8 (UNIT TEST-3 BCI/SCI), Q36] · Easy

(B) Delay variation

Nukta: Jitter = delay variation. Attenuation = signal weakening; distortion = signal shape change; crosstalk = interference from nearby cables.

Jitter = delay variation. Attenuation = signal ɬ shakti ; distortion = signal ę badalna; crosstalk = ɬ cables interference.

Q19. What does jitter refer to in data communication?

Data communication Ø itter Ħ ?

(A) Loss of data / data loss (B) Delay variation in packets / packets Ø delay (variation)

(C) Data synchronization / data synchronization (D) Data encoding / data encoding

[Source: Repo: BCI_SCI Subject Test-01 (UNIT TEST-1 CN & Security), Q1] · Medium

(B) Delay variation in packets

Nukta: Jitter = uneven delay in packet arrival -> distorted audio/video playback.

Jitter = packet õ Ø neven delay -> audio/video playback

Q20. Which of the following is a characteristic of data communication?

ȝ ĸ ğ ɣ Ø data communication ɬ characteristic (ȝ) ?

(A) Timeliness / (timeliness) (B) Delay / деړ (delay)

(C) Speed / ȝ (speed) (D) Size / (size)

[Source: Repo: BCI_SCI Subject Test-01 (UNIT TEST-1 CN & Security), Q6] · Medium

(A) Timeliness

Nukta: 4 characteristics : Delivery, Accuracy, Timeliness, Jitter.

4 characteristics : Delivery, Accuracy, Timeliness, Jitter.

Q21. What is the maximum data rate of a transmission medium known as?

Transmission medium ɬ ɟ data rate Ħ ?

(A) Bandwidth (B) Throughput

(C) Latency (D) Jitter

[Source: Repo: BCI_SCI Subject Test-01 (UNIT TEST-1 CN & Security), Q9] · Medium

(A) Bandwidth

Nukta: Bandwidth = maximum capacity of the channel (in bps). Throughput = actual rate achieved (usually less).

Bandwidth = channel ɬ ɟ capacity (bps Ø Throughput = ľ ȝ ę ğ rate ( ).

Q22. Speed of an internet connection is measured in:

Internet connection ‌ speed ‌ ?

(A) Kilobits per second (kbps) (B) Kilobytes per second (KB/s) only / Kilobytes per second (KB/s)

(C) Metres per second (D) Hertz

[Source: Repo: Unit Test-4 Solution (OCR), p19] · Medium

(A) Kilobits per second (kbps)

Nukta: Speed = bits per second (bps family). KB/s (bytes) is for file-size/download display — exam answer is kbps.

Speed = bits per second (bps family) KB/s (bytes) file-size/download display — exam answer kbps

Q23. Frequency of failure and network recovery time after a failure are measures of the _____ of a network.

Failure ‌ (frequency) failure network recovery time measures

(A) Performance (B) Reliability

(C) Security (D) Feasibility

[Source: Repo: CN Topic Test-01 (UNIT TEST-1), Q39] · Medium

(B) Reliability

Nukta: Reliability = failure frequency + recovery time + robustness. Performance = transit time, response time, throughput, delay.

Reliability = failure frequency + recovery time + robustness. Performance = transit time, response time, throughput, delay.

Q24. What is the criteria to check the reliability of a network? 1. Downtime 2. Failure Frequency 3. Fault Tolerance 4. Security

Network ‌ reliability check criteria

1. Downtime 2. Failure Frequency 3. Fault Tolerance 4. Security

(A) 1, 2, 3 only / 1, 2, 3 (B) 2, 3, 4 only / 2, 3, 4

(C) 1, 2, 4 only / 1, 2, 4 (D) 1, 2, 3, 4 only / 1, 2, 3, 4

[Source: Repo: CN Topic Test-02 (UNIT TEST-2), Q1] · Medium

(A) 1, 2, 3 only

Nukta: 3 network criteria: Performance, Reliability, Security. Reliability = failure frequency + recovery time + fault tolerance.

3 network criteria: Performance, Reliability, Security. Reliability = failure frequency + recovery time + fault tolerance.

Q25. What are the three criteria necessary for an e ective and e cient network?

E ective e cient network criteria

(A) Performance, Reliability, Security (B) Delivery, Accuracy, Timeliness

(C) Speed, Cost, Size (D) Message, Sender, Receiver

[Source: Forouzan Ch1 (Q1-2) — Practice Set] · Medium

(A) Performance, Reliability, Security

Nukta: Network 3 CRITERIA = Performance, Reliability, Security. Data communication 4 CHARACTERISTICS = Delivery, Accuracy, Timeliness, Jitter. N confuse !

Network 3 criteria = Performance, Reliability, Security Data communication 4 characteristics = Delivery, Accuracy, Timeliness, Jitter

Q26. What network property is achieved through the use of the Automatic Repeat reQuest (ARQ) mechanism?

Automatic Repeat reQuest (ARQ) mechanism network ł property ģ ?

(A) Reliability (B) Availability

(C) Confidentiality (D) Scalability

[Source: Repo: Random Test- Rajasthan PYQ (RPSC/RSSB)] · Medium

(A) Reliability

Nukta: ARQ = error control -> Reliability. Flow control = e cient delivery (no overflow).

ARQ = error control -> Reliability. Flow control = e cient delivery (receiver bu er overflow ɓ).

Q27. For e ective data communication, the system must deliver data to the correct destination. This characteristic is called _____.

E ective data communication ģ system data destination ð̄ ε Is characteristic _____ ζ

(A) Accuracy (B) Timeliness

(C) Delivery (D) Jitter

[Source: Extra (Forouzan) - fills syllabus gap] · Medium

(C) Delivery

Nukta: 4 characteristics: Delivery (right destination), Accuracy (error-free), Timeliness (on time), Jitter (even delay).

4 characteristics: Delivery (jagah), Accuracy (error-free), Timeliness (time), Jitter (even delay).

Q28. Data delivered late is useless (especially in real-time audio/video). Delivering data in a timely manner is called _____.

ð̄ data (real-time audio/ video ₡ Time data pahunchane _____ ζ

(A) Delivery (B) Accuracy

(C) Timeliness (D) Jitter

[Source: Extra (Forouzan) - fills syllabus gap] · Medium

(C) Timeliness

Nukta: Timeliness = on-time delivery; late data is useless -> real-time transmission (audio/video).

Timeliness = time ð̄ ; data -> real-time transmission (audio/video).

Q29. The network criterion measured by transit time, response time, throughput and delay is called _____.

Transit time, response time, throughput delay maape jane network criterion _____ ζ

(A) Performance (B) Reliability

(C) Security (D) Delivery

[Source: Extra (Forouzan) - fills syllabus gap] · Medium

(A) Performance

Nukta: Performance = transit time + response time (throughput & delay are related measures).

Performance = transit time + response time (throughput delay isse measures ζ.

Q30. Accuracy in data communication means:

Data communication Ø accuracy ŋ:

(A) Data is sent quickly / data

(B) Data is not corrupted / data corrupted ( )

ß

(C) Data is encrypted / data encrypt (D) Data reaches the receiver / data receiver pahunchta

[Source: Repo: BCI_SCI Subject Test-01 (UNIT TEST-1 CN & Security), Q8] · Hard

(B) Data is not corrupted

Nukta: Accuracy = error-free data; altered/ corrupted data is useless.

Accuracy = error-free data; /corrupt data

Q31. An unauthorized user is a network _____ issue.

An unauthorized user network _____ issue .

(A) Performance (B) Reliability

(C) Security (D) All of the above / ¢

[Source: Repo: CN Topic Test-01 (UNIT TEST-1), Q40] · Hard

(C) Security

Nukta: Security = unauthorized access + confidentiality + integrity (CIA triad).

Security = unauthorized access + confidentiality + integrity (CIA triad).

Q32. Which of the following is NOT one of the three fundamental criteria (properties) that a network must meet?

3 K ĝ ¥ Ø network fundamental criteria (properties) Ø ß ?

(A) Performance (B) Reliability

(C) Security (D) Topology

[Source: Extra (Forouzan) - fills syllabus gap] · Hard

(D) Topology

Nukta: Network criteria = 3: Performance, Reliability, Security.

Network criteria = 3: Performance, Reliability, Security.

Q33. The capacity of a communication channel is measured in:

Communication channel ł w :

(A) Bandwidth (B) Bit capacity

(C) Volume (D) Noise

[Source: KVS PGT/TGT/PRT computer banks 2018 (via jagranjosh)] · Hard

(A) Bandwidth

Nukta: Channel capacity ↔ bandwidth (Hz bps) — standard .

Channel capacity ↔ bandwidth (Hz bps) — standard ŋ.

Q34. Consider a 100 Mbps link between an earth station (sender) and a satellite (receiver) at an altitude of 2100 km. The signal propagates at 3×10^8 m/s. The time taken (in milliseconds, rounded to two decimal places) for the receiver to completely receive a 1000-byte packet transmitted by the sender is _____.

100 Mbps link earth station 2100 km satellite 1000-byte packet ; signal speed 3×10^8 m/s Receiver packet (ms) ?

(A) 7.08 (B) 2.10

(C) 7.00 (D) 21.08

[Source: GATE CS 2022] · Hard

(A) 7.08

Nukta: Round-o trap: 2.10 sirf propagation ka fraction pakadta hai; 7.00 transmission time bhool jata hai.

Round-o jaal: 2.10 sirf propagation; 7.00 mein transmission time chhoot jata hai.

Q35. What is the bit rate for transmitting an uncompressed 800×600 pixel colour frame with 8 bits/pixel at 40 frames per second?

compression 800×600 pixel colour frame (8 bits/pixel) 40 frames per second transmit — bit rate ?

(A) 2.4 Mbps (B) 153.6 Mbps

(C) 15.36 Mbps (D) 1536 Mbps

[Source: KVS PGT Computer Science 2017] · Hard

(B) 153.6 Mbps

Nukta: Decimal shift jaal: 15.36 aur 1536 bas decimal ki galti hain; 2.4 Mbps sirf pixel count se bina bits aur frames ke nikla hai.

Decimal ki galtiyan: 15.36/1536; 2.4 Mbps bits aur frames bhool kar aata hai.

Q36. An analog signal carries 4 bits in each signal unit. If 1000 signal units are sent per second, then the baud rate and bit rate of the signal are _____ and _____.

analog signal signal unit 4 bits 1000 signal units per second signal baud rate bit rate ?

(A) 4000 bauds/sec, 4000 bps (B) 1000 bauds/sec, 1000 bps

(C) 4000 bauds/sec, 1000 bps (D) 1000 bauds/sec, 4000 bps

[Source: KVS PGT Computer Science 2017] · Hard

(D) 1000 bauds/sec, 4000 bps

Nukta: Dono ko same mat samjho: 4000 bauds/4000 bps confusion hai; 4000 bauds/1000 bps ulta ratio hai.

Dono alag hain: 4000/4000 confusion; 4000/1000 ulta hai.

Ώ Samjh — Section C - Transmission Mode (Direction of Data Flow: Simplex, Half-Duplex, Full-Duplex)

Syllabus: Transmission Mode - Simplex, Half-Duplex, Full-Duplex

Transmission mode batata hai ki data kis direction mein flow karta hai — simplex, half-duplex ya full-duplex.

Transmission mode ३ data ३ ६ Ȼ — simplex, half-duplex full-duplex

- Simplex — sirf ek direction; ek device sirf bhejta hai, doosra sirf paata hai (keyboard → monitor; TV/Radio broadcast).
  Simplex — ६ ; ङ ŋ , Ȧ ङ ŋ (keyboard → monitor; TV/Radio)
- Half-Duplex — dono direction, par ek time par sirf ek taraf (walkie-talkie; one LAN, shared medium).
  Half-Duplex — Ȼ , Ȼ (walkie-talkie)
- Full-Duplex — dono taraf ek saath; do do channels chahiye (telephone; modern LAN switches ke ports).
  Full-Duplex — Ȼ ; channels ६ (telephone; switch ports)
- Half-duplex mein capacity waste hoti hai (ek taraf wait karna padta hai); full-duplex mein nahi.
  Half-duplex Ȼapacity (ų w); full-duplex Ȼ ४
- Yaad rakho: Simplex = one way; Half = both ways, one at a time; Full = both ways, together.
  : Simplex = ; Half = Ȼ - ; Full = Ȼ

Mode	Direction	Example
Simplex	(unidirectional)	Keyboard, monitor, radio
Half-Duplex	N ″- ″	Walkie-talkie
Full-Duplex	N	Telephone, switch port

Q37. Communication between a computer and a keyboard involves _____ transmission.

Computer keyboard communication Ȼ _____ transmission

(A) Automatic (B) Half-duplex
(C) Half-duplex (variant) / half-duplex (D) Simplex

[Source: Repo: CN Topic Test-01 (UNIT TEST-1), Q33] · Medium

(D) Simplex

Nukta: Keyboard -> Computer = simplex (one way). Examples: keyboard-monitor, TV broadcast.

Keyboard -> Computer = simplex ( ). Examples: keyboard-monitor, TV broadcast.

Q38. A television broadcast is an example of _____ transmission.

Television broadcast _____ transmission example

(A) simplex (B) half-duplex

(C) full-duplex (D) automatic

[Source: Repo: CN Topic Test-01 (UNIT TEST-1), Q41] · Medium

(A) simplex

Nukta: TV/radio broadcast = simplex (station -> TV only, wapas).

TV/radio broadcast = simplex (station -> TV , wapas).

Q39. What is the mode of transmission where data flows in only one direction?

Wo transmission mode kaunsa data ek direction flow ?

(A) Simplex (B) Half-Duplex

(C) Full-Duplex (D) Multiplex

[Source: Repo: BCI_SCI Subject Test-01 (UNIT TEST-1 CN & Security), Q3] · Medium

(A) Simplex

Nukta: Simplex = one-way only (keyboard to monitor, TV broadcast).

Simplex = ek hi (keyboard to monitor, TV broadcast).

Q40. Which transmission mode is used in walkie-talkie communication?

Walkie-talkie communication me transmission mode use ?

(A) Simplex (B) Half-Duplex

(C) Full-Duplex (D) Multiplex

[Source: Repo: BCI_SCI Subject Test-01 (UNIT TEST-1 CN & Security), Q10] · Medium

(B) Half-Duplex

Nukta: Walkie-talkie = half-duplex (push-to-talk, ek hi bolega).

Walkie-talkie = half-duplex (push-to-talk, me bolega).

Q41. Which of the following statements are TRUE? 1. Hubs operate in Half-Duplex mode. 2. Switches operate in Full-Duplex mode. 3. Routers broadcast by default.

Is me kaunse statements TRUE hain? 1. Hubs Half-Duplex mode me kaam karte hain 2. Switches Full-Duplex mode me kaam karte hain 3. Routers by default broadcast karte hain

(A) 1 and 2 / 1 aur 2 (B) 2 and 3 / 2 aur 3

(C) 1 and 3 / 1 aur 3 (D) 1, 2 and 3 / 1, 2 aur 3

[Source: Repo: BCI Subject Test-9 (BCI/SCI Batch)] · Medium

(A) 1 and 2

Nukta: Hub = half-duplex + collision domain; Switch = full-duplex + per-port collision domain; Router = broadcasts rokta .

Hub = Half-Duplex + collision domain; Switch = Full-Duplex + per-port collision domain; Router = broadcasts

Q42. Which of the following transmission directions listed is not a legitimate channel?

transmission direction legitimate (vaidh) channel ?

(A) Simplex (B) Half Duplex

(C) Full Duplex (D) Double Duplex

[Source: Repo: CN Topic Test-01 (UNIT TEST-1), Q3] · Hard

(D) Double Duplex

Nukta: 3 transmission modes: Simplex, Half-Duplex, Full-Duplex. Double duplex = .

3 transmission modes: Simplex, Half-Duplex, Full-Duplex. Double duplex =

Q43. What is the di erence between half-duplex and full-duplex transmission modes?

Half-Duplex Full-Duplex transmission modes ?

(A) Half-duplex allows only one-way flow; full-duplex also allows only one-way flow / Half-Duplex one-way flow; Full-Duplex one-way flow (B) Half-duplex flows both ways one direction at a time; full-duplex flows both ways at the same time / Half-Duplex data direction; Full-Duplex

(C) Half-duplex is only for wireless; full-duplex only for wired networks / Half-Duplex wireless ; Full-Duplex wired (D) There is no di erence between the two /

[Source: Forouzan Ch1 (Q1-6) — Practice Set] · Hard

(B) Half-duplex flows both ways one direction at a time; full-duplex flows both ways at the same time

Nukta: Half = time side (walkie-talkie). Full = side (telephone).

Half = side (walkie-talkie) Full = side (telephone)

Q44. Data flow between two devices can occur in a _____ way.

Do devices data flow _____ tareeke

(A) simplex (B) half-duplex

(C) full-duplex (D) all of the above /

[Source: Repo: CN Topic Test-01 (UNIT TEST-1), Q45] · Hard

(D) all of the above

Nukta: 3 ways of data flow: simplex, half-duplex, full-duplex.

Data flow 3 tareeke: simplex, half-duplex, full-duplex.

Q45. In a telephone conversation, both persons can speak and hear each other at the same time. This is an example of _____.

Telephone conversation persons bol sun _____ example

(A) Simplex (B) Half-duplex

(C) Full-duplex (D) Unicast

[Source: Extra (Forouzan) - fills syllabus gap] · Hard

(C) Full-duplex

Nukta: Telephone = full-duplex (bol sun).

Telephone = full-duplex (bol sun).

Ώ Samjh — Section D - Network Structure / Topology (Physical: Mesh, Star, Bus, Ring, Others; Logical: IEEE 802)

Syllabus: Physical Topology - Mesh, Star, Bus, Ring, Others (Hybrid, Tree) | Logical Topology - IEEE 802

Network topology batata hai ki devices kis shape mein judi hain. Physical topology = asli wiring; Logical topology = data ka path/frame ka flow (IEEE 802 standards).

Network topology ³ ³ Ȼ ʒPhysical topology = ľ ³ wiring; Logical topology = data ɳ(IEEE 802 standards)

- Mesh topology — har device ka har doosre device se dedicated link: n(n−1)/2 links, n(n−1)/2 ports per device. Robust + private, par costly aur wiring jyada.

  Mesh topology — À dedicated link: n(n−1)/2 links +³ , wiring ɟ

- Star topology — sab devices central hub/switch se jude: easy add/remove, ek link fail hone par sirf woh device; par central device fail = poora network down.

  Star topology — central hub/switch : - ; central device = network

- Bus topology — ek hi backbone cable; saste aur aasan; par backbone fail = poora network down, aur tra c jyada hone par performance girti hai.

  Bus topology — backbone cable; ľ ; backbone = network , Ȼerformance

- Ring topology — devices ek circle mein; data ek direction mein token ke saath chalta hai; equal access milta hai.

  Ring topology — ģ Ȼdata ɛ Ȼoken ;

- Hybrid/Tree — in sab ka combination (star ka star).

  Hybrid/Tree — ³ Ż (star star)

- Logical topology standards: IEEE 802 (Ethernet = bus/logical bus; Token Ring = logical ring) — data ka flow logical topology se tay hota hai.

  Logical topology: IEEE 802 (Ethernet = logical bus; Token Ring = logical ring) — data ų

Topology	Links / Feature	Kamzori (weakness)
Mesh	n(n−1)/2 dedicated links — robust	, ũ wiring
Star	central hub/switch — easy expand	central device = single point of failure
Bus	single backbone — sasta	backbone fail = all down
Ring	token, equal access	ek node fail = ring toot sakti hai

Q46. IEEE 802.2 st d is related to which sub-layer?

IEEE 802.2 st d ³ sub-layer ɟ ?

(A) MAC Sublayer (B) LLC Sublayer

(C) Physical Layer (D) Network Layer

[Source: Repo: BCI Subject Test-8 (BCI/SCI Batch)] · Easy

(B) LLC Sublayer

Nukta: 802.2 = LLC (upper sub-layer of Data Link). 802.3 = Ethernet/MAC, 802.5 = Token Ring, 802.11 = Wi-Fi.

802.2 = LLC 802.3 = Ethernet/MAC, 802.5 = Token Ring, 802.11 = Wi-Fi

Q47. What are the advantages of star topology?

Star topology ₶ advantages ( ) ₹

(A) Less number of cables are required / number ₡ables ε (B) More number of cables are required / ŏ number ₡ables ε

(C) It is a distributed topology / distributed topology (D) Security is high / security ŏ

[Source: Repo: CN Topic Test-01 (UNIT TEST-1), Q1] · Medium

(A) Less number of cables are required

Nukta: Star = hub at centre, dedicated links, fewer cables (vs mesh), easy to install. But if hub fails, whole network goes down.

Star = centre ₡ub, dedicated links, cables (mesh muqable), install . ₃ hub fail toh network band

Q48. Which type of topology is best suited for large businesses which must carefully control and coordinate the operation of distributed branch outlets?

Large businesses ₲ topology ₶ , jahan distributed branch outlets operation carefully control coordinate ?

(A) Ring (B) Local area

(C) Hierarchical / hierarchical (tree) (D) Star

[Source: Repo: CN Topic Test-01 (UNIT TEST-1), Q2] · Medium

(C) Hierarchical

Nukta: Tree = hierarchical star. Large orgs with branches -> hierarchical/tree topology.

Tree = hierarchical star. Badi sansthaon ₡ranches ₲ -> hierarchical/tree topology.

Q49. The repeater is used in _____ topology.

Repeater use _____ topology ₡

(A) Mesh Topology (B) Ring Topology

(C) Bus Topology (D) Star Topology

[Source: Repo: CN Topic Test-01 (UNIT TEST-1), Q50] · Medium

(C) Bus Topology

Nukta: Bus = long backbone -> signal weak -> repeater use . (Ring ɴ device repeater ˣ .)

Bus = backbone -> signal kamzor -> repeater use (Ring ₡ device repeater ɬ .)

Q50. What type of topology connects each node in a closed loop?

topology node closed loop ₡ ?

(A) Star (B) Mesh

(C) Ring (D) Bus

[Source: Repo: BCI_SCI Subject Test-01 (UNIT TEST-1 CN & Security), Q4] · Medium

(C) Ring

Nukta: Ring = closed loop, one direction, token passing, each device = repeater.

Ring = closed loop, direction, token passing, device = repeater.

Q51. In a _____ topology, every device has a dedicated point-to-point link to every other device. For n devices, the number of links is n(n-1)/2 and each device needs n-1 I/O ports.

_____ topology Ȼ device À device dedicated point-to-point link n devices ʃ links ɬ ħ n(n-1)/2 device n-1 I/O ports ɛ

(A) Bus (B) Star

(C) Ring (D) Mesh

[Source: Extra (Forouzan) - fills syllabus gap] · Medium

(D) Mesh

Nukta: Mesh formula (EXAM favourite): links = n(n-1)/2, ports per device = n-1. E.g., 5 devices -> 10 links.

Mesh formula (EXAM favourite): links = n(n-1)/2, device ports = n-1. Jaise 5 devices -> 10 links.

Q52. In a _____ topology, each device has a dedicated point-to-point link only to a central controller, usually called a hub (or switch).

_____ topology Ȼ device ʃ ɳ central controller (hub/switch) dedicated point-to-point link

(A) Mesh (B) Ring

(C) Star (D) Bus

[Source: Extra (Forouzan) - fills syllabus gap] · Medium

(C) Star

Nukta: Star = hub/switch at centre; easy to install & reconfigure; but if hub fails, the WHOLE network goes down.

Star = centre Ȼhub/switch; install reconfigure ; ɜ hub fail network

Q53. Which network topology requires a multipoint connection?

ɜ network topology Ȼmultipoint connection ɬ ¦ ?

(A) Ring (B) Bus

(C) Star (D) Mesh

[Source: Repo: CN Topic Test-02 (UNIT TEST-2)] · Medium

(B) Bus

Nukta: Bus = multipoint (shared backbone). Mesh/Star/Ring = point-to-point links. Yahi di erence exam ɴpoochha .

Bus = multipoint (shared backbone) Mesh/Star/Ring = point-to-point links

Q54. In a _____ topology, each device is connected to its two neighbours, data travels in one direction around the loop, and each device regenerates (repeats) the signal - this is called token passing.

_____ topology Ȼ device neighbours , data loop Ȼ direction Ȼ , device signal regenerate (repeat) - token passing ʐ

(A) Bus (B) Ring

(C) Star (D) Mesh

[Source: Extra (Forouzan) - fills syllabus gap] · Medium

(B) Ring

Nukta: Ring = loop, one direction, token passing, each device = repeater.

Ring = loop, direction, token passing, device = repeater.

Q55. A network in which two or more di erent topologies are combined (e.g., a star connected to another star) is called a _____ topology, while a _____ topology is a hierarchy of stars (a variation of the star topology).

topologies combine (star star), use _____ topology _____ topology stars hierarchy (star topology variation).

(A) Hybrid ; Tree (B) Tree ; Hybrid

(C) Mesh ; Ring (D) Bus ; Star

[Source: Extra (Forouzan) - fills syllabus gap] · Medium

(A) Hybrid ; Tree

Nukta: Hybrid = mix of topologies; Tree = star of stars (hierarchical star).

Hybrid = topologies mix; Tree = star of stars (hierarchical star).

Q56. Hierarchical topology, used to establish a network, is also known as a:

Hierarchical topology, network

?

(A) Bus network (B) Star network

(C) Tree network (D) Mesh network

[Source: Repo: Unit Test-4 Solution (OCR), p17] · Medium

(C) Tree network

Nukta: Hierarchical = Tree = star-of-stars (root + branches). Used in big o ce/campus LANs.

Hierarchical = Tree = star-of-stars (root + branches) o ce/campus LANs se

Q57. The _____ topology of a network defines how the devices are physically laid out/ connected, whereas the _____ topology defines how the data actually flow / how the devices access the network to transmit.

Network _____ topology devices physically jabki _____ topology data asal flow / devices transmit network access

(A) Logical ; Physical (B) Physical ; Logical

(C) Star ; Bus (D) Mesh ; Ring

[Source: Extra (Forouzan) - fills syllabus gap] · Medium

(B) Physical ; Logical

Nukta: Physical = (cables); Logical = data flow (access). Same logical, di erent physical possible.

Physical = (cables); Logical = data flow (access). Same logical, di erent physical

Q58. Which physical topology offers the highest robustness (fault tolerance) because of multiple redundant paths, but is also the most expensive and hardest to install because of the huge amount of cabling?

physical topology ŏ robust (fault tolerant) ismein redundant paths ʒ install ¼¦ cabling ?

(A) Bus (B) Ring

(C) Star (D) Mesh

[Source: Extra (Forouzan) - fills syllabus gap] · Medium

(D) Mesh

Nukta: Mesh = most robust + most costly (n(n-1)/2 cables). Star = cheap & easy but hub is a single point of failure. Bus = cheapest, hardest fault isolation.

Mesh = robust + (n(n-1)/2 cables). Star = sasti hub single point of failure . Bus = sasti, fault isolation ¼¦ .

Q59. Which one of the following is not a network topology?

network topology ?

(A) Star (B) Ring

(C) Bus (D) Peer to Peer

[Source: Repo: CN Topic Test-01 (UNIT TEST-1), Q9] · Hard

(D) Peer to Peer

Nukta: Topologies = mesh, star, bus, ring, hybrid, tree. P2P = architecture (not topology).

Topologies = mesh, star, bus, ring, hybrid, tree. P2P = architecture (topology).

Q60. For n devices in a network, how many cable links are required in a mesh topology?

Network devices mesh topology cable links ?

(A) n (B) n - 1

(C) n(n - 1)/2 (D) n(n - 1)

[Source: Forouzan Ch1 (Q1-8) — Practice Set] · Hard

(C) n(n - 1)/2

Nukta: Mesh cables = n(n-1)/2, ports/device = n-1. Star = n cables. Ring = n. Bus = 1 backbone.

Mesh cables = n(n-1)/2, ports/device = n-1 Star = n Ring = n Bus = 1

Q61. Six devices are arranged in a mesh topology. How many cables are needed, and how many ports does each device need?

devices mesh topology cables device ports ?

(A) 15 cables, 5 ports each / 15 cables, device 5 ports (B) 30 cables, 6 ports each / 30 cables, device 6 ports

(C) 15 cables, 6 ports each / 15 cables, device 6 ports (D) 6 cables, 5 ports each / 6 cables, device 5 ports

[Source: Forouzan Ch1 (P1-3) — Practice Set (Problem)] · Hard

(A) 15 cables, 5 ports each

Nukta: Mesh formula: cables = n(n-1)/2, ports/device = n-1. Numerical yahi do numbers maange .

Mesh: cables = n(n-1)/2, ports/device = n-1 Numerical numbers

Q62. If a computer network cable breaks, the whole network goes down in which topology?

ε computer network cable , ३ topology ∅ network ?

(A) Star (B) Ring

(C) Mesh (D) All of these /

[Source: Repo: Unit Test-4 Solution (OCR), p16] · Hard

(B) Ring

Nukta: Ring = one break downs ALL. Star = only that device drops. Mesh = network survives (many paths).

Ring = break down Star = ∫ ղ device ३ Mesh = network (ղpaths)

Q63. In which topology does the failure of a single cable or device NOT a ect the rest of the network?

३ topology ∅ cable device fail ɬ network ų ३ ʁ ?

(A) Bus (B) Ring

(C) Star (D) Mesh

[Source: Forouzan Ch1 (P1-4) — Practice Set (Problem)] · Hard

(D) Mesh

Nukta: Mesh = most fault-tolerant (redundant paths) — ʁ 'highest robustness' mesh ɴ .

Mesh = ŏ fault-tolerant (redundant paths) — ∫ mesh most robust

Q64. In a _____ topology, one long cable acts as a backbone to link all the devices; devices connect via drop lines and taps, and terminators are needed at both ends of the backbone.

_____ topology ∅ cable backbone ɬ devices ; devices drop lines taps connect ʐ backbone ∅nds terminators ε

(A) Bus (B) Star

(C) Ring (D) Mesh

[Source: Extra (Forouzan) - fills syllabus gap] · Hard

(A) Bus

Nukta: Bus = one backbone + drop lines + taps + terminators at both ends. Fault isolation is di cult; a backbone break stops everything.

Bus = backbone + drop lines + taps + ∅nds terminators. Fault isolation ų¦ ; backbone tootne band

Q65. The IEEE 802 st ds (such as 802.3 for Ethernet, 802.5 for Token Ring, 802.11 for Wi-Fi) primarily define the _____ of LANs/MANs.

IEEE 802 st ds (802.3 Ethernet, 802.5 Token Ring, 802.11 Wi-Fi) primarily LANs/MANs ɬ _____ define ʐ

(A) Physical and Data Link layers (including the logical topology / MAC) / Physical Data Link layers (logical topology / MAC sahit) (B) Only the Application layer / ∫ ղApplication layer

(C) Only IP addressing / ∫ ղIP addressing (D) Only WAN routing / ∫ ղWAN routing

[Source: Extra (Forouzan) - fills syllabus gap] · Hard

(A) Physical and Data Link layers (including the logical topology / MAC)

Nukta: IEEE 802 = LAN/MAN st ds (Physical + Data Link). 802.3 Ethernet, 802.5 Token Ring, 802.11 Wi-Fi.

IEEE 802 = LAN/MAN st ds (Physical + Data Link). 802.3 Ethernet, 802.5 Token Ring, 802.11 Wi-Fi.

Q66. Four network segments in Bus topology are: A1–A2, B1–B2–B3, C1–C2 and D1–D2–D3. If A1, B1, C1 and D1 are interconnected in a mesh, how many mesh connections are required?

Bus topology Ȼ network segments ȥA1–A2, B1–B2–B3, C1–C2 D1–D2–D3 ȇ A1, B1, C1 D1 mesh Ȼ , ȝ mesh connections ȇ ?

(A) 12 (B) 10

(C) 6 (D) 45

[Source: RPSC Programmer 2024 Paper-1 (with o cial key)] · Hard

(C) 6

Nukta: Mesh formula n(n-1)/2 — ȼ n ȣ ķ meshed nodes (4), network hosts (10) ȳ!

Mesh formula n(n-1)/2 — n ȸ ŋmeshed nodes (4), network hosts (10) ȶ!

Ώ Samjh — Section E - Types of Networks (LAN, MAN, WAN, PAN, CAN, SAN; Internet, Intranet, Extranet; Point-to-Point, Multipoint; Client-Server, P2P)

Syllabus: Types of Networks - Geographical (LAN, MAN, WAN, PAN, CAN, SAN) | Accessibility (Internet, Intranet, Extranet) | Architecture (Point-to-Point, Multipoint)

Networks geographical span ke hisaab se LAN, MAN, WAN, PAN, CAN, SAN hote hain; access method se point-to-point ya multipoint; architecture se client-server ya peer-to-peer.

Networks ģ ३ ľ LAN, MAN, WAN, PAN, CAN, SAN ʐ õ point-to-point multipoint; client-server peer-to-peer

- LAN — building/campus tak, high speed, private (Ethernet, Wi-Fi).
  LAN — building/campus , ,३ (Ethernet, Wi-Fi)
- MAN — city tak (cable TV network, DQDB); LANs ko connect karta hai.
  MAN — (cable TV, DQDB); LANs
- WAN — country/continent; public carriers (PSTN, satellite, leased lines); Internet sabse bada WAN.
  WAN — / Â ; public carriers (PSTN, satellite); Internet WAN
- PAN — body ke paas (Bluetooth, Zigbee); CAN — campus area; SAN — storage area network (servers/disks).
  PAN — (Bluetooth); CAN — campus; SAN — storage area network
- Point-to-point = dedicated link do nodes ke beech; Multipoint = ek link par multiple nodes (shared).
  Point-to-point = nodes dedicated link; Multipoint = link ŋnodes (shared)
- Client-Server — server resources deta hai, clients maangte hain (centralized control). P2P — har computer barabar, koi dedicated server nahi.
  Client-Server — server resources , clients ʐP2P — computer , ŋdedicated server ४
- Internet = networks ka network; Intranet = organization ka private internet; Extranet = authorized bahari users ke liye intranet ka extended hissa.
  Internet = networks network; Intranet = ľ private internet; Extranet = ɻ users ģ ३ ľ

Type	Span	Example
PAN	ʔ١Í	Bluetooth headset
LAN	building/campus	o ce Ethernet
MAN	city	cable TV network
WAN	country/continent	Internet
SAN	storage devices	data centre storage

Q67. The term WAN stands for?

WAN (full form) ?

(A) Wide Area Net (B) Wide Access Network

(C) Wide Area Network (D) Wide Access Net

[Source: Repo: CN Topic Test-01 (UNIT TEST-1), Q11 / BCI_SCI Subject Test-01, Q11] · Medium

(C) Wide Area Network

Nukta: WAN = Wide Area Network. Internet WAN .

WAN = Wide Area Network. Internet WAN

Q68. Which network is suitable for a building?

building network (suitable) ?

(A) WAN (B) LAN

(C) MAN (D) PAN

[Source: Repo: CN Topic Test-01 (UNIT TEST-1), Q27] · Medium

(B) LAN

Nukta: LAN = Local Area Network - area (o ce/building/campus), private, high speed.

LAN = Local Area Network - area (o ce/building/campus), private, high speed.

Q69. Bluetooth is an example of:

Bluetooth example :

(A) personal area network / personal area network (PAN) (B) local area network / local area network (LAN)

(C) virtual private network / virtual private network (VPN) (D) none of the mentioned /

[Source: Repo: CN Topic Test-01 (UNIT TEST-1), Q31] · Medium

(A) personal area network

Nukta: PAN = Personal Area Network - person around (Bluetooth: phone <-> earphones).

PAN = Personal Area Network - person around (Bluetooth: phone <-> earphones).

Q70. Which one of the following extends a private network across public networks?

private network public networks paar (across) extend ?

(A) local area network / local area network (LAN) (B) virtual private network / virtual private network (VPN)

(C) enterprise private network (D) storage area network / storage area network (SAN)

[Source: Repo: CN Topic Test-01 (UNIT TEST-1), Q32] · Medium

(B) virtual private network

Nukta: VPN = Virtual Private Network - private network public (Internet) extend , securely.

VPN = Virtual Private Network - private network public (Internet) extend , securely.

Q71. _____ is a network that covers geographic areas that are larger, such as districts or cities.

_____ network geographical areas, districts cities, cover

(A) LAN (B) MAN

(C) WAN (D) None of these / Ø ŋ ɓ

[Source: Repo: CN Topic Test-01 (UNIT TEST-1), Q49] · Medium

(B) MAN

Nukta: MAN = Metropolitan Area Network - city level (cable TV network).

MAN = Metropolitan Area Network - city level (cable TV network).

Q72. Which factor primarily decides whether a communication system is a LAN or a WAN?

- factor ħ ę ɜ communication system LAN WAN?

(A) The geographical area it spans / ɜ geographical area Ø (B) The number of protocols it uses / ɜ protocols use

(C) The colour of the cables / Cables (D) The type of keyboard attached / keyboard ų

[Source: Forouzan Ch1 (Q1-9) — Practice Set] · Medium

(A) The geographical area it spans

Nukta: Chhoti area = LAN, city = MAN, state/country/world = WAN. Faisla ɓ ķarea .

area = LAN, city = MAN, state/country/world = WAN ģ ŋarea

Q73. In terms of the size of the network, the correct order (ascending) is:

Network size hisaab kram (badhta õ / ascending order) Ħ ?

(A) PAN, MAN, LAN, WAN (B) LAN, MAN, WAN, PAN

(C) PAN, LAN, MAN, WAN (D) LAN, PAN, MAN, WAN

[Source: Repo: CN Topic Test-02 (UNIT TEST-2), Q42] · Medium

(C) PAN, LAN, MAN, WAN

Nukta: Remember: PAN < LAN < MAN < WAN (smallest to largest).

ØPAN < LAN < MAN < WAN ( ).

Q74. Which type of network covers an entire city or campus?

network city campus cover ?

(A) LAN (B) MAN

(C) WAN (D) PAN

[Source: Repo: BCI_SCI Subject Test-01 (UNIT TEST-1 CN & Security), Q2] · Medium

(B) MAN

Nukta: MAN = city level. Note: a single building/campus LAN is usually LAN/CAN; an entire city = MAN.

MAN = city level. Note: building/campus LAN LAN/CAN ; city = MAN.

Q75. The Internet is an example of:

Internet example :

(A) LAN (B) MAN

(C) WAN (D) PAN

[Source: Repo: BCI_SCI Subject Test-01 (UNIT TEST-1 CN & Security), Q12] · Medium

(C) WAN

Nukta: Internet = WAN (network of networks).

Internet = WAN (network of networks).

Q76. SAN is mainly used for:

SAN ħ 3 g ?

(A) File Sharing (B) Storage Management

(C) Internet Access (D) Printing Documents / documents print

[Source: Repo: BCI_SCI Subject Test-01 (UNIT TEST-1 CN & Security), Q13] · Medium

(B) Storage Management

Nukta: SAN = Storage Area Network - servers <-> storage devices, data centre.

SAN = Storage Area Network - servers <-> storage devices, data centre.

Q77. PAN is used for connecting:

PAN 3 connect g use ?

(A) Personal devices like mobile, laptop, Bluetooth headset / personal devices mobile, laptop, Bluetooth headset (B) Computers in a company / company computers

(C) Computers in a city / city computers

(D) Global o ces

[Source: Repo: BCI_SCI Subject Test-01 (UNIT TEST-1 CN & Security), Q14] · Medium

(A) Personal devices like mobile, laptop, Bluetooth headset

Nukta: PAN = Personal Area Network - person devices (mobile, laptop, headset).

PAN = Personal Area Network - person devices (mobile, laptop, headset).

Q78. A school has four computer labs in the same building. Each lab has 20 computers connected to a central server. Students can share files and print from any lab. Which type of network is implemented in this school?

Ek school Ø building Ø computer labs ʐ lab Ø20 computers central server ʐ Students 3 lab files share print ʐ School Ø network implemented ?

(A) WAN (B) MAN

(C) LAN (D) CAN

[Source: Repo: BCI_SCI Subject Test-01 (UNIT TEST-1 CN & Security), Q15] · Medium

(C) LAN

Nukta: LAN = single building/campus, private, high speed. Example: school lab, o ce.

LAN = building/campus, private, high speed. Example: school lab, o ce.

Q79. A university has di erent departments in separate buildings like Engineering, Arts, and Commerce. Each department has its own LAN. All these LANs are connected together for communication. Which type of network connects all these LANs?

Ek university Ȼ - departments buildings Ȼȥ Engineering, Arts, Commerce department apna LAN LANs communication ǵ Ȼ ȥ network in LANs ?

(A) LAN (B) CAN

(C) MAN (D) WAN

[Source: Repo: BCI_SCI Subject Test-01 (UNIT TEST-1 CN & Security), Q16] · Medium

(B) CAN

Nukta: CAN = Campus Area Network - campus ˣ multiple buildings LANs .

CAN = Campus Area Network - campus ɬ ŋ buildings LANs

Q80. An international bank connects its main o ce in New York with branches in Delhi, London, and Tokyo. Which network type is used here?

Ek international bank main o ce (New York) Delhi, London, Tokyo ɬ branches network type use ?

(A) MAN (B) WAN

(C) CAN (D) LAN

[Source: Repo: BCI_SCI Subject Test-01 (UNIT TEST-1 CN & Security), Q17] · Medium

(B) WAN

Nukta: WAN = long distance (city/country/ continent). Bank branches worldwide = WAN.

WAN = Ȧ (city/ /continent). Worldwide bank branches = WAN.

Q81. A company has several servers that store and manage large volumes of data. All storage devices are interconnected for high-speed data access. Which type of network is this?

Ek company ŋservers ȥ matra Ȼ data store manage ȥ storage devices high-speed data access ǵ interconnected ȥ network ?

(A) LAN (B) SAN

(C) CAN (D) WAN

[Source: Repo: BCI_SCI Subject Test-01 (UNIT TEST-1 CN & Security), Q18] · Medium

(B) SAN

Nukta: SAN = Storage Area Network - storage devices high-speed access dene ȣ .

SAN = Storage Area Network - storage devices high-speed access dene ǵ

Q82. A supplier accessing a company's order tracking system uses:

Ek supplier company order tracking system access , network use ?

(A) Internet (B) Intranet

(C) Extranet (D) VPN

[Source: Repo: BCI_SCI Subject Test-01 (UNIT TEST-1 CN & Security), Q19] · Medium

(C) Extranet

Nukta: Extranet = intranet + authorized outsiders (suppliers/partners/customers).

Extranet = intranet + authorized outsiders (suppliers/partners/customers).

Q83. Which network type provides maximum security and privacy?

network type ɨ (maximum) security privacy provide ?

(A) Internet (B) Intranet

(C) Extranet (D) None

[Source: Repo: BCI_SCI Subject Test-01 (UNIT TEST-1 CN & Security), Q20] · Medium

(B) Intranet

Nukta: Security order: Intranet (secure/private) > Extranet > Internet (public).

Security order: Intranet (secure/private) > Extranet > Internet (public).

Q84. A software company shares project reports and progress updates with its clients through a secure online portal that requires login credentials. Which network type is this?

Ek software company clients project reports progress updates secure online portal through share ɕ Ȼogin credentials ɛ network type ?

(A) Internet (B) Intranet

(C) Extranet (D) MAN

[Source: Repo: BCI_SCI Subject Test-01 (UNIT TEST-1 CN & Security), Q21] · Medium

(C) Extranet

Nukta: Clients secure portal share = Extranet.

Clients secure portal share = Extranet.

Q85. Three or more devices share a link in _____ connection.

Teen ɨ devices link _____ connection Ȼ share ʐ

(A) Unipoint (B) Multipoint

(C) Point to point (D) None of the mentioned / Ȼ ɳ ʁ

[Source: Repo: CN Topic Test-01 (UNIT TEST-1), Q36] · Medium

(B) Multipoint

Nukta: Multipoint = link, devices; capacity shared. Point-to-point = dedicated link, ʁ ķ2 devices.

Multipoint = link, ɳdevices; capacity shared. Point-to-point = dedicated link, ɠ ɳ2 devices.

Q86. What are the two types of line configuration?

Line configuration ų Ħ ƺ

(A) Simplex and Half-Duplex / Simplex Half-Duplex (B) Serial and Parallel / Serial Parallel

(C) Point-to-Point and Multipoint / Point-to-Point Multipoint (D) Analog and Digital / Analog Digital

[Source: Forouzan Ch1 (Q1-4) — Practice Set] · Medium

(C) Point-to-Point and Multipoint

Nukta: Line configuration ฿ ķ2 type: Point-to-Point & Multipoint. Simplex/Half/Full-Duplex TRANSMISSION MODES ɾ

Line configuration 2 type: Point-to-Point Multipoint Simplex/Half/Full-Duplex transmission modes ƺ

Q87. What is the advantage of a multipoint connection over a point-to-point connection?

Point-to-point connection ɬ Ø multipoint connection Ħ ?

(A) Fewer cables are needed, so the cost is lower / cables ƺ ɠ cost (B) It provides a dedicated link for every pair of devices / device- Ĩ dedicated link ɜ

(C) It is more secure than point-to-point / point-to-point ŏ secure (D) It needs more cables and ports / Ø cables ports ƺ

[Source: Forouzan Ch1 (Q1-3) — Practice Set] · Medium

(A) Fewer cables are needed, so the cost is lower

Nukta: Multipoint = cable = kharch. Point-to-point = dedicated link (ũ cable, ũ privacy).

Multipoint = cable = ŋPoint-to-point = dedicated link (ŏ cable, ŏ privacy)

Q88. A _____ connection provides a dedicated link between two devices.

Ek _____ connection do devices dedicated link provide

(A) point-to-point (B) multipoint

(C) primary (D) secondary

[Source: Repo: CN Topic Test-01 (UNIT TEST-1), Q42] · Medium

(A) point-to-point

Nukta: Point-to-point = dedicated link, ˝ capacity ฿ ķ2 devices ฿ . Example: TV remote (IR), leased line.

Point-to-point = dedicated link, capacity ɠ ŋ2 devices ɠ . Example: TV remote (IR), leased line.

Q89. When a person makes a local telephone call to another person, what type of connection is it?

ŋ ĥĦ À ê ĥĦ local telephone call , ɜ ų connection ?

(A) Multipoint connection (B) Point-to-point connection

(C) Broadcast connection (D) Multicast connection

[Source: Forouzan Ch1 (P1-7) — Practice Set (Problem)] · Medium

(B) Point-to-point connection

Nukta: Telephone call = 2 parties, dedicated link = point-to-point. Bus/Ethernet = multipoint.

Telephone call = 2 parties, dedicated link = point-to-point Bus/Ethernet = multipoint

Q90. Which of the following is an example of Client-Server architecture?

३ к ģ ɣ Ȼ Client-Server architecture example ?

(A) Chat between two Bluetooth devices / do Bluetooth devices chat (B) Web browser accessing a website / web browser website access

(C) File transfer via USB cable / USB cable file transfer (D) Two laptops connected directly / do laptops directly connected

[Source: Repo: BCI_SCI Subject Test-01 (UNIT TEST-1 CN & Security), Q23] · Medium

(B) Web browser accessing a website

Nukta: Client-Server = client request , server service provide . Example: web browser - website.

Client-Server = client request , server service provide Example: web browser - website.

Q91. In a bank, customers use ATMs to check balance or withdraw cash. ATM machines send the request to a central computer for verification and processing. Which type of architecture is used here?

Ek bank Ȼustomers ATMs balance check cash withdraw ʐATM machines request central computer verification processing ģ bhejte ʐ architecture use ?

(A) Peer-to-Peer (B) Client-Server

(C) Multipoint (D) Point-to-Point

[Source: Repo: BCI_SCI Subject Test-01 (UNIT TEST-1 CN & Security), Q25] · Medium

(B) Client-Server

Nukta: ATM -> central server = client-server (client request, server verify/process).

ATM -> central server = client-server (client request, server verify/process).

Q92. In a LAN, connectivity can be created using Wi-Fi or _____.

Ek LAN Ȼonnectivity Wi-Fi _____ ŋ

(A) TV (B) Radio

(C) Mobile (D) Ethernet

[Source: Repo: Random Test- Rajasthan PYQ (RPSC/RSSB)] · Medium

(D) Ethernet

Nukta: LAN connectivity = Wi-Fi (wireless) + Ethernet (wired cable).

LAN connectivity = Wi-Fi (wireless) + Ethernet (wired cable).

Q93. Which network type is built on top of another existing network?

- network type ३ network ?

(A) Overlay network (B) Prime network

(C) Prior network (D) Chief network

[Source: Repo: CN Topic Test-02 (UNIT TEST-2)] · Medium

(A) Overlay network

Nukta: Overlay = network network (VPN classic example). Baaki 3 asli network types ɣ ɾ

Overlay = network network (VPN classic example) ɻ 3 real network types ɓ ʐ

Q94. _____ converts analog voice signals into digital and then transmits them over the broadband line.

_____ analog voice signals digital broadband line .

(A) VoIP (Voice over Internet Protocol) (B) Chatting

(C) Website (D) Video conferencing

[Source: RSMSSB Informatics Assistant 2023 (via examsmantra solved paper)] · Medium

(A) VoIP (Voice over Internet Protocol)

Nukta: VoIP = the new Internet form of the old telephone voice.

VoIP = - internet-

Q95. Which of these is NOT a type of network?

network prakar (type) ?

(A) LAN (B) MAN

(C) WAN (D) TAN

[Source: Repo: BCI_SCI Subject Test-01 (UNIT TEST-1 CN & Security), Q7] · Hard

(D) TAN

Nukta: Network types: PAN, LAN, MAN, WAN (+ CAN, SAN). TAN = .

Network types: PAN, LAN, MAN, WAN (+ CAN, SAN). TAN =

Q96. What is the di erence between an 'internet' and the 'Internet'?

'internet' 'Internet'

(A) There is no di erence; both mean the World Wide Web / ; World Wide Web

(B) An internet is any network of networks; the Internet is the specific global one we use today / internet network of networks ; Internet network of networks

(C) An internet is smaller than a LAN; the Internet is larger than a WAN / internet LAN ; Internet WAN (D) An internet uses cables; the Internet uses only wireless / internet cables use ; Internet wireless use

[Source: Forouzan Ch1 (Q1-10) — Practice Set] · Hard

(B) An internet is any network of networks; the Internet is the specific global one we use today

Nukta: internet = generic network of networks. Internet = THE global network (capital I).

internet = generic network of networks Internet = global network (I)

Q97. Bank employees access internal financial databases, while customers access their accounts through online banking websites. Which networks are being used by employees and customers respectively?

Bank employees internal financial databases access ҙjabki customers online banking websites through accounts access ҙ Employees customers kramashah networks use ҙ?

(A) Employees - Intranet; Customers - Extranet
(B) Employees - Internet; Customers - Intranet
(C) Employees - Extranet; Customers - Internet
(D) Both - Internet / ¢Internet

[Source: Repo: BCI_SCI Subject Test-01 (UNIT TEST-1 CN & Security), Q22] · Hard

(A) Employees - Intranet; Customers - Extranet

Nukta: Employees -> Intranet (internal). Outsiders (customers/suppliers) -> Extranet. Public -> Internet.

Employees -> Intranet (internal). Outsiders (customers/suppliers) -> Extranet. Public -> Internet.

Q98. How many point-to-point WANs are needed to connect n LANs so that every LAN can directly communicate with every other LAN?

n LANs ɡ ɜ point-to-point WANs ɛ ɜ LAN À LAN communicate ?

(A) n (B) n - 1
(C) n(n - 1)/2 (D) n²

[Source: Forouzan Ch1 (Q1-13) — Practice Set] · Hard

(C) n(n - 1)/2

Nukta: Full mesh of LANs = n(n-1)/2 WAN links. Formula mesh .

LANs full mesh = n(n-1)/2 WAN links Formula mesh

Q99. Which of the following uses Peer-to-Peer (P2P) architecture, not Client-Server?

ɜ ĸ ɡ ɣ ¢ Peer-to-Peer (P2P) architecture use , Client-Server ʁ?

(A) BitTorrent file sharing (B) Google search
(C) Gmail (D) YouTube

[Source: Repo: BCI_SCI Subject Test-01 (UNIT TEST-1 CN & Security), Q24] · Hard

(A) BitTorrent file sharing

Nukta: P2P = peers ɴdirectly share r (BitTorrent). Client-Server = server service (Google, Gmail, YouTube).

P2P = peers ¢directly share ҙ(BitTorrent). Client-Server = server service (Google, Gmail, YouTube).

Ώ Samjh — Section F - Network History (ARPANET to Today's Network)

Syllabus: Network History - ARPANET to today's network

Aaj ka Internet ARPANET se nikla hai — packet switching ka pehla bada network, jo research se hokar aam internet bana.

Internet ARPANET 3 — packet switching network, research ij internet

- ARPANET (1969, DARPA/US Defense) — packet switching ka pehla operational network; 4 nodes se shuruaat.
  ARPANET (1969, DARPA) — packet switching network; 4 nodes ę
- Pehle NCP (Network Control Protocol) chala, phir 1983 mein TCP/IP adopt hua — is tarikh ko Internet ka janm manate hain.
  NCP , 3 1983 ØTCP/IP — Internet ij ʐ
- NSFNET (1986) — universities ko joda; phir 1990s mein commercial use shuru hua.
  NSFNET (1986) — universities ⁵; 3 1990s Œe 3 ę
- World Wide Web (1991) — Tim Berners-Lee ne CERN mein banaya (HTML, URL, HTTP); Internet par information rakhne ka aasan tareeka.
  World Wide Web (1991) — Tim Berners-Lee CERN Ø (HTML, URL, HTTP)
- Evolution: ARPANET → NSFNET → Internet → WWW → broadband/Wi-Fi/mobile Internet.
  3 : ARPANET → NSFNET → Internet → WWW → broadband/Wi-Fi/mobile Internet

Saal (Year)	Ghatna (Event)
1969	ARPANET — packet switching network (4 nodes)
1983	TCP/IP adopt — Internet ka janm
1986	NSFNET — university backbone
1991	WWW (Tim Berners-Lee, CERN)
1990s+	commercial Internet, broadband, Wi-Fi

Q100. ARPANET stands for _____.
ARPANET _____.

(A) Advanced Recheck Projects Agency Internet
(B) Advanced Recheck Projects Agency Network
(C) Advanced Research Projects Agency Network
(D) Advanced Research Projects Agency Internet

[Source: Repo: CN Topic Test-01 (UNIT TEST-1), Q26] · Medium

(C) Advanced Research Projects Agency Network

Nukta: ARPANET = Advanced Research Projects Agency Network (U.S. Dept. of Defense).

ARPANET = Advanced Research Projects Agency Network (U.S. Dept. of Defense).

Q101. The first network was:

network :

(A) CNNET (B) NSFNET

(C) ASAPNET (D) ARPANET

[Source: Repo: CN Topic Test-01 (UNIT TEST-1), Q34] · Medium

(D) ARPANET

Nukta: First network = ARPANET (1969). NSFNET = 1986 (backbone).

First network = ARPANET (1969). NSFNET = 1986 (backbone).

Q102. In the original ARPANET, _____ were directly connected together.

Original ARPANET _____ directly connected the

(A) IMPs / IMPs (Interface Message Processors)

(B) host computers

(C) networks (D) routers

[Source: Repo: CN Topic Test-01 (UNIT TEST-1), Q43] · Medium

(A) IMPs

Nukta: Original ARPANET: IMPs (packet switches) directly connected; hosts attached to IMPs.

Original ARPANET: IMPs (packet switches) directly connected the; hosts IMPs attach the

Q103. The first operational computer network in the world was the _____ for the United States Department of Defense.

operational computer network, United States Department of Defense _____

(A) ARPANET (B) ERNET

(C) SKYNET (D) DARPANET

[Source: Repo: CN Topic Test-01 (UNIT TEST-1), Q48] · Medium

(A) ARPANET

Nukta: First operational network = ARPANET (1969, U.S. DoD).

First operational network = ARPANET (1969, U.S. DoD).

Q104. ARPANET - the precursor of today's Internet - was created in _____ by ARPA (Advanced Research Projects Agency) of the U.S. Department of Defense, starting with four nodes (UCLA, SRI, UC Santa Barbara, University of Utah).

ARPANET - aaj Internet purvaj - _____ ARPA (Advanced Research Projects Agency), U.S. Department of Defense , nodes shuruaat (UCLA, SRI, UC Santa Barbara, University of Utah).

(A) 1957 (B) 1969

(C) 1983 (D) 1991

[Source: Extra (Forouzan) - fills syllabus gap] · Medium

(B) 1969

Nukta: ARPANET = 1969, 4 nodes (UCLA, SRI, UCSB, Utah); first message 29 Oct 1969 (LO of LOGIN).

ARPANET = 1969, 4 nodes (UCLA, SRI, UCSB, Utah); message 29 Oct 1969 (LOGIN LO).

Q105. TCP/IP (developed by Vint Cerf and Bob Kahn) was adopted by ARPANET as its o cial protocol suite on 1 January _____ - an event known as the ag day.

TCP/IP (Vint Cerf Bob Kahn Â develop) ARPANET apna o cial protocol suite 1 January _____ - is event ag day

(A) 1969 (B) 1972

(C) 1983 (D) 1995

[Source: Extra (Forouzan) - fills syllabus gap] · Medium

(C) 1983

Nukta: Fflag day = 1 Jan 1983 (ARPANET -> TCP/IP). TCP/IP developed by Cerf & Kahn in 1970s (demo 1972). Syllabus TCP/IP model 1972 = development era.

Fflag day = 1 Jan 1983 (ARPANET -> TCP/IP). TCP/IP Cerf & Kahn 1970s develop (demo 1972). Syllabus TCP/IP model 1972 = development era.

Q106. NSFNET, created by the U.S. National Science Foundation in _____, became a major backbone of the Internet, connecting universities and research centres.

NSFNET, U.S. National Science Foundation Â _____ , Internet major backbone , universities research centres

(A) 1969 (B) 1983

(C) 1986 (D) 1991

[Source: Extra (Forouzan) - fills syllabus gap] · Medium

(C) 1986

Nukta: NSFNET = 1986 (National Science Foundation), T1 backbone -> Internet backbone.

NSFNET = 1986 (National Science Foundation), T1 backbone -> Internet backbone.

Q107. The World Wide Web (WWW) was invented in 1991 at CERN by _____, and it made the Internet popular among the general public.

World Wide Web (WWW) invention 1991 CERN _____ , isne Internet public popular

(A) Vint Cerf (B) Tim Berners-Lee

(C) Bob Kahn (D) Marc Andreessen

[Source: Extra (Forouzan) - fills syllabus gap] · Medium

(B) Tim Berners-Lee

Nukta: WWW = 1991, Tim Berners-Lee @ CERN. Mosaic browser 1993. Vint Cerf & Bob Kahn = fathers of the Internet (TCP/IP).

WWW = 1991, Tim Berners-Lee @ CERN. Mosaic browser 1993. Vint Cerf & Bob Kahn = Internet fathers (TCP/IP).

Ω Samjh — Section G - General Network Basics (WWW, E-mail, Protocols, Servers, Switching - from repo intro tests)

Syllabus: Introduction of DC - general basics asked in BCI/RPSC exams (Specific Address/URL, Application services, Switching)

General basics: WWW aur Internet alag cheezein hain; e-mail protocols, servers, aur switching ke tareeke — yeh sab chhote exams mein bahut puche jaate hain.

General basics: WWW Internet ₵ʒe-mail protocols, servers, switching — w ₵₵ ō ʒ

- WWW = Internet par rakhi linked information ka sangrah (web pages + hyperlinks); Internet khud network infrastructure hai.
  WWW = Internet linked information ŧ ; Internet network infrastructure
- E-mail: SMTP se bhejte hain, POP3/IMAP se lete hain; mail user agent + mail transfer agent kaam karte hain.
  E-mail: SMTP ʒPOP3/IMAP ʒ
- Server = resources/services dene wala computer; Client = service maangne wala.
  Server = resources/services ; Client =
- Switching ke 3 tareeke: Circuit switching (dedicated path — telephone), Packet switching (packets alag-alag routes se — Internet), Message switching (store-and-forward).
  Switching 3 : Circuit switching (dedicated path), Packet switching (packets - ɪ), Message switching (store-and-forward)
- URL = address of web page: scheme://host/path (jaise https://example.com/page).
  URL = web page : scheme://host/path

Switching	Path	Example
Circuit	dedicated, reserved	telephone call
Packet	har packet alag route	Internet
Message	store-and-forward messages	telegraph/early networks

Q108. A collection of hyperlinked documents on the internet forms the?
Internet hyperlinked documents sangrah (collection) Ħ banata ?

(A) World Wide Web (WWW) (B) E-mail system
(C) Mailing list (D) Hypertext markup language / hypertext markup language (HTML)

[Source: Repo: CN Topic Test-01 (UNIT TEST-1), Q5] · Medium

(A) World Wide Web (WWW)

Nukta: WWW = hyperlinked documents collection. HTML = un documents ˣ language (WWW ʸ).

WWW = hyperlinked documents collection. HTML = un documents ɻ language (WWW ʁ).

Q109. Which one of the following is a valid email address?

३ к ङ ४ ∅ valid email address ?

(A) computech@academy.com (B) gmail.com

(C) compu@.com (D) computech@books

[Source: Repo: CN Topic Test-01 (UNIT TEST-1), Q6] · Medium

(A) computech@academy.com

Nukta: Valid email = local@domain.TLD (e.g., name@site.com).

Valid email = local@domain.TLD (name@site.com).

Q110. Which of the following best describes uploading information?

३ к ङ ४ ∅ uploading information achhe describe ?

(A) Sorting data on a disk drive / disk drive data sort (B) Sending information to a host computer / information host computer

(C) Receiving information from a host computer / host computer information receive

(D) Sorting data on a hard drive / hard drive data sort

[Source: Repo: CN Topic Test-01 (UNIT TEST-1), Q7] · Medium

(B) Sending information to a host computer

Nukta: Uploading = local remote/host . Downloading = remote local .

Uploading = local remote/host . Downloading = remote local .

Q111. Which one of the following is the most common internet protocol?

Nimnlikhis ∅ common internet protocol ?

(A) HTML (B) NetBEUI

(C) TCP/IP (D) IPX/SPX

[Source: Repo: CN Topic Test-01 (UNIT TEST-1), Q8] · Medium

(C) TCP/IP

Nukta: Internet core protocol = TCP/IP. HTML = language, protocol ४.

Internet core protocol = TCP/IP. HTML = language , protocol ४.

Q112. The dial-up connections to access the internet use ___ protocol.

Internet access ङ dial-up connections ___ protocol use ८

(A) PPP (Point to Point protocol) (B) X.25 protocol

(C) PLIP protocol (D) None of the above / ∅ ŋ ४

[Source: Repo: CN Topic Test-01 (UNIT TEST-1), Q18] · Medium

(A) PPP (Point to Point protocol)

Nukta: Dial-up Internet = PPP (Point-to-Point Protocol).

Dial-up Internet = PPP (Point-to-Point Protocol).

Q113. When a resident uses a dial-up or DSL service to connect to the Internet, what is the role of the telephone company?

ŋ̊ Internet ʃ dial-up DSL service use , telephone company ɬ ɜ ʜ ?

(A) It provides the switching/connection path from the user to the ISP / user ISP switching/connection path (B) It hosts every website the user visits / user ɬ website host

(C) It assigns IP addresses to every website / website IP address (D) It encrypts all the user's data / user data encrypt

[Source: Forouzan Ch1 (Q1-14) — Practice Set] · Medium

(A) It provides the switching/connection path from the user to the ISP

Nukta: Telephone company ɓ ḳ'road' (local loop + switching). Website hosting/IP assignment ISP/DNS .

Telephone company ʃ ɳ ľ (local loop + switching) Website hosting/IP assignment ISP/DNS

Q114. Which of the following servers allows LAN users to share data?

ɜ ĸ ʃ ɣ Ȼ server LAN users data share ɬ ɜ ?

(A) Data server (B) Point server

(C) File server (D) Communication server

[Source: Repo: CN Topic Test-01 (UNIT TEST-1), Q22] · Medium

(C) File server

Nukta: File server = files store + LAN users share .

File server = files store + LAN users share

Q115. EDI stands for ___.

EDI ___.

(A) Electronic Data Interchange (B) Electronic Digital Internet

(C) Electronic Digital Interchange (D) Electronic Data Internet

[Source: Repo: CN Topic Test-01 (UNIT TEST-1), Q25] · Medium

(A) Electronic Data Interchange

Nukta: EDI = Electronic Data Interchange (business documents e-exchange).

EDI = Electronic Data Interchange (business documents e-exchange).

Q116. The term UPnP is a short form of:

UPnP term (full form) ʜ ?

(A) Uniform Plug and Play (B) Unit Plug and Play

(C) Useful Plug and Play (D) Universal Plug and Play

[Source: Repo: Unit Test-4 Solution (OCR), p11] · Medium

(D) Universal Plug and Play

Nukta: UPnP = Universal Plug and Play (device auto-discovery). Compare: DHCP = auto IP configuration.

UPnP = Universal Plug and Play (device auto-discovery) : DHCP = auto IP configuration

Q117. Consider the switching statements: 1. Circuit Switching establishes a dedicated path. 2. Packet Switching routes data in packets on di erent paths. 3. Message Switching uses store-and-forward. Which statements are correct?

Switching statements ɣ : 1. Circuit Switching dedicated path 2. Packet Switching data packets ₡ paths 3. Message Switching store-and-forward use statements ₹

(A) 1 and 2 only / ₫ ɳ1 2 (B) 2 and 3 only / ₫ ɳ2 3

(C) 1 and 3 only / ₫ ɳ1 3 (D) 1, 2 and 3 / 1, 2 3

[Source: Repo: BCI Subject Test-9 (BCI/SCI Batch)] · Medium

(D) 1, 2 and 3

Nukta: 3 switching types: Circuit (dedicated path), Packet (packets, paths), Message (store-and-forward) — teeno statements correct.

3 switching types: Circuit (dedicated path), Packet (packets, paths), Message (store-and-forward) — ₡

Q118. All the information on the internet is passed using _____ switching method.

Internet saari information _____ switching method pass

(A) Portable (B) Packets

(C) Pouch (D) Primary

[Source: Repo: Random Test- Rajasthan PYQ (RPSC/RSSB)] · Medium

(B) Packets

Nukta: Internet = packet switching. Circuit switching = dedicated path (old telephone).

Internet = packet switching. Circuit switching = dedicated path (telephone).

Q119. Which technique breaks data into packets for transmission?

Transmission ₫ data packets ₡ ɬ technique - ?

(A) Circuit switching (B) Packet switching

(C) Frequency division multiplexing (D) Time division multiplexing

[Source: Repo: CN Topic Test 6 (UNIT TEST-6)] · Medium

(B) Packet switching

Nukta: Internet packet switching — data packets ɴtootta .

Internet packet switching — data packets ₡

Q120. Home broadband connections that work on DSL (Digital Subscriber Line) use ___ protocol.

DSL (Digital Subscriber Line) home broadband connections ___ protocol use ₹

(A) PPPoE (PPP over Ethernet) (B) PPPoA (PPP over ATM)

(C) Bridge protocol (D) All of the above / ɛ

[Source: Repo: CN Topic Test-01 (UNIT TEST-1), Q19] · Hard

(A) PPPoE (PPP over Ethernet)

Nukta: Home DSL broadband = PPPoE (PPP over Ethernet).

Home DSL broadband = PPPoE (PPP over Ethernet).

Q121. A computer network permits sharing of:

computer network ... sharing ...?

(A) Resources / resources (printer, disk, etc.)
(B) Information / information (data)
(C) Both a & b / a b (D) None of these / ...

[Source: Repo: CN Topic Test-01 (UNIT TEST-1), Q47 / CN Topic Test-02 (UNIT TEST-2), Q47] · Hard

(C) Both a & b

Nukta: Network = resources + information sharing.

Network = resources + information sharing.

Q122. Which of the following is a benefit of networking?

Networking - (benefit) ?

(A) File sharing (B) Easier access to resources / resources ...
(C) Easier backups / backups (D) All of the above / ...

[Source: InstrumentationTools 150 CN MCQs] · Hard

(D) All of the above

Nukta: Networking benefits: sharing (files/resources/peripherals) + communication + backup + cost saving. Exam favourite: "All of the above".

Networking : sharing (files/resources/peripherals) + communication + backup + cost saving Exam favourite: "All of the above"

Q123. How many digits of the Data Network Identification Code (DNIC) identify the country?

Data Network Identification Code (DNIC) kitne digits (country) ...?

(A) first three / (B) first four /
(C) first five / (D) first six /

[Source: Repo: CN Topic Test-01 (UNIT TEST-1), Q15] · Hard

(A) first three

Nukta: DNIC: first 3 digits = country code, 4th digit = network code.

DNIC: 3 digits = country code, digit = network code.

Q124. Which of the following is true for Circuit Switching handling multiple simultaneous connections?

connections Circuit Switching ...?

(A) A single path is used and shared for all data / path data ... share (B) All connections are multiplexed into a single channel / connections channel ...multiplex ...
(C) It encodes all data into a single channel / data channel ...ncode (D) Separate physical paths are allocated for each channel / channel ... physical paths ...

[Source: RPSC Programmer 2024 Paper-1 (with o cial key)] · Hard

(D) Separate physical paths are allocated for each channel

Nukta: Circuit switching = connection apna path. Packet switching = shared medium, packets paths .

Circuit switching = connection path
Packet switching = shared medium, packets paths

TOPIC 2 — Network Models (OSI & TCP/IP)

82 MCQs • Theory → Questions (Easy → Medium → Hard)

Ω Samjh — Section A — OSI Model: Basics, Layer Order & Support Layers

OSI = Open Systems Interconnection (ISO, 1984): 7 layers, layer order, network vs user support layers, address types.

OSI (Open Systems Interconnection) model ISO ne 1984 mein banaya — communication ko 7 layers mein baanta taaki har layer ka kaam alag ho sake.

OSI (Open Systems Interconnection) model ISO 1984 ∅ — 7 layers ∅ ३ layer

- 7 layers ka kram upar se neeche: Application, Presentation, Session, Transport, Network, Data Link, Physical.

  7 layers ť : Application, Presentation, Session, Transport, Network, Data Link, Physical

- Yad rakhne ka mantra: 'Please Do Not Throw Sausage Pizza Away' (Physical, Data Link, Network, Transport, Session, Presentation, Application).

  Ů: 'Please Do Not Throw Sausage Pizza Away'

- Har layer sirf apne upar-wali aur neeche-wali layer se baat karti hai (interfaces); do systems ki SAME layer peer-to-peer protocol se baat karti hai.

  layer ģ ŋ / layer ; systems ↿ SAME layer peer protocol

- Layers 1–4 = support layers (data transport ke liye); Layers 5–7 = user support layers (user ke kaam ke liye).

  Layers 1–4 = support layers; Layers 5–7 = user support layers

- Model ek reference framework hai — khud koi data carry nahi karta; real implementations isse map hote hain.

  Model reference framework — data carry ४

- Har layer apni header (data link trailer bhi) add karti jaati hai — isse kehte hain encapsulation.

  layer header — encapsulation ट

Layer #	Naam	PDU
7	Application	Data/Message
6	Presentation	Data/Message
5	Session	Data/Message
4	Transport	Segment (TCP) / Datagram (UDP)
3	Network	Packet / Datagram
2	Data Link	Frame
1	Physical	Bits

Q125. The OSI model was developed by:

OSI model ३ develop ३ ?

(A) IEEE (B) ISO

(C) IETF (D) ITU

[Source: Repo: BCI_SCI Subject Test-7 (BCI/SCI Batch)] · Easy

(B) ISO

Nukta: OSI = ISO ka (1984). TCP/IP = DoD/ARPA ka. IEEE = 802 LAN standards. IETF = TCP/IP standards.

OSI = ISO (1984) TCP/IP = DoD/ARPA IEEE = 802 LAN standards IETF = TCP/IP standards

Q126. OSI stands for:

OSI (full form) ?

(A) Open Systems Interconnection (B) Operating System Interface

(C) Optical Service Implementation (D) None of these

[Source: Repo: CN Topic Test-02 (UNIT TEST-2)] · Medium

(A) Open Systems Interconnection

Nukta: OSI = Open Systems Interconnection (ISO, 1984, 7 layers). Ye sabse favourite one-liner hai.

OSI = Open Systems Interconnection (ISO, 1984, 7 layers) favourite one-liner

Q127. The OSI model has how many layers?

OSI model layers ?

(A) 5 (B) 6

(C) 7 (D) 8

[Source: Repo: BCI_SCI Subject Test-7 (BCI/SCI Batch)] · Medium

(C) 7

Nukta: Mnemonic: Please Do Not Throw Sausage Pizza Away (Physical, Data-link, Network, Transport, Session, Presentation, Application).

Mnemonic: Please Do Not Throw Sausage Pizza Away (Physical, Data-link, Network, Transport, Session, Presentation, Application)

Q128. Why was the OSI model developed?

OSI model develop ?

(A) Manufacturers disliked the TCP/IP protocol suite / Manufacturers TCP/IP protocol suite (B) The rate of data transfer was increasing exponentially / Data transfer

(C) Standards were needed to allow any two systems to communicate / systems communication standards (D) None of the above /

[Source: Repo: CN Topic Test-02 (UNIT TEST-2)] · Medium

(C) Standards were needed to allow any two systems to communicate

Nukta: OSI = OPEN systems — matlab koi bhi do systems jud sakein. Isliye 'standards were needed' sahi jawab hai.

OSI = OPEN systems — systems 'standards were needed'

Q129. In the OSI model, when data is sent from device A to device B, which is the 5th layer to receive data at B (counting from the bottom)?

OSI model data device A device B , B data 5 layer (- ?

(A) Application layer (B) Transport layer

(C) Link layer (D) Session layer

[Source: Repo: CN Topic Test-02 (UNIT TEST-2)] · Medium

(D) Session layer

Nukta: Layer order bottom-up: Physical(1), Data-link(2), Network(3), Transport(4), Session(5), Presentation(6), Application(7).

Layer (bottom-up): Physical(1), Data-link(2), Network(3), Transport(4), Session(5), Presentation(6), Application(7)

Q130. Network support layers are:

Network support layers - ?

(A) 1, 2, 3 (B) 4, 5

(C) 5, 6, 7 (D) 4 only

[Source: Repo: CN Topic Test-02 (UNIT TEST-2)] · Medium

(A) 1, 2, 3

Nukta: 1,2,3 = network support; 5,6,7 = user support; 4 = bridge. Exam mein ye grouping bar-bar aata hai.

1,2,3 = network support; 5,6,7 = user support; 4 = bridge

Q131. Which layer connects the network support layers and the user support layers?

- layer network support layers user support layers ?

(A) Transport layer (B) Network layer

(C) Data Link layer (D) Session layer

[Source: Repo: CN Topic Test-02 (UNIT TEST-2)] · Medium

(A) Transport layer

Nukta: Transport = bridge layer (na upar wale, na neeche wale — dono se judta hai).

Transport = bridge layer ( )

Q132. In the ISO-OSI reference model, the Session layer o ers dialog control, token management and _____ as services.

ISO-OSI reference model Session layer dialog control, token management _____ services

(A) Synchronization (B) Asynchronization

(C) Errors (D) Flow control

[Source: Repo: CN Topic Test-03 (UNIT TEST-3)] · Medium

(A) Synchronization

Nukta: Session ki 3 services: Dialog control + Token management + Synchronization (checkpoint).

Session 3 services: Dialog control + Token management + Synchronization (checkpoint)

Q133. How many address types (identifiers) are used from sender to receiver in OSI/TCP-IP communication?

OSI/TCP-IP communication Sender receiver 3 address types (identifiers) use ?

(A) 2 (B) 3

(C) 4 (D) 5

[Source: Repo: BCI_SCI Subject Test-7 (BCI/SCI Batch)] · Medium

(C) 4

Nukta: 4 addresses: Physical(48-bit) / IP(32-bit) / Port(16-bit) / Specific (user-friendly).

4 addresses: Physical(48-bit) / IP(32-bit) / Port(16-bit) / Specific

Q134. Who developed the standards for the OSI reference model?

OSI reference model standards ३ develop ३ ?

(A) ANSI - American National Standards Institute (B) ISO - International Organization for Standardization

(C) IEEE - Institute of Electrical and Electronics Engineers (D) ACM - Association for Computing Machinery

[Source: Repo: CN Topic Test-02 (UNIT TEST-2)] · Medium

(B) ISO - International Organization for Standardization

Nukta: OSI ke standards = ISO ke. ANSI/IEEE/ACM ke alag kaam hain (US standards / LAN / computing society).

OSI standards = ISO ANSI/IEEE/ACM ☰

Q135. The session, presentation, and application layers are the _____ support layers.

Session, Presentation Application layers _____ support layers ☰

(A) user (B) network

(C) both (a) and (b) (D) neither (a) nor (b)

[Source: Repo: CN Topic Test-02 (UNIT TEST-2)] · Hard

(A) user

Nukta: User support = 5,6,7. Network support = 1,2,3. Transport(4) = dono ka bridge.

User support = 5,6,7 Network support = 1,2,3 Transport(4) = ☐ bridge

Q136. Which of the following is NOT a network support layer?

३ ĸ ğ ɣ ☐ network support layer ʁ ?

(A) Transport Layer (B) Network Layer

(C) Data Link Layer (D) Physical Layer

[Source: Repo: CN Topic Test-02 (UNIT TEST-2)] · Hard

(A) Transport Layer

Nukta: Yaad rakho: NOT-network-support = Transport. (1,2,3 = network support; 5,6,7 = user support.)

: NOT-network-support = Transport (1,2,3 = network support; 5,6,7 = user support.)

Ώ Samjh — Section B — OSI Layer Functions (Layer 1 → Layer 7)

Which layer does what: Physical, Data Link (LLC/MAC, framing, FCS), Network (logical addressing/routing), Transport (segmentation, end-to-end, ports), Session (dialog/checkpoint), Presentation (translation/encryption/compression), Application (services & protocols).

Har layer ka apna kaam hai — yeh exam ka sabse favourite hissa hai: kaun si layer kya karti hai.

layer — exam ȝų : - layer Ħ

- Physical (L1) — bits ko signals mein badal kar medium par bhejna; voltage, cables, connectors, topology, data rate.

  Physical (L1) — bits signals Ȼ medium ; voltage, cables, topology, data rate

- Data Link (L2) — framing, physical (MAC) addressing, error control, flow control; FCS (error check) trailer; LLC + MAC sub-layers.

  Data Link (L2) — framing, MAC addressing, error/flow control; FCS; LLC + MAC sub-layers

- Network (L3) — logical (IP) addressing, routing (best path), fragmentation; protocols: IPv4/IPv6, ARP, ICMP, IGMP.

  Network (L3) — logical addressing, routing, fragmentation; IPv4/IPv6, ARP, ICMP, IGMP

- Transport (L4) — end-to-end delivery, port addressing, segmentation, reliability, flow/congestion control; TCP (reliable) vs UDP (fast).

  Transport (L4) — end-to-end delivery, ports, segmentation, flow/congestion control; TCP vs UDP

- Session (L5) — sessions/dialog control, synchronization (checkpoints), simplex/half/full dialogs.

  Session (L5) — sessions/dialog control, synchronization (checkpoints)

- Presentation (L6) — translation (ASCII/EBCDIC), encryption/decryption, compression — 'data ka format'.

  Presentation (L6) — translation, encryption/decryption, compression

- Application (L7) — user ko services: HTTP, FTP, SMTP, DNS, SNMP — user applications yahin se service lete hain.

  Application (L7) — ŋ : HTTP, FTP, SMTP, DNS

- Yaad karo: 'Some People Fear To Buy Stupid Appliances' ulte kram mein (7→1) bhi poochha jaata hai!

  : 7→1 ť

Layer	Kaam (key work)	Address / PDU
Physical	bits → signals	no address / bits
Data Link	framing, MAC, FCS	MAC (48-bit) / frame
Network	routing, logical address	IP (32/128-bit) / packet
Transport	end-to-end, ports	port (16-bit) / segment
Session	dialog + checkpoints	—
Presentation	translate, encrypt, compress	—
Application	user services	—

Q137. Fiber optic cables and coaxial cables function at which OSI layer?

Fiber optic cables coaxial cables OSI layer ?

(A) Data Link Layer (B) Physical Layer

(C) Network Layer (D) Transport Layer

[Source: Repo: BCI Subject Test-8 (BCI/SCI Batch)] · Easy

(B) Physical Layer

Nukta: Media (cable, fiber, radio) = Physical layer ka kaam.

Media (cable, fiber, radio) = Physical layer

Q138. The Physical layer deals with:

Physical layer ?

(A) Logical addressing (B) Media, signals and binary transmission / Media, signals binary transmission

(C) Error correction (D) Segmentation

[Source: Repo: BCI Subject Test-8 (BCI/SCI Batch)] · Easy

(B) Media, signals and binary transmission

Nukta: Physical = bits + signals + media. Baaki sab upar ki layers ka kaam hai.

Physical = bits + signals + media layers

Q139. The sub-layer of the Physical layer which directly interfaces with the MAC sub-layer is called:

Physical layer sub-layer MAC sub-layer :

(A) Physical signalling sub-layer (B) Physical data sub-layer

(C) Physical address sub-layer (D) Physical transport sub-layer

[Source: Repo: BCI Subject Test-8 (BCI/SCI Batch)] · Easy

(A) Physical signalling sub-layer

Nukta: PLS (Physical Layer Signalling) = Physical layer ka upper sub-layer jo MAC se directly judta hai.

PLS (Physical Layer Signalling) = Physical layer upper sub-layer MAC

Q140. Which layer creates the 'collision domain'?

'Collision domain' - layer ?

(A) Physical Layer (hubs) (B) Data Link Layer (switches/bridges)

(C) Network Layer (D) Transport Layer

[Source: Repo: BCI Subject Test-8 (BCI/SCI Batch)] · Easy

(A) Physical Layer (hubs)

Nukta: Hub network = ONE collision domain (Physical). Switch = per-port collision domain (Data Link). Router = broadcast domain boundary (Network).

Hub network = collision domain (Physical) Switch = per-port collision domain (Data Link) Router = broadcast domain boundary (Network)

Q141. Data Link Layer is divided into:

Data Link Layer ?

(A) LLC and MAC (B) Physical and Network

(C) Transport and Session (D) Application and Presentation

[Source: Repo: BCI Subject Test-8 (BCI/SCI Batch)] · Easy

(A) LLC and MAC

Nukta: Data Link = LLC (upper, 802.2) + MAC (lower, 802.3). Ye hamesha saath aate hain.

Data Link = LLC (, 802.2) + MAC (, 802.3)

Q142. Physical (hardware) addressing is added at:

Physical (hardware) addressing ३ layer ?

(A) Network (B) Transport

(C) Data Link (D) Session

[Source: Repo: BCI_SCI Subject Test-7 (BCI/SCI Batch)] · Easy

(C) Data Link

Nukta: Physical/MAC address = Data Link. IP/Logical = Network. Port = Transport.

Physical/MAC address = Data Link IP/Logical = Network Port = Transport

Q143. 'Transparent transmission' is the function of which layer?

'Transparent transmission' ३ layer ?

(A) Data link (B) Network

(C) Session (D) Transport

[Source: Repo: BCI_SCI Subject Test-7 (BCI/SCI Batch)] · Easy

(A) Data link

Nukta: Data Link = 'transparent service' — Network layer ko pata hi nahi chalta ki neeche kitni mehnat hui.

Data Link = 'transparent service' — Network layer ५ ४

Q144. Which layer handles 'hop-to-hop' flow control?

'Hop-to-hop' flow control - layer ?

(A) Network (B) Data Link

(C) Transport (D) Physical

[Source: Repo: BCI_SCI Subject Test-7 (BCI/SCI Batch)] · Easy

(B) Data Link

Nukta: Hop-to-hop = Data Link. End-to-end = Transport. Dono flow control karte hain — scope alag hai.

Hop-to-hop = Data Link End-to-end = Transport

Q145. Which OSI layer is responsible for logical addressing?

Logical addressing ई - OSI layer ६ ⊼ ?

(A) Data Link (B) Network

(C) Transport (D) Application

[Source: Repo: BCI_SCI Subject Test-7 (BCI/SCI Batch)] · Easy

(B) Network

Nukta: Logical address = IP = Network layer (L3). Physical address = MAC = Data Link (L2).

Logical address = IP = Network layer (L3) Physical address = MAC = Data Link (L2)

Q146. Which layer performs segmentation?

Segmentation - layer ?

(A) Data link (B) Transport

(C) Network (D) Session

[Source: Repo: BCI_SCI Subject Test-7 (BCI/SCI Batch)] · Easy

(B) Transport

Nukta: Segmentation + Reassembly = Transport. (Packetisation Network ki, Framing Data Link ki.)

Segmentation + Reassembly = Transport (Packetisation Network ५, Framing Data Link ५)

Q147. Which layer provides end-to-end communication?

End-to-end communication - layer ?

(A) Network (B) Transport

(C) Session (D) Physical

[Source: Repo: BCI_SCI Subject Test-7 (BCI/SCI Batch)] · Easy

(B) Transport

Nukta: End-to-end = Transport. Source-to-destination = Network. Hop-to-hop = Data Link. Bit-by-bit = Physical.

End-to-end = Transport Source-to-destination = Network Hop-to-hop = Data Link Bit-by-bit = Physical

Q148. In the OSI model, which layer(s) participate in 'end-to-end' delivery? (1) Network (2) Transport (3) Session (4) Data link

OSI model Ȼend-to-end' delivery Ȼ - layer(s) ʔ (1) Network (2) Transport (3) Session (4) Data link

(A) 1, 2 only (B) 2 only

(C) 1, 2, 3 (D) 2, 3, 4

[Source: Repo: BCI_SCI Subject Test-7 (BCI/SCI Batch)] · Easy

(B) 2 only

Nukta: End-to-end = Transport only. (Network = host-to-host, Data Link = hop-to-hop.)

End-to-end = ǵ ɲTransport

Q149. Which layer performs 'service point addressing'?

'Service point addressing' - layer ?

(A) Network (B) Data Link

(C) Transport (D) Session

[Source: Repo: BCI_SCI Subject Test-7 (BCI/SCI Batch)] · Easy

(C) Transport

Nukta: Service point = Port. Port addressing = Transport (16-bit port numbers).

Service point = Port Port addressing = Transport (16-bit port numbers)

Q150. Which OSI layer handles message boundaries (knowing where one message ends and the next begins)?

- OSI layer message boundaries (message ĵ ę)?

(A) Presentation (B) Session

(C) Transport (D) Network

[Source: Repo: BCI_SCI Subject Test-7 (BCI/SCI Batch)] · Easy

(C) Transport

Nukta: Message boundary = transport-level concept (UDP message = poora message).

Message boundary = transport-level concept

Q151. Determine the OSI layer responsible for dialogue control (half-duplex or full-duplex communication).

Dialogue control (half-duplex full-duplex communication) ǵ - OSI layer ǫ ĸ ?

(A) Session Layer (B) Transport Layer

(C) Network Layer (D) Data Link Layer

[Source: Repo: BCI Subject Test-8 (BCI/SCI Batch)] · Easy

(A) Session Layer

Nukta: Dialog control + Token management + Synchronization = Session layer ka kaam.

Dialog control + Token management + Synchronization = Session layer

Q152. Which layer supports 'checkpointing' (synchronisation for recovery)?

- layer 'checkpointing' (recovery ǵ synchronization) support ?

(A) Transport (B) Application

(C) Session (D) Data link

[Source: Repo: BCI_SCI Subject Test-7 (BCI/SCI Batch)] · Easy

(C) Session

Nukta: Checkpointing = Session (synchronization). Ye exam ka classic sawaal hai.

Checkpointing = Session

Q153. Which layer translates data from a format used by the application into a common format (e.g., ASCII to EBCDIC)?

- layer data application format common format Ȼ (ASCII EBCDIC)?

(A) Application Layer (B) Presentation Layer

(C) Session Layer (D) Transport Layer

[Source: Repo: BCI Subject Test-8 (BCI/SCI Batch)] · Easy

(B) Presentation Layer

Nukta: Presentation = Translation + Encryption + Compression (TEC). ASCII<->EBCDIC isi layer ka classic example hai.

Presentation = Translation + Encryption + Compression

Q154. Which layer is responsible for encryption?

Encryption ǵ - layer ɢ ĸ ?

(A) Session (B) Presentation

(C) Application (D) Transport

[Source: Repo: BCI_SCI Subject Test-7 (BCI/SCI Batch)] · Easy

(B) Presentation

Nukta: Encryption + Compression + Translation = Presentation (L6).

Encryption + Compression + Translation = Presentation (L6)

Q155. Which OSI layer ensures interoperability between di erent data formats?

- OSI layer - data formats interoperability ɜ ǵ Ì ?

(A) Presentation (B) Application

(C) Session (D) Network

[Source: Repo: BCI_SCI Subject Test-7 (BCI/SCI Batch)] · Easy

(A) Presentation

Nukta: Common format = Presentation. Isi liye ise 'syntax layer' bhi kehte hain.

Common format = Presentation 'syntax layer' ƺ

Q156. Which OSI layer decides the type of compression?

- OSI layer compression ų ?

(A) Application (B) Presentation

(C) Transport (D) Data link

[Source: Repo: BCI Subject Test-8 (BCI/SCI Batch)] · Easy

(B) Presentation

Nukta: Compression = Presentation. (Encryption aur translation bhi isi ke saath.)

Compression = Presentation

Q157. The term 'Baud Rate' is associated with which layer?

'Baud Rate' ķ ३ layer ?

(A) Network Layer (B) Data Link Layer

(C) Physical Layer (D) Application Layer

[Source: Repo: BCI Subject Test-8 (BCI/SCI Batch)] · Medium

(C) Physical Layer

Nukta: Baud rate = signalling rate = Physical layer. Bit rate ≠ baud rate (bits per symbol yaad rakho).

Baud rate = signalling rate = Physical layer Bit rate ≠ baud rate

Q158. The layer used to deal with mechanical and electrical specifications is:

Mechanical electrical specifications ३ layer - ?

(A) Physical Layer (B) Data Link Layer

(C) Network Layer (D) Transport Layer

[Source: Repo: CN Topic Test-02 (UNIT TEST-2)] · Medium

(A) Physical Layer

Nukta: Mechanical + Electrical + Functional + Procedural = Physical layer ke 4 specifications.

Mechanical + Electrical + Functional + Procedural = Physical layer ।4 specifications

Q159. The Physical layer is responsible for which of the following operations?

Physical layer ३ ĸ ģ ɣ Ȼ ३ operations ģ ६ ĸ ?

(A) Line coding (B) Channel coding

(C) Modulation (D) All of the above

[Source: Repo: BCI Subject Test-8 (BCI/SCI Batch)] · Medium

(D) All of the above

Nukta: Physical layer = Line coding + Channel coding + Modulation + signals/media — 'All of the above' yaad rakho.

Physical layer = Line coding + Channel coding + Modulation 'All of the above'

Q160. Transmission data rate is decided by:

Transmission data rate ?

(A) Network layer (B) Physical layer

(C) Data Link layer (D) Transport layer

[Source: Repo: CN Topic Test-01 (UNIT TEST-1)] · Medium

(B) Physical layer

Nukta: Data rate = Physical layer. (Throughput/ delay performance bhi isi layer ke signals par depend karti hai.)

Data rate = Physical layer

Q161. The physical layer translates logical communication requests from the _____ into hardware-specific operations.

Physical layer, _____ logical communication requests hardware-specific operations Ȼ

(A) Network layer (B) Transport layer

(C) Data link layer (D) Application layer

[Source: InstrumentationTools 150 CN MCQs (key corrected vs Forouzan)] · Medium

(C) Data link layer

Nukta: Each layer serves the layer ABOVE it — physical serves data link. Exam banks sometimes mis-key this as 'network layer'; the correct neighbour is the data link layer.

layer layer ।4 — physical, data link ।4 exam banks 'network layer' ģ ȥ ģ data link layer

Q162. The Application layer is implemented in:

Application layer implement ?

(A) End system (B) NIC

(C) Ethernet (D) Packet transport

[Source: Sanfoundry CN MCQs — Reference Models] · Medium

(A) End system

Nukta: End systems (hosts) implement layers up to the application; switches implement only Layers 1-2, routers only Layers 1-3.

End systems (hosts) application layers implement switches Layers 1-2, routers Layers 1-3

Q163. Which statement best describes the di erence between the LLC and MAC sub-layers?

LLC MAC sub-layers ?

(A) LLC handles physical addressing, MAC handles flow control / LLC physical addressing , MAC flow control (B) LLC provides the service interface to the Network layer; MAC controls access to the medium / LLC, Network layer service interface ; MAC medium access control

(C) LLC works only in wireless networks, MAC only in wired networks / LLC wireless MAC wired (D) LLC performs framing, MAC performs routing / LLC framing , MAC routing

[Source: Repo: BCI Subject Test-9 (BCI/SCI Batch)] · Medium

(B) LLC provides the service interface to the Network layer; MAC controls access to the medium

Nukta: LLC = upar wala sub-layer (network layer se link). MAC = neeche wala (medium access + physical address).

LLC = sub-layer MAC = (medium access + physical address)

Q164. A frame is received at the Data Link layer with an incorrect FCS. What will happen?

Data Link layer frame FCS ?

(A) Frame discarded (B) Frame corrected

(C) Sent to network layer (D) Re-transmitted automatically

[Source: Repo: BCI_SCI Subject Test-7 (BCI/SCI Batch)] · Medium

(A) Frame discarded

Nukta: Bad FCS = frame drop (discard). Correction/Data Link khud re-transmit nahi karti — wo ARQ/higher layer ka decision hai.

FCS = frame discard

Q165. The _____ layer is responsible for delivering data units from one station to the next without errors.

_____ layer data units station station errors

(A) Transport (B) Network

(C) Data link (D) Physical

[Source: Repo: CN Topic Test-02 (UNIT TEST-2)] · Medium

(C) Data link

Nukta: 'Station to next station' = Data Link (hop-to-hop). 'Source to destination' = Network.

'Station to next station' = Data Link 'Source to destination' = Network

Q166. _____ Layer is responsible for framing, error detection and physical addressing.

_____ Layer framing, error detection physical addressing ∫ G K

(A) Network (B) Transport

(C) Application (D) Data Link

[Source: Repo: CN Topic Test-02 (UNIT TEST-2)] · Medium

(D) Data Link

Nukta: Framing + Error detection + MAC addressing = Data Link ka 'trio'.

Framing + Error detection + MAC addressing = Data Link trio

Q167. From which layer does the Data Link layer collect packets and encapsulate them into frames for transmission?

Data Link layer 3 layer packets ij Ø transmission ∫ frames Øncapsulate ?

(A) Transport layer (B) Application layer

(C) Network layer (D) Physical layer

[Source: Repo: CN Topic Test-02 (UNIT TEST-2)] · Medium

(C) Network layer

Nukta: Encapsulation: Network packet -> Data Link frame -> Physical bits. Upar se neeche.

Encapsulation: Network packet -> Data Link frame -> Physical bits

Q168. IPv4 and IPv6 work at which OSI layer?

IPv4 IPv6 3 OSI layer ?

(A) Data link layer (B) Network layer

(C) Transport layer (D) Application layer

[Source: Repo: CN Topic Test 6 (UNIT TEST-6)] · Medium

(B) Network layer

Nukta: IP = Internet Protocol = Network/ Internet layer (L3). Hamesha!

IP = Network/Internet layer (L3)

Q169. The _____ layer is responsible for the source-to-destination delivery of a packet across multiple network links.

_____ layer packet ↓ multiple network links source-to-destination delivery ∫ G K

(A) Transport (B) Network

(C) Data link (D) Physical

[Source: Repo: CN Topic Test-02 (UNIT TEST-2)] · Medium

(B) Network

Nukta: Source-to-destination = Network. Process-to-process = Transport. Station-to-station = Data Link.

Source-to-destination = Network Process-to-process = Transport Station-to-station = Data Link

Q170. Which layer is responsible for reliable delivery in the OSI model?

OSI model Øeliable delivery ∫ - layer G K ?

(A) Session (B) Transport

(C) Network (D) Application

[Source: Repo: CN Topic Test 5 (UNIT TEST-5)] · Medium

(B) Transport

Nukta: Reliable = Transport. (OSI Transport class 4 = error recovery wali service.)

Reliable = Transport

Q171. Which layer is responsible for process-to-process delivery?

Process-to-process delivery ∫ - layer ∫ K ?

(A) Network layer (B) Transport layer

(C) Session layer (D) Data link layer

[Source: Repo: CN Topic Test-02 (UNIT TEST-2)] · Medium

(B) Transport layer

Nukta: Process-to-process = Transport (ports ke saath). Host-to-host = Network (IP ke saath).

Process-to-process = Transport Host-to-host = Network

Q172. Which OSI layer is responsible for encryption and compression?

Encryption compression ∫ - OSI layer ∫ K ?

(A) Presentation Layer (B) Session Layer

(C) Transport Layer (D) Network Layer

[Source: Repo: CN Topic Test 6 (UNIT TEST-6)] · Medium

(A) Presentation Layer

Nukta: Trio yaad rakho: Translation, Encryption, Compression = Presentation.

Trio: Translation, Encryption, Compression = Presentation

Q173. Which layer helps to present data in one form on a host to another host in its native representation?

- layer host 3 form Ø data À host 4 native representation Ø Ql' Ø ?

(A) Application Layer (B) Presentation Layer

(C) Session Layer (D) Transport Layer

[Source: Repo: CN Topic Test-02 (UNIT TEST-2)] · Medium

(B) Presentation Layer

Nukta: Native format <-> common format = Presentation (translation).

Native format <-> common format = Presentation

Q174. _____ layer defines how data is formatted, presented, encoded, and converted for use on the network.

_____ layer 3 data network Ø ∫ format, present, encode convert

(A) Transport (B) Presentation

(C) Application (D) Physical

[Source: Central exams — SSC/Bank/Insurance Computer Awareness (BYJU'S)] · Medium

(B) Presentation

Nukta: 'Formatted, presented, encoded, converted' = Presentation ke liye mnemonic-type line.

'Formatted, presented, encoded, converted' = Presentation

Q175. Which layer provides services to the user?

- layer user services ?

(A) Application layer (B) Session layer

(C) Presentation layer (D) Physical layer

[Source: Repo: CN Topic Test-02 (UNIT TEST-2)] · Medium

(A) Application layer

Nukta: User ke sabse kareeb = Application layer (L7).

User = Application layer (L7)

Q176. A server at a company is using TCP/IP to connect to the network and is running multiple applications. At which OSI layer is this server operating?

company server network ग TCP/IP use applications server ३ OSI layer ?

(A) Application Layer (B) Physical Layer

(C) Network Layer (D) Transport Layer

[Source: Repo: CN Topic Test-02 (UNIT TEST-2)] · Medium

(A) Application Layer

Nukta: Server + running applications = Application layer.

Server + applications = Application layer

Q177. Simple Network Management Protocol (SNMP) runs on:

Simple Network Management Protocol (SNMP) ३ layer ?

(A) Application layer (B) Transport layer

(C) Network layer (D) Data link layer

[Source: Repo: CN Topic Test-03 (UNIT TEST-3)] · Medium

(A) Application layer

Nukta: SNMP/DNS/HTTP/FTP/SMTP = Application layer protocols. UDP/TCP = Transport. IP = Network.

SNMP/DNS/HTTP/FTP/SMTP = Application layer protocols

Q178. Consider the statements: S1: Network layer handles both logical addressing and routing. S2: Transport layer handles segmentation and reassembly. Which is correct?

Statements ɣ : S1: Network layer logical addressing routing Ȼ S2: Transport layer segmentation reassembly ?

(A) Only S1 (B) Only S2

(C) Both S1 and S2 (D) Neither S1 nor S2

[Source: Repo: BCI_SCI Subject Test-7 (BCI/SCI Batch)] · Medium

(C) Both S1 and S2

Nukta: Network = IP addressing + Routing. Transport = Segmentation + Reassembly. Dono sahi.

Network = IP addressing + Routing Transport = Segmentation + Reassembly

Q179. Repeaters function in:

Repeaters ३ layer ?

(A) Physical layer (B) Data link layer

(C) Network layer (D) Both (A) and (B)

[Source: Repo: CN Topic Test-03 (UNIT TEST-3)] · Hard

(A) Physical layer

Nukta: Repeater/Hub = Physical (L1). Bridge/Switch = Data Link (L2). Router = Network (L3).

Repeater/Hub = Physical (L1) Bridge/Switch = Data Link (L2) Router = Network (L3)

Q180. The physical layer is responsible for:

Physical layer ३ ग ह ?

(A) Bit-by-bit delivery (B) Port to port delivery

(C) Process to process delivery (D) Application to application delivery

[Source: RPSC Programmer 2024 Paper-1 (with o cial key)] · Hard

(A) Bit-by-bit delivery

Nukta: Bit-by-bit = Physical. Hop-to-hop = Data Link. Source-to-destination = Network. Process-to-process = Transport.

Bit-by-bit = Physical Hop-to-hop = Data Link Source-to-destination = Network Process-to-process = Transport

Q181. Which one of the following is a data link (Layer 2) protocol?

३ क ☐ data link (Layer 2) protocol ?

(A) Point-to-Point Protocol (PPP) (B) HDLC

(C) Ethernet (D) All of these

[Source: InstrumentationTools 150 CN MCQs] · Hard

(D) All of these

Nukta: Layer 2 protocols = PPP, HDLC, Ethernet, Frame Relay, ATM (cell-based). Network layer = IP, ICMP, ARP.

Layer 2 protocols = PPP, HDLC, Ethernet, Frame Relay, ATM (cell-based) Network layer = IP, ICMP, ARP

Q182. Which layer of the OSI model is responsible for the transmission and reception of unstructured raw data between a device and a physical transmission medium?

OSI model ४ - layer device physical transmission medium unstructured raw data transmission reception ग ह ?

(A) Transport layer (B) Data-Link layer

(C) Network layer (D) Physical layer

[Source: Central exams — SSC/Railway/UPSC Computer Awareness (Exampundit)] · Hard

(D) Physical layer

Nukta: 'Raw data + physical medium' = Physical layer. SSC/Railway mein ye one-liner common hai.

'Raw data + physical medium' = Physical layer

Q183. Bridge works in which layer of the OSI model?

Bridge OSI model ४३ layer ?

(A) Application layer (B) Transport layer

(C) Network layer (D) Data link layer

[Source: Repo: CN Topic Test-02 (UNIT TEST-2) · also asked in KVS PGT CS 2018 O cial Paper] · Hard

(D) Data link layer

Nukta: Bridge/Switch = L2. Router = L3. Hub/Repeater = L1. Gateway = L4+.

Bridge/Switch = L2 Router = L3 Hub/Repeater = L1 Gateway = L4+

Q184. The reliable data transfer is implemented by:

Reliable data transfer - layer implement ?

(A) Network layer (B) Physical layer

(C) Presentation layer (D) Transport layer

[Source: RPSC Programmer 2024 Paper-1 (with o cial key)] · Hard

(D) Transport layer

Nukta: Reliable delivery = Transport (TCP). IP (Network) unreliable hai — best-e ort only.

Reliable delivery = Transport (TCP) IP (Network) unreliable

Q185. If a port number is 16 bits (2 bytes), what is the MINIMUM size of the transport-layer header needed to carry the port addresses?

ह port number 16 bits (2 bytes) , port addresses ढ transport-layer header ने MINIMUM size है ?

(A) 2 bytes (B) 4 bytes

(C) 8 bytes (D) 20 bytes

[Source: Forouzan Ch1 (Q1-23) — Practice Set] · Hard

(B) 4 bytes

Nukta: Min transport header (ports only) = 4 bytes. UDP = 8, TCP = 20 — teeno numbers exam mein aate hain.

Min transport header (ports only) = 4 bytes UDP = 8, TCP = 20

Q186. Application layer protocol defines:

Application layer protocol ☐ define ?

(A) Types of messages exchanged / - messages -ų ☐ (B) Rules for when and how processes send and respond to messages / processes messages ☐ ☐ rules

(C) Message format, syntax and semantics / Message format, syntax semantics (D) All of the above / ☐

[Source: Repo: CN Topic Test-02 (UNIT TEST-2)] · Hard

(D) All of the above

Nukta: Application protocol = Message types + Rules + Format. 'All of the above' hi jawab hai.

Application protocol = Message types + Rules + Format

Q187. SMTP, FTP and DNS are the protocols of which layer of the OSI model?

SMTP, FTP DNS OSI model ने layer protocols ☐

(A) Network layer (B) Transport layer

(C) Application layer (D) Data Link layer

[Source: Central exams — SSC/Railway/UPSC Computer Awareness (Exampundit)] · Hard

(C) Application layer

Nukta: User-facing protocols (HTTP, FTP, SMTP, DNS, POP3, IMAP) = Application layer.

User-facing protocols = Application layer

Ω Samjh — Section C — TCP/IP Model

TCP/IP (DoD, 1970s): 4/5 layers, layer mapping to OSI, Internet layer protocols, address levels.

TCP/IP model (DoD model) practical model hai jo aaj Internet par chalta hai — 4 layers (kuch jagah 5) aur OSI ka simplified version.

TCP/IP model (DoD model) ê ∂ model Internet — 4 layers (∫5) OSI ę

- TCP/IP ke 4 layers: Network Access (Physical + Data Link), Internet, Transport, Application (Session+Presentation+Application mil kar).
  TCP/IP ∫4 layers: Network Access, Internet, Transport, Application
- 5-layer version: Physical, Data Link, Network, Transport, Application — books mein yeh zyada padhaya jata hai.
  5-layer version: Physical, Data Link, Network, Transport, Application
- Internet layer: IP, ARP, ICMP, IGMP — packet delivery ke liye.
  Internet layer: IP, ARP, ICMP, IGMP
- Transport layer: TCP (connection-oriented, reliable) aur UDP (connectionless, fast).
  Transport layer: TCP (connection-oriented) UDP (connectionless)
- Application layer: HTTP, FTP, SMTP, DNS, DHCP, SNMP — OSI ke top 3 layers ka kaam.
  Application layer: HTTP, FTP, SMTP, DNS — OSI ∫top 3 layers
- Network Access layer: Ethernet, Wi-Fi, PPP — framing + physical transmission.
  Network Access layer: Ethernet, Wi-Fi, PPP

TCP/IP Layer	OSI Layers	Protocols
Application	Application + Presentation + Session	HTTP, FTP, SMTP, DNS
Transport	Transport	TCP, UDP
Internet	Network	IP, ARP, ICMP, IGMP
Network Access	Data Link + Physical	Ethernet, Wi-Fi, PPP

Q188. Which is the standard protocol suite of the Internet?
Internet ∫ protocol suite - ?
(A) TCP/IP (B) UDP
(C) FTP (D) WWW
[Source: IBPS RRB PO Computer Awareness PYQ (via meritnotes) + SSC MTS computer PYQ (via Testbook)] · Easy

(A) TCP/IP
Nukta: UDP/FTP suite protocols ę WWW , protocol ¥.
UDP/FTP suite protocols ęWWW , protocol ∫.

Q189. The TCP/IP model was developed _____ the OSI model.
TCP/IP model, OSI model _____ develop õ
(A) prior to (B) after
(C) simultaneous to (D) none of these
[Source: Repo: CN Topic Test-02 (UNIT TEST-2)] · Medium

(A) prior to
Nukta: TCP/IP (1970s, adopted 1983) PEHLE; OSI (1984) BAAD mein. 'Prior to' yaad rakho.
TCP/IP (1970s) ; OSI (1984) Ȼ

Q190. The TCP/IP _____ layer is equivalent to the combined Session, Presentation and Application layers of the OSI model.

TCP/IP ↓ _____ layer, OSI model ↓ Session, Presentation Application layers w

(A) application (B) network

(C) data link (D) physical

[Source: Repo: CN Topic Test-02 (UNIT TEST-2)] · Medium

(A) application

Nukta: Mapping: TCP/IP Application = OSI Session+Presentation+Application; TCP/IP Internet = OSI Network; TCP/IP Network Access = OSI Data Link+Physical.

Mapping: TCP/IP Application = OSI 5+6+7; Internet = Network; Network Access = Data Link+Physical

Q191. The number of layers in the Internet protocol stack (reference model) is:

Internet protocol stack (reference model) ∅ayers ↓ ħ ʒ ?

(A) 5 (B) 7

(C) 6 (D) 3

[Source: Repo: CN Topic Test-02 (UNIT TEST-2)] · Medium

(A) 5

Nukta: 5-layer Internet stack vs 4-layer TCP/IP suite vs 7-layer OSI — teeno numbers exam mein aate hain!

5-layer Internet stack, 4-layer TCP/IP suite, 7-layer OSI — ∅umbers !

Q192. The Data Link layer and Physical layer of the OSI model combine to form the _____ layer of the TCP/IP model.

OSI model ↓ Data Link Physical layers ʒ TCP/IP model ↓ _____ layer ʐ

(A) Network Access (B) Internet

(C) Host to Host (D) Application

[Source: Central exams — SSC/Bank Computer Awareness (BYJU'S)] · Medium

(A) Network Access

Nukta: OSI 1+2 = TCP/IP Network Access. OSI 5+6+7 = TCP/IP Application. Bas ye 2 mappings yaad rakho.

OSI 1+2 = Network Access OSI 5+6+7 = Application

Q193. IP (Internet Protocol) is a protocol of the _____ layer of the TCP/IP model.

IP (Internet Protocol) TCP/IP model ↓ _____ layer protocol .

(A) Network Access (B) Internet

(C) Transport (D) Application

[Source: Central exams — SSC/Bank Computer Awareness (BYJU'S)] · Medium

(B) Internet

Nukta: IP = Internet layer. TCP/UDP = Transport. Ethernet = Network Access.

IP = Internet layer TCP/UDP = Transport Ethernet = Network Access

Q194. In TCP/IP, which layer uses specific (user-friendly) addresses — e.g. a URL like example.com?

TCP/IP ∅ - layer specific (user-friendly) addresses use — example.com URL?

(A) Application layer (B) Network layer

(C) Data link layer (D) Physical layer

[Source: Forouzan Ch1 (Q1-24) — Practice Set] · Medium

(A) Application layer

Nukta: Address levels: Application = specific (URL), Transport = port (16-bit), Network = IP (32/128-bit), Data Link = MAC (48-bit).

Address levels: Application = specific, Transport = port, Network = IP, Data Link = MAC

Q195. In the TCP/IP model, a link-layer switch operates at which layers?

TCP/IP model Ø link-layer switch ३ layers ?

(A) Network and Transport layers (B) Physical and Data Link layers

(C) Application layer only (D) All five layers

[Source: Forouzan Ch1 (Q1-16) — Practice Set] · Medium

(B) Physical and Data Link layers

Nukta: Switch = L1+L2 (Network Access). Router = L3 tak. Gateway = L4+.

Switch = L1+L2 Router = L3 Gateway = L4+

Q196. TCP/IP model does not have _____ layer but the OSI model has it.

TCP/IP model Ø____ layer ४ OSI model Ø

(A) Session layer (B) Presentation layer

(C) Application layer (D) Both (a) and (b)

[Source: Repo: CN Topic Test-02 (UNIT TEST-2)] · Hard

(D) Both (a) and (b)

Nukta: TCP/IP mein Session + Presentation nahi hoti — dono Application mein merge ho jaati hain.

TCP/IP Ø Session + Presentation ४ — Ø Application Ø merge ८

Q197. Which of the following is NOT a part of the TCP/IP model?

३ K ∫ ɣ Ø TCP/IP model ६ ľ ४ ?

(A) Application (B) Transport

(C) Data link (D) Session

[Source: Repo: CN Topic Test 6 (UNIT TEST-6)] · Hard

(D) Session

Nukta: TCP/IP ke 4 layers: Application, Transport, Internet, Network Access. Session sirf OSI mein.

TCP/IP ५ 4 layers: Application, Transport, Internet, Network Access Session ∫ ɲ OSI Ø

Q198. How many layers are there in the TCP/IP model (original 4-layer suite)?

TCP/IP model (4-layer suite) Ø layers ट

(A) Seven (B) Five

(C) Four (D) Nine

[Source: Central exams — SSC/Bank Computer Awareness (BYJU'S)] · Hard

(C) Four

Nukta: 4-layer TCP/IP: Network Access + Internet + Transport (Host-to-Host) + Application.

4-layer TCP/IP: Network Access + Internet + Transport + Application

Q199. Which of the following is NOT a layer of the TCP/IP model? (Host-to-Host / Internet / Network Access / Physical)

३ K ∫ ɣ Ø TCP/IP model ५ layer ४ ?

(Host-to-Host / Internet / Network Access / Physical)

(A) Host to Host (B) Internet

(C) Network Access (D) Physical

[Source: Central exams — SSC/Bank Computer Awareness (BYJU'S)] · Hard

(D) Physical

Nukta: TCP/IP ki 4 layers: Network Access, Internet, Host-to-Host, Application. 'Physical' alag layer nahi hai.

TCP/IP ५ 4 layers: Network Access, Internet, Host-to-Host, Application

Ω Samjh — Section D — Comparison, Encapsulation & PDUs

OSI vs TCP/IP, encapsulation order & direction, PDUs per layer, numericals.

OSI aur TCP/IP mein kya fark hai, aur data kaise packets/frames mein badal kar chalta hai (encapsulation) — yeh concepts numericals ke liye zaroori hain.

OSI TCP/IP ₹ ŋ , data packets/frames ₡ (encapsulation)

- OSI = reference model (7 layers, theory); TCP/IP = practical model (4 layers, Internet par chalta hai).
  OSI = reference model (7 layers); TCP/IP = practical model (4 layers)
- Encapsulation: sender par har layer header add karti hai — data → segment → packet → frame → bits.
  Encapsulation: sender layer header — data → segment → packet → frame → bits
- De-encapsulation: receiver par ulta — bits → frame → packet → segment → data; har layer apni header utarti hai.
  De-encapsulation: receiver Í ; layer header
- PDU: Physical = bits; Data Link = frame; Network = packet/datagram; Transport = segment; upper layers = message.
  PDU: Physical = bits; Data Link = frame; Network = packet; Transport = segment; ɬ layers = message
- Address bhi har level par milte hain: MAC (hop-to-hop), IP (source-to-destination), port (process-to-process).
  Address level : MAC (hop-to-hop), IP (source-to-destination), port (process-to-process)
- Data link header + trailer add karti hai (FCS), baaki layers sirf header.
  Data link header + trailer (FCS), ɬ ŋ header

Stage	Kya banta hai	Header kaun lagata hai
Data	message	Application/Presentation/Session
Segment	data + port info	Transport
Packet	segment + IP addresses	Network
Frame	packet + MAC + FCS	Data Link
Bits	signals	Physical

Q200. Identify the correct order of data encapsulation from Layer 7 to Layer 1:
Layer 7 Layer 1 data encapsulation ť ɜ :

(A) Data → Segment → Packet → Frame → Bits
(B) Data → Packet → Segment → Frame → Bits
(C) Bits → Frame → Packet → Segment → Data
(D) Data → Frame → Packet → Segment → Bits

[Source: Repo: BCI Subject Test-8 (BCI/SCI Batch)] · Easy

(A) Data → Segment → Packet → Frame → Bits

Nukta: Encapsulation order: Data -> Segment -> Packet -> Frame -> Bits. Mnemonic: 'Do Some People Fear Barking?'

Encapsulation: Data -> Segment -> Packet -> Frame -> Bits

Q201. Encapsulation happens from:

Encapsulation ३ ६ ☐ ?

(A) Top to bottom (B) Bottom to top

(C) Random (D) Parallel

[Source: Repo: BCI_SCI Subject Test-7 (BCI/SCI Batch)] · Easy

(A) Top to bottom

Nukta: Sender = encapsulation (top->bottom, headers add). Receiver = de-encapsulation (bottom->top, headers remove).

Sender = encapsulation (top->bottom) Receiver = de-encapsulation (bottom->top)

Q202. In the layer hierarchy, as the data packet moves from the upper to the lower layers, headers are:

Layer hierarchy ☐ data packet ↓ layers ↓ , headers:

(A) Added (B) Removed

(C) Rearranged (D) Modified

[Source: Repo: CN Topic Test-02 (UNIT TEST-2)] · Medium

(A) Added

Nukta: Down the stack = headers ADD. Up the stack = headers REMOVE.

↓ = headers ADD ↓ = headers REMOVE

Q203. In TCP/IP, which data unit is encapsulated INSIDE a frame?

TCP/IP ☐ - data unit frame encapsulate ?

(A) A segment (B) A datagram

(C) A bit (D) A message

[Source: Forouzan Ch1 (Q1-20) — Practice Set] · Medium

(B) A datagram

Nukta: Frame ke andar DATAGRAM hota hai. Segment -> Datagram -> Frame -> Bits.

Frame DATAGRAM

Q204. In the TCP/IP model, the unit of data at the transport layer is called a:

TCP/IP model ☐ransport layer data ↓ unit :

(A) Packet (B) Segment

(C) Frame (D) Stream

[Source: Forouzan Ch1 (Q1-20/21) — Practice Set] · Medium

(B) Segment

Nukta: PDU pairs: Transport = Segment, Network = Datagram/Packet, Data Link = Frame, Physical = Bits.

PDU: Transport = Segment, Network = Datagram, Data Link = Frame, Physical = Bits

Q205. A system uses five protocol layers. The application creates a 100-byte message and EVERY layer (including the 5th and the 1st) adds a 10-byte header. What is the e ciency (application bytes / total transmitted bytes)?

system protocol layers Application 100-byte message layer (5 1) 10-byte header E ciency (application bytes / transmitted bytes)?

(A) 50% (B) 66.7%

(C) 75% (D) 80%

[Source: Forouzan Ch1 (P1-11) — Practice Set (Problem)] · Medium

(B) 66.7%

Nukta: Encapsulation overhead numerical: e ciency = data / (data + headers). 100/ (100+50) = 66.7%.

E ciency = data / (data + headers) = 100/150 = 66.7%

Q206. A source S and destination D are connected through ONE intermediate router R. How many times does a packet visit the Network layer and the Data Link layer during the journey?

Source S destination D intermediate router R packet Network layer Data Link layer visit ?

(A) Network layer - 4 times, Data link layer - 4 times (B) Network layer - 4 times, Data link layer - 6 times

(C) Network layer - 2 times, Data link layer - 4 times (D) Network layer - 3 times, Data link layer - 4 times

[Source: Repo: CN Topic Test-03 (UNIT TEST-3)] · Hard

(D) Network layer - 3 times, Data link layer - 4 times

Nukta: Formula: n routers -> Network layer visits = n+2, Data Link visits = 2(n+1). Yahan n=1 -> NL=3, DL=4.

Formula: n routers -> Network layer = n+2, Data Link = 2(n+1) n=1 -> NL=3, DL=4

TOPIC 3 — Addressing (MAC, IPv4, IPv6, Port, DNS)

93 MCQs • Theory → Questions (Easy → Medium → Hard)

Ω Samjh — Section A — Address Levels & MAC Address (Physical Address)

Forouzan's four address levels; the 48-bit burned-in MAC address, unicast/multicast/broadcast MAC.

Forouzan ke hisaab se communication mein 4 levels ke addresses lagte hain — physical (MAC), logical (IP), port, aur application. MAC address data link layer ka physical address hai.

Forouzan communication ∅ addresses ← physical (MAC), logical (IP), port, application

- Physical address (MAC) — node ka data link address; NIC par burned-in hota hai isliye 'burned-in address' bhi; 48 bits (6 bytes) ka.
  Physical address (MAC) — node data link address; NIC burned-in; 48 bits (6 bytes)
- MAC flat address hota hai (hierarchy nahi) — har hop par badalta hai: frame ka source/dest MAC har link ke hisaab se.
  MAC flat address — hop (frame source/dest MAC)
- Unicast (ek node), Multicast (group), Broadcast (sab) — MAC ke 3 types; broadcast MAC = FF-FF-FF-FF-FF-FF.
  Unicast ( ), Multicast ( ), Broadcast ( ); broadcast MAC = FF-FF-FF-FF-FF-FF
- Logical address (IP) — network layer ka address, poore network mein unique, routing ke liye; har hop par nahi, end-to-end same rehta hai.
  Logical address (IP) — network layer address, end-to-end same
- Port address — transport layer, process ko identify karti hai; socket = IP + port.
  Port address — transport layer; socket = IP + port
- Application address (jaise e-mail address) — user ki pehchaan, sabse upar ki layer.
  Application address (e-mail address) —

Address Level	Layer	Size / Example
Physical (MAC)	Data Link	48-bit, FF-FF-FF-FF-FF-FF broadcast
Logical (IP)	Network	32-bit IPv4 / 128-bit IPv6
Port	Transport	16-bit (0–65535)
Application	Application	e-mail address, URL

Q207. Which of the following is a static hardware address?

- static hardware address ?

(A) IP address (B) MAC address
(C) Port number (D) Domain name

[Source: Repo: BCI_SCI Subject Test-7 (BCI/SCI Batch)] · Easy

(B) MAC address

Nukta: Static = burned into hardware (MAC). Dynamic/logical = configured (IP via DHCP, port per connection, name via DNS).

Static = hardware ∅ burned (MAC) Dynamic/logical = configure (IP via DHCP, port per connection, name via DNS)

Q208. A MAC address is also called a _____ address.

MAC address _____ address

(A) Physical address (B) Logical address

(C) Source address (D) Destination address

[Source: Repo: CN Topic Test-01 (UNIT TEST-1) · also BCI_SCI ST-7] · Medium

(A) Physical address

Nukta: MAC = physical/hardware address (burned-in, 48-bit). IP = logical address (configured). Both change hands at di erent layers.

MAC = physical/hardware address (burned-in, 48-bit) IP = logical address (configured)

Q209. The address used in the Data Link Layer is called:

Data Link Layer ∅ address Ħ ₹

(A) Logical address (B) Port address

(C) MAC address (D) IP address

[Source: Repo: CN Topic Test-04 (UNIT TEST-4)] · Medium

(C) MAC address

Nukta: Layer-address map again: Data link→MAC, Network→IP, Transport→Port, Application→socket.

Layer-address map: Data link→MAC, Network→IP, Transport→Port, Application→socket

Q210. Which address identifies a unique device on a local network segment?

Local network segment - address unique device ł ?

(A) IP address (B) MAC address

(C) Gateway (D) Subnet mask

[Source: Repo: BCI_SCI Subject Test-7 (BCI/SCI Batch)] · Medium

(B) MAC address

Nukta: L2 delivery (same network) = MAC. L3 delivery (across networks) = IP. Exam loves this pair.

L2 delivery (same network) = MAC L3 delivery (across networks) = IP Exam õ

Q211. What type of MAC address is FF:FF:FF:FF:FF:FF?

FF:FF:FF:FF:FF:FF ३ ų MAC address ?

(A) Unicast (B) Multicast

(C) Broadcast (D) Loopback

[Source: Repo: CN Topic Test 5 (UNIT TEST-5)] · Medium

(C) Broadcast

Nukta: MAC broadcast = FF:FF:FF:FF:FF:FF (all 48 bits 1). Unicast = first byte's last bit 0; multicast = first byte's last bit 1.

MAC broadcast = FF:FF:FF:FF:FF:FF (48 bits 1) Unicast = first byte ३ bit 0; multicast = first byte ३ bit 1

Q212. Which addresses are used in an internet employing the TCP/IP protocols?

TCP/IP protocols internet ∅ - addresses ₹

(A) Physical address and logical address (B) Port address

(C) Specific address (D) All of these

[Source: LMT MCQ bank (Computer Communication Networks)] · Hard

(D) All of these

Nukta: Forouzan's 4 address levels: Physical(48-bit MAC) at data link, IP(32-bit) at network, Port(16-bit) at transport, Specific (socket) at application.

Forouzan ł 4 address levels: Physical(48-bit MAC) data link , IP(32-bit) network , Port(16-bit) transport , Specific (socket) application

Q213. What is the size (length) of a MAC address?

MAC address size ( ) ?

(A) 42 bits (B) 48 bits

(C) 24 bits (D) 36 bits

[Source: RPSC Programmer 2024 Paper-1 (o cial key) · also BCI_SCI ST-7 / Sanfoundry / InstrumentationTools] · Hard

(B) 48 bits

Nukta: MAC = 48 bits (6 bytes / 12 hex digits). IPv4 = 32 bits, IPv6 = 128 bits, Port = 16 bits. Ye char sizes exam N - r

MAC = 48 bits (6 bytes / 12 hex digits) IPv4 = 32 bits, IPv6 = 128 bits, Port = 16 bits sizes exam - z

Q214. An Ethernet destination address 0A-01-12-03-04-05 is a:

Ethernet destination address 0A-01-12-03-04-05 - address ?

(A) Unicast address (B) Multicast address

(C) Broadcast address (D) All of these

[Source: Repo: CN Topic Test-04 (UNIT TEST-4)] · Hard

(A) Unicast address

Nukta: I/G bit (last bit of first octet): 0 = unicast, 1 = multicast. All-ones (FF:FF...) = broadcast.

I/G bit (octet bit): 0 = unicast, 1 = multicast All-ones (FF:FF...) = broadcast

Q215. Consider the statements: S1: Every computer or Internet device has a unique MAC address. S2: Every computer or Internet device has a unique IP address. Which is correct?

: S1: computer Internet device unique MAC address S2: computer Internet device unique IP address - ?

(A) S1 is true and S2 is false (B) Both S1 and S2 are false

(C) S1 is false and S2 is true (D) Both S1 and S2 are true

[Source: Repo: CN Topic Test-01 (UNIT TEST-1)] · Hard

(A) S1 is true and S2 is false

Nukta: MAC = permanent + globally unique. IP = logical + can repeat across private networks (NAT handles the overlap).

MAC = + globally unique IP = logical + private networks (NAT overlap)

Q216. Which header field is rewritten (not removed) when an IP packet passes through a router?

IP packet router - header field (rewrite) ?

(A) Source IP address (B) Destination IP address

(C) MAC (Media Access Control) address (D) Subnet mask

[Source: Repo: CN Topic Test-03 (UNIT TEST-3)] · Hard

(C) MAC (Media Access Control) address

Nukta: Per hop: MAC header changes. End-to-end: IP addresses stay the same. TTL also decrements (a router function).

Per hop: MAC header End-to-end: IP addresses same TTL (router)

Ω Samjh — Section B — IPv4 Addressing (Format, Classes & Special Addresses)

32-bit dotted-decimal, octets, classes A-E, default masks, loopback/APIPA/private/special addresses, NAT.

IPv4 address 32 bits ka hota hai jo dotted-decimal mein 4 numbers (octets) ke roop mein likha jaata hai — classes A se E tak, aur kuch special addresses har exam mein poochhe jaate hain.

IPv4 address 32 bits dotted-decimal ∅4 numbers (octets) ∅₵ — classes A E , special addresses

- 32 bits = 4 octets (har octet 8 bits, 0–255); dotted-decimal: jaise 192.168.1.1.
  32 bits = 4 octets (octet 8 bits, 0–255); dotted-decimal: 192.168.1.1
- Class A: first bit 0 (0–127), mask /8 — bahut bade networks; Class B: 10 (128–191), mask /16; Class C: 110 (192–223), mask /24.
  Class A: bit 0 (0–127), mask /8; Class B: (128–191), /16; Class C: (192–223), /24
- Class D = multicast (224–239); Class E = reserved/research (240–255).
  Class D = multicast (224–239); Class E = reserved (240–255)
- Loopback = 127.x.x.x (127.0.0.1 = localhost); APIPA = 169.254.x.x (jab DHCP fail ho).
  Loopback = 127.x.x.x (127.0.0.1); APIPA = 169.254.x.x (DHCP fail hone)
- Private addresses (Internet par nahi chalte): 10.x.x.x, 172.16–31.x.x, 192.168.x.x — NAT se bahar jaate hain.
  Private addresses: 10.x, 172.16–31.x, 192.168.x — NAT ɀ
- Special: 0.0.0.0 = default/unknown; 255.255.255.255 = limited broadcast; network ID aur host ID bits class se tay hote hain.
  Special: 0.0.0.0 = default; 255.255.255.255 = broadcast; network/host bits class
- Classful addressing ka rule: network ID bits fixed (class se), host ID bits network ke andar hosts ko dete hain.
  Classful addressing: network ID bits class fix, host ID bits hosts

Class	First bits	Range (1st octet)	Default mask	Networks
A	0	0–127	255.0.0.0 (/8)	2^7 (0–127)
B	10	128–191	255.255.0.0 (/16)	2^{14}
C	110	192–223	255.255.255.0 (/24)	2^{21}
D	1110	224–239	—	multicast
E	1111	240–255	—	reserved

Q217. The first octet of a Class A IP address ranges from:

Class A IP address octet 3 range ∅ ?

(A) 0-127 (B) 128-191

(C) 192-223 (D) 224-239

[Source: Repo: BCI_SCI Subject Test-7 (BCI/SCI Batch)] · Easy

(A) 0-127

Nukta: Class ranges by first octet: A 0-127, B 128-191, C 192-223, D 224-239 (multicast), E 240-255 (experimental).

octet Class ranges: A 0-127, B 128-191, C 192-223, D 224-239 (multicast), E 240-255 (experimental)

Q218. Which class of IP address is used for multicasting?

Multicasting ģ ȝ class IP address ?

(A) Class A (B) Class B

(C) Class C (D) Class D

[Source: Repo: BCI_SCI Subject Test-7 (BCI/SCI Batch) · also CN Topic Test-6] · Easy

(D) Class D

Nukta: D = Multicast (one-to-group). E = Experimental (240-255). Broadcast (255.255.255.255) is separate.

D = Multicast (one-to-group) E = Experimental (240-255) Broadcast (255.255.255.255)

Q219. Which class of address is reserved for experimental purposes?

ų ȝ (experimental) ǹļ Ȼ ģ ȝ class address reserved ?

(A) Class C (B) Class B

(C) Class D (D) Class E

[Source: Repo: BCI_SCI Subject Test-7 (BCI/SCI Batch)] · Easy

(D) Class E

Nukta: Class E = Experimental (240-255). Remember: A-ny (0), B-128, C-192, D-224 (multicast), E-240 (experiment).

Class E = Experimental (240-255) ȻA-ny (0), B-128, C-192, D-224 (multicast), E-240 (experiment)

Q220. The IP address 127.0.0.1 is used for:

IP address 127.0.0.1 ȝ Ȼ ?

(A) Broadcast (B) Class D network

(C) Loopback testing (D) APIPA

[Source: Repo: BCI Subject Test-8 (BCI/SCI Batch) · also CN Topic Test-03 / CN Topic Test-04] · Easy

(C) Loopback testing

Nukta: Whole 127.0.0.0/8 block = loopback. `ping 127.0.0.1` tests the local IP stack.

127.0.0.0/8 block = loopback `ping 127.0.0.1` local IP stack ȥ

Q221. What is the maximum number of unique IPv4 addresses?

Unique IPv4 addresses ɬ ɟ ħ ȝ ?

(A) 2^16 (B) 2^24

(C) 2^32 (D) 2^64

[Source: IP Addressing quiz (Studocu bank, with keys)] · Medium

(C) 2^32

Nukta: 2^32 ≈ 4.29 billion IPv4 addresses. IPv6 = 2^128 ≈ 3.4 × 10^38.

2^32 ≈ 4.29 billion IPv4 addresses IPv6 = 2^128 ≈ 3.4 × 10^38

Q222. An IPv4 address is divided into how many parts?

IPv4 address ȝ ȻȻ ?

(A) 2 (B) 3

(C) 4 (D) 5

[Source: Repo: BCI_SCI Subject Test-7 (BCI/SCI Batch)] · Medium

(A) 2

Nukta: 2 parts = network + host. IPv4 CLASSIC division; subnetting borrows bits from host to make sub-networks.

2 parts = network + host Subnetting Ȼost bits bits sub-networks ȥ

Q223. The first octet of a Class B IP address falls between:

Class B IP address octet ३ range ☐ ?

(A) 0-127 (B) 128-191

(C) 192-223 (D) 224-239

[Source: Repo: CN Topic Test-03 (UNIT TEST-3)] · Medium

(B) 128-191

Nukta: B = 128-191 (default /16). Trick: A starts 0, B starts 128, C starts 192, D starts 224, E starts 240.

B = 128-191 (default /16) Trick: A 0 , B 128 , C 192 , D 224 , E 240

Q224. How many bits are used for the network portion in a Class C IP address?

Class C IP address ☐etwork ङ ३ bits ₹?

(A) 8 (B) 16

(C) 24 (D) 32

[Source: Repo: BCI_SCI Subject Test-7 (BCI/SCI Batch)] · Medium

(C) 24

Nukta: Class C = /24 network (24 bits) + 8 host bits. Class B = /16, Class A = /8.

Class C = /24 network (24 bits) + 8 host bits Class B = /16, Class A = /8

Q225. Which IP address class provides the largest number of hosts per network?

- IP address class ु३ network ɻ hosts ?

(A) Class A (B) Class B

(C) Class C (D) Class D

[Source: Repo: CN Topic Test-03 (UNIT TEST-3)] · Medium

(A) Class A

Nukta: Hosts per class: A = 2^24−2, B = 2^16−2 (65,534), C = 2^8−2 (254). D/E have no hosts (multicast/experimental).

ु३ class hosts: A = 2^24−2, B = 2^16−2 (65,534), C = 2^8−2 (254) D/E ☐osts ६ (multicast/experimental)

Q226. Which class of IP addresses supports 65,534 hosts per network?

- IP address class ु३ network 65,534 hosts support ?

(A) Class A (B) Class B

(C) Class C (D) Class D

[Source: Repo: CN Topic Test-03 (UNIT TEST-3)] · Medium

(B) Class B

Nukta: 65,534 = 2^16−2 = Class B hosts. 254 = 2^8−2 = Class C. 16,777,214 = 2^24−2 = Class A.

65,534 = 2^16−2 = Class B hosts 254 = 2^8−2 = Class C 16,777,214 = 2^24−2 = Class A

Q227. What is the maximum number of IP addresses in a Class C network?

Class C network ☐P addresses ↳ ɻ ħ ३ ?

(A) 254 (B) 256

(C) 1024 (D) 65536

[Source: Repo: CN Topic Test 5 (UNIT TEST-5)] · Medium

(B) 256

Nukta: Total addresses = 2^host-bits. Usable hosts = total − 2. Exam asks BOTH — read 'addresses' vs 'hosts' carefully!

addresses = 2^host-bits hosts = −2 Exam ☐ — 'addresses' 'hosts' ☐ ŋ ☐

Q228. Which of the following is a Class D address?

৩ ক ৫ - Class D address ?

(A) 224.1.1.1 (B) 233.0.0.10

(C) 239.5.5.5 (D) All of the above

[Source: Repo: BCI Subject Test-8 (BCI/SCI Batch)] · Medium

(D) All of the above

Nukta: Class D = 224-239 (multicast). Any address in this range is Class D — exam often gives mixed octets like these.

Class D = 224-239 (multicast) range ন্ address Class D — exam ৫ ম ৩ গ octets ৪ ২

Q229. Which IP address is reserved for future use and experimental purposes?

- IP address ৩ ল ু ৩ ন৷ ৫ গ reserved ?

(A) 127.0.0.1 (B) 224.0.0.1

(C) 240.0.0.0 (D) 192.168.0.1

[Source: Repo: CN Topic Test-03 (UNIT TEST-3)] · Medium

(C) 240.0.0.0

Nukta: 240.0.0.0/4 = Class E = future use. 127.x = loopback, 224.x = multicast, 192.168.x = private.

240.0.0.0/4 = Class E = future use 127.x = loopback, 224.x = multicast, 192.168.x = private

Q230. Which of the following is a Class B host address?

৩ ক ৫ - Class B host address ?

(A) 230.0.0.0 (B) 130.4.5.6

(C) 230.7.6.5 (D) 30.4.5.6

[Source: Repo: CN Topic Test-03 (UNIT TEST-3)] · Medium

(B) 130.4.5.6

Nukta: 230.x = Class D (multicast). 30.x = Class A. 130.x = Class B — look at the FIRST octet.

230.x = Class D (multicast) 30.x = Class A 130.x = Class B — octet ৫

Q231. Which of the following is a valid IPv4 address?

৩ ক ৫ - IPv4 address ?

(A) 192.256.1.1 (B) 172.31.0.1

(C) 300.100.50.25 (D) 500.10.5.2

[Source: 30 important CN MCQs (Medium blog bank)] · Medium

(B) 172.31.0.1

Nukta: Octet range check: 0-255 only. 172.16-172.31.x is also the private Class B range (bonus!).

Octet range check: 0-255 172.16-172.31.x private Class B range (bonus!)

Q232. The address 255.255.255.255 is used for:

Address 255.255.255.255 ৩ গ ?

(A) A public IP (B) A default gateway

(C) A network ID (D) Broadcast to all nodes

[Source: Repo: CN Topic Test-03 (UNIT TEST-3)] · Medium

(D) Broadcast to all nodes

Nukta: Limited broadcast = 255.255.255.255 (this network only). Directed broadcast = last address of a subnet (e.g., 192.168.1.255).

Limited broadcast = 255.255.255.255 (network) Directed broadcast = subnet ৩ address (192.168.1.255)

Q233. What is the purpose of the reserved IP address 0.0.0.0?

Reserved IP address 0.0.0.0 ṅ¦ ₹ ?

(A) Default gateway (B) Used for DHCP servers

(C) Default route or 'any network' (D) Multicast address

[Source: Repo: CN Topic Test-03 (UNIT TEST-3)] · Medium

(C) Default route or 'any network'

Nukta: 0.0.0.0/0 = default route (gateway of last resort). As source = 'not yet assigned an IP'.

0.0.0.0/0 = default route (gateway of last resort) Source ę ₡ ' IP ঃ '

Q234. The IP address used by a host's packets as the source while the host is being booted (before it gets an IP) is:

host boot (IP ঃ)

packets source IP address - ?

(A) 0.0.0.0 (B) 1.0.0.0

(C) 1.1.1.1 (D) 255.255.255.255

[Source: Repo: CN Topic Test-04 (UNIT TEST-4)] · Medium

(A) 0.0.0.0

Nukta: Boot/DHCP phase: source 0.0.0.0, destination 255.255.255.255 (broadcast). Ye DHCP handshake ka classic pattern hai.

Boot/DHCP : source 0.0.0.0, destination 255.255.255.255 (broadcast) DHCP handshake classic pattern

Q235. What is the range of addresses for APIPA (Automatic Private IP Addressing)?

APIPA (Automatic Private IP Addressing) ɬ address range ₹ ?

(A) 169.254.0.1 to 169.254.255.254 (B) 10.0.0.0 to 10.255.255.255

(C) 192.168.0.0 to 192.168.255.255 (D) 172.16.0.0 to 172.31.255.255

[Source: Repo: CN Topic Test-03 (UNIT TEST-3)] · Medium

(A) 169.254.0.1 to 169.254.255.254

Nukta: APIPA = 169.254.x.x (link-local, auto). Private blocks: 10.x / 172.16-31.x / 192.168.x (RFC 1918).

APIPA = 169.254.x.x (link-local, auto) Private blocks: 10.x / 172.16-31.x / 192.168.x (RFC 1918)

Q236. Which IP address can be used in a WAN (public network)?

- IP address WAN (public network) ₡ ?

(A) 256.0.0.1 (B) 172.16.0.10

(C) 15.1.5.6 (D) 127.256.0.118

[Source: Repo: CN Topic Test-03 (UNIT TEST-3)] · Medium

(C) 15.1.5.6

Nukta: WAN needs public IPs. Private ranges (10.x, 172.16-31.x, 192.168.x) + loopback + invalid octets — sab hatana seekho.

WAN ʃ public IPs ɛ Private ranges (10.x, 172.16-31.x, 192.168.x) + loopback + invalid octets — ₡

Q237. Each computer connected to the internet must:

Internet ụị computer Ħ ɛ ?

(A) Be an IBM PC (B) Have a unique IP address

(C) Be internet compatible (D) Have a modem connection

[Source: SSC CGL Internet & Networking (via EduRev bank)] · Medium

(B) Have a unique IP address

Nukta: Unique IP = the fundamental Internet requirement (IPv4 32-bit or IPv6 128-bit).

Unique IP = Internet ɬ ị (IPv4 32-bit IPv6 128-bit)

Q238. Which organization is responsible for managing IP address allocation globally?

gỒ ľ IP address allocation ụ - ?

(A) IEEE (B) ICANN

(C) IETF (D) ISO

[Source: Repo: CN Topic Test-03 (UNIT TEST-3)] · Medium

(B) ICANN

Nukta: ICANN/IANA = addresses + names. IETF = protocols (RFCs). IEEE = LAN standards (802.x). ISO = OSI model.

ICANN/IANA = addresses + names IETF = protocols (RFCs) IEEE = LAN standards (802.x) ISO = OSI model

Q239. What is the size of each octet in an IPv4 address?

IPv4 address ụị octet size ɜ ?

(A) 4 bits (B) 8 bits

(C) 16 bits (D) 32 bits

[Source: IP Addressing quiz (Studocu bank, with keys)] · Hard

(B) 8 bits

Nukta: Octet = 8 bits (0-255 in decimal). 4 octets = 32 bits = dotted-decimal notation.

Octet = 8 bits (decimal Ø-255) 4 octets = 32 bits = dotted-decimal notation

Q240. An IPv4 address is of size:

IPv4 address ɜ bits ?

(A) 8 bit (B) 16 bit

(C) 32 bit (D) 64 bit

[Source: Networking MCQ bank (MQuestions blog)] · Hard

(C) 32 bit

Nukta: Size shortcut: MAC 48, IPv4 32, IPv6 128, Port 16 (bits).

Size shortcut: MAC 48, IPv4 32, IPv6 128, Port 16 (bits)

Q241. If the number of networks and number of hosts in Class B are 2^m and (2^n − 2) respectively, then the relation between m and n is:

ɛ Class B Ønetworks hosts ɬ ħ ť 2^m (2^n − 2) , m n Ħ ?

(A) 3m = 2n (B) 7m = 8n

(C) 8m = 7n (D) 2m = 3n

[Source: Repo: CN Topic Test-03 (UNIT TEST-3)] · Hard

(C) 8m = 7n

Nukta: Class B classful counts: networks 2^14, hosts 2^16−2. So m=14, n=16 → 8m=7n. (Class A: 2^7 nets, 2^24−2 hosts; Class C: 2^21 nets, 2^8−2 hosts.)

Class B classful counts: networks 2^14, hosts 2^16−2 m=14, n=16 → 8m=7n (Class A: 2^7 nets, 2^24−2 hosts; Class C: 2^21 nets, 2^8−2 hosts)

Q242. Which subnet mask is the default for the Class B IP address range?

Class B IP address range ģ default subnet mask - ?

(A) 255.255.255.0 (B) 255.0.0.0

(C) 255.255.0.0 (D) 255.255.255.255

[Source: 30 important CN MCQs (Medium blog bank) · also CN Topic Test-6] · Hard

(C) 255.255.0.0

Nukta: Default masks: A 255.0.0.0 (/8), B 255.255.0.0 (/16), C 255.255.255.0 (/24).

Default masks: A 255.0.0.0 (/8), B 255.255.0.0 (/16), C 255.255.255.0 (/24)

Q243. What is the primary purpose of NAT (Network Address Translation) in IPv4?

IPv4 ØNAT (Network Address Translation) ħ ǹ¦ Ħ ?

(A) To map private IP addresses to a public IP address (B) To increase the security of IP addresses

(C) To facilitate communication between di erent subnets (D) All of the above

[Source: IP Addressing quiz (Studocu bank, with keys)] · Hard

(A) To map private IP addresses to a public IP address

Nukta: NAT core = private→public mapping. Bonus e ects: hides internal IPs (security), helps address exhaustion. Some banks key 'All of these' — the PRIMARY purpose is the mapping.

NAT core = private→public mapping ɜ ɞ Ħ ų : internal IPs ɞ ƶ(security), address exhaustion banks 'All of these' key ƶ— ħ ǹ¦ mapping

Q244. A host is connected to a department network which is part of a university network; the university network in turn is part of the Internet. Which is the LARGEST network among these?

host department network university network ɛ ľ ; university network Ø Internet ɛ ľ Ø network - ?

(A) The subnet to which the host belongs (B) The department network

(C) The university network (D) The Internet

[Source: Repo: CN Topic Test-02 (UNIT TEST-2)] · Hard

(D) The Internet

Nukta: Hierarchy: subnet department LAN university CAN Internet. Addressing mirrors this (network prefix lengths).

ť : subnet department LAN university CAN Internet Addressing (network prefix lengths)

Q245. Which IP address type is used to communicate with all devices on a subnet?

Subnet devices ģ ɜ type IP address ?

(A) Unicast (B) Broadcast

(C) Multicast (D) Anycast

[Source: Repo: CN Topic Test-03 (UNIT TEST-3)] · Hard

(B) Broadcast

Nukta: IP communication types: Unicast (one-to-one), Broadcast (one-to-all-on-subnet), Multicast (one-to-group), Anycast (one-to-nearest).

IP communication types: Unicast (one-to-one), Broadcast (one-to-all-on-subnet), Multicast (one-to-group), Anycast (one-to-nearest)

Q246. Which of the following is NOT a valid private (RFC-1918) IP address?

३ K Ȼ - private (RFC-1918) IP ४ ?

(A) 10.0.0.1 (B) 172.16.5.4

(C) 192.168.1.10 (D) 11.0.0.1

[Source: CN Topic Test 5.pdf] · Hard

(D) 11.0.0.1

Nukta: Private ranges : 10.x (Class A), 172.16–172.31.x (Class B), 192.168.x (Class C) — ˣ public.

Private ranges : 10.x (Class A), 172.16–172.31.x (Class B), 192.168.x (Class C) — ɬ public.

Q247. In the IPv4 addressing format, the number of networks allowed under Class C addresses is:

IPv4 addressing Ȼlass C addresses ɳ३ networks allowed ₹

(A) 2^{14} (B) 2^{7}

(C) 2^{21} (D) 2^{24}

[Source: GATE CS 2012] · Hard

(C) 2^{21}

Nukta: 2^{24} poore network field ka size hai (fixed 3 bits bhool kar); 2^{7} aur 2^{14} host/other class ke figures hain.

2^{24} poora field (3 fixed bits bhool kar); 2^{7}/2^{14} doosre figures hain.

Q248. Which one of the following CIDR prefixes exactly represents the range of IP addresses 10.12.2.0 to 10.12.3.255?

CIDR prefix Ȼ - 10.12.2.0 10.12.3.255 ɬIP range ɻ - ɻ ɳ ?

(A) 10.12.2.0/24 (B) 10.12.2.0/23

(C) 10.12.0.0/22 (D) 10.12.3.0/25

[Source: GATE CS 2024 Set-2] · Hard

(B) 10.12.2.0/23

Nukta: /24 sirf 2.0–2.255 cover karta hai; /22 bahut bada (4 blocks); /25 aadha block.

/24 sirf 2.x; /22 bahut bada; /25 aadha block.

Q249. Three machines M, N and P have IP addresses 100.10.5.2, 100.10.5.5 and 100.10.5.6 respectively. The subnet mask is set to 255.255.255.252 for all three machines. Which one of the following is true?

machines M, N, P IP addresses ť 100.10.5.2, 100.10.5.5 100.10.5.6 ʐ Ȼ subnet mask 255.255.255.252 ३ K Ȼ - ?

(A) All three belong to the same subnet / Ȼ subnet Ȼʐ (B) M and N belong to the same subnet / M N subnet Ȼʐ

(C) N and P belong to the same subnet / N P subnet Ȼʐ (D) M and P belong to the same subnet / M P subnet Ȼʐ

[Source: GATE CS 2019] · Hard

(C) N and P belong to the same subnet

Nukta: Adjacent numbers ka matlab same subnet nahi — /30 par block boundary .3/.4 par hoti hai.

numbers = subnet ४; /30 Ȼoundary .3/.4

Ω Samjh — Section C — Subnetting & CIDR

Subnet masks, /n notation, usable hosts & subnet counts, block arithmetic — basic to advanced calculations.

Subnetting se ek network ko chhote hisson mein baante hain; CIDR se addresses ko class se azaad kar aggregate karte hain. Yeh numericals ka favourite area hai.

Subnetting network CIDR addresses class aggregate

- Subnet mask batata hai ki address mein kitne bits network (aur subnet) ke hain aur kitne host ke — 1 = network, 0 = host.

  Subnet mask bits network host — 1 = network, 0 = host

- /n notation: mask ke pehle n bits 1 hote hain (jaise /24 = 255.255.255.0).

  /n notation: mask n bits 1 /24)

- Subnetting: host bits mein se borrow kar subnet banate hain; usable hosts = $2^h - 2$ (network aur broadcast chhod kar).

  Subnetting: host bits borrow subnet usable hosts = $2^h - 2$

- Subnets ki sankhya = 2^s (s = borrowed bits); CIDR mein 'subnet zero' allowed hone par 2^s hi rehta hai.

  Subnets = 2^s; CIDR subnet zero' allowed 2^s

- CIDR (Classless Inter-Domain Routing) — /n prefix se address blocks; route aggregation (summarization) se routing table chhoti hoti hai.

  CIDR — /n prefix address blocks; route aggregation routing table

- VLSM (Variable Length Subnet Mask) — alag-alag subnets ko alag size dene ke liye mask badalna.

  VLSM — - subnets size mask

- NAT — private IPs ko public IP mein badalna (static, dynamic, overload/PAT); ek public IP se bahut sare hosts chalte hain.

  NAT — private IPs public IP (static, dynamic, overload)

- Numerical tip: network address = IP AND mask; broadcast = network + sab host bits 1; first host = network + 1, last host = broadcast − 1.

  Numerical tip: network = IP AND mask; broadcast = network + host bits 1; first host = network+1

Mask	/n	Host bits (h)	Usable hosts ($2^h - 2$)
255.255.255.0	/24	8	254
255.255.255.128	/25	7	126
255.255.255.192	/26	6	62
255.255.255.224	/27	5	30
255.255.255.252	/30	2	2

Q250. A subnet /30 is MOST commonly used for:

Subnet /30 ɟ ɜ ʃ ?

(A) Two-host networks (B) Four-host networks

(C) Loopback only (D) VPN networks

[Source: Repo: BCI Subject Test-8 (BCI/SCI Batch)] · Medium

(A) Two-host networks

Nukta: Link between 2 routers = /30. WAN links ya tunnel endpoints ke liye standard.

2 routers link = /30 WAN links tunnel endpoints ʃ standard

Q251. Which part of the IP address 172.16.5.4/24 identifies the network?

IP address 172.16.5.4/24 - network ?

(A) 172 (B) 172.16

(C) 172.16.5 (D) 172.16.5.4

[Source: Repo: CN Topic Test-03 (UNIT TEST-3)] · Medium

(C) 172.16.5

Nukta: Visual trick: /8 → 1 octet, /16 → 2 octets, /24 → 3 octets network.

Visual trick: /8 → 1 octet, /16 → 2 octets, /24 → 3 octets network

Q252. A network ID (address with all host bits = 0) cannot be assigned to:

Network ID (host bits = 0 address) ɜ assign ʁɜ ?

(A) A router (B) A switch

(C) A host device (D) The entire network

[Source: Repo: CN Topic Test-03 (UNIT TEST-3)] · Medium

(C) A host device

Nukta: Network ID = first address of the block (host bits 0). Broadcast = last (host bits 1). Both reserved; usable hosts = middle addresses only.

Network ID = block address (host bits 0) Broadcast = ɜ (host bits 1) Øeserved; hosts = addresses

Q253. A host has IP 192.168.10.45/27. Which statement is TRUE?

host IP 192.168.10.45/27 - ?

(A) Broadcast address = 192.168.10.63 (B) Network ID = 192.168.10.32

(C) First usable host = 192.168.10.33 (D) All of the above

[Source: Repo: BCI Subject Test-8 (BCI/SCI Batch) · also CN Topic Test-03] · Medium

(D) All of the above

Nukta: Block arithmetic: 45 ÷ 32 → block 32. Network = block start; broadcast = next block − 1.

Block arithmetic: 45 ÷ 32 → block 32 Network = block ɻ Ę ; broadcast = block − 1

Q254. What does the term 'subnetting' refer to in networking?

Networking Øsubnetting' ʞ ʜ ɳ?

(A) Dividing a large network into smaller logical networks (B) Allocating IP addresses to devices

(C) Filtering network tra c (D) Assigning MAC addresses

[Source: 30 important CN MCQs (Medium blog bank) · also CN Topic Test-03] · Hard

(A) Dividing a large network into smaller logical networks

Nukta: Subnetting = big network → smaller networks (better performance, security, less broadcast).

Subnetting = network → networks (performance, security, broadcast)

Q255. When IPv4 addressing is manually configured, which property identifies the network and host portions of the address?

IPv4 addressing manually configure ,
- property address network host
?

(A) DNS server address (B) Subnet mask
(C) Default gateway (D) DHCP server address

[Source: IP Addressing quiz (Studocu bank, with keys)] · Hard

(B) Subnet mask

Nukta: Mask = boundary between network and host. /24 = 255.255.255.0 = 24 network bits + 8 host bits.

Mask = network host /24 = 255.255.255.0 = 24 network bits + 8 host bits

Q256. In a subnet mask, the number of 1's represents:

Subnet mask 1's

(A) Host bits (B) Network bits
(C) Broadcast bits (D) Subnet bits

[Source: Repo: CN Topic Test-03 (UNIT TEST-3)] · Hard

(B) Network bits

Nukta: Count of 1s = prefix length (/n). Example: 255.255.255.0 has 24 ones = /24.

1s = prefix length (/n) : 255.255.255.0 24 ones = /24

Q257. In CIDR notation, what does /16 represent?

CIDR notation 16

(A) 16 host bits (B) 16 network bits
(C) 16 bytes (D) 16 subnets

[Source: Repo: CN Topic Test-03 (UNIT TEST-3)] · Hard

(B) 16 network bits

Nukta: CIDR /n = n network bits. Host bits = 32 − n. /16 → 2^16 − 2 hosts.

CIDR /n = n network bits Host bits = 32 − n /16 → 2^16 − 2 hosts

Q258. What does /24 denote in CIDR?

CIDR 24

(A) 24 bits are used for the host portion (B) 24 bits are used for the network portion
(C) 24 subnets are created (D) 24 addresses are available in the subnet

[Source: RPSC Programmer 2024 Paper-1 (o cial key)] · Hard

(B) 24 bits are used for the network portion

Nukta: O cial RPSC 2024 key: /24 = 24 network bits. Same idea: /27 = 27 network bits, /30 = 30.

O cial RPSC 2024 key: /24 = 24 network bits : /27 = 27 network bits, /30 = 30

Q259. How many USABLE hosts are there in a /24 subnet?

/24 subnet (usable) hosts

(A) 128 (B) 256
(C) 254 (D) 512

[Source: Repo: CN Topic Test 5 (UNIT TEST-5) · also CN Topic Test-03] · Hard

(C) 254

Nukta: Formula: usable hosts = 2^host_bits − 2. /24 → 254, /25 → 126, /26 → 62, /27 → 30, /28 → 14, /29 → 6, /30 → 2.

: hosts = 2^host_bits − 2 /24 → 254, /25 → 126, /26 → 62, /27 → 30, /28 → 14, /29 → 6, /30 → 2

Q260. How many usable hosts are there in a /30 subnet?

/30 subnet ¢ hosts ?

(A) 2 (B) 4

(C) 6 (D) 14

[Source: Repo: CN Topic Test-03 (UNIT TEST-3)] · Hard

(A) 2

Nukta: /30 = 2 hosts (router-to-router links). /29 = 6 hosts (small o ce).

/30 = 2 hosts (router-to-router links) /29 = 6 hosts (o ce)

Q261. A subnet mask of 255.255.255.248 allows how many usable hosts?

Subnet mask 255.255.255.248 hosts ?

(A) 2 (B) 6

(C) 14 (D) 30

[Source: Repo: CN Topic Test 5 (UNIT TEST-5)] · Hard

(B) 6

Nukta: .248 = /29 = 6 hosts. Remember the /30-/29-/28 ladder: 2, 6, 14.

.248 = /29 = 6 hosts /30-/29-/28 ladder ¢2, 6, 14

Q262. The subnet mask 255.255.255.128 provides how many subnets for a Class C network?

Subnet mask 255.255.255.128, Class C network subnets ?

(A) 1 (B) 2

(C) 4 (D) 8

[Source: Repo: CN Topic Test-03 (UNIT TEST-3)] · Hard

(B) 2

Nukta: Borrowed bits n → 2^n subnets. .128 = 1 bit borrowed → 2 subnets.

bits n → 2^n subnets .128 = 1 bit → 2 subnets

Q263. A network needs 5 subnets, each with at least 16 hosts. Which subnet mask (Class C) should be used?

network 5 subnets , ¢ 16 hosts - subnet mask (Class C) ¢

(A) 255.255.255.192 (B) 255.255.255.248

(C) 255.255.255.240 (D) 255.255.255.224

[Source: Repo: CN Topic Test-03 (UNIT TEST-3)] · Hard

(D) 255.255.255.224

Nukta: Subnetting trade-o : more subnets → fewer hosts each. Borrow just enough bits (3 here).

Subnetting trade-o : ŏ subnets → ¢ hosts bits ¢ 3)

Q264. A subnet mask for Class C can have how many 1's by default (remaining bits 0)?

Class C subnet mask ¢ 1's bits 0)?

(A) 10 (B) 24

(C) 12 (D) 7

[Source: Repo: CN Topic Test-03 (UNIT TEST-3)] · Hard

(B) 24

Nukta: Class C default = /24 = 24 ones. (With subnetting it can go up to /30 practically.)

Class C default = /24 = 24 ones (Subnetting /30)

Q265. What is the valid host range for the subnet 192.168.1.0/25?

Subnet 192.168.1.0/25 ǵ host range Ħ ?

(A) 192.168.1.0 - 192.168.1.255 (B) 192.168.1.1 - 192.168.1.126

(C) 192.168.1.1 - 192.168.1.254 (D) 192.168.1.0 - 192.168.1.127

[Source: Repo: CN Topic Test-03 (UNIT TEST-3)] · Hard

(B) 192.168.1.1 - 192.168.1.126

Nukta: Host range = network+1 to broadcast−1. For /25: .1-.126 (first subnet) and .129-.254 (second).

Host range = network+1 broadcast−1 /25 ǵ : .1-.126 (subnet) .129-.254 (Ȧ)

Q266. In subnetting, reducing the number of host bits does what?

Subnetting Ø host bits ɬ ħ Ħ ?

(A) Increases the broadcast domain (B) Decreases the number of subnets

(C) Increases the number of subnets (D) Increases the number of hosts

[Source: Repo: CN Topic Test 5 (UNIT TEST-5)] · Hard

(C) Increases the number of subnets

Nukta: More subnet bits ⇨ more subnets ⇨ fewer hosts each. Ye subnetting ka basic trade-o hai.

ŏ subnet bits ⇨ ŏ subnets ⇨ Ȼ hosts subnetting trade-o

Q267. In a class B subnet: IP address 125.134.112.66, mask 255.255.224.0. What is the first address (network address)?

Class B subnet ØIP address 125.134.112.66, mask 255.255.224.0 address (network address) Ħ ?

(A) 125.134.96.0 (B) 125.134.112.0

(C) 125.134.112.66 (D) 125.134.0.0

[Source: Repo: CN Topic Test-03 (UNIT TEST-3)] · Hard

(A) 125.134.96.0

Nukta: Block size = 256 − last non-255 octet (256−224 = 32). Floor(112/32)×32 = 96.

Block size = 256 − ɜ -255 octet (256−224 = 32) Floor(112/32)×32 = 96

Q268. A Class B network is to be split into subnets with a 3-bit subnet number. What is the maximum number of subnets and hosts in each subnet?

Class B network 3-bit subnet number Ȼ ųİ subnet Ø subnets hosts ɬ ɟ ħ ɜ ?

(A) 8 subnets and 262141 hosts (B) 6 subnets and 262141 hosts

(C) 6 subnets and 1022 hosts (D) None of the above

[Source: Repo: CN Topic Test-03 (UNIT TEST-3)] · Hard

(D) None of the above

Nukta: Class B /16 + 3 subnet bits = /19: 8 subnets × 8190 hosts each. Since options are unusual, the correct choice is 'None of the above'.

Class B /16 + 3 subnet bits = /19: 8 subnets × 8190 hosts ɜ Í ij ƺ ǵ ɜ Í 'None of the above'

Q269. IP address 172.16.123.213 has subnet mask 255.255.255.240. What is its network (first) address?

IP address 172.16.123.213 subnet mask 255.255.255.240 network ( ) address ₹ ?

(A) 172.16.123.240 (B) 172.16.123.208

(C) 172.16.123.213 (D) 172.16.123.245

[Source: Repo: CN Topic Test-03 (UNIT TEST-3)] · Hard

(B) 172.16.123.208

Nukta: /28 = 16-address blocks: .208-.223 here. Network = .208, broadcast = .223, hosts = .209-.222 (14 hosts).

/28 = 16-address blocks: .208-.223 Network = .208, broadcast = .223, hosts = .209-.222 (14 hosts)

Q270. An organization has a Class B network and wishes to form subnets for 64 departments. The subnet mask would be:

Class B network 64 departments ∉ subnets Subnet mask ₹ ?

(A) 255.255.128.0 (B) 255.255.0.0

(C) 255.255.64.0 (D) 255.255.252.0

[Source: RPSC Programmer 2024 Paper-1 (o cial key)] · Hard

(D) 255.255.252.0

Nukta: Borrow 6 bits: 16+6 = 22 network bits = 255.255.252.0. RPSC 2024 o cial key.

6 bits : 16+6 = 22 network bits = 255.255.252.0 RPSC 2024 o cial key

Q271. A Class B network has a subnet mask of 255.255.248.0. What is the maximum number of hosts per subnet?

Class B network subnet mask 255.255.248.0 ų subnet hosts ɬ ɟ ₹ ?

(A) 2046 (B) 1022

(C) 1023 (D) 1024

[Source: RPSC Programmer 2024 Paper-1 (o cial key) · Rajasthan PYQ Q56] · Hard

(A) 2046

Nukta: O cial key (RPSC 2024 + Rajasthan PYQ): 2046 hosts. /21 → 2^11−2 = 2046.

O cial key (RPSC 2024 + Rajasthan PYQ): 2046 hosts /21 → 2^11−2 = 2046

Q272. An IP address 156.233.42.56 is configured with a 7-bit subnet mask (7 bits borrowed from the host part of the classful network). How many hosts and subnets are possible?

IP address 156.233.42.56 7-bit subnet mask (classful network host 7 bits) configure ₹ hosts subnets ₹

(A) 126 hosts, 510 subnets (B) 128 hosts, 512 subnets

(C) 510 hosts, 126 subnets (D) 512 hosts, 128 subnets

[Source: Unit Test -2 Solutions (OCR), p5] · Hard

(A) 126 hosts, 510 subnets

Nukta: Bank key pattern: 7 host bits → 126 hosts; 9 subnet bits → 510 usable subnets (both −2).

Bank key pattern: 7 host bits → 126 hosts; 9 subnet bits → 510 subnets (∉2)

Q273. If a Class B network on the internet has a subnet mask of 255.255.248.0, what is the maximum number of hosts per subnet?

ε internet ȝ Class B network ɬ subnet mask 255.255.248.0 , subnet Ȼ ɟ hosts ȝ Ȼ?

(A) 1023 (B) 1022

(C) 1024 (D) 2046

[Source: RPSC Programmer 2024 Paper-1 (via Suraku Academy o cial-key solutions)] · Hard

(D) 2046

Nukta: 248 = 11111000 → octet ₦5 subnet bits. 2^hostbits − 2 (network broadcast).

248 = 11111000 → octet Ø5 subnet bits. 2^hostbits − 2 (network broadcast).

Q274. The address of a Class B host is to be split into subnets with a 6-bit subnet number. What is the maximum number of subnets and the maximum number of hosts in each subnet?

Class B host 6-bit subnet number subnets Ȼ . ɟ subnets subnet Ȼ ɟ hosts ȝ Ȼ?

(A) 62 subnets and 262142 hosts (B) 64 subnets and 262142 hosts

(C) 62 subnets and 1022 hosts (D) 64 subnets and 1024 hosts

[Source: RPSC Programmer 2011 Paper-1 (via pyqp.in) + GATE CS 2005] · Hard

(C) 62 subnets and 1022 hosts

Nukta: RPSC 2011 GATE 2005 ₦₦ ₭Ṛ — o cial key (C) 62 subnets, 1022 hosts.

RPSC 2011 GATE 2005 ȻȻ ų Î — o cial key (C) 62 subnets, 1022 hosts.

Q275. Suppose that in an IP-over-Ethernet network, a machine X wishes to find the MAC address of another machine Y in its subnet. Which one of the following techniques can be used for this?

IP-over-Ethernet network Ø machine X subnet ɬ machine Y MAC address ǧ ȝ ĸ Ȼ - technique ų ?

(A) ARP request broadcast (B) RARP request broadcast

(C) DNS query (D) ICMP query

[Source: GATE CS 2019] · Hard

(A) ARP request broadcast

Nukta: RARP ulta (MAC→IP) hai; DNS naam→IP; ICMP error/control — koi bhi MAC nahi deta.

RARP Í (MAC→IP); DNS →IP; ICMP error — MAC ŋ ʁ

Q276. Node X has a TCP connection open to node Y. Packets from X to Y go through an intermediate IP router R; Ethernet switch S is the first switch on the path. For a packet sent from X, which of the following is TRUE about its destination IP and MAC addresses at the time it leaves X?

Node X ↓ node Y TCP connection ; packets router R ↙ Ethernet switch S X ३ packet destination IP MAC addresses ☎ ?

(A) Destination IP = Y's IP; destination MAC = R's MAC / Destination IP = Y IP; destination MAC = R MAC (B) Destination IP = R's IP; destination MAC = R's MAC / Destination IP = R IP; destination MAC = R MAC

(C) Destination IP = Y's IP; destination MAC = Y's MAC / Destination IP = Y IP; destination MAC = Y MAC (D) Destination IP = R's IP; destination MAC = S's MAC / Destination IP = R IP; destination MAC = S MAC

[Source: GATE CS 2025 Set-2] · Hard

(A) Destination IP = Y's IP; destination MAC = R's MAC

Nukta: IP ko hop-by-hop badalna common galti hai; switch MAC forwarding karta hai par packet ka destination MAC router ka hi rehta hai.

IP hop-by-hop ; packet destination MAC router

Ώ Samjh — Section D — IPv6 Addressing

128-bit addresses, address types (no broadcast), IPv6-only header changes, transition (tunneling), AAAA records.

IPv6 addresses IPv4 ki khatam hote addresses ka solution hai — 128 bits, hexadecimal colon notation, aur kuch naye rules.

IPv6, IPv4 Ĵ addresses — 128 bits, hexadecimal colon notation, ³

- IPv6 = 128 bits = 16 bytes; hexadecimal ke 8 groups (har group 4 hex digits), colon se alag: 2001:0db8::1.
  IPv6 = 128 bits = 16 bytes; 8 hex groups, colon : 2001:0db8::1
- Broadcast NAHI hota IPv6 mein — multicast, unicast aur anycast use hote hain.
  IPv6 Œroadcast ˠ — multicast, unicast, anycast
- IPv6 header fixed 40 bytes ka, simplified — checksum header nahi (error handling neeche/upper layers karte hain).
  IPv6 header 40 bytes, — checksum ˠ
- Address types: global unicast, link-local (fe80::), loopback (::1), unspecified (::), multicast.
  Address types: global unicast, link-local (fe80::), loopback (::1), multicast
- Transition techniques: Dual stack (dono IP), Tunneling (IPv6 ko IPv4 mein), Translation (NAT-PT).
  Transition: Dual stack, Tunneling, Translation
- SLAAC (Stateless Address Autoconfiguration) — host khud address bana sakta hai (router advertisement se).
  SLAAC — host address
- IPv4 = 32-bit dotted-decimal; IPv6 = 128-bit hex — yehi fark exam ka classic sawaal hai.
  IPv4 = 32-bit dotted-decimal; IPv6 = 128-bit hex

Fakt (Feature)	IPv4	IPv6
Size	32 bits	128 bits
Notation	dotted-decimal	hex colon
Broadcast	haan	nahi (multicast/anycast)
Header	variable (20–60 B)	fixed 40 B
Checksum	header mein	nahi

Q277. Which IPv6 address type is globally unique and globally routable?

- IPv6 address type globally unique globally routable ?

(A) Link-Local (B) Multicast
(C) Unique Local (D) Global Unicast

[Source: Repo: CN Topic Test 5 (UNIT TEST-5)] · Medium

(D) Global Unicast

Nukta: IPv6 unicast scopes: Link-Local (fe80::/10) = this link only; Unique-Local (fc00::/7) = like private; Global (2000::/3) = Internet.

IPv6 unicast scopes: Link-Local (fe80::/10) = link; Unique-Local (fc00::/7) = private ; Global (2000::/3) = Internet

Q278. IPv6 was developed by:

IPv6 ३ develop ३ ?

(A) IETF (B) ANSI

(C) ISO (D) IEEE

[Source: Repo: CN Topic Test-03 (UNIT TEST-3)] · Medium

(A) IETF

Nukta: IPv4/IPv6 = IETF (RFCs). OSI = ISO. LAN standards = IEEE (802). ANSI = US standards body.

IPv4/IPv6 = IETF (RFCs) OSI = ISO LAN standards = IEEE (802) ANSI = US standards body

Q279. Two IPv6 nodes must interoperate but are connected through intervening IPv4 routers. The best solution is:

IPv6 nodes interoperable ३ ☐ IPv4 routers ☐ ☐ ? ?

(A) Use the dual-stack approach (B) Tunneling

(C) No solution (D) Replace the system

[Source: Sanfoundry CN MCQs — IPv6] · Medium

(B) Tunneling

Nukta: Transition mechanisms: Tunneling (IPv6 over IPv4), Dual-stack (both), Translation (NAT64).

Transition mechanisms: Tunneling (IPv6 over IPv4), Dual-stack (☐ Translation (NAT64)

Q280. Which DNS resource record should be queried to retrieve the IPv6 address associated with a name?

३ name IPv6 address ∫ - DNS resource record query ६ ?

(A) The AAAA record (B) The A record

(C) The NS record (D) The MX record

[Source: cnp3 MCQ (Bonaventure CN textbook), DNS1] · Medium

(A) The AAAA record

Nukta: DNS records: A = name→IPv4, AAAA = name→IPv6, PTR = IP→name, MX = mail, NS = name server.

DNS records: A = name→IPv4, AAAA = name→IPv6, PTR = IP→name, MX = mail, NS = name server

Q281. IPv6 addresses are _____ bit addresses.

IPv6 addresses _____ bit ☐

(A) 32 (B) 64

(C) 16 (D) 128

[Source: SBI/IBPS Computer Awareness Capsule 2018 (IBT India)] · Medium

(D) 128

Nukta: IPv4 = 32-bit (4 numbers); IPv6 = 128-bit (8 groups).

IPv4 = 32-bit (4 numbers); IPv6 = 128-bit (8 groups).

Q282. In IPv6, how many bits are in the address?

IPv6 ☐address ३ bits ?

(A) 32 (B) 64

(C) 128 (D) 256

[Source: Repo: CN Topic Test-6 (UNIT TEST-6) · Sanfoundry / Medium / MQuestions banks] · Hard

(C) 128

Nukta: Size line: IPv4 32, IPv6 128, MAC 48, Port 16. 128 bits = 2^128 ≈ 3.4 × 10^38 addresses.

Size line: IPv4 32, IPv6 128, MAC 48, Port 16 128 bits = 2^128 ≈ 3.4 × 10^38 addresses

Q283. IPv6 does NOT use which type of address?

IPv6 type address ?

(A) Broadcast (B) Multicast

(C) Anycast (D) None of the mentioned

[Source: Sanfoundry CN MCQs — IPv6] · Hard

(A) Broadcast

Nukta: IPv4: unicast + broadcast + multicast. IPv6: unicast + multicast + anycast (NO broadcast).

IPv4: unicast + broadcast + multicast IPv6: unicast + multicast + anycast (broadcast)

Q284. Which features are present in IPv4 but NOT in IPv6?

- features IPv4 IPv6 ?

(A) Fragmentation by routers (B) Header checksum

(C) Options (D) All of the mentioned

[Source: Sanfoundry CN MCQs — IPv6] · Hard

(D) All of the mentioned

Nukta: IPv6 header = fixed 40 bytes, 8 fields, NO checksum / NO router fragmentation / NO options field.

IPv6 header = fixed 40 bytes, 8 fields, checksum / router fragmentation / options field

Q285. Which of the following is TRUE for IPv6?

IPv6 ?

(A) Uses 32-bit addresses (B) Includes a header checksum

(C) Supports anycast addressing (D) Does not support multicast

[Source: Repo: CN Topic Test 5 (UNIT TEST-5)] · Hard

(C) Supports anycast addressing

Nukta: IPv6 address types: Unicast, Multicast, Anycast. No broadcast, no header checksum.

IPv6 address types: Unicast, Multicast, Anycast Broadcast , header checksum

Ω Samjh — Section E — Port Numbers, DNS/URL & Address Resolution (ARP)

16-bit ports & ranges, socket addressing, DNS records & lookups, ARP/RARP, URL layer.

Process-to-process delivery ke liye ports, web ka address URL, naam se IP nikalne ka DNS, aur IP se MAC nikalne ka ARP — yeh teeno addressing ki finishing touch hain.

Process-to-process delivery ʃ ports, web URL, IP DNS, IP MAC ARP

- Port = 16-bit number (0–65535): Well-known 0–1023 (HTTP 80, FTP 21, DNS 53), Registered 1024–49151, Ephemeral 49152–65535.
  Port = 16-bit (0–65535): Well-known 0–1023 (HTTP 80, DNS 53), Registered, Ephemeral
- Socket address = IP address + port number — yeh process ko uniquely identify karta hai.
  Socket address = IP + port — process ɬ unique
- DNS — hierarchical naam→IP system: Root servers → TLD (.com, .in) → Authoritative; records: A (IPv4), AAAA (IPv6), MX (mail), CNAME (alias), PTR (IP→naam).
  DNS — →IP ų : Root → TLD → Authoritative; records: A, AAAA, MX, CNAME, PTR
- Reverse lookup = IP se naam = PTR record; forward lookup = naam se IP = A/AAAA.
  Reverse lookup = IP (PTR); forward = IP (A/AAAA)
- URL parts: scheme (protocol)://host[:port]/path — jaise https://www.example.com/index.html.
  URL : scheme://host/path
- ARP (Address Resolution Protocol) — IP se MAC nikalna: ARP request broadcast, ARP reply unicast.
  ARP — IP MAC: request broadcast, reply unicast
- RARP (Reverse ARP) — MAC se IP (boot time); DHCP ne ise zyada tar replace kar diya.
  RARP — MAC IP; DHCP ɻ replace ɜ

Port	Service	Port	Service
20/21	FTP data/control	110	POP3
22	SSH	143	IMAP
23	Telnet	161/162	SNMP
25	SMTP	443	HTTPS
53	DNS	67/68	DHCP
80	HTTP	123	NTP

Q286. Which of the following sets of port numbers belongs to the 'well-known' ports category?

ɜ ĸ Ȼ - port numbers 'well-known' ports category Ȼ ʔ

(A) 80, 443, 8080 (B) 49152, 110, 53

(C) 22, 25, 80 (D) 1024, 3306, 8080

[Source: Repo: BCI Subject Test-8 (BCI/SCI Batch)] · Easy

(C) 22, 25, 80

Nukta: Port ranges: Well-known 0-1023, Registered 1024-49151, Ephemeral (dynamic) 49152-65535.

Port ranges: Well-known 0-1023, Registered 1024-49151, Ephemeral (dynamic) 49152-65535

Q287. What is the typical range of ephemeral (dynamic) ports?

Ephemeral (dynamic) ports ɬ ij range ⊞ ?

(A) 0-1023 (B) 1024-49151

(C) 49152-65535 (D) 100-200

[Source: Repo: BCI Subject Test-8 (BCI/SCI Batch) · also CN Topic Test 5] · Easy

(C) 49152-65535

Nukta: Well-known 0-1023 (services), Registered 1024-49151 (apps), Ephemeral 49152-65535 (clients).

Well-known 0-1023 (services), Registered 1024-49151 (apps), Ephemeral 49152-65535 (clients)

Q288. The transport layer uses which type of addressing?

Transport layer ȝ ų ɬ addressing ?

(A) IP address (B) MAC address

(C) Port number (D) Domain name

[Source: Repo: BCI_SCI Subject Test-7 (BCI/SCI Batch)] · Easy

(C) Port number

Nukta: Layer addressing: Data link = MAC (48), Network = IP (32), Transport = Port (16), Application = specific/socket.

Layer addressing: Data link = MAC (48), Network = IP (32), Transport = Port (16), Application = specific/socket

Q289. A URL (like www.example.com) belongs to which layer?

URL (www.example.com) ȝ layer ɻ ?

(A) Presentation (B) Application

(C) Transport (D) Network

[Source: Repo: BCI_SCI Subject Test-7 (BCI/SCI Batch)] · Easy

(B) Application

Nukta: Address levels at layers: specific (app) → port (transport) → IP (network) → MAC (data link).

Layers address levels: specific (app) → port (transport) → IP (network) → MAC (data link)

Q290. DNS works at which layer of the OSI model?

DNS, OSI model ɬȝ layer ?

(A) Transport (B) Application

(C) Network (D) Data link

[Source: Repo: BCI_SCI Subject Test-7 (BCI/SCI Batch)] · Easy

(B) Application

Nukta: Application layer examples: DNS, HTTP, FTP, SMTP, SNMP, DHCP. Transport carries them (UDP 53 for DNS).

Application layer : DNS, HTTP, FTP, SMTP, SNMP, DHCP Transport ij Ȼ (DNS ɡ UDP 53)

Q291. Which of the following uniquely identifies a process on a host?

Host ȝ process ɬ unique ?

(A) IP address only (B) Port number only

(C) IP address + Port number (D) MAC address

[Source: Repo: BCI Subject Test-9 (BCI/SCI Batch)] · Medium

(C) IP address + Port number

Nukta: IP = the host (machine). IP + Port = the specific process (socket/specific address).

IP = host (machine) IP + Port = ȝ ɡ Ô process (socket/specific address)

Q292. What is the default port number for DNS?

DNS default port number ⊬ ?

(A) 80 (B) 21

(C) 53 (D) 110

[Source: Repo: CN Topic Test 6 (UNIT TEST-6)] · Medium

(C) 53

Nukta: Port one-liners: DNS 53, HTTP 80, HTTPS 443, FTP 21/20, SMTP 25, POP3 110, Telnet 23, SSH 22, DHCP 67/68.

Port one-liners: DNS 53, HTTP 80, HTTPS 443, FTP 21/20, SMTP 25, POP3 110, Telnet 23, SSH 22, DHCP 67/68

Q293. DNS is the abbreviation of:

DNS (full form) ⊬ ?

(A) Dynamic Name System (B) Dynamic Network System

(C) Domain Name System (D) Domain Network Service

[Source: Networking MCQ bank (MQuestions blog)] · Medium

(C) Domain Name System

Nukta: DNS = Domain Name System (port 53). Hierarchical: root → TLD → authoritative servers.

DNS = Domain Name System (port 53) t' : root → TLD → authoritative servers

Q294. Which DNS record type maps a domain name to an IPv4 address?

- DNS record type domain name IPv4 address ?

(A) CNAME (B) A

(C) MX (D) PTR

[Source: Repo: CN Topic Test 6 (UNIT TEST-6) · also cnp3 DNS2] · Medium

(B) A

Nukta: Records quick map: A = name→IPv4, AAAA = name→IPv6, PTR = IP→name, CNAME = alias, MX = mail.

Records quick map: A = name→IPv4, AAAA = name→IPv6, PTR = IP→name, CNAME = alias, MX = mail

Q295. What is the role of DNS? It is best described as:

DNS ⊬ ?

(A) Maps a domain name to an IP address (B) Translates a host's private IP to a public IP

(C) Translates a host's IPv6 address to an IPv4 address (D) Maps a MAC address to an IP address

[Source: RPSC Programmer 2024 Paper-1 (o cial key) · also CN Topic Test-02 / Medium bank] · Hard

(A) Maps a domain name to an IP address

Nukta: O cial RPSC 2024 key. Reverse: PTR records map IP→name. DHCP = auto IP config; NAT = private→public.

O cial RPSC 2024 key Reverse: PTR records IP→name map ζDHCP = auto IP config; NAT = private→public

Q296. A reverse lookup in DNS means a lookup to _____ using _____ records.

DNS Ȼeverse lookup ŋ _____ records _____ ɬ lookup

(A) Identify mail servers, SOA (B) Map an IP address to a domain name, PTR

(C) Map domain names to IP addresses, A (D) Determine the authoritative name server, NS

[Source: RPSC Programmer 2024 Paper-1 (o cial key) · also CN Topic Test 5] · Hard

(B) Map an IP address to a domain name, PTR

Nukta: Forward: name → IP (A / AAAA). Reverse: IP → name (PTR, in-addr.arpa).

Forward: name → IP (A / AAAA) Reverse: IP → name (PTR, in-addr.arpa)

Q297. The Address Resolution Protocol (ARP) is used for:

Address Resolution Protocol (ARP) ȝ Ȼ ?

(A) Finding the IP address that corresponds to a MAC address (B) Finding the MAC address that corresponds to an IP address

(C) Finding the IP address from the DNS name

(D) Finding the IP address of the default gateway

[Source: RPSC Programmer 2024 Paper-1 (o cial key)] · Hard

(B) Finding the MAC address that corresponds to an IP address

Nukta: ARP = IP→MAC (Address Resolution). RARP = MAC→IP (obsolete — replaced by BOOTP/DHCP). DNS = name→IP.

ARP = IP→MAC (Address Resolution) RARP = MAC→IP (ų ɠ — BOOTP/DHCP) DNS = name→IP

Q298. Which one of the following protocols is NOT used to resolve one form of address into another?

ȝ ĸ Ȼ - protocol ų À Ȼ ɠ ʁ ?

(A) DNS (B) ARP

(C) DHCP (D) RARP

[Source: GATE CS PYQ (via Computer Science Junction / examprev)] · Hard

(C) DHCP

Nukta: Address resolvers: DNS, ARP, RARP. Address distributor: DHCP.

Address resolvers: DNS, ARP, RARP. Address distributor: DHCP.

Q299. A reverse lookup in DNS means a lookup to _____ using _____ records.

DNS Ȼeverse lookup ŋ _____ records _____ ɬ .

(A) map domain names to IP address, A (B) map IP address to domain name, PTR

(C) determine the authoritative name servers, CNAME (D) identify mail servers, SOA

[Source: RPSC Programmer 2024 Paper-1 (via Suraku Academy o cial-key solutions)] · Hard

(B) map IP address to domain name, PTR

Nukta: Normal: name→IP = A record; Reverse: IP→name = PTR record. ˝ ˝w ɴ - .

Normal: name→IP = A record; Reverse: IP→name = PTR record. w Ȼ - .

TOPIC 4 — Networking Devices

38 MCQs • Theory → Questions (Easy → Medium → Hard)

Ώ Samjh — Section A — Physical-Layer Devices (Hub, Repeater, Modem & Simple Devices)

Layer-1 dumb devices — hub, repeater, modem, multiplexer, file server; collision-domain basics.

Physical layer ke devices 'dumb' devices kehlate hain — signal ko regenerate ya combine karte hain, intelligence nahi hoti.

Physical layer devices 'dumb devices' ζ— signal regenerate/combine ζintelligence ϐ

- Repeater — weak signal ko copy karke original strength tak regenerate karta hai (amplify nahi karta!); distance badhane ke liye; Layer 1.

  Repeater — signal regenerate (amplify ϐ!); Ȧ ʃ ; Layer 1

- Hub — multiport repeater; ek port par jo aata hai sab ports par bhej deta hai; sab ports = ek collision domain.

  Hub — multiport repeater; port data ports ; ports = collision domain

- Hub par shared bandwidth — jitne zyada devices, utna kam speed har device ko.

  Hub shared bandwidth — ϑ ɟ devices, speed

- Modem (Modulator-Demodulator) — digital ↔ analog conversion; phone line par data.

  Modem — digital ↔ analog ę

- Multiplexer — ek hi line par kai signals ko combine karta hai (MUX/DEMUX).

  Multiplexer — line ɲignals

- File server — files share karne wala dedicated computer (LAN mein).

  File server — files dedicated computer

- Yaad: Repeater/Hub = Layer 1; koi filtering/intelligence nahi — bas signal level ka kaam.

  : Repeater/Hub = Layer 1; ɲiltering ϐ

Device	Layer	Kaam
Repeater	1	signal regenerate (distance badhana)
Hub	1	multiport repeater — sabko bhejo
Modem	1	digital ↔ analog
Multiplexer	1	kai signals ek line par

Q300. Which device operates only at the Physical layer (Layer 1) of the OSI model?

OSI model ɬ Physical layer (Layer 1) - device ?

(A) Switch (B) Hub

(C) Router (D) Bridge

[Source: Repo: BCI_SCI Subject Test-7 (BCI/SCI Batch) · also CN Topic Test-6] · Easy

(B) Hub

Nukta: Layer 1 = Hub, Repeater, Modem, cables. Layer 2 = Switch, Bridge, NIC. Layer 3 = Router. Gateway = all 7 (protocol converter).

Layer 1 = Hub, Repeater, Modem, cables Layer 2 = Switch, Bridge, NIC Layer 3 = Router Gateway = 7 (protocol converter)

Q301. Which of the following pairs lists the network devices used at the physical layer?

३ K Ȼ - physical layer network devices ?

(A) Gateway, Bridge (B) Router, Repeater

(C) Hub, Switch (D) Repeater, Hub

[Source: Random Test- Rajasthan PYQ (OCR), p12 — Q93 (options reconstructed from screenshot)] · Medium

(D) Repeater, Hub

Nukta: Rajasthan PYQ (verified key: D). Physical-layer pair = Repeater + Hub. Switch/Bridge = L2, Gateway = L7.

Rajasthan PYQ (verified key: D) Physical-layer = Repeater + Hub Switch/Bridge = L2, Gateway = L7

Q302. Which of the following are network connecting devices? 1. Repeater 2. Bridge 3. Hub 4. Bluetooth

३ K Ȼ network connecting devices ₹ 1. Repeater 2. Bridge 3. Hub 4. Bluetooth

(A) 1, 2 and 3 (B) 1, 2 and 4

(C) 2, 3 and 4 (D) 1, 3 and 4

[Source: Unit Test -2 Solutions (OCR), p11] · Medium

(A) 1, 2 and 3

Nukta: Devices vs technologies: repeater/bridge/hub/switch/router = devices. Bluetooth/Wi-Fi/Ethernet = technologies/standards.

Devices technologies: repeater/bridge/hub/switch/router = devices Bluetooth/Wi-Fi/Ethernet = technologies/standards

Q303. Which of the following statements are TRUE? 1. Switch reduces collision domain. 2. Hub creates a separate collision domain for each port. 3. Router separates broadcast domains.

३ K Ȼ - ₹ 1. Switch collision domain 2. Hub port ɡ collision domain 3. Router broadcast domains

(A) 1 and 2 only (B) 2 and 3 only

(C) 1 and 3 only (D) 1, 2 and 3

[Source: Repo: BCI Subject Test-9 (BCI/SCI Batch) — Q26] · Medium

(C) 1 and 3 only

Nukta: Switch: collision ↓ (per-port). Hub: 1 collision domain. Router: broadcast domains separated. Ye teeno golden statements hain.

Switch: collision ↓ (per-port) Hub: 1 collision domain Router: broadcast domains Ȼ golden statements ɀ

Q304. Which device combines transmission from various I/O devices into one line?

- device ३ ɡ È I/O devices transmission line Ȼ ?

(A) Modifier (B) Multiplexer

(C) Concentrator (D) Gateway

[Source: Unit Test-4 Solution (OCR), p22] · Medium

(B) Multiplexer

Nukta: MUX = many→one line (FDM/TDM/WDM). Concentrator = statistical sharing of a line. Gateway = protocol converter.

MUX = ɲ line (FDM/TDM/WDM) Concentrator = line statistical sharing Gateway = protocol converter

Q305. A file server is a _____ which is used to store files received from various client machines.

File server _____ client machines files

(A) Specialized computer system (B) Modem

(C) Router (D) Transponder

[Source: Unit Test-4 Solution (OCR), p24] · Medium

(A) Specialized computer system

Nukta: Servers (file/print/mail/web) = specialised computers. Modem = signal converter, Router = packet forwarder, Transponder = satellite relay.

Servers (file/print/mail/web) = computers Modem = signal converter, Router = packet forwarder, Transponder = satellite relay

Q306. In a data-entry office, computer centres on the 1st and 4th floors are connected by LANs, but signals take time to travel. Which hardware can improve the data transmission?

data-entry office 4 computer centres LAN signals . transmission - hardware ?

(A) Repeater (B) Router

(C) Computer (D) Modem

[Source: RSMSSB Informatics Assistant 2023 (via examsmantra solved paper)] · Medium

(A) Repeater

Nukta: → Repeater; → Router; analog↔digital → Modem.

→ Repeater; → Router; analog↔digital → Modem.

Q307. Which of the following devices operates only in the physical layer?

device physical layer ?

(A) Two-layer switch (B) Router

(C) Repeater (D) Bridge

[Source: Kerala PSC repeated PYQs (via pscarivukal)] · Hard

(C) Repeater

Nukta: Repeater = signal regenerator (L1). Hub = multi-port repeater (L1). Both are 'dumb' devices.

Repeater = signal regenerator (L1) Hub = multi-port repeater (L1) 'dumb' devices

Q308. A device that operates at the physical layer of the OSI model is:

OSI model physical layer device :

(A) Bridge (B) Router

(C) Repeater (D) All of these

[Source: SBI/IBPS Computer Awareness Capsule 2018 (IBT India)] · Hard

(C) Repeater

Nukta: Layer-device : L1 = Repeater/Hub, L2 = Bridge/Switch, L3 = Router, L4+ = Gateway.

Layer-device : L1 = Repeater/Hub, L2 = Bridge/Switch, L3 = Router, L4+ = Gateway.

Ω Samjh — Section B — Switch / NIC / Bridge

CAM table, per-port collision domains, flooding behaviour, VLANs & broadcast domains, wireless AP as bridge.

Switch, NIC aur bridge Layer 2 ke devices hain — frame level par MAC addresses dekh kar kaam karte hain.

Switch, NIC bridge Layer 2 devices ← frame level MAC addresses ←

- Switch = intelligent hub: MAC address table (CAM table) banata hai aur frame sirf usi port par bhejta hai jahan destination hai.
  Switch = intelligent hub: MAC table (CAM) , frame port
- Switch ke har port ka apna collision domain — full-duplex, collision khatam.
  Switch port collision domain — collision
- Par switch ka ek hi broadcast domain (jab tak VLAN na banao).
  switch broadcast domain (VLAN)
- Switching modes: Store-and-forward (poora frame lekar FCS check), Cut-through (sirf destination MAC padh kar aage — fast, no error check).
  Switching modes: Store-and-forward (FCS check), Cut-through (error check)
- VLAN = virtual LAN — ek switch par alag-alag broadcast domains; security + organisation.
  VLAN = virtual LAN — switch - broadcast domains
- NIC (Network Interface Card) — computer ko network se jodta hai; apna MAC address rakhta hai; Layer 1–2 par kaam.
  NIC — computer network ; MAC
- Bridge = 2-port switch: do network segments ko MAC addresses se connect/filter karta hai; single collision domain ko split karta hai.
  Bridge = 2-port switch: segments MAC /filter

Device	Kya dekhta hai	Collision domain	Broadcast domain
Hub	— (sirf signal)	sab ek	ek
Switch	MAC table	har port alag	ek (VLAN se zyada)
Bridge	MAC filter	2 hisse	ek

Q309. Which device stores MAC addresses in a CAM table?
- device MAC addresses CAM table ?

(A) Hub (B) Router
(C) Switch (D) Modem

[Source: Repo: BCI Subject Test-8 (BCI/SCI Batch)] · Easy

(C) Switch

Nukta: Switch = MAC table (CAM). Router = routing table (IP). Hub = no table (floods). ARP cache = IP→MAC mapping (host).

Switch = MAC table (CAM) Router = routing table (IP) Hub = table (floods) ARP cache = IP→MAC mapping (host)

Q310. A router is connected between two switches. If a broadcast frame is sent on Switch-A, what will Switch-B receive?

switches router ঃ Switch-A broadcast frame , Switch-B ꢀ ꢁ ?

(A) The broadcast (B) A unicast copy

(C) Nothing (D) A multicast

[Source: Repo: BCI Subject Test-8 (BCI/SCI Batch)] · Easy

(C) Nothing

Nukta: Router = broadcast firewall. This is why routers split broadcast domains and switches don't.

Router = broadcast firewall ꢂ router broadcast domains ꢃ switches ꢄ

Q311. What does a MAC address represent?

MAC address ꢅ ꢆ ꢅ ꢇ ꢈ ?

(A) The location of the device in the network

(B) The unique identity of the NIC

(C) Internet connectivity (D) The routing path

[Source: Repo: BCI_SCI Subject Test-7 (BCI/SCI Batch)] · Easy

(B) The unique identity of the NIC

Nukta: MAC = NIC identity (who). IP = location in network (where). Routing path = router's job.

MAC = NIC identity ( ) IP = network ꢉ ( ) Routing path = router

Q312. Which device reduces collision domains (each port has its own collision domain) but does NOT break broadcast domains?

- device collision domains (port collision domain) ꢅ broadcast domains ꢄ?

(A) Switch (B) Router

(C) Hub (D) Gateway

[Source: Repo: BCI Subject Test-8 (BCI/SCI Batch) · also CN Topic Test-6] · Medium

(A) Switch

Nukta: Switch = collision domains many, broadcast 1. Router = broadcast domains many. Hub = 1 collision + 1 broadcast. VLAN = splits broadcast on a switch.

Switch = collision domains ꢊ broadcast 1 Router = broadcast domains ꢊ Hub = 1 collision + 1 broadcast VLAN = switch broadcast

Q313. A switch receives a frame whose destination MAC is NOT in its MAC table. What does the switch do?

switch frame ꢋ ꢌ destination MAC MAC table ꢉ ꢄ Switch ꢀ ?

(A) Send the frame to the router only (B) Flood the frame out of all its ports (same broadcast domain)

(C) Drop the frame (D) Broadcast it on the entire internet

[Source: Repo: BCI Subject Test-8 (BCI/SCI Batch)] · Medium

(B) Flood the frame out of all its ports (same broadcast domain)

Nukta: Unknown unicast → flood. Known unicast → single port. Broadcast FF:FF:... → flood. Source MAC → learn/refresh in CAM.

Unknown unicast → flood Known unicast → port Broadcast FF:FF:... → flood Source MAC → CAM ꢉ learn/refresh

Q314. A switch has 4 VLANs configured. How many broadcast domains exist?

switch 4 VLANs configure broadcast domains ?

(A) 1 (B) 2

(C) 4 (D) Depends on the router

[Source: Repo: BCI Subject Test-8 (BCI/SCI Batch)] · Medium

(C) 4

Nukta: VLAN = virtual switch = its own broadcast domain. Inter-VLAN communication needs a router (or L3 switch).

VLAN = virtual switch = broadcast domain Inter-VLAN router (L3 switch)

Q315. In a router-on-a-stick setup with VLANs, which device handles VLAN tagging and which handles inter-VLAN routing?

VLANs router-on-a-stick setup VLAN tagging inter-VLAN routing ?

(A) Switch – VLAN tags, Router – routing (B) Router – VLAN tags, Switch – routing

(C) Both switch and router do VLAN tagging

(D) Neither does VLAN tagging

[Source: Repo: BCI Subject Test-8 (BCI/SCI Batch)] · Medium

(A) Switch – VLAN tags, Router – routing

Nukta: 802.1Q tagging = switch trunk job. Inter-VLAN routing = router (router-on-a-stick) or L3 switch.

802.1Q tagging = switch trunk Inter-VLAN routing = router (router-on-a-stick) L3 switch

Q316. A packet travels: wired PC → Switch → Access Point → Wireless laptop. Which devices operate at Layer 2 only?

packet : wired PC → Switch → Access Point → Wireless laptop - devices Layer 2

(A) Switch only (B) Access Point only

(C) Both switch and access point (D) Neither

[Source: Repo: BCI Subject Test-8 (BCI/SCI Batch)] · Medium

(C) Both switch and access point

Nukta: AP = wireless bridge (L2). Switch = L2. Router = L3. WAPs don't route unless configured as routers.

AP = wireless bridge (L2) Switch = L2 Router = L3 WAPs routing (router mode

Q317. Which of the following component is responsible for network communication in a computer system?

network communication - component ?

(A) UPS (B) Ethernet card (NIC)

(C) SVGA card (D) SMPS

[Source: Unit Test-4 Solution.pdf (OCR), p18] · Medium

(B) Ethernet card (NIC)

Nukta: NIC = computer network — MAC address .

NIC = computer network — MAC address .

Q318. LANs can be connected by which of the following devices that operate on the data link layer?

LANs data link layer ३ ĸ ₡ ३ device ?

(A) Hub (B) Bridges

(C) HDLC (D) Tunnel

[Source: SBI PO Computer Awareness PYQ (via bankersadda)] · Medium

(B) Bridges

Nukta: Hub = L1, Bridge/Switch = L2, Router = L3, Gateway = L4+.

Hub = L1, Bridge/Switch = L2, Router = L3, Gateway = L4+.

Ω Samjh — Section C — Router / Brouter

Router functions & layers, per-hop packet changes, broadcast-domain boundaries, congestion (RED), routing basics.

Router Layer 3 ka device hai — packets ko best route par aage bhejta hai aur broadcast domains ko todta hai.

Router Layer 3 device — packets best route broadcast domains

- Router IP addresses dekh kar routing table se best path chunta hai; har hop par forwarding decision.
  Router IP addresses routing table best path
- Router har packet ko aage bhejte waqt: TTL 1 kam, checksum naya, MAC addresses naye hop ke hisaab se — source IP same rehta hai.
  Router: TTL 1 , checksum , MAC — source IP
- Router broadcast domain ko todta hai — ek taraf ka broadcast doosri taraf nahi jaata.
  Router broadcast domain
- Router fragmentation kar sakta hai (MTU ke hisaab se) par reassembly sirf destination karta hai.
  Router fragmentation reassembly ∫ ndestination
- Static routing (manual) vs Dynamic routing (protocols: RIP, OSPF, BGP).
  Static routing (manual) vs Dynamic (RIP, OSPF, BGP)
- Brouter = bridge + router: kuch tra c bridge ki tarah, kuch router ki tarah.
  Brouter = bridge + router ৯ Ż

Device	Layer	Address	Kaam
Switch	2	MAC	frames ko sahi port par
Router	3	IP	packets ko best path par
Gateway	5–7 / sab	—	protocol conversion

Q319. A router performs which of the following? (1) Packet forwarding (2) Logical addressing (3) Collision detection (4) Routing decision

Router ৩ ĸ Ȼ - ŋ ? (1) Packet forwarding (2) Logical addressing (3) Collision detection (4) Routing decision

(A) 1, 2 and 4 (B) 2, 3 and 4

(C) 1 and 3 only (D) 1, 2, 3 and 4

[Source: Repo: BCI_SCI Subject Test-7 (BCI/SCI Batch) — Q28] · Easy

(A) 1, 2 and 4

Nukta: Router = forward + route + logical addressing. NOT collision detection (that's Ethernet L2 on switches/NICs).

Router = forward + route + logical addressing Collision detection ৬ (switches/NICs Ethernet L2)

Q320. A router drops some packets deliberately due to congestion. What is this technique called?

Router congestion packets

(A) Delay policing (B) Redundancy

(C) Random Early Detection (RED) (D) Reverse flooding

[Source: Repo: BCI_SCI Subject Test-7 (BCI/SCI Batch)] · Easy

(C) Random Early Detection (RED)

Nukta: Congestion tools: RED (early random drops), Queuing (FIFO/PQ), Tra c shaping. Dropping ALL at overflow = tail drop.

Congestion tools: RED (early random drops), Queuing (FIFO/PQ), Tra c shaping Overflow = tail drop

Q321. Which device creates a separate broadcast domain?

- device broadcast domain ?

(A) Switch (B) Bridge

(C) Router (D) Repeater

[Source: Repo: BCI Subject Test-8 (BCI/SCI Batch)] · Easy

(C) Router

Nukta: Broadcast domain boundary = router (or VLAN). Collision domain boundary = switch port/bridge.

Broadcast domain = router (VLAN) Collision domain = switch port/bridge

Q322. Which device operates at Layer 3 of the OSI model?

OSI model Layer 3 - device ?

(A) Hub (B) Switch

(C) Router (D) Repeater

[Source: SSC CGL Tier-2 repeated Qs (via Oliveboard)] · Medium

(C) Router

Nukta: Device-layer ladder: Repeater/Hub 1, Switch/Bridge/NIC 2, Router 3, Gateway 7.

Device-layer ladder: Repeater/Hub 1, Switch/Bridge/ NIC 2, Router 3, Gateway 7

Q323. Consider: S1: The TTL field decreases at each router. S2: The MAC address changes at each hop. Which is correct?

: S1: TTL field router S2: MAC address hop - ?

(A) Only S1 (B) Only S2

(C) Both S1 and S2 (D) Neither S1 nor S2

[Source: Repo: BCI_SCI Subject Test-7 (BCI/SCI Batch)] · Medium

(C) Both S1 and S2

Nukta: Per-hop changes: TTL + MAC rewritten. End-to-end: IP addresses unchanged (except NAT).

Per-hop : TTL + MAC End-to-end: IP addresses (NAP)

Q324. Which hardware forwards data packets between computer networks?

- hardware computer networks data packets ?

(A) Switch (B) Router

(C) Cable (D) Nibble

[Source: KVS Librarian 2023 O cial Paper (via Testbook) · also CN Topic Test-03] · Hard

(B) Router

Nukta: O cial KVS key. Router = between networks (L3). Switch = within a network (L2). Cable = medium. Nibble = 4 bits!

O cial KVS key Router = networks (L3) Switch = network (L2) Cable = medium Nibble = 4 bits!

Q325. Among these network devices — i. Repeater ii. Hub iii. Switch iv. Router — which are considered intelligent devices?

network devices Ȼ— i. Repeater ii. Hub iii. Switch iv. Router — - intelligent devices ₹

(A) i and iv only (B) i and ii only

(C) iii and iv only (D) ii and iii only

[Source: Kerala PSC repeated PYQs (via pscarivukal)] · Hard

(C) iii and iv only

Nukta: Intelligent = reads addresses + filters/ forwards selectively (switch, router, bridge, gateway). Dumb = hub, repeater.

Intelligent = addresses + selectively filter/ forward (switch, router, bridge, gateway) Dumb = hub, repeater

Q326. Consider the statements about an IP router: I. A router does not modify the IP packets during forwarding. II. It is not necessary for a router to implement any routing protocol. III. A router should reassemble IP fragments if the MTU of the outgoing link is larger than the incoming fragment. Which is correct?

IP router Ȼ ɣ : I. Forwarding router IP packets ʁ II. Router ǥ ɳ routing protocol ļ ʁ III. ɛ outgoing link MTU fragment router IP fragments ३ ɛ - ?

(A) I and II only (B) I only

(C) II and III only (D) II only

[Source: Unit Test-4 Solution (OCR), p15] · Hard

(D) II only

Nukta: Advanced but classic: routers modify IP header (TTL/checksum), may run static-only, never reassemble fragments (destination does).

È ३ classic: router IP header (TTL/ checksum), static , fragments ʁ (destination)

Ω Samjh — Section D — Gateway, Firewall & NAT Devices

Gateway as protocol converter, firewall types (packet/stateful/proxy), NAT types, device-vs-software identification.

Gateway, firewall aur NAT — network ko connect aur secure karne wale special devices.

Gateway, firewall NAT — network GW 3 devices

- Gateway = protocol converter: alag architecture/network ko jodta hai (jaise e-mail gateway, VoIP gateway); sabse 'intelligent' device.
  Gateway = protocol converter: networks ; 'intelligent' device
- Firewall = tra c ko rules se filter karne wala guard: Packet-filter (headers dekhta hai), Stateful (connection ka state yaad rakhta hai), Proxy/Application gateway (poora message check).
  Firewall = rules tra c filter: Packet-filter, Stateful, Proxy/Application gateway
- DMZ (Demilitarized Zone) — firewall ke andar/bahar ka beech ka area jahan public servers (web/mail) rakhte hain.
  DMZ — public servers wŮ
- NAT — private ↔ public IP translation: Static (1-1 fixed), Dynamic (pool se), Overload/PAT (ek public IP, alag-alag ports).
  NAT — private ↔ public: Static, Dynamic, Overload/PAT
- NAT jaate packet mein source IP (aur zaroorat par port) badalta hai — destination nahi.
  NAT packet Ȼource IP — destination ᵾ
- Firewall = network security ki pehli deewar; antivirus/IDS alag hote hain.
  Firewall = security ɬ

Device	Kaam	Layer
Gateway	protocol conversion	5–7 (ya sab)
Packet-filter firewall	header rules (IP, port)	3–4
Stateful firewall	connection state tracking	3–4
Proxy firewall	poora message inspect	7
NAT	private ↔ public IP	3

Q327. Which type of firewall inspects packets at the network layer and decides based on IP addresses?

- firewall network layer packets ɬ IP addresses 3 ŋ ?

(A) Packet Filtering Firewall (B) Stateful Firewall

(C) Proxy Firewall (D) Application Firewall

[Source: Repo: BCI Subject Test-9 (BCI/SCI Batch)] · Easy

(A) Packet Filtering Firewall

Nukta: Firewall types: Packet filtering (L3/L4, stateless) → Stateful (tracks sessions) → Proxy/Application gateway (L7) → Circuit-level gateway (L5).

Firewall types: Packet filtering (L3/L4, stateless) → Stateful (sessions) → Proxy/Application gateway (L7) → Circuit-level gateway (L5)

Q328. Which firewall acts as an intermediary between the client and server at the application layer?

- firewall application layer client server (intermediary) ?

(A) Packet Filtering Firewall (B) Stateful Firewall

(C) Proxy Firewall (D) Circuit-level Gateway

[Source: Repo: BCI Subject Test-9 (BCI/SCI Batch)] · Easy

(C) Proxy Firewall

Nukta: Proxy = L7 intermediary (deep inspection, hides internals). Circuit gateway = L5 (session/relay).

Proxy = L7 intermediary (, internals) Circuit gateway = L5 (session/relay)

Q329. If two systems use DIFFERENT protocols, which device is used to link them?

systems protocols - device ?

(A) Repeater (B) Gateway

(C) Bridge (D) Hub

[Source: Repo: CN Topic Test-01 (UNIT TEST-1)] · Medium

(B) Gateway

Nukta: Gateway = protocol converter (L7). Bridge = similar LANs, same protocols (L2). Repeater/Hub = signal level (L1).

Gateway = protocol converter (L7) Bridge = LANs, protocols (L2) Repeater/Hub = signal level (L1)

Q330. Which device can operate across all 7 layers of the OSI model?

- device OSI model 7 layers ?

(A) Router (B) Hub

(C) Gateway (D) Switch

[Source: Repo: CN Topic Test 6 (UNIT TEST-6)] · Medium

(C) Gateway

Nukta: Gateway = 'all 7 layers' device. Hub 1, switch/bridge 2, router 3 — exam one-liner.

Gateway = ' 7 layers' device Hub 1, switch/bridge 2, router 3 — exam one-liner

Q331. Devices on one network can communicate with devices on another network via:

network devices network devices

(A) File server (B) Utility server

(C) Printer server (D) Gateway

[Source: SSC CGL Internet & Networking (via EduRev bank)] · Medium

(D) Gateway

Nukta: Gateway = network-to-network doorway (also default gateway = router IP used to exit the LAN).

Gateway = network-to-network (default gateway = LAN router IP)

Q332. What is the role of a firewall in computer networks?

Computer networks Øirewall ɬ ɜ Ħ ?

(A) Managing IP addresses (B) Filtering and controlling network tra c

(C) Resolving domain names (D) Assigning MAC addresses dynamically

[Source: 30 important CN MCQs (Medium blog bank)] · Medium

(B) Filtering and controlling network tra c

Nukta: Firewall = tra c filter between trusted & untrusted networks (screening router, DMZ...). DHCP = IP management; DNS = names.

Firewall = trusted untrusted networks tra c filter (screening router, DMZ...) DHCP = IP management; DNS = names

Q333. What is the purpose of a proxy server in networking?

Networking Øroxy server ṅ Ħ ?

(A) Managing IP addresses (B) Filtering and forwarding network tra c

(C) Assigning MAC addresses dynamically (D) Encrypting network communication

[Source: 30 important CN MCQs (Medium blog bank)] · Medium

(B) Filtering and forwarding network tra c

Nukta: Proxy = filter + forward + cache + hide client. Firewall = block/allow by rules. Both are security/tra c devices.

Proxy = filter + forward + cache + client ɓ Firewall = rules block/allow Øecurity/tra c devices ʐ

Q334. Which is a network security system designed to protect a trusted private network from an untrusted outside network to which it is connected?

- network security system trusted private network õ untrusted outside network ɠ ?

(A) Bridge (B) Cookies

(C) Firewall (D) SSL

[Source: Random Test- Rajasthan PYQ (OCR), p6 — Q110 (options reconstructed from screenshot)] · Medium

(C) Firewall

Nukta: Rajasthan PYQ (verified key: C) — the classic firewall definition. Compare: bridge = L2 link, cookie = web state, SSL = encryption.

Rajasthan PYQ (verified key: C) — firewall ɬ classic ɓ : bridge = L2 link, cookie = web state, SSL = encryption

Q335. What kind of NAT maps one private IP to exactly one public IP?

ɜ ų NAT private IP ɻ public IP ?

(A) Dynamic NAT (B) Static NAT

(C) PAT (Port Address Translation) (D) Port Forwarding

[Source: Repo: CN Topic Test 6 (UNIT TEST-6)] · Medium

(B) Static NAT

Nukta: NAT types: Static = 1↔1 permanent. Dynamic = pool mapping. PAT/NAT-overload = many↔1 using ports.

NAT types: Static = 1↔1 ľ Dynamic = pool mapping PAT/NAT-overload = ports many↔1

Q336. Which of the following is NOT a networking device?

३ K Ȼ - networking device ४ ?

(A) Gateway (B) Linux

(C) Router (D) Firewall

[Source: InstrumentationTools 150 CN MCQs] · Hard

(B) Linux

Nukta: Device vs OS: hardware devices (hub/switch/router) vs software (OS, antivirus) vs firmware-based security devices (firewall).

Device OS: hardware devices (hub/switch/router) software (OS, antivirus) firmware-based security devices (firewall)

Q337. Which of the following is NOT a networking device? (Hub / Switch / Bridge / None of these)

३ K Ȼ - networking device ४ ? (Hub / Switch / Bridge / Ȼ ŋ ४)

(A) Hub (B) Switch

(C) Bridge (D) None of these

[Source: Repo: CN Topic Test-01 (UNIT TEST-1)] · Hard

(D) None of these

Nukta: Mirror question of the 'Linux' variant — read the option SET carefully: here everything listed IS a device.

'Linux' ų Î ŋ — option SET Ⅱ Ȼ á devices ʐ

TOPIC 5 — Transmission Medium

48 MCQs • Theory → Questions (Easy → Medium → Hard)

Ω Samjh — Section A — Signals & Performance (Encoding, Noise, Bandwidth, Data Rate)

Line coding (Manchester, 8B/10B, scramblers), impairments (attenuation/crosstalk/noise), signal parameters, data-rate & Shannon/Nyquist calculations.

Signal kya hota hai (analog/digital), data kaise encode hota hai, noise kya bigadta hai, aur bandwidth/datarate ke formulas — yeh topic ke core concepts hain.

Signal ₹ (analog/digital), data encode , noise ₹ ³ , bandwidth/data rate formulas

- Analog signal continuous (sine wave), Digital signal discrete (0/1 levels); dono analog ya digital medium par chal sakte hain.
  Analog signal (sine wave), Digital signal discrete (0/1); Ø medium ২
- Line coding: Manchester = har bit ke beech transition (self-synchronizing, Ethernet); NRZ = level constant; 8B/10B, scrambler = DC balance ke liye.
  Line coding: Manchester = bit Ø transition (self-synchronizing); NRZ; 8B/10B
- Impairments: Attenuation (signal kamzor), Distortion (shape badalna), Noise (thermal, crosstalk, impulse).
  Impairments: Attenuation ( ), Distortion (³), Noise (thermal, crosstalk, impulse)
- Bandwidth = frequency range (Hz) — medium ki capacity; data rate = bits per second (bps).
  Bandwidth = frequency range (Hz); data rate = bps
- Nyquist: $2 \times B \times \log_2(L)$ bps (noiseless channel, L signal levels).
  Nyquist: $2 \times B \times \log_2(L)$ bps (noiseless channel)
- Shannon: $C = B \times \log_2(1 + S/N)$ bps (noisy channel); $SNR(dB) = 10 \times \log_{10}(S/N)$.
  Shannon: $C = B \times \log_2(1 + S/N)$; $SNR(dB) = 10 \times \log_{10}(S/N)$
- Attenuation loss dB mein: $10 \log_{10}(P_{out}/P_{in})$; repeater se signal restore hota hai.
  Attenuation loss dB Ø $10 \log_{10}(P_{out}/P_{in})$; repeater restore

Formula	Kab use	Formula
Nyquist	noiseless channel	$2B \log_2(L)$
Shannon	noisy channel	$B \log_2(1+S/N)$
SNR (dB)	signal-to-noise	$10 \log_{10}(S/N)$
Attenuation (dB)	signal loss	$10 \log_{10}(P_2/P_1)$

Q338. 'Manchester Encoding' and 'Di erential Manchester Encoding' are signal encoding techniques used at which layer?

'Manchester Encoding' 'Di erential Manchester Encoding' signal encoding techniques ³ layer ২

(A) Physical Layer (B) Data Link Layer
(C) Network Layer (D) Presentation Layer

[Source: Repo: BCI Subject Test-8 (BCI/SCI Batch)] · Easy

(A) Physical Layer

Nukta: Physical layer = signalling & encoding (Manchester = mid-bit transition, self-clocking).

Physical layer = signalling encoding (Manchester = mid-bit transition, self-clocking)

Q339. A channel has high bandwidth but su ers from high noise. Which encoding scheme gives better error resistance?

channel ɬ bandwidth ɻ ɜ noise õ - encoding scheme error resistance ?

(A) NRZ (B) Manchester

(C) AMI (D) ASK

[Source: Repo: BCI Subject Test-8 (BCI/SCI Batch)] · Easy

(B) Manchester

Nukta: Manchester = self-clocking + DC-free + strong error visibility. NRZ = long runs → clock loss in noise.

Manchester = self-clocking + DC-free + error visibility NRZ = runs → noise Ȼlock loss

Q340. Reduction in signal strength over distance is called:

Ȧ signal strength Ȼ Ħ ʔ

(A) Distortion (B) Noise

(C) Attenuation (D) Interference

[Source: Repo: BCI Subject Test-9 (BCI/SCI Batch) · also CN Topic Test-04] · Easy

(C) Attenuation

Nukta: Impairments: Attenuation (strength loss), Distortion (shape change), Noise (unwanted signals). Attenuation ↑ with distance & frequency.

Impairments: Attenuation (strength loss), Distortion (shape change), Noise (unwanted signals) Attenuation ↑ distance frequency

Q341. In a copper cable, attenuation increases with frequency. Which change will REDUCE attenuation?

Copper cable Ȼttenuation frequency - attenuation ?

(A) Increase cable length (B) Use lower-category UTP

(C) Use thicker copper core (D) Increase signal frequency

[Source: Repo: BCI Subject Test-8 (BCI/SCI Batch)] · Easy

(C) Use thicker copper core

Nukta: Reduce attenuation: shorter cable, thicker/better copper, higher category cable, lower frequency, or add repeaters/amplifiers.

Attenuation : cable, / copper, Ī Ż cable, frequency, repeaters/amplifiers Ȼ

Q342. Which noise is caused by electromagnetic coupling between nearby cables?

ɬ cables electromagnetic coupling - noise ?

(A) Thermal noise (B) Crosstalk

(C) Impulse noise (D) Quantization noise

[Source: Repo: BCI Subject Test-9 (BCI/SCI Batch)] · Easy

(B) Crosstalk

Nukta: Noise types: Thermal (random electron motion), Crosstalk (pair-to-pair coupling), Impulse (sudden spikes), Quantization (PCM sampling).

Noise types: Thermal (random electron motion), Crosstalk (pair-to-pair coupling), Impulse (sudden spikes), Quantization (PCM sampling)

Q343. Which impairment cannot be completely eliminated but only minimised?

- impairment Ĵ ʁ3 , 3 ?

(A) Attenuation (B) Distortion

(C) Noise (D) Delay

[Source: Repo: BCI Subject Test-9 (BCI/SCI Batch)] · Easy

(C) Noise

Nukta: Attenuation/distortion can be fixed with repeaters/equalisers; noise is fundamental (thermal floor) — only minimised.

Attenuation/distortion repeaters/equalisers ɹ ʐnoise (thermal floor) —

Q344. Which parameter of an analog signal represents the number of cycles per second?

Analog signal - parameter ɥ3 cycles ɻ ħ ɳ ?

(A) Amplitude (B) Phase

(C) Frequency (D) Wavelength

[Source: Repo: BCI Subject Test-9 (BCI/SCI Batch)] · Easy

(C) Frequency

Nukta: Signal parameters: Amplitude (height), Frequency (cycles/sec), Phase (start position), Wavelength (distance/cycle = speed/frequency).

Signal parameters: Amplitude (ɲ Frequency (cycles/sec), Phase (Ę ħl' 3), Wavelength (Ȧ / cycle = speed/frequency)

Q345. Which of the following Ethernet encodings has the highest redundancy?

3 ĸ Ȼ - Ethernet encoding ɻ redundancy ɟ ?

(A) 1B2B (B) 4B5B

(C) 5B6B (D) 8B10B

[Source: Unit Test-4 Solution (OCR), p9 (options recovered from source)] · Medium

(A) 1B2B

Nukta: Redundancy (overhead): 1B2B 100% > 4B5B/8B10B 25% > 5B6B 20%. More redundancy = better clocking/error properties, more bandwidth.

Redundancy (overhead): 1B2B 100% > 4B5B/8B10B 25% > 5B6B 20% ŏ redundancy = clocking/ error properties, ŏ bandwidth

Q346. Which encoding technique is used to transmit signals in Gigabit Ethernet over fiber optic medium?

Fiber optic medium Gigabit Ethernet Ø signal ʃ - encoding technique ?

(A) Di erential Manchester encoding (B) Non-Return-to-Zero (NRZ)

(C) 4B/5B (D) 8B/10B

[Source: Unit Test-4 Solution (OCR), p30 (options recovered from source)] · Medium

(D) 8B/10B

Nukta: 100BASE-TX uses 4B/5B + MLT-3; 1000BASE-X uses 8B/10B; 10GBASE-R uses 64B/66B. Ye exam favourite ladder hai.

100BASE-TX Ø4B/5B + MLT-3; 1000BASE-X Ø8B/10B; 10GBASE-R Ø64B/66B exam favourite ladder

Q347. _____ signals are discrete and are generated by digital modulation.

_____ signals discrete ʐ digital modulation ʐ

(A) Analog (B) Digital

(C) Noise (D) Bandwidth

[Source: Random Test- Rajasthan PYQ (OCR), p5 — Q108 (options recovered)] · Medium

(B) Digital

Nukta: Rajasthan PYQ verified key (B). Analog = continuous; Digital = discrete. Ye sabse basic signal distinction hai.

Rajasthan PYQ verified key (B) Analog = continuous; Digital = discrete signal 3

Q348. Quantizing noise occurs in:

Quantizing noise ३ ₡ ?

(A) PCM (Pulse Code Modulation) (B) TDM

(C) FDM (D) PPM

[Source: Random Test- Rajasthan PYQ (OCR), p10 — Q39 (options recovered)] · Medium

(A) PCM (Pulse Code Modulation)

Nukta: Rajasthan PYQ verified key (A). PCM = sampling + quantizing + encoding → quantizing error = quantization noise.

Rajasthan PYQ verified key (A) PCM = sampling + quantizing + encoding → quantizing error = quantization noise

Q349. The e ective bandwidth of a communication channel is primarily limited by:

Communication channel ɬe ective bandwidth ħ ę ३ ३ ?

(A) Protocol overhead (B) Noise and attenuation

(C) File size (D) MAC address filtering

[Source: Repo: CN Topic Test-04 (UNIT TEST-4)] · Medium

(B) Noise and attenuation

Nukta: E ective bandwidth is a physics/SNR limit. Protocol overhead reduces goodput but is not the channel's fundamental limit.

E ective bandwidth physics/SNR Protocol overhead goodput channel ɬ ४

Q350. What technique allows multiple signals to share the same communication channel?

- technique ŋsignals communication channel share ?

(A) Multiplexing (B) Phase modulation

(C) Analog modulation (D) Digital modulation

[Source: Repo: CN Topic Test-02 (UNIT TEST-2)] · Medium

(A) Multiplexing

Nukta: Multiplexing = many share one channel. FDM = frequency bands, TDM = time slots, WDM = light wavelengths, CDM = codes.

Multiplexing = ŋ channel share ₡FDM = frequency bands, TDM = time slots, WDM = light wavelengths, CDM = codes

Q351. Communication using a modulated carrier to apply a data signal to a transmission medium in analog form is:

Modulated carrier data signal analog ę ₡ransmission medium ₦ ?

(A) Broadband networking (B) Baseband networking

(C) Modem (D) Local area network

[Source: IBPS RRB PO Computer Awareness PYQ (via Meritnotes)] · Medium

(A) Broadband networking

Nukta: Baseband = digital signalling (Manchester etc.). Broadband = analog carriers + modulation (FDM channels).

Baseband = digital signalling (Manchester ६) Broadband = analog carriers + modulation (FDM channels)

Q352. The amount of data transmitted in a fixed amount of time is called:

data

(A) Refresh rate (B) Resolution

(C) Bandwidth (D) Frequency

[Source: Unit Test-4 Solution (OCR), p23] · Medium

(C) Bandwidth

Nukta: Strictly: THROUGHPUT = data actually sent per second; BANDWIDTH = maximum capacity. Banks often call throughput 'bandwidth' — read options carefully.

: THROUGHPUT = data; BANDWIDTH = capacity Banks throughput 'bandwidth' options

Q353. _____ signals are discrete and are generated by digital modulation.

_____ signals discrete digital modulation

(A) Digital (B) Noise

(C) Bandwidth (D) Analog

[Source: RSMSSB Informatics Assistant 2023 (via examsmantra solved paper)] · Medium

(A) Digital

Nukta: Discrete = (digital); Continuous = (analog).

Discrete = (digital); Continuous = (analog).

Q354. Communication using a modulated carrier to apply a data signal to a transmission medium in analog form is called:

Modulated carrier data signal analog transmission medium :

(A) Broadband communication (B) Baseband communication

(C) Modem (D) Local area network

[Source: IBPS RRB PO Computer Awareness PYQ (via meritnotes)] · Medium

(A) Broadband communication

Nukta: Baseband = digital, ; Broadband = analog, modulated.

Baseband = digital, ; Broadband = analog, modulated.

Q355. Identify the odd one out: Optical fibre, Twisted pair wire, Microwaves, Hertz

Optical fibre, Twisted pair wire, Microwaves, Hertz

(A) Optical fibre (B) Twisted pair wire

(C) Microwaves (D) Hertz

[Source: RRB NTPC previous-year computer banks (via adda247 50-PYQ PDF)] · Medium

(D) Hertz

Nukta: Media = the path; Hertz = the frequency/speed measure. This distinction is what is being asked.

Media = ; Hertz = / .

Q356. The primary use of scramblers in line coding schemes is to:

Line coding schemes ⌀scramblers ħ ⌂ ?

(A) Make the data look random (B) Compress transmission data

(C) Increase signal bandwidth (D) Decrease signal bandwidth

[Source: RPSC Programmer 2024 Paper-1 (o cial key)] · Hard

(A) Make the data look random

Nukta: O cial RPSC 2024 key. Scrambler = random-looking line signal (clock recovery). Not compression, not encryption.

O cial RPSC 2024 key Scrambler = random-looking line signal (clock recovery) Compression ⌘, encryption ⌘

Q357. The period of a signal is 100 ms. What is its frequency in kilohertz?

signal ↓period 100 ms ↓frequency kilohertz ⌀ ?

(A) 10 ¹ kHz (B) 10 ² kHz

(C) 10 ³ kHz (D) 10 ⁴ kHz

[Source: Repo: CN Topic Test-04 (UNIT TEST-4)] · Hard

(B) 10 ² kHz

Nukta: f = 1/T. 100 ms = 0.1 s → 10 Hz. Convert Hz→kHz by dividing by 1000.

f = 1/T 100 ms = 0.1 s → 10 Hz Hz→kHz ∫ 1000 ⌀

Q358. If the channel bandwidth is limited to 6 kHz and the signal-to-noise ratio is 16, what is the channel capacity?

ε channel bandwidth 6 kHz ∋ signal-to-noise ratio 16 , channel capacity ∃ ?

(A) 15.15 kbps (B) 24.74 kbps

(C) 30.12 kbps (D) 52.18 kbps

[Source: Random Test- Rajasthan PYQ (OCR), p30 — Q45 (options recovered)] · Hard

(B) 24.74 kbps

Nukta: Rajasthan PYQ verified key (B). Shannon: $C = B \log_2(1+SNR)$. Nyquist: $C = 2B \log_2(L)$ (noiseless).

Rajasthan PYQ verified key (B) Shannon: $C = B \log_2(1+SNR)$ Nyquist: $C = 2B \log_2(L)$ (noiseless)

Q359. Given a bit rate of b bits/sec, the time required to send 16 bits is:

ε bit rate b bits/sec , 16 bits ⌀ ?

(A) 16 × b sec (B) 16/b sec

(C) 16b sec (D) b/16 sec

[Source: Repo: CN Topic Test-04 (UNIT TEST-4)] · Hard

(B) 16/b sec

Nukta: Time = size/rate. 1000 bits at 100 bps = 10 s. Units: bits ÷ (bits/sec) = sec.

Time = size/rate 1000 bits at 100 bps = 10 s Units: bits ÷ (bits/sec) = sec

Q360. An analog signal carries 4 bits per signal unit. If 1000 signal units are sent per second, what are the baud rate and bit rate?

analog signal ų signal unit 4 bits ę ų 1000 signal units , baud rate bit rate ħ ?

(A) 1000 bauds/sec, 4000 bps (B) 1000 bauds/sec, 500 bps

(C) 4000 bauds/sec, 1000 bps (D) 2000 bauds/sec, 1000 bps

[Source: Repo: CN Topic Test-04 (UNIT TEST-4)] · Hard

(A) 1000 bauds/sec, 4000 bps

Nukta: Bit rate = Baud rate × bits per signal unit. Baud = signalling speed; bps = data speed.

Bit rate = Baud rate × ų signal unit bits Baud = signalling speed; bps = data speed

Q361. Consider a noiseless channel with a bandwidth of 5000 Hz transmitting a signal with two signal levels. The maximum bit rate is:

noiseless channel ę ł bandwidth 5000 Hz , signal levels transmission . ɟ bit rate :

(A) 2500 bps (B) 10000 bps

(C) 5000 bps (D) 20000 bps

[Source: UGC NET Computer Science Jan 2025 (via samagracs solved paper)] · Hard

(B) 10000 bps

Nukta: UGC NET Jan 2025 o cial key (ii).

: 2B $\log_2$L (noiseless) B $\log_2$(1+S/N) (noisy — Shannon).

UGC NET Jan 2025 o cial key (ii). : 2B $\log_2$L (noiseless) B $\log_2$(1+S/N) (noisy — Shannon).

Q362. Which of the following is NOT a type of broadband internet connection?

ɜ ĸ - broadband internet connection ų ʁ ?

(A) Cable (B) DSL

(C) Satellite (D) Dial-up

[Source: SBI PO Computer Awareness PYQ (via bankersadda)] · Hard

(D) Dial-up

Nukta: Broadband = fast internet (Cable/DSL/Fibre/Satellite); Dial-up = Ř .

Broadband = fast internet (Cable/DSL/Fibre/Satellite); Dial-up = ľ .

Ώ Samjh — Section B — Guided Media (UTP, Coaxial, Fiber Optic)

Twisted pair, coaxial and fiber — properties, connectors, distances, termination, EMI behaviour, total internal reflection.

Guided media = wired media: twisted pair, coaxial, fiber optic — teeno ki properties exam mein classic.

Guided media = medium: twisted pair, coaxial, fiber optic

- UTP (Unshielded Twisted Pair) — 4 wire pairs twisted; sasta, aasan; 100 m tak; Cat5e/Cat6/Cat7/Cat8 grades; crosstalk ka issue.

  UTP — 4 twisted pairs; ľ ; 100 m; Cat5e–Cat8; crosstalk ł ľ

- STP = shielded twisted pair — extra metal shield, EMI kam, mehenga.

  STP — extra shield, EMI ,

- Coaxial — central copper + insulation + shield (BNC connector); 50 ohm (thin/thick Ethernet) aur 75 ohm (cable TV).

  Coaxial — central copper + shield (BNC); 50 Ω (Ethernet), 75 Ω (cable TV)

- Fiber optic — glass/plastic core mein light pulses: sabse tez, EMI immune, door tak; Single-mode (ek ray, lambi doori) vs Multi-mode (kai rays, chhoti doori).

  Fiber optic — ł pulses: , EMI immune; Single-mode vs Multi-mode

- Fiber connectors: SC, LC, ST; fiber mehenga hai aur termination mushkil.

  Fiber connectors: SC, LC, ST; fiber termination ł

- Comparison: cost — UTP < coax < fiber; speed/distance — fiber sabse aage; noise immunity — fiber best.

  : UTP < coax < fiber; ł /Å — fiber ŧg

Medium	Speed/Distance	Feature
UTP	100 m, 1–10 Gbps	Ř , crosstalk
Coaxial	~500 m	shielded, cable TV
Fiber (multi-mode)	~2 km	light, tez
Fiber (single-mode)	100 km+	long haul, sabse tez

Q363. Optical fiber uses reflection to guide light through a channel. At total internal reflection, the angle of incidence is _____ the critical angle.

Optical fiber ų channel Ȼ ǥ reflection Total internal reflection Ȼ angle of incidence, critical angle _____

(A) Equal to (B) Less than

(C) Greater than (D) Less than or equal to

[Source: Repo: CN Topic Test-04 (UNIT TEST-4) — recovered from BCI T8 Q33 options] · Easy

(C) Greater than

Nukta: TIR rule: angle of incidence > critical angle (denser → rarer medium). Ye fiber propagation ka base hai.

TIR rule: angle of incidence > critical angle (denser → rarer medium) fiber propagation

Q364. A network uses coaxial cable but su ers from signal reflection. What is the most likely cause?

network coaxial cable ³ signal reflection ⁶ ³ Ħ ?

(A) EMI (B) Incorrect termination resistance

(C) Crosstalk (D) Low bandwidth

[Source: Repo: BCI Subject Test-8 (BCI/SCI Batch)] · Easy

(B) Incorrect termination resistance

Nukta: Coax rule: proper termination (BNC/terminator) prevents reflections. Un-terminated bus = standing waves & echoes.

Coax rule: termination (BNC/terminator) reflections Un-terminated bus = standing waves echoes

Q365. Which transmission medium is primarily used in modern internet backbone networks?

³ internet backbone networks Ȼ ħ ę - transmission medium ?

(A) Twisted pair cable (B) Coaxial cable

(C) Single mode fiber optic cable (D) Microwave links

[Source: Repo: BCI Subject Test-8 (BCI/SCI Batch)] · Easy

(C) Single mode fiber optic cable

Nukta: Backbone = single-mode fiber (undersea/terrestrial cables). LAN = UTP. Old cable TV = coax. Remote areas = microwave/satellite.

Backbone = single-mode fiber (undersea/terrestrial cables) LAN = UTP cable TV = coax Ȧ ľ wŮ = microwave/satellite

Q366. Which cable has the maximum bandwidth?

³ cable ⁴ bandwidth ¦ ?

(A) Coaxial (B) Twisted pair

(C) Optical fiber (D) USB cable

[Source: Repo: BCI_SCI Subject Test-7 (BCI/SCI Batch)] · Easy

(C) Optical fiber

Nukta: Bandwidth order: Fiber >> Coax > Twisted pair. USB is a peripheral bus, not a network cable type here.

Bandwidth ť : Fiber >> Coax > Twisted pair USB peripheral bus , network cable type ⁸

Q367. Assertion (A): Fiber optic cable is immune to electromagnetic interference (EMI). Reason (R): It transmits data in the form of light pulses rather than electrical signals. Which option is correct?

Assertion (A): Fiber optic cable electromagnetic interference (EMI) Ħ Reason (R): data electrical signals light pulses ę Ȼ - ³ Í ?

(A) Both (A) and (R) are true, and (R) is the correct explanation of (A) (B) Both (A) and (R) are true, but (R) is NOT the correct explanation

(C) (A) is true, but (R) is false (D) (A) is false, but (R) is true

[Source: Repo: BCI Subject Test-8 (BCI/SCI Batch) — Assertion-Reason] · Medium

(A) Both (A) and (R) are true, and (R) is the correct explanation of (A)

Nukta: Fiber = light pulses → EMI-proof + tap-proof + huge bandwidth. Copper = electrical → EMI/crosstalk prone.

Fiber = light pulses → EMI-proof + tap-proof + ³ bandwidth Copper = electrical → EMI/crosstalk ų

Q368. Which statements are TRUE for fiber optic cables? I. Immune to tapping II. High bandwidth III. Requires electrical grounding IV. Works using total internal reflection

Fiber optic cables ∫ - ƺ I. Tapping Ħ II. Ī bandwidth III. Electrical grounding ε IV. Total internal reflection

(A) I, II and IV (B) II and III

(C) I, III and IV (D) I and II

[Source: Repo: BCI Subject Test-8 (BCI/SCI Batch) — Q33] · Medium

(A) I, II and IV

Nukta: Fiber truths: immune to tapping & EMI, highest bandwidth, TIR propagation. Myths: grounding (NO), a ected by lightning (NO).

Fiber truths: tapping EMI Ħ , Ł bandwidth, TIR propagation Myths: grounding ( ), lightning ( )

Q369. A company needs long-distance communication (5-10 km), high data rate and immunity from electromagnetic interference. Which medium is best?

À (5-10 km) , Ī data rate EMI ħĦ ε - medium ɬg ?

(A) UTP (B) Coaxial cable

(C) Single-mode fiber (D) Multi-mode fiber

[Source: Repo: BCI Subject Test-8 (BCI/SCI Batch)] · Medium

(C) Single-mode fiber

Nukta: Fiber choices: Multi-mode = short (LANs, few hundred m). Single-mode = long haul (km) + highest rate. Copper = short & EMI-prone.

Fiber ɜ Í : Multi-mode = À (LANs, m) Single-mode = (km) + Ī rate Copper = EMI-ų

Q370. A UTP cable used in Gigabit Ethernet shows high delay skew. What is the cause?

Gigabit Ethernet Ȼ UTP cable Ȼdelay skew ɻ Ħ ?

(A) Di erent twist rates in the cable pairs (B) Excessive EMI

(C) Fiber splice error (D) Wrong wireless channel

[Source: Repo: CN Topic Test-04 (UNIT TEST-4)] · Medium

(A) Di erent twist rates in the cable pairs

Nukta: Delay skew = pair-to-pair propagation time di erence (tight twist → shorter wire → less skew). Cat6 has tighter specs for Gigabit.

Delay skew = pair-to-pair propagation time (tight twist → wire → skew) Cat6 ȻGigabit ∫ specs ƺ

Q371. Which of the following best explains why fiber optic cables have a lower bit error rate than copper cables?

Fiber optic cables ɬcopper cables ɬ Ȼ bit error rate ɬ Ğʳ ê ħ - ?

(A) Higher signal power (B) Immunity to electromagnetic interference

(C) Simpler modulation techniques (D) Larger physical size

[Source: Repo: CN Topic Test-04 (UNIT TEST-4)] · Medium

(B) Immunity to electromagnetic interference

Nukta: Fiber wins on: EMI immunity, low attenuation, huge bandwidth, tap security.

Fiber ȻŘ : EMI immunity, attenuation, ɜ bandwidth, tap security

Q372. Which transmission medium works on the principle of total internal reflection?

- transmission medium total internal reflection ʃ á ?

(A) Optical fiber cable (B) Shielded twisted pair

(C) Unshielded twisted pair (D) Coaxial cable

[Source: Repo: CN Topic Test-04 (UNIT TEST-4)] · Medium

(A) Optical fiber cable

Nukta: Fiber = TIR. Copper media use electrical signals (electromagnetic fields), not light reflection.

Fiber = TIR Copper media electrical signals (electromagnetic fields) ʐų reflection ʁ

Q373. Crosstalk in twisted pair cables is typically minimised by:

Twisted pair cables Ȼrosstalk ɜ ?

(A) Increasing the power level (B) Increasing the twist rate

(C) Adding shielding to the core (D) Using longer cables

[Source: Repo: CN Topic Test-04 (UNIT TEST-4)] · Medium

(B) Increasing the twist rate

Nukta: Anti-crosstalk tricks: tighter twists, pair separation, shielding (STP), lower power.

Anti-crosstalk tricks: tighter twists, pair separation, shielding (STP), power

Q374. Which connector is commonly used for coaxial cable in physical networks?

Physical networks Ȼoaxial cable ʃ ij - connector ?

(A) RJ-45 (B) USB-C

(C) BNC (D) HDMI

[Source: Repo: CN Topic Test-04 (UNIT TEST-4)] · Medium

(C) BNC

Nukta: Connectors: RJ-45 (UTP Ethernet), BNC (coax/Thinnet), SC/ST/LC (fiber), RJ-11 (telephone).

Connectors: RJ-45 (UTP Ethernet), BNC (coax/Thinnet), SC/ST/LC (fiber), RJ-11 (telephone)

Q375. Which transmission medium is best for a network exposed to heavy electrical interference?

electrical interference network ʃ - transmission medium ʁg ?

(A) Unshielded twisted pair (B) Optical fiber

(C) Coaxial cable (D) Microwave

[Source: Repo: CN Topic Test-01 (UNIT TEST-1)] · Medium

(B) Optical fiber

Nukta: EMI-heavy areas (factories, power plants) → fiber. 'Immune to EMI' = keyword for fiber.

EMI- wŮ (, power plants) → fiber 'Immune to EMI' = fiber ʃ keyword

Q376. Which of the following CANNOT be used as a medium for 802.3 Ethernet?

ɜ ĸ Ȼ - 802.3 Ethernet ʃ medium ʁɜ ?

(A) Thin coaxial cable (B) Twisted pair cable

(C) Microwave link (D) Fiber optic cable

[Source: Repo: CN Topic Test-01 (UNIT TEST-1)] · Medium

(C) Microwave link

Nukta: 802.3 = Ethernet (wired: 10Base2/5, 10Base-T, 100Base-TX, 1000Base-X...). 802.11 = Wi-Fi (wireless).

802.3 = Ethernet (wired: 10Base2/5, 10Base-T, 100Base-TX, 1000Base-X...) 802.11 = Wi-Fi (wireless)

Q377. What is the typical maximum length of a UTP cable in Ethernet before signal degradation?

Signal degradation Ethernet UTP cable

?

(A) 50 meters (B) 100 meters

(C) 150 meters (D) 200 meters

[Source: Repo: CN Topic Test-04 (UNIT TEST-4)] · Hard

(B) 100 meters

Nukta: Copper distance limits: UTP Ethernet 100 m, Thinnet 185 m, Thicknet 500 m, Fiber (multi-mode) ~550 m-2 km, (single-mode) tens of km.

Copper distance limits: UTP Ethernet 100 m, Thinnet 185 m, Thicknet 500 m, Fiber (multi-mode) ~550 m-2 km, (single-mode)

Q378. Which of the following is NOT a type of transmission medium?

transmission medium ?

(A) Coaxial cable (B) Fiber optics

(C) NIC (D) Twisted pair

[Source: Repo: CN Topic Test-04 (UNIT TEST-4)] · Hard

(C) NIC

Nukta: Media = guided (twisted pair, coax, fiber) + unguided (radio, microwave, satellite, infrared). NIC = adapter.

Media = guided (twisted pair, coax, fiber) + unguided (radio, microwave, satellite, infrared) NIC = adapter

Ω Samjh — Section C — Unguided Media (Wireless)

Wi-Fi standards, infrared/microwave/Bluetooth, line-of-sight, CDMA sharing and its true/false statements.

Unguided media = wireless: radio, microwave, infrared, satellite — air mein signals.

Unguided media = wireless: radio, microwave, infrared, satellite

- Radio waves — omnidirectional, buildings se guzar sakte hain; AM/FM, wireless LAN.
  Radio waves — omnidirectional, Ȼ ʐ
- Microwave — unidirectional, line-of-sight (towers beech 50 km); satellite = space ka microwave repeater (geostationary 36,000 km).
  Microwave — line-of-sight; satellite = ə w repeater (36,000 km)
- Infrared — short range, walls se block ho jata hai (remote controls).
  Infrared — Ȧ , Ȼ Ȩ
- Bluetooth (IEEE 802.15) — PAN, 2.4 GHz, ~10 m; Zigbee — low power IoT.
  Bluetooth (802.15) — PAN, 2.4 GHz
- Wi-Fi (IEEE 802.11): a = 5 GHz/54 Mbps; b = 2.4/11; g = 2.4/54; n = 2.4+5/600 (MIMO); ac = 5 GHz/1+ Gbps; ax = Wi-Fi 6.
  Wi-Fi (802.11): a/b/g/n/ac/ax — 2.4 5 GHz bands
- Wireless sharing: FDMA, TDMA, CDMA — ek channel par kai users alag tareekon se.
  Wireless sharing: FDMA, TDMA, CDMA
- Wireless mein CSMA/CD kaam nahi karta (collision detect mushkil) — CSMA/CA use hota hai (Topic 6 mein).
  Wireless ȻSMA/CD ɓ — CSMA/CA (Topic 6)

Wireless Type	Range/Feature
Radio	omnidirectional, buildings se guzarta hai
Microwave	line-of-sight, towers ~50 km
Satellite	geostationary 36,000 km — long distance
Infrared	short range, walls block
Bluetooth	PAN, ~10 m, 2.4 GHz

Q379. Which of the following is a wireless networking technology?

ɜ ĸ Ȼ - wireless networking technology ?

(A) Ethernet (B) Wi-Fi

(C) Fiber Optic (D) DSL

[Source: Repo: BCI_SCI Subject Test-01 (CN and Security)] · Medium

(B) Wi-Fi

Nukta: Wireless family: Wi-Fi (802.11), Bluetooth (802.15), WiMAX (802.16), mobile (GSM/LTE/5G).

Wireless family: Wi-Fi (802.11), Bluetooth (802.15), WiMAX (802.16), mobile (GSM/LTE/5G)

Q380. Which wireless standard provides the highest data rate?

- wireless standard ₹ data rate ?

(A) 802.11b (B) 802.11g

(C) 802.11n (D) 802.11ac

[Source: Repo: CN Topic Test 6 (UNIT TEST-6)] · Medium

(D) 802.11ac

Nukta: Wi-Fi rates: b = 11 Mbps, g = 54 Mbps, n = 600 Mbps, ac = multi-Gbps. Order: b < g < n < ac.

Wi-Fi rates: b = 11 Mbps, g = 54 Mbps, n = 600 Mbps, ac = multi-Gbps ť : b < g < n < ac

Q381. Which type of wireless transmission requires line-of-sight?

३ ų ५ wireless transmission ङ line-of-sight ļ ?

(A) Wi-Fi (B) Infrared

(C) Radio (D) Bluetooth

[Source: Repo: CN Topic Test-04 (UNIT TEST-4)] · Medium

(B) Infrared

Nukta: IR = line-of-sight (TV remotes, IrDA). Radio/RF (Wi-Fi, Bluetooth) passes through walls (with loss).

IR = line-of-sight (TV remotes, IrDA) Radio/RF (Wi-Fi, Bluetooth) Ȼ ȥ(loss)

Q382. Which transmission medium is used in Bluetooth technology?

Bluetooth technology Ȼ - transmission medium ?

(A) Radio links (B) Microwave links

(C) VSAT communication (D) Fiber optic

[Source: Repo: CN Topic Test-04 (UNIT TEST-4)] · Medium

(A) Radio links

Nukta: Bluetooth = 2.4 GHz radio, WPAN (802.15), ~10-100 m range, frequency hopping.

Bluetooth = 2.4 GHz radio, WPAN (802.15), ~10-100 m range, frequency hopping

Q383. Which physical-layer technique allows multiple users to share the same bandwidth simultaneously?

- physical-layer technique ŋusers bandwidth - share ?

(A) TDMA (B) FDMA

(C) CDMA (D) ARQ

[Source: Repo: CN Topic Test-04 (UNIT TEST-4)] · Medium

(C) CDMA

Nukta: Sharing methods: FDMA = frequency bands, TDMA = time slots, CDMA = codes (same band+time).

Sharing methods: FDMA = frequency bands, TDMA = time slots, CDMA = codes (same band+time)

Q384. Which of the following statements is NOT true about microwaves?

Microwaves Ȼ ĸ Ȼ - ঌ ?

(A) They are electromagnetic waves with frequencies from 300 GHz to 400 THz (B) Propagation is line-of-sight

(C) Very high-frequency waves cannot penetrate walls (D) Use of certain portions of the band requires permission from authorities

[Source: Repo: CN Topic Test-04 (UNIT TEST-4) — options cleaned from source] · Hard

(A) They are electromagnetic waves with frequencies from 300 GHz to 400 THz

Nukta: Microwaves: 300 MHz-300 GHz, line-of-sight, no wall penetration, licensed bands. Exam asks the odd range out.

Microwaves: 300 MHz-300 GHz, line-of-sight, ঌ , licensed bands Exam range

Q385. Select the FALSE statement about CDMA technology: I. CDMA optimises the use of available bandwidth as it transmits over the entire frequency range. II. Each user in a CDMA system uses a di erent code to modulate their signal. III. CDMA o ers less security compared to GSM technology. IV. A SIM card is always required for a CDMA device to work.

CDMA technology Ȼ 3 : I. CDMA Ê bandwidth ɡ Ħ Ȼ frequency range transmit II. CDMA system Ȼ user signal modulate ɡ code III. CDMA, GSM ɬ Ȼ security IV. CDMA device ɡ SIM card ļ

(A) Only (I) (B) Only (II)

(C) (I) and (II) (D) (III) and (IV)

[Source: Random Test- Rajasthan PYQ (OCR), p11 — Q83 (statements recovered from screenshot)] · Hard

(D) (III) and (IV)

Nukta: Rajasthan PYQ verified key (D). True: I (spread over full band) and II (unique codes per user). False: III (CDMA more secure) and IV (SIM not mandatory).

Rajasthan PYQ verified key (D) : I (band spread) II (user unique code) : III (CDMA ɟ ɢw) IV (SIM 3 ŋ ᵬ)

TOPIC 6 — OSI Layers in Detail

193 MCQs • Theory → Questions (Easy → Medium → Hard)

Ω Samjh — Section A — Data Link Layer: Flow & Error Control

LLC/MAC sub-layers, flow control, ARQ protocols (Stop-and-Wait, Go-Back-N, Selective Repeat), error detection & correction (parity, CRC, checksum, Hamming) — basic to advanced.

Data link layer do sub-layers mein banti hai — LLC aur MAC; is layer par flow control aur error control ke protocols lagte hain.

Data link layer sub-layers Ȼ — LLC MAC; layer flow control error control protocols ȥ

- LLC (Logical Link Control) — upper layers ke saath interface, flow control; MAC (Media Access Control) — medium access + physical addressing.
  LLC — upper layers interface, flow control; MAC — medium access + MAC addressing
- Flow control = receiver ki capacity ke hisaab se data bhejna: Stop-and-Wait (ek frame, phir ACK) aur Sliding Window (kai frames ek saath).
  Flow control = receiver ɨw : Stop-and-Wait Sliding Window
- ARQ protocols: Stop-and-Wait ARQ; Go-Back-N (n tak frames, error par sab dobara); Selective Repeat (sirf galat frame dobara).
  ARQ: Stop-and-Wait; Go-Back-N ( ); Selective Repeat (ǧ ŋ)
- Error detection: Parity bit (single bit check), CRC (polynomial division — best), Internet checksum.
  Error detection: Parity, CRC (polynomial division), checksum
- Error correction: Hamming code (Hamming distance ≥ 3 for single-bit error); FEC (forward error correction) vs ARQ.
  Error correction: Hamming code; FEC vs ARQ
- Hamming distance = do codewords mein alag bits ki sankhya; error detect ke liye d ≥ e+1, correct ke liye d ≥ 2e+1.
  Hamming distance = ǧ Ȅ bits ɨ ħ ; detect: d ≥ e+1; correct: d ≥ 2e+1
- CRC: generator polynomial G(x) se data ko divide karna; remainder = CRC bits; receiver Ȼemainder 0 ɛ
- CRC x+1 factor hone par error detect hota hai (odd-bit conditions) — GATE ka classic.
  CRC x+1 factor error detect — GATE Ħ ǧ

Protocol	Window	Error par kya hota hai
Stop-and-Wait	1 frame	wahi frame dobara
Go-Back-N	n frames	error se aage ke sab dobara
Selective Repeat	n frames	sirf galat frame dobara

Q386. Which sub-layer of the Data Link Layer is responsible for flow control and error control?

Data Link Layer - sub-layer flow control error control ?

(A) MAC (B) LLC (Logical Link Control)

(C) PVC (D) None of the above

[Source: BCI Subject Test-8.pdf] · Easy

(B) LLC (Logical Link Control)

Nukta: DLL = MAC + LLC — : MAC = addressing + medium access, LLC = flow + error + interface.

DLL = MAC + LLC — : MAC = addressing + medium access, LLC = flow + error + interface.

Q387. Which of the following functions are performed by the Data Link Layer? (1) Framing (2) Error Detection (3) Flow Control (4) Path Determination

- Data Link Layer ? (1) Framing (2) Error Detection (3) Flow Control (4) Path Determination

(A) 1, 2 and 3 only (B) 1 and 4 only

(C) 2 and 4 only (D) 1, 2, 3 and 4

[Source: BCI Subject Test-9.pdf] · Easy

(A) 1, 2 and 3 only

Nukta: : DLL F-E-F (Framing, Error, Flow) + addressing; (routing) Network Layer .

: DLL F-E-F (Framing, Error, Flow) + addressing; (routing) Network Layer .

Q388. The Frame Check Sequence (FCS) field in an Ethernet frame is used for:

Ethernet frame Frame Check Sequence (FCS) field ?

(A) Flow control (B) Address resolution

(C) Error detection using CRC (D) Medium access

[Source: BCI Subject Test-9.pdf] · Easy

(C) Error detection using CRC

Nukta: FCS = CRC = error detection. Ethernet error correct , detect frame drop .

FCS = CRC = error detection. Ethernet error correct , detect frame drop .

Q389. A Stop-and-Wait ARQ channel has high latency and large bandwidth. What problem occurs?

Stop-and-Wait ARQ channel latency bandwidth ?

(A) High throughput (B) Low channel utilization

(C) CRC failure (D) Frame duplication

[Source: BCI Subject Test-8.pdf] · Easy

(B) Low channel utilization

Nukta: Bandwidth × Delay (BDP) , Stop-and-Wait .

Bandwidth × Delay (BDP) , Stop-and-Wait .

Q390. A Go-Back-N sender uses window size 4. Frame 2 is lost. What happens after the receiver receives frames 3 and 4?

Go-Back-N sender window size 4 . Frame 2 . Receiver frames 3 4 ?

(A) Receiver stores frames 3 and 4 in bu er
(B) Receiver sends ACK for frame 4
(C) Receiver discards frames 3 and 4 (D) Receiver acts like Selective Repeat

[Source: BCI Subject Test-8.pdf] · Easy

(C) Receiver discards frames 3 and 4

Nukta: GBN receiver dumb (bu er); SR receiver smart (bu er).

GBN receiver dumb (bu er); SR receiver smart (bu er).

Q391. A noisy channel uses a Hamming code with minimum distance 3. What can it guarantee?

noisy channel minimum distance 3 Hamming code guarantee ?

(A) Detect 2-bit errors (B) Correct 2-bit errors
(C) Correct 3-bit errors (D) Detect 4-bit errors

[Source: BCI Subject Test-8.pdf] · Easy

(A) Detect 2-bit errors

Nukta: d=3 → detect 2 correct 1 — 'correct 2' d ≥ 5 .

d=3 → detect 2 correct 1 — 'correct 2' d ≥ 5 .

Q392. Which of the following verifies the integrity of a message?

message integrity ?

(A) Message Encryption (B) Message Checksum
(C) Message Digest (D) None of these

[Source: BCI Subject Test-9.pdf] · Easy

(C) Message Digest

Nukta: Digest = - ; Encryption = , integrity .

Digest = - ; Encryption = , integrity .

Q393. Which of the following statements about MAC and LLC sublayers is TRUE? (1) MAC sublayer handles physical addressing (2) LLC sublayer provides error control and flow control (3) MAC sublayer performs routing

MAC LLC sub-layers - ?
(1) MAC sub-layer physical addressing
(2) LLC sub-layer error control flow control
(3) MAC sub-layer routing

(A) 1 and 2 only (B) 2 and 3 only
(C) 1 and 3 only (D) 1, 2 and 3

[Source: BCI Subject Test-9.pdf] · Medium

(A) 1 and 2 only

Nukta: MAC = addressing + channel access; LLC = flow + error + interface; routing = Network Layer.

MAC = addressing + channel access; LLC = flow + error + interface; routing = Network Layer.

Q394. Which of the following is NOT a responsibility of the LLC sublayer?

ȝ K Ȼ - LLC sub-layer ɬG K ɮ ?

(A) Flow control (B) Error control

(C) Service interface to Network layer (D) Physical addressing

[Source: BCI Subject Test-9.pdf] · Medium

(D) Physical addressing

Nukta: MAC = Media Access Control = addressing; LLC = Logical Link Control = flow/error.

MAC = Media Access Control = addressing; LLC = Logical Link Control = flow/error.

Q395. Which of the following is NOT a function of the Data Link Layer?

ȝ K Ȼ - Data Link Layer function ɮ ?

(A) Framing (B) Error detection

(C) Error correction (D) Routing

[Source: BCI Subject Test-9.pdf + CN Topic Test-04.pdf] · Medium

(D) Routing

Nukta: DLL = next-hop delivery (frame); Network = end-to-end delivery (packet + route).

DLL = next-hop delivery (frame); Network = end-to-end delivery (packet + route).

Q396. Error control is the process of detecting and correcting errors, while _____ is a mechanism to ensure the e cient delivery of data.

Error control, errors detect correct ɬ ų ȝ ť , ȝ _____ data delivery mechanism .

(A) Error control, Flow control (B) Process control, Network control

(C) Session control, Flow control (D) OSI Control, Data control

[Source: CN Topic Test-02.pdf] · Medium

(A) Error control, Flow control

Nukta: Error = putting right; Flow = maintaining balance. Both are separate jobs.

Error = ; Flow = . Ȼ - ʐ

Q397. _____ control refers to a set of procedures used to restrict the amount of data that the sender can send before waiting for acknowledgment.

_____ control procedures sender Â acknowledgment ɬų w data ɬ Ů ȝ ʐ

(A) Flow (B) Error

(C) Transmission (D) Access control

[Source: Unit Test -2 Solutions.pdf (OCR), p17] · Medium

(A) Flow

Nukta: Flow control = the balance between sender and receiver; this definition is asked again and again in exams.

Flow control = sender ↔ receiver ; definition w Ȼ - .

Q398. What is the main purpose of the Sliding Window in TCP?

TCP Sliding Window ...

(A) Address resolution (B) Flow control

(C) Error detection (D) Encryption

[Source: CN Topic Test 5.pdf] · Medium

(B) Flow control

Nukta: TCP window = receiver capacity advertisement; ...

TCP window = receiver capacity advertisement; ...

Q399. Flow control prevents which of the following issues?

Flow control ...

(A) Deadlock (B) Congestion collapse

(C) Overflow of the receiver's bu er (D) Packet routing error

[Source: CN Topic Test-04.pdf] · Medium

(C) Overflow of the receiver's bu er

Nukta: Flow control = ... ; Congestion control = network ...

Flow control = ... ; Congestion control = network ...

Q400. In Stop-and-Wait ARQ, after sending one frame, the sender:

Stop-and-Wait ARQ ... frame ... sender:

(A) Waits for acknowledgment (B) Sends the next frame immediately

(C) Stops transmission permanently (D) Sends a NAK

[Source: CN Topic Test-04.pdf] · Medium

(A) Waits for acknowledgment

Nukta: Because of this waiting, its name became 'Stop-and-Wait'.

... 'Stop-and-Wait' ...

Q401. In Stop-and-Wait protocol, the sender sends:

Stop-and-Wait protocol ...

(A) Multiple frames at once (B) One frame at a time

(C) Acknowledgments only (D) Data without waiting

[Source: CN Topic Test-04.pdf] · Medium

(B) One frame at a time

Nukta: Window size = 1 — ... - frame.

Window size = 1 — ... - frame.

Q402. What is the window size in Stop-and-Wait ARQ?

Stop-and-Wait ARQ ...

(A) 0 (B) 1

(C) N (D) 2

[Source: CN Topic Test-04.pdf] · Medium

(B) 1

Nukta: Window = 1 ... frame at a time — ...

Window = 1 ... frame at a time — ...

Q403. What is the main disadvantage of Stop-and-Wait flow control?

Stop-and-Wait flow control ↓ ħ ⊞ ?

(A) Complexity (B) Low e ciency due to waiting

(C) High memory usage (D) Frame loss

[Source: CN Topic Test-04.pdf] · Medium

(B) Low e ciency due to waiting

Nukta: Waiting = idle time = utilization. ß Sliding Window .

Waiting = idle time = utilization. ∫ Sliding Window .

Q404. In Stop-and-Wait ARQ, what happens when a frame is lost?

Stop-and-Wait ARQ Ø rame ⊞ ?

(A) Receiver sends NAK (B) Sender retransmits after timeout

(C) Receiver retransmits (D) No retransmission

[Source: CN Topic Test-04.pdf] · Medium

(B) Sender retransmits after timeout

Nukta: Lost frame + timeout = retransmission. Receiver retransmit ¥ .

Lost frame + timeout = retransmission. Receiver retransmit ¥ .

Q405. In Go-Back-N ARQ, if an error is detected in a frame:

Go-Back-N ARQ Ø frame Ø error ⁄ :

(A) Only that frame is retransmitted (B) All frames are ignored

(C) That frame and all subsequent outstanding frames are retransmitted (D) No frame is retransmitted

[Source: CN Topic Test-04.pdf] · Medium

(C) That frame and all subsequent outstanding frames are retransmitted

Nukta: GBN = 'from where you slipped, start again' — in Selective Repeat only the wrong frame is resent.

GBN = ' , 3 ' — Selective Repeat Ø ŋ frame .

Q406. In a Go-Back-N protocol, if the window size = 63, how many sequence number bits are required?

Go-Back-N protocol Ø window size = 63 sequence number 3 bits ε ?

(A) 5 (B) 6

(C) 7 (D) 8

[Source: BCI Subject Test-8.pdf] · Medium

(B) 6

Nukta: The reverse of the question above — memorise the formula both ways.

ų Î Í — formula Ø .

Q407. A Selective Repeat sliding window protocol uses a window size of 7 with 3-bit sequence numbers. Is it valid?

Selective Repeat sliding window protocol ⍰3-bit sequence numbers window size 7 ₶ ?

(A) Valid (B) Invalid – violates the maximum window rule

(C) Valid only for piggybacking (D) Valid only with Selective Repeat

[Source: BCI Subject Test-8.pdf] · Medium

(B) Invalid – violates the maximum window rule

Nukta: SR window ≤ 2ⁿ ¹; GBN window ≤ 2ⁿ − 1. ķ .

SR window ≤ 2ⁿ ¹; GBN window ≤ 2ⁿ − 1. ŋ .

Q408. Which protocol has better bandwidth utilization?

3 protocol bandwidth utilization ?

(A) Stop-and-Wait (B) Selective Repeat

(C) Go-Back-N (D) None of the above

[Source: CN Topic Test-04.pdf] · Medium

(B) Selective Repeat

Nukta: E ciency Ь : Selective Repeat > Go-Back-N > Stop-and-Wait.

E ciency t' : Selective Repeat > Go-Back-N > Stop-and-Wait.

Q409. What is the Hamming distance between the two codes 10010010 and 11011001?

codes 10010010 11011001 Hamming distance 3 ?

(A) 3 (B) 4

(C) 6 (D) 2

[Source: Unit Test -2 Solutions.pdf (OCR), p28] · Medium

(B) 4

Nukta: Hamming distance = XOR 1s ˣ (.

Hamming distance = XOR 1s ₰3 .

Q410. The Hamming distance between two code words can easily be found if we apply the _____ operation.

code words Hamming distance 3 ε _____ operation .

(A) XOR (B) XNOR

(C) OR (D) AND

[Source: Unit Test -2 Solutions.pdf (OCR), p35] · Medium

(A) XOR

Nukta: XOR = di erence counter. ɜ operations ɣ .

XOR = di erence counter. Ȧ operations ɮ .

Q411. To detect or correct errors, we send extra bits with the data. These extra bits are called:

Errors detect correct ɡ data 3 ə ₶ bits ʐ

(A) redundancy (B) correction

(C) detection (D) parity

[Source: Unit Test -2 Solutions.pdf (OCR), p36] · Medium

(A) redundancy

Nukta: Redundancy = 'extra bits sent on purpose' — this is what makes error control possible.

Redundancy = ' - 3 ə ₶ ' — error control .

Q412. Which error detection method uses one extra bit added to the end of a byte?

- error detection method Ø byte Ø bit ?

(A) CRC (B) Checksum

(C) Hamming code (D) Parity bit

[Source: CN Topic Test-04.pdf] · Medium

(D) Parity bit

Nukta: Parity = 1 bit; CRC = remainder; Checksum = ; Hamming = parity bits.

Parity = 1 bit; CRC = remainder; Checksum = ; Hamming = parity bits.

Q413. Which error detection method consists of a parity bit for each data unit as well as an entire data unit of parity bits?

error detection method Ø data unit parity bit - parity bits data unit ?

(A) Simple parity check (B) CRC

(C) Two-dimensional parity check (D) Checksum

[Source: Unit Test -2 Solutions.pdf (OCR), p7] · Medium

(C) Two-dimensional parity check

Nukta: 2D parity = rows + columns parity; simple parity = bit.

2D parity = rows + columns Ø parity; simple parity = bit.

Q414. A simple parity-check code can detect:

simple parity-check code detect :

(A) odd number of errors (B) even number of errors

(C) positive number of errors (D) negative number of errors

[Source: Unit Test -2 Solutions.pdf (OCR), p34] · Medium

(A) odd number of errors

Nukta: Parity catches odd errors and misses even ones — that is its limit.

Parity = odd errors ; even — .

Q415. Which of the following is an error detection technique?

Ø - error detection technique ?

(A) ARQ (B) Parity Bit

(C) Stop-and-Wait (D) Sliding Window

[Source: CN Topic Test-04.pdf] · Medium

(B) Parity Bit

Nukta: Detection techniques: Parity, CRC, Checksum. Control protocols: SW, GBN, SR.

Detection techniques: Parity, CRC, Checksum. Control protocols: SW, GBN, SR.

Q416. High-level Data Link Control (HDLC) is a protocol for communication over point-to-point and multipoint links. HDLC is a _____ protocol.

High-level Data Link Control (HDLC) point-to-point multipoint links communication protocol . HDLC _____ protocol .

(A) byte-oriented (B) character-oriented

(C) bit-oriented (D) none of the above

[Source: Unit Test -2 Solutions.pdf (OCR), p15] · Medium

(C) bit-oriented

Nukta: HDLC/SDLC = bit-oriented; PPP = byte-oriented. .

HDLC/SDLC = bit-oriented; PPP = byte-oriented. .

Q417. The most common protocol for point-to-point access is the Point-to-Point Protocol (PPP), which is a _____ protocol.

Point-to-point access ij protocol Point-to-Point Protocol (PPP) , _____ protocol .

(A) bit-oriented (B) byte-oriented

(C) character-oriented (D) none of the above

[Source: Unit Test -2 Solutions.pdf (OCR), p16] · Medium

(B) byte-oriented

Nukta: PPP = byte-oriented (flag = 0x7E); HDLC = bit-oriented. ″w N ″ x .

PPP = byte-oriented (flag = 0x7E); HDLC = bit-oriented. w Ø ŋ ł .

Q418. Which protocol allows the sender to transmit multiple frames before needing an acknowledgment?

- protocol sender acknowledgment ł ę nframes ?

(A) Stop-and-Wait (B) Go-Back-N

(C) ALOHA (D) CSMA/CD

[Source: CN Topic Test-04.pdf] · Medium

(B) Go-Back-N

Nukta: Sliding window = outstanding frames; Stop-and-Wait = .

Sliding window = ŋoutstanding frames; Stop-and-Wait = .

Q419. Which of the following techniques uses both acknowledgments and sequence numbers?

3 K Ø - technique acknowledgments sequence numbers Ø ?

(A) Go-Back-N (B) Stop-and-Wait

(C) Selective Repeat (D) All of the above

[Source: Unit Test -2 Solutions.pdf (OCR) + CN Topic Test-04.pdf] · Hard

(D) All of the above

Nukta: ARQ = Automatic Repeat reQuest — (sequence/ack ARQ ¥ .

ARQ = Automatic Repeat reQuest — 3 sequence/ack nARQ ¥ .

Q420. In the TCP header, the acknowledgment field value represents:

TCP header Øacknowledgment field 3 ŋ ?

(A) Number of previous bytes received (B) Total number of bytes to receive

(C) Sequence number of the next byte expected

(D) Sequence of zeros and ones

[Source: CN Topic Test-04.pdf] · Hard

(C) Sequence number of the next byte expected

Nukta: TCP bytes (, packets ¥ — ACK = 'next byte '.

TCP bytes 3 , packets ¥ — ACK = 'next byte '.

Q421. For Stop-and-Wait ARQ, for 10 data packets sent, ... acknowledgements are needed.

Stop-and-Wait ARQ Ø10 data packets ..... acknowledgements ¢ Ƶ

(A) 9 (B) 20

(C) 11 (D) 10

[Source: RPSC Programmer 2024 Paper-1 (with o cial key)] · Hard

(D) 10

Nukta: RPSC o cial key (d) — Stop-and-Wait ɴ ack:packet 1:1.

RPSC o cial key (d) — Stop-and-Wait Øack:packet 1:1.

Q422. In the _____ protocol, if no acknowledgment for a frame has arrived, we resend all outstanding frames.

_____ protocol Ø frame acknowledgment outstanding frames Ƶ

(A) Stop-and-Wait ARQ (B) Go-Back-N ARQ

(C) Selective-Repeat ARQ (D) Both B and C

[Source: Unit Test -2 Solutions.pdf (OCR), p8] · Hard

(B) Go-Back-N ARQ

Nukta: GBN = Go Back N — everything up to N is resent; SR = only the wrong one.

GBN = Go Back N — N Ø ; SR = ɡ ɳ .

Q423. In Go-Back-N ARQ, if 6 is the number of bits for the sequence number, then the maximum size of the send window must be:

Go-Back-N ARQ Øequence number 6 bits Ø send window ɟ 3 ?

(A) 63 (B) 61

(C) 31 (D) 64

[Source: Unit Test -2 Solutions.pdf (OCR), p9] · Hard

(A) 63

Nukta: GBN/SW window formula SR — GBN: $2^n - 1$, SR: 2^{n-1}.

GBN/SW window formula SR — GBN: $2^n - 1$, SR: 2^{n-1}.

Q424. The maximum number of outstanding frames in Selective Repeat (with N sequence numbers) is:

Selective Repeat Øutstanding frames ɬ ɟ ɦ (N sequence numbers) :

(A) N (B) N − 1

(C) N/2 (D) 2N

[Source: CN Topic Test-04.pdf] · Hard

(C) N/2

Nukta: SR: window N/2 (windows), GBN: window N − 1 (window).

SR: window N/2 (windows), GBN: window N−1 (window).

Q425. Any coding scheme for error control needs to have at least the parameters of:

Error control ग coding scheme क parameters ह :

(A) codeword size (B) dataword size

(C) minimum Hamming distance (D) All of them

[Source: Unit Test -2 Solutions.pdf (OCR), p37] · Hard

(D) All of them

Nukta: Code = (n, k, d) — n = codeword, k = dataword, d = distance. न code .

Code = (n, k, d) — n = codeword, k = dataword, d = distance. क code .

Q426. The message 1001 is to be transmitted using the CRC polynomial $x^3 + x + 1$ to protect it from errors. The codeword that should be transmitted is:

1001 CRC ō $x^3 + x + 1$ errors ग . codeword :

(A) 1001100 (B) 1001011

(C) 1001110 (D) 1001000

[Source: RPSC Programmer 2024 Paper-1 (with o cial key)] · Hard

(C) 1001110

Nukta: CRC: degree n polynomial n zeros append → remainder . degree 3 → 3 zeros.

CRC: degree n polynomial n zeros append → remainder . degree 3 → 3 zeros.

Q427. The checksum is used in the Internet by several protocols although not at the _____ layer.

Internet क checksum protocols ज 3 _____ layer .

(A) data link (B) transport

(C) network (D) none of the above

[Source: Unit Test -2 Solutions.pdf (OCR), p33] · Hard

(A) data link

Nukta: DLL = CRC; Network/Transport = checksum. ध .

DLL = CRC; Network/Transport = checksum. ध .

Q428. Which of the following is NOT a responsibility of the network layer?

क - network layer ने ?

(A) Congestion control (B) Error control

(C) Routing (D) Inter-networking

[Source: CN Topic Test-02.pdf] · Hard

(B) Error control

Nukta: Network layer = Ř + congestion; error control .

Network layer = ľ + congestion; error control .

Q429. The types of data services o ered by an LLC (Logical Link Control) layer are:

LLC (Logical Link Control) layer Â data services ų ʐ

(A) Connectionless (fast and unreliable) (B) Connection-oriented (slow and reliable)

(C) Both A and B (D) None of the above

[Source: Unit Test -2 Solutions.pdf (OCR), p32] · Hard

(C) Both A and B

Nukta: LLC types: 1 (connectionless), 2 (connection-oriented), 3 (acknowledged connectionless).

LLC types: 1 (connectionless), 2 (connection-oriented), 3 (acknowledged connectionless).

Q430. The message 11001001 is to be transmitted using the CRC polynomial $x^3 + 1$ to protect it from errors. The codeword that should be transmitted is:

11001001 CRC ō $x^3 + 1$ errors ģ . codeword :

(A) 11001001000 (B) 11001001011

(C) 11001010 (D) 110010010011

[Source: GATE CS 2007] · Hard

(B) 11001001011

Nukta: GATE o cial key (B). ˝ RPSC x^3+x+1 ĶṚ ɴ — degree n n zeros, remainder .

GATE o cial key (B). RPSC x^3+x+1 ų Î Ȼ — degree n n zeros, remainder .

Q431. Let G(x) be the generator polynomial used for CRC checking. What condition must G(x) satisfy to detect an odd number of bits in error?

ɢ G(x), CRC Ȼ generator polynomial . Odd ħ Ȼ -Ů₰ ģ G(x) Ħ ɳ₀ ɛ ?

(A) G(x) contains more than two terms (B) G(x) does not divide 1 + x^k for any k within the frame length

(C) 1 + x is a factor of G(x) (D) G(x) has an odd number of terms

[Source: GATE CS 2009] · Hard

(C) 1 + x is a factor of G(x)

Nukta: GATE o cial key (C). x+1 factor ↔ even parity code ↔ odd errors detect.

GATE o cial key (C). x+1 factor ↔ even parity code ↔ odd errors detect.

Q432. In CRC coding, if the data word is 111111, the divisor is 1010 and the remainder is 110, which of the following is the transmitted code?

CRC coding Ȼ ɛ data word 111111 , divisor 1010 remainder 110 , ɜ ĸ Ȼ - code ?

(A) 011111101 (B) 001010110

(C) 111111110 (D) 110111111

[Source: UGC NET Computer Science Dec 2022 (via samagracs)] · Hard

(C) 111111110

Nukta: CRC ɴcodeword = data remainder. ˣ options remainder Ȼ ɣ .

CRC Ȼcodeword = data remainder. ɬ options remainder ɛ ʁ .

Q433. A 2 km long broadcast LAN has 10^7 bps bandwidth and uses CSMA/CD. The signal travels along the wire at 2×10^8 m/s. What is the minimum packet size that can be used on this network?

2 km broadcast LAN ला bandwidth 10^7 bps CSMA/CD . Signal ₹ 2×10^8 m/s . network ij packet ३ ?

(A) 50 bytes (B) 100 bytes

(C) 200 bytes (D) None of the above

[Source: GATE CS 2003 + RPSC Programmer 2011 Paper-1 (via pyqp.in)] · Hard

(D) None of the above

Nukta: GATE 2003 RPSC 2011 ∩ o cial key (D). : min frame = 2 × propagation delay × bandwidth (round-trip).

GATE 2003 RPSC 2011 ₹ o cial key (D). : min frame = 2 × propagation delay × bandwidth (round-trip).

Q434. A network using CSMA/CD has a bandwidth of 20 Mbps. If the maximum propagation time is 25 μs, what is the minimum size of the frame?

CSMA/CD network ला bandwidth 20 Mbps . ह ू propagation time 25 μs , frame ij ₹ ?

(A) 500 bits (B) 1000 bits

(C) 1500 bits (D) 2000 bits

[Source: UGC NET Computer Science Jan 2025 (via samagracs solved paper)] · Hard

(B) 1000 bits

Nukta: UGC NET Jan 2025 key (ii). formula GATE ₹ ∩ — units .

UGC NET Jan 2025 key (ii). formula GATE ू ₹ — units .

Q435. A CSMA/CD network transmits data at 100 Mbps over a 1 km cable with no repeaters. If the minimum frame size required is 1250 bytes, what is the signal speed in the cable (in km/sec)?

CSMA/CD network ३ repeaters 1 km cable 100 Mbps data . ह ू ij frame 1250 bytes , cable ₹ signal ला ३ (km/sec ₹ ३ ?

(A) 8000 (B) 10000

(C) 16000 (D) 20000

[Source: GATE CSE 2015 Set 3] · Hard

(D) 20000

Nukta: GATE 2015 key (d) 20000 km/s. Getting the speed by inverting the formula — that is the trick of this question.

GATE 2015 key (d) 20000 km/s. Formula speed ३ — ू ₹ .

Q436. In the starting phase (slow start) of the TCP congestion control algorithm, the congestion window size:

TCP congestion control algorithm Ę (slow start) Ȼongestion window :

(A) Increases linearly (B) Does not increase

(C) Increases exponentially (D) Remains constant at threshold

[Source: GATE CS 2008] · Hard

(C) Increases exponentially

Nukta: GATE 2008 key: exponential. Ь : Slow start (exponential) → Congestion avoidance (linear).

GATE 2008 key: exponential. ť : Slow start (exponential) → Congestion avoidance (linear).

Q437. Consider the following statements: I. TCP connections are full duplex. II. TCP has no option for selective acknowledgment. III. TCP connections are message streams. Which of the following is correct?

ɜ ĸ Ȼ ɜ ØI. TCP connections full duplex ʐII. TCP Ȼelective acknowledgment ŋ Í ʁ . III. TCP connections message streams ʐɜ ĸ Ȼ Ħ ?

(A) Only I is correct (B) Only I and II are correct

(C) Only II and III are correct (D) All of I, II and III are correct

[Source: GATE CS 2012] · Hard

(A) Only I is correct

Nukta: GATE 2012 o cial key (a). : TCP = full-duplex byte-stream + SACK Ő ; UDP = message-oriented.

GATE 2012 o cial key (a). : TCP = full-duplex byte-stream + SACK Ê; UDP = message-oriented.

Ω Samjh — Section B — Data Link Layer: Framing & MAC Sub-layer

Framing, bit/byte stu ng, Ethernet frame format, CSMA/CD-CA, ALOHA, token & reservation access, client-server basics.

Data link layer data ko frames mein baant-ti hai; medium access ke liye CSMA/CD (wired), CSMA/CA (wireless), ALOHA, token passing — yeh sab MAC methods hain.

Data link layer data frames ; medium access CSMA/CD, CSMA/CA, ALOHA, token passing

- Framing methods: Byte count, Byte stu ng (DLE-ESC), Bit stu ng (0 after five 1s), Physical layer coding violation.
  Framing: Byte count, Byte stu ng, Bit stu ng (1 0), Coding violation
- Ethernet frame: Preamble | Destination MAC | Source MAC | Type | Data (46–1500 B) | FCS — minimum frame 64 bytes.
  Ethernet frame: Preamble | Dest MAC | Src MAC | Type | Data | FCS — 64 bytes
- CSMA/CD (wired Ethernet): medium sense karke bhejo; collision par jamming + backo ; 1-persistent = sunne par turant bhejo.
  CSMA/CD: sense ; collision jamming + backo
- Collision window = 2 × propagation time; minimum frame size isi se tay hota hai (2 km/10^7 bps → 25 B jaisa classic).
  Collision window = 2 × propagation time; minimum frame size
- CSMA/CA (wireless): collision detect nahi hota — RTS/CTS, ACK, inter-frame spacing se collision avoid; hidden terminal problem.
  CSMA/CA (wireless): RTS/CTS, ACK; hidden terminal problem
- ALOHA: Pure (jab mann kare bhejo) vs Slotted (time slots) — random access ke sabse purane tareeke.
  ALOHA: Pure vs Slotted — random access
- Token passing (Token Ring/FDDI) — controlled access: token milne par hi bhejo; collision nahi.
  Token passing — controlled access; collision
- Access methods: Random (ALOHA, CSMA) vs Controlled (token, polling).
  Access: Random vs Controlled

Method	Medium	Collision handling
CSMA/CD	wired Ethernet	detect + jam + backo
CSMA/CA	wireless	avoid (RTS/CTS, ACK)
ALOHA	radio	retransmit after collision
Token passing	ring	collision nahi hota

Q438. Why is bit stu ng required in bit-oriented protocols?

Bit-oriented protocols Ø it stu ng H Ø ę ?

(A) To increase bandwidth (B) To prevent confusion between data and the flag

(C) To reduce collisions (D) To detect errors

[Source: BCI Subject Test-9.pdf] · Easy

(B) To prevent confusion between data and the flag

Nukta: Bit stu ng = data flag ♪ .

Bit stu ng = data flag ε .

Q439. The correct order of fields in an Ethernet frame is:

Ethernet frame Ø ields t :

(A) Preamble → Destination MAC → Source MAC → Type/Length → Data → FCS (B) Destination MAC → Source MAC → FCS → Data → Preamble

(C) Source MAC → Destination MAC → Data → CRC → Type (D) Preamble → Source MAC → Destination MAC → Data

[Source: BCI Subject Test-9.pdf] · Easy

(A) Preamble → Destination MAC → Source MAC → Type/Length → Data → FCS

Nukta: Remember: after the Preamble come the RECEIVER's address (Destination) first, then the sender's — the rest later.

: Preamble RECEIVER (Destination), ३ — ४ Ø

Q440. Collision detection and avoidance mechanisms are implemented in:

Collision detection avoidance mechanisms ?

(A) LLC Sublayer (B) Network Layer

(C) MAC Sublayer (D) Physical Layer

[Source: BCI Subject Test-9.pdf] · Easy

(C) MAC Sublayer

Nukta: MAC = Medium Access Control — N 'medium access' ß .

MAC = Medium Access Control — Ø 'medium access' ģ .

Q441. Which protocol requires time synchronization among nodes?

३ protocol Ø odes time synchronization ε ?

(A) Pure ALOHA (B) Slotted ALOHA

(C) CSMA (D) CSMA/CD

[Source: BCI Subject Test-9.pdf] · Easy

(B) Slotted ALOHA

Nukta: Pure ALOHA = send whenever you like; Slotted = send only when the slot clock says so.

Pure ALOHA = ; Slotted = slot ४ .

Q442. The vulnerable period of Pure ALOHA is:

Pure ALOHA vulnerable period :

(A) T (B) 2T

(C) T/2 (D) 4T

[Source: BCI Subject Test-9.pdf] · Easy

(B) 2T

Nukta: Pure ALOHA: 2T G·e ^{2G} (~18.4%); Slotted ALOHA: T G·e ^{G} (~36.8%).

Pure ALOHA: 2T G·e ^{2G} (~18.4%); Slotted ALOHA: T G·e ^{G} (~36.8%).

Q443. Which protocol has the highest e ciency under heavy load?

load ३ protocol ↓e ciency ↓ ?

(A) Pure ALOHA (B) Slotted ALOHA
(C) CSMA (D) CSMA/CD

[Source: BCI Subject Test-9.pdf] · Easy

(D) CSMA/CD

Nukta: ь : CSMA/CD > CSMA > Slotted ALOHA > Pure ALOHA.

t' : CSMA/CD > CSMA > Slotted ALOHA > Pure ALOHA.

Q444. Which protocol detects collisions while transmission is in progress?

- protocol transmission collisions detect ?

(A) CSMA (B) Slotted ALOHA
(C) CSMA/CD (D) CSMA/CA

[Source: BCI Subject Test-9.pdf] · Easy

(C) CSMA/CD

Nukta: CD = detect-and-jam; CA = avoid (RTS/CTS) — wireless ন

CD = detect-and-jam; CA = avoid (RTS/CTS) — wireless Ȼ

Q445. Which mechanism is used by CSMA/CA to reduce collisions?

CSMA/CA collisions ∫ - mechanism ?

(A) Collision detection (B) Slot restriction
(C) RTS/CTS handshake (D) Jam signal

[Source: BCI Subject Test-9.pdf] · Easy

(C) RTS/CTS handshake

Nukta: CA = Collision Avoidance — ,

CA = Collision Avoidance — , ৪.

Q446. Bit stu ng refers to:

Bit stu ng ন:

(A) Inserting a '0' in the user stream to di erentiate it from the flag (B) Inserting a '0' in the flag stream to avoid ambiguity
(C) Appending a nibble to the flag sequence
(D) Appending a nibble to the user data stream

[Source: Unit Test-4 Solution.pdf (OCR), p26] · Medium

(A) Inserting a '0' in the user stream to di erentiate it from the flag

Nukta: Stu ng is always done in DATA, never in the flag; in byte stu ng 0x7D follows 0x7E.

Stu ng DATA Ȼ , flag Ȼ ৪; byte stu ng Ȼ 0x7E 0x7D .

Q447. In Ethernet CSMA/CD, the special sequence transmitted by media access management to handle a collision is called:

Ethernet CSMA/CD Ȼcollision ∫ media access management Â ३ sequence :

(A) Preamble (B) Postamble
(C) Jam signal (D) None of these

[Source: Unit Test-4 Solution.pdf (OCR), p28] · Medium

(C) Jam signal

Nukta: Jam = the siren shouting 'a collision happened!'. Preamble = announcement that a frame is starting.

Jam = ' !' . Preamble = frame ę ↓ .

Q448. The minimum size of an Ethernet frame (excluding preamble) is:

Ethernet frame ij (preamble) :

(A) 46 bytes (B) 64 bytes

(C) 72 bytes (D) 1518 bytes

[Source: BCI Subject Test-9.pdf] · Medium

(B) 64 bytes

Nukta: 64 = minimum frame; 1518 = maximum frame; 46 = ß ķdata minimum.

64 = minimum frame; 1518 = maximum frame; 46 = ģ ŋdata minimum.

Q449. The data field size in an Ethernet frame ranges from:

Ethernet frame Ød ata field :

(A) 64–1518 bytes (B) 46–1500 bytes

(C) 40–1480 bytes (D) 32–1500 bytes

[Source: BCI Subject Test-9.pdf] · Medium

(B) 46–1500 bytes

Nukta: The whole frame ranges 64–1518 bytes; the data 46–1500. This pair is asked often.

frame range 64–1518; data 46–1500. Ħ r .

Q450. In Ethernet, the minimum frame size is fixed mainly to:

Ethernet Ø frame ħ ę 3 ģ 3 ?

(A) Improve throughput (B) Reduce overhead

(C) Detect collisions using CSMA/CD (D) Increase security

[Source: BCI Subject Test-9.pdf] · Medium

(C) Detect collisions using CSMA/CD

Nukta: Round-trip time (slot time) frame ß Γ — collision ˝ .

Round-trip time (slot time) frame ŋ - — collision .

Q451. What is the primary function of the MAC sublayer?

MAC sublayer ħ Ħ ?

(A) Routing (B) Error checking

(C) Addressing (D) Channel access

[Source: CN Topic Test-04.pdf] · Medium

(D) Channel access

Nukta: MAC = addressing + channel access; ˝w ɴprimary function' = channel access.

MAC = addressing + channel access; w Ø 'primary function' = channel access.

Q452. What is the primary role of CSMA/CD in Ethernet?

Ethernet ØCSMA/CD ɬ ħ ȝ Ħ ?

(A) Error correction (B) Collision detection

(C) Address resolution (D) Flow control

[Source: CN Topic Test 5.pdf] · Medium

(B) Collision detection

Nukta: ɴ Collision Detection ß .

Ø Collision Detection ģ .

Q453. Why is CSMA/CD not suitable for wireless networks?

CSMA/CD wireless networks ∫ Ħ Ħ Ȼ ɮ ?

(A) No random access (B) Collision detection is not feasible in wireless

(C) High bandwidth requirement (D) Requires slot synchronization

[Source: BCI Subject Test-9.pdf] · Medium

(B) Collision detection is not feasible in wireless

Nukta: Wired = CSMA/CD; Wireless = CSMA/CA. ″ Í ˣ .

Wired = CSMA/CD; Wireless = CSMA/CA. Ħ ɬ .

Q454. What is multiple access?

Multiple access Ħ ?

(A) Accessing a computer from multiple locations (B) A kind of network (MAN)

(C) Sharing of one physical link by more than one station (D) None of the above

[Source: SSC CGL Internet & Networking (via EduRev bank)] · Medium

(C) Sharing of one physical link by more than one station

Nukta: Shared medium protocols Multiple Access Protocols ɾ

Shared medium protocols Multiple Access Protocols ʐ

Q455. A Token Ring implementation works on the _____ mode of medium access.

Token Ring implementation medium access _____ mode .

(A) Contention (B) Reservation

(C) Round Robin (D) None of the above

[Source: Unit Test -2 Solutions.pdf (OCR), p26] · Medium

(C) Round Robin

Nukta: MAC ″ : Contention (ALOHA/CSMA), Reservation, Round Robin (Token).

MAC : Contention (ALOHA/CSMA), Reservation, Round Robin (Token).

Q456. Which of the following is an example of Client–Server architecture?

ɜ ĸ Ȼ - Client–Server architecture ?

(A) Chat between two Bluetooth devices (B) Web browser accessing a website

(C) File transfer via USB cable (D) Two laptops connected directly

[Source: BCI_SCI Subject Test-01 (CN and Security)] · Medium

(B) Web browser accessing a website

Nukta: Client–Server: , ɜ . Bluetooth chat = peer-to-peer.

Client–Server: , Ȧ . Bluetooth chat = peer-to-peer.

Q457. What is the MTU size for Ethernet?

Ethernet ∫ MTU ɜ ?

(A) 1024 bytes (B) 1500 bytes

(C) 512 bytes (D) 2048 bytes

[Source: CN Topic Test 5.pdf] · Hard

(B) 1500 bytes

Nukta: MTU = the data limit; 1518 = the whole frame limit. Don't confuse the two.

MTU = data ɬ ; 1518 = frame ɬ . Ŷ Ȼ .

Q458. In a slotted ALOHA with G stations attempting to transmit in one slot, the throughput is equal to:

Slotted ALOHA ∅ slot ∅G stations transmit ų ∅ throughput :

(A) G × e ⁻ᴳ (B) 2G × e ⁻ᴳ

(C) e ⁻ᴳ (D) G × e ⁻²ᴳ

[Source: RPSC Programmer 2024 Paper-1 (with o cial key)] · Hard

(A) G × e ⁻ᴳ

Nukta: RPSC o cial key (a). Pure ALOHA formula G·e ⁻²ᴳ — ◊ .

RPSC o cial key (a). Pure ALOHA formula G·e ⁻²ᴳ — ३ .

Q459. Which of the following Multiple Access Protocols is not suitable for wired networks?

३ ĸ ∅ - Multiple Access Protocol wired networks ģ Ħ ঙ ?

(A) CSMA/CD (B) TDMA

(C) Token passing (D) CSMA/CA

[Source: RPSC Programmer 2024 Paper-1 (with o cial key)] · Hard

(D) CSMA/CA

Nukta: RPSC o cial key (d) — wired = CSMA/CD, wireless = CSMA/CA.

RPSC o cial key (d) — wired = CSMA/CD, wireless = CSMA/CA.

Q460. Network Allocation Vector (NAV) is associated with which of the following MAC technologies?

Network Allocation Vector (NAV) ३ ĸ ∅ ३ MAC technology ?

(A) Wi-Fi (CSMA/CA) (B) CSMA/CD

(C) Bluetooth (D) Ethernet

[Source: RPSC Programmer 2024 Paper-1 (with o cial key)] · Hard

(A) Wi-Fi (CSMA/CA)

Nukta: RPSC o cial key (a) — NAV = virtual carrier sensing (Ř .

RPSC o cial key (a) — NAV = virtual carrier sensing ε ľ .

Q461. In Reservation mode of MAC, each station in the network _____ a time slot for a finite or infinite amount of time to access the shared medium.

MAC Reservation mode ∅network station shared medium õ ģ time slot _____ .

(A) should reserve (B) need not reserve

(C) Can't say (D) None of the above

[Source: Unit Test -2 Solutions.pdf (OCR), p25] · Hard

(A) should reserve

Nukta: Reservation = book first, travel later — contention (ALOHA/CSMA) has no booking.

Reservation = ł - , ३ Ů — contention (ALOHA/CSMA) ∅ł - ঙ.

Q462. Which one of the following is not used in media access control?

३ K Ȼ - media access control Ȼ ɓ ?

(A) Ethernet (B) Digital subscriber line (DSL)
(C) FDDI (D) None of the mentioned

[Source: CN Topic Test-01.pdf] · Hard

(B) Digital subscriber line (DSL)

Nukta: MAC = LAN ˣ ; DSL = telephone line broadband.

MAC = LAN ɬ ; DSL = telephone line broadband.

Q463. Which of the following is NOT true with respect to a transparent bridge and a router?

Transparent bridge router Ȼ K Ȼ - ɓ ?

(A) Both bridge and router selectively forward data packets (B) A bridge uses IP addresses while a router uses MAC addresses
(C) A bridge builds its table by inspecting incoming packets (D) A router can connect between a LAN and a WAN

[Source: RPSC Programmer 2011 Paper-1 (via pyqp.in)] · Hard

(B) A bridge uses IP addresses while a router uses MAC addresses

Nukta: RPSC 2011 o cial key (2). : Bridge = MAC (L2), Router = IP (L3).

RPSC 2011 o cial key (2). : Bridge = MAC (L2), Router = IP (L3).

Q464. Which of the following is NOT a function of the AAL (ATM Adaptation Layer)?

३ K Ȼ - AAL (ATM Adaptation Layer) ɓ ?

(A) Cell header generation (B) Handling of lost and misinserted cells
(C) Handling of cell delay variation (D) Segmentation and reassembly of user information

[Source: RPSC Programmer 2011 Paper-1 (via pyqp.in)] · Hard

(A) Cell header generation

Nukta: RPSC 2011 o cial key (1). ATM ɴ ATM layer = cell header/switching; AAL = segmentation+reassembly.

RPSC 2011 o cial key (1). ATM ȻATM layer = cell header/switching; AAL = segmentation+reassembly.

Ω Samjh — Section C — Network Layer: IP, ICMP, ARP & Routing Protocols

IPv4 header & fragmentation, IPv6 fields, logical addressing, ARP/ICMP/IGMP, routing (RIP/OSPF/BGP/MPLS/STP), traceroute, DHCP & probability problems.

Network layer = IP addressing + routing + fragmentation + error reporting (ICMP). Routing protocols RIP/OSPF/BGP aur distance-vector vs link-state — exam ke favourites.

Network layer = IP addressing + routing + fragmentation + ICMP Routing protocols RIP/OSPF/BGP

- IPv4 header (20–60 B): IHL, Type of Service, Total Length, TTL, Protocol, Flags, Fragment O set, Source/Dest IP, Checksum.

  IPv4 header: TTL, Protocol, Flags, Fragment O set, Source/Dest IP, Checksum

- Fragmentation: MTU se bada packet tootta hai; o set = 8-byte units; M bit (More fragments); reassembly sirf destination par.

  Fragmentation: MTU packet ; o set = 8-byte units; M bit; reassembly ∮ ŋdestination

- TTL = loop prevention: har router par 1 kam, 0 par drop (ICMP Time Exceeded).

  TTL = loop prevention: router 1

- Routing: Static (manual) vs Dynamic; Distance Vector (RIP — hop count, periodic update, slow converge) vs Link State (OSPF — topology map, Dijkstra, fast converge).

  Routing: Static vs Dynamic; Distance Vector (RIP) vs Link State (OSPF)

- BGP = inter-domain routing (autonomous systems ke beech); EGP family.

  BGP = inter-domain routing

- ICMP = error + control messages: Echo Request/Reply (ping), Destination Unreachable, Time Exceeded, Redirect.

  ICMP: Echo (ping), Destination Unreachable, Time Exceeded, Redirect

- ARP — IP→MAC (broadcast request); IGMP — multicast group management.

  ARP — IP→MAC; IGMP — multicast

- Congestion control: choke points, backpressure; open-loop vs closed-loop.

  Congestion control: choke points, backpressure

Protocol	Type	Convergence	Metric
RIP	distance vector	slow (count-to-infinity)	hop count (max 15)
OSPF	link state	fast	cost/bandwidth
BGP	path vector	—	policy/AS path

Q465. What is the function of ARP?

ARP Ħ ?

(A) Resolve IP address to MAC address (B) Resolve domain name to IP

(C) Encrypt packets (D) Determine the route

[Source: CN Topic Test 6.pdf + Unit Test -2 Solutions.pdf (OCR)] · Medium

(A) Resolve IP address to MAC address

Nukta: ARP = IP→MAC; DNS = name→IP; RARP = MAC→IP (obsolete).

ARP = IP→MAC; DNS = name→IP; RARP = MAC→IP (obsolete).

Q466. Which tool is used to view the MAC address of a system?

ȝ system MAC address ǥ - tool ?

(A) ping (B) ipconfig /all

(C) netstat (D) arp -a

[Source: BCI_SCI Subject Test-7.pdf] · Medium

(B) ipconfig /all

Nukta: Windows: ipconfig /all; Linux: ip link / ifconfig. ɴ NIC ʖ ɾ

Windows: ipconfig /all; Linux: ip link / ifconfig. Ȼ NIC ȣ ȥ

Q467. Which protocol is used for path MTU discovery?

Path MTU discovery ǥ - protocol ?

(A) TCP (B) UDP

(C) ICMP (D) ARP

[Source: BCI_SCI Subject Test-7.pdf] · Medium

(C) ICMP

Nukta: ICMP = network — ping, traceroute, PMTU .

ICMP = network — ping, traceroute, PMTU .

Q468. What is the function of IGMP in networking?

Networking ȻGMP Ħ ?

(A) Manage IP address allocation (B) Route datagrams

(C) Manage multicast group membership (D) Handle device discovery

[Source: CN Topic Test 5.pdf] · Medium

(C) Manage multicast group membership

Nukta: IGMP = multicast ˣ member-list; Network layer protocol.

IGMP = multicast ɬ member-list; Network layer protocol.

Q469. What is the role of the TTL field in an IP packet?

IP packet ȻTTL field ɬ ʖ Ħ ?

(A) Authentication (B) Packet size control

(C) Packet lifespan control (D) Encryption

[Source: CN Topic Test 6.pdf] · Medium

(C) Packet lifespan control

Nukta: TTL = loop-stopper. It is also what runs traceroute (di erent TTLs map the path).

TTL = loop- . traceroute (- TTL path).

Q470. Which field in the IPv4 header indicates the priority of the packet?

IPv4 header Ȼ - field packet ɬɥ ʖ ?

(A) Protocol (B) TTL

(C) Type of Service (ToS) (D) Flags

[Source: CN Topic Test 6.pdf] · Medium

(C) Type of Service (ToS)

Nukta: ToS ɴDi Serv/DSCP — : priority/class.

ToS ȻDi Serv/DSCP — : priority/class.

Q471. Which of the following uses a logical addressing scheme?

ȝ ĸ Ȼ - logical addressing scheme ?

(A) MAC address (B) IP address

(C) ARP (D) DNS

[Source: CN Topic Test 5.pdf] · Medium

(B) IP address

Nukta: Forouzan ȼ : Data link = physical, Network = logical, Transport = port, Application = specific (name).

Forouzan ȝ : Data link = physical, Network = logical, Transport = port, Application = specific (name).

Q472. Which algorithm is used by OSPF to compute the shortest path?

OSPF shortest path ȝ ɠ - algorithm ?

(A) Bellman-Ford (B) Dijkstra

(C) A* (D) Kruskal

[Source: CN Topic Test 6.pdf] · Medium

(B) Dijkstra

Nukta: Link-state = Dijkstra; Distance-vector = Bellman-Ford. ˝ Í ˣ.

Link-state = Dijkstra; Distance-vector = Bellman-Ford. Ħ ɬ.

Q473. Which one is a distance-vector routing protocol?

ȝ ĸ Ȼ - distance-vector routing protocol ?

(A) RIP (B) OSPF

(C) BGP (D) IS-IS

[Source: CN Topic Test 6.pdf] · Medium

(A) RIP

Nukta: RIP = DV (Bellman-Ford); OSPF/IS-IS = Link-state (Dijkstra); BGP = Path-vector.

RIP = DV (Bellman-Ford); OSPF/IS-IS = Link-state (Dijkstra); BGP = Path-vector.

Q474. What is a major drawback of distance vector routing?

Distance vector routing ɬ Ħ ?

(A) High complexity (B) Count-to-infinity problem

(C) Slow convergence (D) Lack of security

[Source: CN Topic Test 5.pdf] · Medium

(B) Count-to-infinity problem

Nukta: Count-to-infinity is the signature of DV; link-state is free of it (but uses more memory).

DV ɬ count-to-infinity; link-state Ħ (ŏ memory).

Q475. BGP is mainly used for which type of routing, and it belongs to which algorithm family?

BGP ħ ę ȝ ų routing ɠ ȝ algorithm ȣ ?

(A) Intra-AS routing; link-state (B) Inter-AS routing; path-vector (distance-vector family)

(C) Static routing; flooding (D) Host routing; Dijkstra

[Source: CN Topic Test 5.pdf + CN Topic Test 6.pdf] · Medium

(B) Inter-AS routing; path-vector (distance-vector family)

Nukta: AS = Autonomous System. (AS) OSPF/RIP; ɴ(Inter-AS) BGP — Internet glue protocol.

AS = Autonomous System. (AS) OSPF/RIP; Ȼ (Inter-AS) BGP — Internet glue protocol.

Q476. In the early ARPANET, which routing method was used?

ų ğ ARPANET Ø - routing method ?

(A) Distance vector routing (B) Link state routing

(C) Least cost routing (D) Flooding

[Source: Unit Test-4 Solution.pdf (OCR), p8] · Medium

(A) Distance vector routing

Nukta: ◊ : ARPANET = distance-vector ɣ, (link-state (OSPF).

ɜ : ARPANET = distance-vector ę, ɜ link-state (OSPF).

Q477. Which of these protocols allows loop-free switching in Layer 2 networks?

ɜ ĸ Ø - protocol Layer 2 networks Øoop-free switching ?

(A) RIP (B) OSPF

(C) STP (Spanning Tree Protocol) (D) EIGRP

[Source: CN Topic Test 6.pdf] · Medium

(C) STP (Spanning Tree Protocol)

Nukta: STP = Layer 2 loop- ; routing protocols Layer 3 ɾ

STP = Layer 2 loop- ; routing protocols Layer 3 ƺ

Q478. MPLS operates between which two layers?

MPLS ɜ layers ?

(A) Layer 1 and 2 (B) Layer 2 and 3

(C) Layer 3 and 4 (D) Layer 4 and 7

[Source: CN Topic Test 5.pdf] · Medium

(B) Layer 2 and 3

Nukta: MPLS = 'Layer 2.5' — label-based forwarding, IP header (.

MPLS = 'Layer 2.5' — label-based forwarding, IP header ɜ .

Q479. What does the 'traceroute' command do?

'traceroute' command Ħ ?

(A) Lists all devices on a network (B) Tracks the path a packet takes to the destination

(C) Resolves domain names (D) Opens ports on the firewall

[Source: CN Topic Test 5.pdf] · Medium

(B) Tracks the path a packet takes to the destination

Nukta: tracert (Windows) ICMP; traceroute (Linux) UDP — : Ř ȣ .

tracert (Windows) ICMP; traceroute (Linux) UDP — : Ĭ ɛ .

Q480. Which protocol does traceroute use on UNIX/Linux systems?

UNIX/Linux systems traceroute - protocol ?

(A) TCP (B) UDP

(C) ICMP Echo (D) IGMP

[Source: CN Topic Test 5.pdf] · Medium

(B) UDP

Nukta: Windows tracert ICMP ; Linux traceroute UDP. ɴ .

Windows tracert ICMP ; Linux traceroute UDP. Ø .

Q481. Which command is used to see the routing table in Linux?

Linux ₡outing table ₫ - command ?

(A) ifconfig (B) netstat -r

(C) ping (D) ipconfig

[Source: CN Topic Test 5.pdf] · Medium

(B) netstat -r

Nukta: Windows: route print; Linux: netstat -r / ip route.

Windows: route print; Linux: netstat -r / ip route.

Q482. Which protocol is used for dynamic IP address assignment?

Dynamic IP address assignment ₫ - protocol ?

(A) DNS (B) ARP

(C) DHCP (D) FTP

[Source: CN Topic Test-03.pdf + CN Topic Test 6.pdf] · Medium

(C) DHCP

Nukta: DHCP = 'Plug and Play' networking — zeroconf ₭ tool.

DHCP = 'Plug and Play' networking — zeroconf ų tool.

Q483. ICMP is a ..... layer protocol.

ICMP ..... layer protocol .

(A) Data link (B) Transport

(C) Application (D) Network

[Source: RPSC Programmer 2024 Paper-1 (with o cial key) + Random Test- Rajasthan PYQ.pdf (OCR)] · Hard

(D) Network

Nukta: RPSC o cial key (d). ICMP/IP/ARP/IGMP Network layer ; TCP/UDP Transport.

RPSC o cial key (d). ICMP/IP/ARP/IGMP Network layer ; TCP/UDP Transport.

Q484. Which of the following is NOT a valid ICMP error message?

३ ₭ ₡ - ICMP error message ४ ?

(A) Destination Unreachable (B) Time Exceeded

(C) Echo Request (D) Port Unreachable

[Source: CN Topic Test 5.pdf] · Hard

(C) Echo Request

Nukta: ICMP errors: Destination Unreachable, Time Exceeded, Source Quench, Redirect. Queries: Echo, Timestamp.

ICMP errors: Destination Unreachable, Time Exceeded, Source Quench, Redirect. Queries: Echo, Timestamp.

Q485. Which field in the IP header helps in fragmentation?

IP header ₡ - field fragmentation ₡ ?

(A) TTL (B) Header Length

(C) Identification (D) Protocol

[Source: CN Topic Test 6.pdf] · Hard

(C) Identification

Nukta: Fragmentation tool: Identification + Flags (MF) + Fragment O set.

Fragmentation tool: Identification + Flags (MF) + Fragment O set.

Q486. How many bits are there in the 'Fragment O set' field of the IPv4 header?

IPv4 header 'Fragment O set' field bits ?

(A) 16 bits (B) 10 bits

(C) 13 bits (D) 32 bits

[Source: RPSC Programmer 2024 Paper-1 (with o cial key)] · Hard

(C) 13 bits

Nukta: RPSC o cial key (c). IPv4 header: Version(4)+HLen(4)+ToS(8)+TotalLen(16)+ID(16)+Fl

RPSC o cial key (c). IPv4 header: Version(4)+HLen(4)+ToS(8)+TotalLen(16)+ID(16)+Flags(3

Q487. In IPv4, packet fragmentation and reassembly is handled by:

IPv4 packet fragmentation reassembly ?

(A) the More Fragments flag and Fragment O set field (B) the MAC Header

(C) the Header Checksum (D) encapsulating packets in larger frames

[Source: RPSC Programmer 2024 Paper-1 (with o cial key)] · Hard

(A) the More Fragments flag and Fragment O set field

Nukta: RPSC o cial key (a). : fragmentation sender/routers reassembly destination host.

RPSC o cial key (a). : fragmentation sender/routers reassembly destination host.

Q488. What is fragmentation in IP networking?

IP networking fragmentation ?

(A) Splitting data into packets (B) Splitting large packets so they fit a smaller MTU link

(C) Combining multiple packets (D) Encrypting packets

[Source: CN Topic Test 6.pdf] · Hard

(B) Splitting large packets so they fit a smaller MTU link

Nukta: Packet segmentation ; packet fragmentation. IPv6 routers fragmentation .

Packet segmentation ; packet fragmentation. IPv6 routers fragmentation .

Q489. The size of the 'total length field' of an IPv4 datagram is:

IPv4 datagram 'total length field' :

(A) 16 bits (B) 32 bits

(C) 8 bits (D) 4 bits

[Source: RPSC Programmer 2024 Paper-1 (with o cial key)] · Hard

(A) 16 bits

Nukta: RPSC o cial key (a). 16 bits = $2^{16} - 1 = 65535$.

RPSC o cial key (a). 16 bits = $2^{16} - 1 = 65535$.

Q490. What is the minimum header size of an IPv4 packet?

IPv4 packet header ?

(A) 16 bytes (B) 10 bytes

(C) 20 bytes (D) 32 bytes

[Source: Networking MCQ bank (MQuestions)] · Hard

(C) 20 bytes

Nukta: 20 bytes = 5 words × 4 bytes — Header Length field .

20 bytes = 5 words × 4 bytes — Header Length field .

Q491. In the IPv6 header, the tra c class field is similar to which field in the IPv4 header?

IPv6 header Ctra c class field, IPv4 header 3 field ?

(A) Fragmentation field (B) Flow label

(C) Type of Service (ToS) field (D) Options field

[Source: Sanfoundry CN MCQs — IPv6] · Hard

(C) Type of Service (ToS) field

Nukta: IPv6 Nfragmentation fields Y r— routers Y .

IPv6 Cfragmentation fields 6 z— routers 6 .

Q492. How many maximum hops are allowed in RIP?

RIP C 3 hops 4 3 ?

(A) 8 (B) 16

(C) 15 (D) 32

[Source: CN Topic Test 6.pdf] · Hard

(C) 15

Nukta: RIP: max 15 hops, metric = hop count, 30 sec update.

RIP: max 15 hops, metric = hop count, 30 sec update.

Q493. How does RIP prevent routing loops? (A) Poison reverse (B) Sequence numbers (C) Split horizon

RIP routing loops ? (A) Poison reverse (B) Sequence numbers (C) Split horizon

(A) A only (B) C only

(C) B only (D) Both A and C

[Source: CN Topic Test 6.pdf] · Hard

(D) Both A and C

Nukta: Loop ″ : split horizon, poison reverse, hold-down timers.

Loop : split horizon, poison reverse, hold-down timers.

Q494. Dynamic routing protocols enable routers to:

Dynamic routing protocols routers w z

(A) Dynamically discover and maintain routes

(B) Distribute routing updates to other routers

(C) Reach agreement with other routers about network topology (D) All of the above

[Source: Unit Test-4 Solution.pdf (OCR), p27] · Hard

(D) All of the above

Nukta: Static vs Dynamic: static Nadmin 8 ; dynamic Nprotocol .

Static vs Dynamic: static Cadmin g ; dynamic C protocol .

Q495. Which of the following protocols may be used as zeroconf and plug-and-play?

3 K C - protocol zeroconf plug-and-play e C ?

(A) TCP (B) DHCP

(C) NAT (D) ICMP

[Source: RPSC Programmer 2024 Paper-1 (with o cial key)] · Hard

(B) DHCP

Nukta: RPSC o cial key (b). Zeroconf = (admin automatic network settings.

RPSC o cial key (b). Zeroconf = 3 admin automatic network settings.

Q496. There are n stations in a slotted LAN. Each station attempts to transmit with a probability p in each time slot. The probability that exactly one station transmits in a slot is:

slotted LAN Ø n stations ʐ time slot Ø station p ų ʒ transmit ų . ʒ slot Ø ʃ station transmit ɬ ų ʒ :

(A) $n \times p \times (1 - p)^{n-1}$ (B) $p \times (1 - p)^{n-1}$

(C) $1 - (1 - p)^{n-1}$ (D) $(1 - p)^{n-1}$

[Source: RPSC Programmer 2024 Paper-1 (with o cial key)] · Hard

(A) $n \times p \times (1 - p)^{n-1}$

Nukta: RPSC o cial key (a). binomial 'exactly one success' formula .

RPSC o cial key (a). binomial 'exactly one success' formula .

Q497. One of the header fields in an IP datagram is the Time to Live (TTL). Which statement best explains the need for this field?

IP datagram header fields Ø Time to Live (TTL) .ʒ ĸ Ø - field ɬ ļ Ğ ?

(A) It can be used to prioritize packets (B) It can be used to reduce delays

(C) It can be used to optimize throughput (D) It can be used to prevent packet looping

[Source: RPSC Programmer 2011 Paper-1 (via pyqp.in)] · Hard

(D) It can be used to prevent packet looping

Nukta: RPSC 2011 o cial key (4). TTL = loop- ; priority ToS , delay/throughput ɣ.

RPSC 2011 o cial key (4). TTL = loop- ; priority ToS , delay/throughput ɮ.

Q498. Suppose the stop-and-wait protocol is used on a link with a bit rate of 64 kilobits per second and 20 milliseconds propagation delay. Acknowledgement transmission time and processing times are negligible. The minimum frame size in bytes to achieve a link utilisation of at least 50% is _____.

64 kbps ɬ link stop-and-wait protocol , propagation delay 20 ms (ACK transmission processing ĵ) 50% link utilisation ģ ij frame size (bytes Ø ʒ ?

(A) 160 (B) 320

(C) 640 (D) 1280

[Source: GATE CS 2015 Set-1] · Hard

(B) 320

Nukta: 160 bas ek Tp ka hissa hai; 640/1280 utilisation formula ke ratio bhool jane se aate hain.

160 Tp; 640/1280 ratio ʐ

Q499. A link has a transmission speed of 10^6 bits/sec. It uses data packets of size 1000 bytes each. Acknowledgements have negligible transmission delay and the same propagation delay as data; processing delays are negligible. The e ciency of the stop-and-wait protocol is exactly 25%. The one-way propagation delay (in milliseconds) is _____.

link ↓speed 10^6 bits/sec packets 1000 bytes ↙ACK transmission delay ¡ , propagation delay data Stop-and-wait ↓e ciency ↓ 25% - propagation delay (ms ₡3 ?

(A) 8 (B) 10

(C) 12 (D) 20

[Source: GATE CS 2015 Set-2] · Hard

(C) 12

Nukta: 8 ms Tt hai Tp nahi; 10/20 round numbers ke jaal hain.

8 ms Tt ; 10/20 ₡

Q500. Consider the following statements regarding the slow start phase of TCP congestion control (cwnd = congestion window, MSS = maximum segment size): (i) cwnd increases by 2 MSS on every successful acknowledgement. (ii) cwnd approximately doubles on every successful acknowledgement. (iii) cwnd increases by 1 MSS every round trip time. (iv) cwnd approximately doubles every round trip time. Which one of the following is correct?

TCP congestion control slow start (cwnd = congestion window, MSS = maximum segment size) ₡ ↓ : (i) successful acknowledgement cwnd 2 MSS (ii) successful acknowledgement cwnd (iii) round trip time ₡wnd 1 MSS (iv) round trip time ₡wnd ₩ ?

(A) (i) and (iii) only / (i) (iii) (B) (ii) and (iv) only / (ii) (iv)

(C) (iv) only / (iv) (D) (ii) only / (ii)

[Source: GATE CS 2018] · Hard

(C) (iv) only

Nukta: Per-ACK doubling (ii) bahut tez hota; (iii) per-RTT sirf 1 MSS congestion avoidance jaisa hai.

Per-ACK doubling (ii) - ; (iii) congestion avoidance

Ω Samjh — Section D — Transport Layer: TCP/UDP, Ports & Sockets

Connection-oriented vs connectionless services, sockets, TCP header & states, 3-way handshake, flow & congestion control, security (SYN flood, hijacking).

Transport layer end-to-end delivery karta hai — TCP (reliable) aur UDP (fast). TCP states, handshake aur congestion control numericals ke favourites hain.

Transport layer end-to-end delivery — TCP (reliable) UDP (fast)

- TCP = connection-oriented, reliable, ordered, full-duplex, byte-stream, flow + congestion control; UDP = connectionless, unreliable, fast, message-stream.

  TCP = connection-oriented, reliable; UDP = connectionless, , unreliable

- TCP header: Source/Dest port, Sequence number, Acknowledgement number, Window size, Flags (SYN, ACK, FIN, RST, PSH, URG), Checksum.

  TCP header: ports, seq, ack, window, flags (SYN/ACK/FIN/RST), checksum

- Three-way handshake: SYN → SYN+ACK → ACK; sequence numbers random (X, Y); data X+1 se shuru.

  Three-way handshake: SYN → SYN+ACK → ACK

- TCP connection close: FIN → ACK → FIN → ACK; states: ESTABLISHED, FIN_WAIT_1/2, TIME_WAIT, CLOSE_WAIT, LAST_ACK.

  TCP close: FIN → ACK → FIN → ACK; states: FIN_WAIT_2, TIME_WAIT...

- Flow control = sliding window (receiver advertised window); unacked bytes ≤ advertised window.

  Flow control = sliding window

- Congestion control: Slow start (cwnd 1 MSS se, per RTT double), Congestion avoidance (AIMD — +1 MSS per RTT), timeout par ssthresh = cwnd/2, Fast retransmit/recovery (3 dup ACKs).

  Congestion control: Slow start (per RTT double), AIMD, ssthresh = cwnd/2

- TCP sequence numbers BYTES count karte hain — wrap-around time = 2^{32} bytes / bandwidth (bytes/sec).

  TCP sequence numbers BYTES ; — wrap time = 2^{32} / bandwidth

- UDP header sirf 8 bytes (ports + length + checksum); DNS, DHCP, VoIP, streaming UDP use karte hain.

  UDP header ; 8 bytes; DNS, VoIP, streaming

- Socket = IP + port; well-known ports service identify karte hain.

  Socket = IP + port

Fakt	TCP	UDP
Connection	oriented (handshake)	connectionless
Reliability	haan (ACK, retransmit)	nahi
Order	maintain hota hai	nahi
Header	20 B + options	8 B
Use	HTTP, FTP, SMTP	DNS, DHCP, VoIP, streaming

Q501. A client opens a TCP connection to a web server on port 80 using source port 49152. What is the complete socket pair?

client source port 49152 port 80 web server TCP connection . socket pair ₶ ?

(A) 192.168.1.10:49152 ↔ 172.16.0.5:80 (B) 192.168.1.10:80 ↔ 172.16.0.5:49152

(C) 49152 ↔ 80 (D) 80 ↔ 49152

[Source: BCI Subject Test-8.pdf] · Easy

(A) 192.168.1.10:49152 ↔ 172.16.0.5:80

Nukta: : client port ephemeral (number), server port well-known (80).

: client port ephemeral (number), server port well-known (80).

Q502. Which port is used by a DHCP client when requesting an IP address?

IP address DHCP client - port ?

(A) UDP 67 (B) UDP 68

(C) TCP 68 (D) TCP 67

[Source: BCI Subject Test-8.pdf] · Easy

(B) UDP 68

Nukta: ″ : server 67, client 68 — DHCP (UDP).

: server 67, client 68 — DHCP (UDP).

Q503. Which of the following is a connectionless transport service?

ɜ ĸ ₡ - connectionless transport service ?

(A) TCP (B) ARP

(C) UDP (D) FTP

[Source: BCI_SCI Subject Test-7.pdf + Oliveboard SSC CGL bank] · Medium

(C) UDP

Nukta: Transport layer Ř : TCP (, connection-oriented) UDP (, connectionless).

Transport layer Ľ : TCP (, connection-oriented) UDP (, connectionless).

Q504. What is a socket in networking?

Networking ₲ocket ₶ ?

(A) IP address + MAC address (B) IP address + Port number

(C) MAC address + Port number (D) IP address + DNS name

[Source: CN Topic Test 6.pdf] · Medium

(B) IP address + Port number

Nukta: Socket pair = (Source IP:Port, Destination IP:Port) — TCP connection .

Socket pair = (Source IP:Port, Destination IP:Port) — TCP connection .

Q505. What term is used for an endpoint in an inter-process communication across a network?

Network inter-process communication ₡ endpoint ₲ - term ?

(A) Port (B) Machine

(C) Socket (D) Pipe

[Source: CN Topic Test-02.pdf] · Medium

(C) Socket

Nukta: Port = address ₵ Ř ; Socket = endpoint (IP+Port).

Port = address ₹ Ľ ; Socket = endpoint (IP+Port).

Q506. What is the purpose of the SYN flag in TCP?

TCP ØSYN flag ṅḷ ⱶ ?

(A) To end a session (B) To initiate (open) a connection

(C) To reset a connection (D) For data transmission

[Source: CN Topic Test 6.pdf] · Medium

(B) To initiate (open) a connection

Nukta: Remember the flags: SYN = start; FIN = end; RST = abort; ACK = confirmation.

Flags : SYN = Ḛ ; FIN = ; RST = ṅ; ACK = ȝÔ.

Q507. In which scenario will TCP's three-way handshake fail?

ȝ ḧl' ȝ ØTCP three-way handshake ȝ ?

(A) Firewall blocks the SYN-ACK (B) High latency

(C) ACK is slightly delayed (D) Server is busy

[Source: CN Topic Test 5.pdf] · Medium

(A) Firewall blocks the SYN-ACK

Nukta: The immediate cause of a failed handshake: the messages are not getting through (firewall/filter).

Handshake : Ø õ (firewall/filter).

Q508. In TCP, which state follows the SYN-SENT state in the normal case?

TCP Ø ij ḧl' ȝ ØSYN-SENT - state ?

(A) SYN-RECEIVED (B) LISTEN

(C) ESTABLISHED (D) CLOSED

[Source: CN Topic Test 6.pdf] · Medium

(C) ESTABLISHED

Nukta: ŏ Ь : CLOSED → SYN-SENT → ESTABLISHED. (Simultaneous open ꞑSYN-RECEIVED ꞑ .)

ij t' : CLOSED → SYN-SENT → ESTABLISHED. (Simultaneous open ØSYN-RECEIVED Ø .)

Q509. Which one of the following attacks exploits TCP's three-way handshake?

ȝ ĸ Ø - attack TCP three-way handshake ?

(A) SYN Flood (B) DNS Spoofing

(C) Man in the Middle (D) ARP Poisoning

[Source: CN Topic Test 5.pdf] · Medium

(A) SYN Flood

Nukta: Handshake = SYN flood; sequence = session hijacking.

Handshake = SYN flood; sequence = session hijacking.

Q510. TCP initially uses which congestion control technique when a connection starts?

Connection ę TCP ę Ø - congestion control technique ?

(A) Congestion avoidance (B) Fast recovery

(C) Slow start (D) Leaky bucket

[Source: CN Topic Test 6.pdf] · Medium

(C) Slow start

Nukta: TCP Ь : Slow start → Congestion avoidance; loss → Fast recovery.

TCP t' : Slow start → Congestion avoidance; loss → Fast recovery.

Q511. What is the purpose of TCP's congestion window (cwnd)?

TCP congestion window (cwnd) ṅḷ Ħ ?

(A) Track unacknowledged packets (B) Control the sending rate to avoid network congestion

(C) Handle encryption (D) Prioritize packets

[Source: CN Topic Test 5.pdf] · Medium

(B) Control the sending rate to avoid network congestion

Nukta: TCP ˣe ective window = min(rwnd, cwnd) — flow congestion ɴ .

TCP ɬe ective window = min(rwnd, cwnd) — flow congestion Ȼ .

Q512. What kind of delay does TCP's window size help handle on a long, high-bandwidth path?

, high-bandwidth path TCP window size ȝ ų ɬ Ȼ ?

(A) Transmission delay (B) Queuing delay

(C) Propagation delay (D) Bandwidth-delay product

[Source: CN Topic Test 5.pdf] · Medium

(D) Bandwidth-delay product

Nukta: Window ≥ BDP link ɣ — high-speed long links Ǥ .

Window ≥ BDP link ɮ — high-speed long links Ů .

Q513. In which protocol is 'port knocking' used as a security mechanism?

ȝ protocol Ȼport knocking' security mechanism ę Ȼ ?

(A) TCP (B) UDP

(C) SSH (D) IP

[Source: CN Topic Test 6.pdf] · Medium

(C) SSH

Nukta: Port knocking = 'secret door knock' — (SSH) .

Port knocking = 'secret door knock' — (SSH) .

Q514. Which of the following supports QoS (Quality of Service) through label switching?

ȝ ĸ Ȼ - label switching ȣ QoS (Quality of Service) ?

(A) TCP (B) UDP

(C) MPLS (D) ARP

[Source: CN Topic Test 6.pdf] · Medium

(C) MPLS

Nukta: MPLS = fast path + QoS classes; Di Serv/IntServ QoS ˝ ɾ

MPLS = fast path + QoS classes; Di Serv/IntServ QoS ȥ

Q515. Virtual circuits are associated with which type of service?

Virtual circuits ȝ ų ɬ service ȥ

(A) Connectionless service (B) Error-free service

(C) Segmentation (D) Connection-oriented service

[Source: CN Topic Test-02.pdf] · Medium

(D) Connection-oriented service

Nukta: Virtual circuit = connection, ʗ data (connection-oriented); Datagram = packet - (connectionless).

Virtual circuit = connection, ȝ data (connection-oriented); Datagram = packet - (connectionless).

Q516. Transport layer multiplexing allows:

Transport layer multiplexing ɬ ȝ :

(A) Multiple hosts to share one IP (B) Multiple processes to share one network connection

(C) Multiple routers to share a routing table

(D) Domain name resolution

[Source: BCI_SCI Subject Test-7.pdf] · Medium

(B) Multiple processes to share one network connection

Nukta: Ports multiplexing ◊ ŕ— browser 443, mail 25, .

Ports multiplexing ɟ ƶ— browser 443, mail 25, .

Q517. Which protocol must use both TCP and UDP for di erent operations?

- protocol - Ȼ ɠ TCP UDP Ȼ ?

(A) DNS (B) HTTP

(C) POP3 (D) Telnet

[Source: BCI Subject Test-8.pdf + CN Topic Test 5.pdf] · Medium

(A) DNS

Nukta: DNS = port 53; UDP for small queries, TCP for big/zone transfers.

DNS = port 53; UDP for small queries, TCP for big/zone transfers.

Q518. Which transport-layer protocol is used to support electronic mail delivery between machines?

Machines electronic mail delivery ɠ - transport-layer protocol ?

(A) SNMP (B) IP

(C) SMTP (D) TCP

[Source: CN Topic Test-03.pdf] · Medium

(D) TCP

Nukta: ЌŖ 'transport-layer protocol' — SMTP application layer , TCP transport .

ųÎ 'transport-layer protocol' — SMTP application layer , TCP transport .

Q519. How many bits are used for port addressing at the transport layer?

Transport layer port addressing ɠ ȝ bits ƺ

(A) 8 (B) 10

(C) 16 (D) 32

[Source: BCI_SCI Subject Test-7.pdf] · Hard

(C) 16

Nukta: Addresses Ř : Data link = 48-bit MAC; Network = 32-bit IP; Transport = 16-bit port.

Addresses Ľ : Data link = 48-bit MAC; Network = 32-bit IP; Transport = 16-bit port.

Q520. The three-way handshake mechanism in TCP aims to:

TCP three-way handshake mechanism ३ ṅḷ ?

(A) Secure the communication (B) Send bits from one device to another

(C) Perform error detection and correction (D) Establish a connection and synchronize sequence numbers

[Source: RPSC Programmer 2024 Paper-1 (with o cial key)] · Hard

(D) Establish a connection and synchronize sequence numbers

Nukta: RPSC o cial key (d). SYN, SYN-ACK, ACK — .

RPSC o cial key (d). SYN, SYN-ACK, ACK — .

Q521. What type of attack is prevented by using random initial sequence numbers in TCP?

TCP Ȼandom initial sequence numbers ३ ų ɬattack Ę ?

(A) Smurf attack (B) SYN Flooding

(C) TCP session hijacking (D) Ping of Death

[Source: CN Topic Test 5.pdf] · Hard

(C) TCP session hijacking

Nukta: Predictable sequence numbers = session hijacking .

Predictable sequence numbers = session hijacking .

Q522. Which field in the TCP header is used for flow control?

TCP header Ȼlow control ģ - field ?

(A) Window Size (B) Sequence Number

(C) Acknowledgment Number (D) Checksum

[Source: CN Topic Test 6.pdf] · Hard

(A) Window Size

Nukta: TCP (Ǵ : Window (flow control) cwnd (congestion control).

TCP ३ Ů : Window (flow control) cwnd (congestion control).

Q523. Which TCP header field ensures that data arrives in order?

TCP header - field ३ ģỈ ३ data ť Ȼõ ?

(A) Checksum (B) Flags

(C) Sequence Number (D) Port Number

[Source: CN Topic Test 5.pdf] · Hard

(C) Sequence Number

Nukta: UDP has no sequence numbers — that is why it gives no ordering guarantee.

UDP Ȼequence ʁ — ģ ť ɬguarantee ʁ.

Q524. Which field in the TCP header ensures data integrity?

TCP header - field data integrity ३ ģỈ ?

(A) Urgent Pointer (B) Sequence Number

(C) Checksum (D) Acknowledgment Number

[Source: CN Topic Test 5.pdf] · Hard

(C) Checksum

Nukta: Integrity = Checksum (TCP/UDP/IP); reliability = ACK + retransmission.

Integrity = Checksum (TCP/UDP/IP); reliability = ACK + retransmission.

Q525. Which of the following is NOT a property of TCP?

३ K ₡ - TCP ʁ ?

(A) Reliable (B) Ordered delivery

(C) Connection-oriented (D) Broadcast

[Source: CN Topic Test 5.pdf] · Hard

(D) Broadcast

Nukta: TCP = reliable + ordered + connection-oriented; ˣ (broadcast) UDP ˣ .

TCP = reliable + ordered + connection-oriented; ɬ (broadcast) UDP ɬ .

Q526. Which of the following provides reliable, connection-oriented communication?

३ K ₡ - , connection-oriented communication ?

(A) TCP (B) IP

(C) UDP (D) All of the above

[Source: MQuestions DCN bank + Oliveboard SSC CGL bank] · Hard

(A) TCP

Nukta: : TCP = phone call (connection + guarantee); UDP = postcard (fast, no promise).

: TCP = phone call (connection + guarantee); UDP = postcard (fast, no promise).

Q527. Which of the following statements about TCP/IP is INCORRECT?

TCP/IP ₡ K ₡ - ?

(A) TCP/IP has high latency (B) Data packets sent over TCP/IP are private

(C) TCP is used by Secure Shell (SSH) (D) TCP/IP uses checksums to check errors

[Source: DSSSB TGT CS (via Testbook solved)] · Hard

(B) Data packets sent over TCP/IP are private

Nukta: DSSSB o cial key (b). : Internet = open postcard, (encryption) .

DSSSB o cial key (b). : Internet = open postcard, (encryption) .

Q528. Why are port numbers included in the TCP header of a segment?

Segment TCP header ₱ort numbers ₶ ₡ ₹

(A) To indicate the correct router interface for forwarding (B) To identify which switch ports should receive the frame

(C) To determine which Layer 3 protocol encapsulates the data (D) To enable a receiving host to forward the data to the appropriate application

[Source: RPSC Programmer 2011 Paper-1 (via pyqp.in)] · Hard

(D) To enable a receiving host to forward the data to the appropriate application

Nukta: RPSC 2011 o cial key (4). transport-layer multiplexing .

RPSC 2011 o cial key (4). transport-layer multiplexing .

Q529. Which one of the following fields of an IP header is NOT modified by a typical IP router?

IP header - field ij IP router Â ¼ ¥³ ?

(A) Checksum (B) Source address

(C) Time to Live (TTL) (D) Length

[Source: GATE CS 2015 Set-1] · Hard

(B) Source address

Nukta: Checksum aur TTL router ke haath badalte hain; length fragmentation se badal sakti hai.

Checksum/TTL ¸length fragmentation

Q530. Which of the following fields of an IP header is/are ALWAYS modified by any router before it forwards the IP packet?

IP packet forward router IP header - fields ?

(A) TTL only / TTL (B) Checksum only / Checksum

(C) TTL and header checksum / TTL header checksum (D) Source address and TTL / Source address TTL

[Source: GATE CS 2024 Set-2] · Hard

(C) TTL and header checksum

Nukta: Checksum ko optional samajhna galti hai — TTL badalne par checksum invalid ho jata hai.

Checksum optional ¥ — TTL checksum ij

Q531. Which of the following fields is/are modified in the IP header of a packet going out of a Network Address Translation (NAT) device from an internal network to an external network?

NAT device internal network external network ¼ packet IP header Ç - fields ¸

(A) Source IP address only / Source IP address (B) Destination IP address only / Destination IP address

(C) Source IP address and header checksum / Source IP address header checksum (D) TTL and destination address / TTL destination address

[Source: GATE CS 2024 Set-1] · Hard

(C) Source IP address and header checksum

Nukta: Destination address wahi rehta hai (jana to server ko hi hai); TTL alag cheez hai.

Destination ; TTL

Q532. Consider sending an IP datagram of size 1420 bytes (including a 20-byte IP header) over a path of two links with a router between them. The first link has MTU 542 bytes and the second link has MTU 360 bytes. The number of fragments delivered at the receiver is _____.

1420 bytes IP datagram (20-byte header ε) links ľ , Ȼ router link MTU 542 bytes Ȧ 360 bytes Receiver ȝ fragments ȭ Ȼ?

(A) 4 (B) 5

(C) 6 (D) 8

[Source: GATE CS 2024 Set-1] · Hard

(C) 6

Nukta: 5/8 fragmentation ko sirf ek baar maante hain; 4 total data ko 360 se seedha bhaag dete hain.

5/8 fragmentation ʐ4 360 ʐ

Q533. In an IPv4 datagram, the M bit is 0, the value of HLEN is 10, the value of total length is 400 and the fragment o set value is 300. The position of the datagram, and the sequence numbers of the first and the last bytes of the payload, respectively, are:

IPv4 datagram ØM bit = 0, HLEN = 10, total length = 400 fragment o set = 300 datagram ɬ ĥľ ȝ payload ɣ bytes sequence numbers ť ʐ

(A) Last fragment, 2400 and 2759 / Last fragment, 2400 2759 (B) First fragment, 2400 and 2759 / First fragment, 2400 2759

(C) Last fragment, 300 and 659 / Last fragment, 300 659 (D) Last fragment, 2400 and 2779 / Last fragment, 2400 2779

[Source: GATE CS 2013] · Hard

(A) Last fragment, 2400 and 2759

Nukta: O set ko byte nahi 8-byte units mein count karna zaroori hai; 2779 header ko do baar jod deta hai.

O set 8-byte units Ȼ ; 2779 header

Q534. Consider the following three statements about link state and distance vector routing protocols for a large network with 500 nodes and 4000 links: S1: The computational overhead in link state protocols is higher than in distance vector protocols. S2: A distance vector protocol (with split horizon) avoids persistent routing loops, but a link state protocol does not. S3: After a topology change, a link state protocol converges faster than a distance vector protocol. Which one is correct?

500 nodes 4000 links network ģ link state distance vector routing protocols Ȼ : S1: Link state Ȼcomputational overhead distance vector ɟ S2: Distance vector (split horizon ɛ) persistent loops , link state ɮ S3: Topology link state, distance vector converge - ?

(A) S1, S2 and S3 are all true / S1, S2, S3 Ȼ (B) S1, S2 and S3 are all false / S1, S2, S3 Ȼ

(C) S1 and S2 are true, S3 is false / S1 S2 , S3 (D) S1 and S3 are true, S2 is false / S1 S3 , S2

[Source: GATE CS 2014 Set-1] · Hard

(D) S1 and S3 are true, S2 is false

Nukta: S2 ulta hai — split horizon count-to-infinity se poora bachata bhi nahi.

S2 Í — split horizon count-to-infinity ɮ

Q535. Which one of the following is TRUE about the interior gateway routing protocols RIP (Routing Information Protocol) and OSPF (Open Shortest Path First)?

Interior gateway routing protocols RIP (Routing Information Protocol) OSPF (Open Shortest Path First) Ȼ - ?

(A) RIP uses distance vector routing and OSPF uses link state routing / RIP distance vector OSPF link state routing ų (B) OSPF uses distance vector routing and RIP uses link state routing / OSPF distance vector RIP link state routing ų

(C) Both RIP and OSPF use link state routing / Ȼink state routing ų ʐ (D) Both RIP and OSPF use distance vector routing / Ȼ distance vector routing ų ʐ

[Source: GATE CS 2014 Set-2] · Hard

(A) RIP uses distance vector routing and OSPF uses link state routing

Nukta: Dono ko ek hi category mein daalna common galti hai.

Ȼ Ż Ȼ

Q536. Consider the following statements about the functionality of an IP based router: I. A router does not modify the IP packets during forwarding. II. It is not necessary for a router to implement any routing protocol. III. A router should reassemble IP fragments if the MTU of the outgoing link is larger than the size of the incoming IP packet. Which of the above statement(s) is/are TRUE?

IP ə router ɬ ɳ Ȼ : I. Router forwarding IP packets ɟ ʁ II. Router ɠ ɳouting protocol ɻ ʁ III. ɛ outgoing link MTU packet router fragments ɛ Ȼ - ?

(A) I only / I (B) II only / II

(C) I and III only / I III (D) None of the above / ɳ ʁ

[Source: GATE CS 2020] · Hard

(B) II only

Nukta: Fragmentation ke baad re-fragmentation ka kaam router ka hai, jodna nahi.

Router re-fragment , ʁ

Q537. Host A (on TCP/IPv4 network A) sends an IP datagram D to host B (on TCP/IPv4 network B) with no transmission error. When D reaches B, which of the following IP header field(s) may be di erent from that of the original datagram D? I. TTL II. Checksum III. Fragment O set

Host A (TCP/IPv4 network A) ɜ ɜ Ůɸ IP datagram D, host B D B ɵ̄ ɜ ĸ Ȼ - IP header fields datagram ɠ È ʔ I. TTL II. Checksum III. Fragment O set

(A) I only / I (B) II and III only / II III

(C) I and II only / I II (D) I, II and III / I, II III Ȼ

[Source: GATE CS 2014 Set-3] · Hard

(D) I, II and III

Nukta: O set ko 'sirf fragmentation ke time' samajh kar chhod dena galti hai — puchha hai 'ho sakta hai'.

O set — ' '

Ω Samjh — Section E — Session, Presentation & Application Layer

Presentation-layer functions, application protocols & ports (HTTP/HTTPS/FTP/SMTP/DNS/SSH/Telnet/NFS/LDAP), TCP/IP layer-protocol matching.

Upar ki 3 layers: sessions sambhalna, data ka format, aur user ki services. Application layer ke protocols aur unke ports — exam ka constant area.

ɬ 3 layers: sessions, data format, user services Application protocols ports

- Session layer — dialog control (simplex/half/full dialogs), synchronization checkpoints (sync points) se recovery.

  Session layer — dialog control, synchronization checkpoints

- Presentation layer — translation (ASCII/EBCDIC), encryption/decryption, compression — 'data ki presentation'.

  Presentation layer — translation, encryption, compression

- Application layer — user ke liye services: HTTP (web), FTP (files), SMTP/POP3/IMAP (mail), DNS (naam), DHCP (auto IP), SNMP (management).

  Application layer — HTTP, FTP, SMTP/POP3/IMAP, DNS, DHCP, SNMP

- Ports: HTTP 80, HTTPS 443, FTP 20/21, SSH 22, Telnet 23, SMTP 25, DNS 53, DHCP 67/68, POP3 110, IMAP 143, SNMP 161/162, RIP 520.

  Ports: HTTP 80, HTTPS 443, FTP 20/21, SMTP 25, DNS 53, POP3 110...

- E-mail: SMTP se bhejte hain (push), POP3 download karta hai, IMAP server par hi manage karta hai.

  E-mail: SMTP ( ), POP3 ( ), IMAP (server manage)

- FTP — control (21) + data (20) connections; active/passive modes.

  FTP — control + data connections

- HTTP — stateless, request/response (GET, POST); HTTPS = HTTP + SSL/TLS encryption.

  HTTP — stateless; HTTPS = HTTP + SSL/TLS

- Domain name hierarchy: root → TLD (.com/.in) → domain → host; DNS resolution recursive/iterative.

  Domain hierarchy: root → TLD → domain; DNS resolution

Protocol	Port	Kaam
HTTP / HTTPS	80 / 443	web pages
FTP	20/21	file transfer
SMTP / POP3 / IMAP	25 / 110 / 143	e-mail
DNS	53	naam → IP
DHCP	67/68	auto IP configuration
SNMP	161/162	network management
SSH / Telnet	22 / 23	remote login

Q538. A server receives a packet with destination port 443. This implies:

server destination port 443 packet ३ . ŋ:

(A) HTTP request (B) HTTPS tra c

(C) FTP command (D) Email tra c

[Source: BCI Subject Test-8.pdf] · Easy

(B) HTTPS tra c

Nukta: Ports: 80=HTTP, 443=HTTPS, 21/20=FTP, 22=SSH, 23=Telnet, 25=SMTP, 53=DNS, 67/68=DHCP.

Ports: 80=HTTP, 443=HTTPS, 21/20=FTP, 22=SSH, 23=Telnet, 25=SMTP, 53=DNS, 67/68=DHCP.

Q539. For what application is the HTTP protocol used?

HTTP protocol ३ ग ?

(A) To increase upload speed (B) To develop a website

(C) To deliver data on the World Wide Web

(D) To download images only

[Source: SSC MTS computer PYQ (via Testbook solved)] · Easy

(C) To deliver data on the World Wide Web

Nukta: SSC MTS o cial key (3). HTTP = web ×
(port 80).

SSC MTS o cial key (3). HTTP = web ↓ (port 80).

Q540. The structure or format of data is called:

Data structure format :

(A) Syntax (B) Struct

(C) Semantic (D) None of the above

[Source: Unit Test-4 Solution.pdf (OCR), p14] · Medium

(A) Syntax

Nukta: Rule: Syntax = how it looks; Semantics = what it means. Both must be agreed for communication.

३ : Syntax = ε ; Semantics = ₹ ŋ . Communication Ø Ø ε .

Q541. Which protocol layer uses protocols such as WWW, HTTP, FTP, SMTP and e-mail?

- protocol layer WWW, HTTP, FTP, SMTP e-mail protocols ?

(A) Application layer (B) Transport layer

(C) Internet layer (D) Hardware layer

[Source: CN Topic Test-02.pdf + CN Topic Test-01.pdf] · Medium

(A) Application layer

Nukta: : Layer 7 = (HTTP/FTP/SMTP/DNS); Layer 4 = TCP/UDP; Layer 3 = IP.

: Layer 7 = (HTTP/FTP/SMTP/DNS); Layer 4 = TCP/UDP; Layer 3 = IP.

Q542. Which layer protocols provide user and application programme support such as passwords, resource sharing and file transfer?

३ layer protocols ŋ application programme passwords, resource sharing file transfer ?

(A) Layer 2 protocols (B) Layer 5 protocols

(C) Layer 7 protocols (D) Layer 10 protocols

[Source: CN Topic Test-02.pdf] · Medium

(C) Layer 7 protocols

Nukta: OSI has exactly 7 layers — there can be no 'Layer 10'.

OSI Ø layers ← 'Layer 10' ध .

Q543. Which of the following protocols is primarily used for secure web browsing?

ȝ к Ȼ - protocol ħ ę ɯ web browsing ɠ ?

(A) FTP (B) HTTP

(C) HTTPS (D) SMTP

[Source: CN Topic Test-01.pdf] · Medium

(C) HTTPS

Nukta: HTTPS port 443; HTTP 80. (lock) site = HTTPS.

HTTPS port 443; HTTP 80. (lock) site = HTTPS.

Q544. Which protocol is generally used to encrypt data between a web browser and a web server?

Web browser web server data encrypt ɠ - protocol ?

(A) SSL/TLS (B) Digital Signature

(C) IPSec (D) TCP

[Source: CN Topic Test-01.pdf] · Medium

(A) SSL/TLS

Nukta: HTTPS = HTTP over SSL/TLS. IPSec network-level VPN ɓ .

HTTPS = HTTP over SSL/TLS. IPSec network-level VPN ɠ .

Q545. Which protocol is used for secure shell (remote) login?

Secure shell (remote) login ɠ - protocol ?

(A) FTP (B) SSH

(C) Telnet (D) SNMP

[Source: CN Topic Test 6.pdf + Medium 30 CN MCQ bank] · Medium

(B) SSH

Nukta: SSH = port 22 (encrypted); Telnet = port 23 (, ɯ).

SSH = port 22 (encrypted); Telnet = port 23 (, ɯ).

Q546. A firewall blocks all ports except 22. What service remains accessible?

firewall port 22 . - ?

(A) FTP (B) SSH

(C) HTTPS (D) SMTP

[Source: BCI Subject Test-8.pdf] · Medium

(B) SSH

Nukta: Identify the port, identify the service — that is the trick of this question.

Port , — ų Î .

Q547. Which port does Telnet use by default?

Telnet ɛ Í ę - port ?

(A) 21 (B) 23

(C) 22 (D) 25

[Source: CN Topic Test 6.pdf] · Medium

(B) 23

Nukta: : 22 = SSH, 23 = Telnet, 25 = SMTP, 21 = FTP.

: 22 = SSH, 23 = Telnet, 25 = SMTP, 21 = FTP.

Q548. Which of the following uses port number 25?

३ K Ȼ - port number 25 ?

(A) FTP (B) HTTP

(C) SMTP (D) Telnet

[Source: CN Topic Test 6.pdf] · Medium

(C) SMTP

Nukta: Mail ports: 25 (SMTP), 110 (POP3), 143 (IMAP).

Mail ports: 25 (SMTP), 110 (POP3), 143 (IMAP).

Q549. Which protocol uses di erent ports for control and data connections?

- protocol control data connections ʃ
- ports ?

(A) FTP (B) HTTP

(C) DNS (D) Telnet

[Source: BCI_SCI Subject Test-7.pdf] · Medium

(A) FTP

Nukta: FTP = double connection (control + data); HTTP/SMTP/DNS N ¥.

FTP = double connection (control + data); HTTP/SMTP/DNS Ȼ ¥.

Q550. Which protocol is used for file transfer over a network?

Network file transfer ʃ - protocol ?

(A) FTP (B) HTTP

(C) DNS (D) ARP

[Source: CN Topic Test 6.pdf + Medium 30 CN MCQ bank] · Medium

(A) FTP

Nukta: HTTP files ɒ ɾ 'file transfer' protocol FTP .

HTTP files ɜ ʐ 'file transfer' protocol FTP .

Q551. Which application allows a user to access and use remote files without actually transferring them?

- application ɳ remote files ३ l' Ȼ ə ३ ?

(A) DNS (B) FTP

(C) NFS (D) Telnet

[Source: SSC CGL Internet & Networking (via EduRev bank)] · Medium

(C) NFS

Nukta: NFS = remote files local ; FTP = files x copy .

NFS = remote files local ; FTP = files ɬ copy .

Q552. Which protocol translates domain names to IP addresses?

- protocol domain names IP addresses Ȼ ?

(A) DHCP (B) DNS

(C) HTTP (D) FTP

[Source: CN Topic Test 5.pdf + CN Topic Test-01.pdf] · Medium

(B) DNS

Nukta: DNS = name→IP; ARP = IP→MAC; DHCP = IP .

DNS = name→IP; ARP = IP→MAC; DHCP = IP .

Q553. What does a 'ping' command test?

'ping' command Ħ ?

(A) IP configuration (B) Host reachability

(C) DNS resolution (D) Port scanning

[Source: CN Topic Test 6.pdf] · Medium

(B) Host reachability

Nukta: ping = 'Í ?' — reachability + round-trip time.

ping = 'Ħ ?' — reachability + round-trip time.

Q554. Which HTTP status code means 'Forbidden'?

- HTTP status code ң 'Forbidden'?

(A) 200 (B) 301

(C) 403 (D) 404

[Source: Oliveboard SSC CGL Tier-2 bank] · Medium

(C) 403

Nukta: : 200 OK, 301 Moved, 403 Forbidden, 404 Not Found, 500 Server Error.

: 200 OK, 301 Moved, 403 Forbidden, 404 Not Found, 500 Server Error.

Q555. Lightweight Directory Access Protocol (LDAP) is used for:

Lightweight Directory Access Protocol (LDAP) ३ ?

(A) Routing the packets (B) Authentication and directory access

(C) Obtaining an IP address (D) Domain name resolving

[Source: Unit Test-4 Solution.pdf (OCR), p29] · Medium

(B) Authentication and directory access

Nukta: LDAP = the network's 'address book + identity-desk counter'.

LDAP = network ' - ŕ' + - Ů '.

Q556. Which protocol is used for receiving an email?

Email ų Ĵ ģ - protocol ?

(A) HTTP (B) FTP

(C) POP3 (D) SSH

[Source: RRB NTPC previous-year computer banks (via adda247 50-PYQ PDF)] · Medium

(C) POP3

Nukta: Mail ˣ ˝: SMTP (25), POP3 (110) IMAP (143).

Mail ɬ : SMTP (25), POP3 (110) IMAP (143).

Q557. Which of the following is the odd one out?

३ K Ȼ - ?

(A) IMAP (B) POP

(C) SNMP (D) SMTP

[Source: RRB NTPC previous-year computer banks (via adda247 50-PYQ PDF)] · Medium

(C) SNMP

Nukta: Mail family: SMTP/POP3/IMAP; Management family: SNMP.

Mail family: SMTP/POP3/IMAP; Management family: SNMP.

Q558. ICMP is a network-layer protocol used by devices to diagnose network issues. ICMP stands for:

ICMP network-layer protocol devices network l' Ȼ . ICMP :

(A) Internet Control Manage Process (B) Internet Comfort Mass Protocol

(C) Information Control Manage Protocol (D) Internet Control Message Protocol

[Source: IBPS PO Computer Awareness PYQ (via Testbook solved)] · Medium

(D) Internet Control Message Protocol

Nukta: The other names are made up — the real name contains 'Control Message'.

ɻ ʐ— ØControl Message' .

Q559. POP3 stands for:

POP3 :

(A) Peer to Peer Protocol (B) Peer to Post Protocol

(C) Peer O ce Protocol (D) Post O ce Protocol

[Source: SBI/IBPS Computer Awareness Capsule 2018 (IBT India)] · Medium

(D) Post O ce Protocol

Nukta: POP = Post O ce Protocol. : POP3 mail , SMTP .

POP = Post O ce Protocol. : POP3 mail , SMTP .

Q560. SLIP stands for:

SLIP :

(A) System Line Internet Protocol (B) Serial Line Internet Protocol

(C) Signal Line Internet Protocol (D) Signal Internet Protocol

[Source: IBPS/banking computer banks (via tpointtech MCQ)] · Medium

(B) Serial Line Internet Protocol

Nukta: SLIP Ÿ ʭ ″ PPP (byte-oriented) — ɴpoint-to-point.

SLIP ǵ ɟ PPP (byte-oriented) — Ø point-to-point.

Q561. The functionalities of the presentation layer include:

Presentation layer ƬȻ ʒ ʐ

(A) Data compression (B) Data encryption

(C) Data description (translation) (D) All of the mentioned

[Source: CN Topic Test-02.pdf] · Hard

(D) All of the mentioned

Nukta: Presentation = data (ɣ , , w); Application = .

Presentation = data (ę , , w); Application = .

Q562. Which of the following protocols is NOT secure?

३ K Ȼ - protocol ɢw ʁ ?

(A) HTTPS (B) FTP

(C) SSH (D) SFTP

[Source: CN Topic Test 6.pdf] · Hard

(B) FTP

Nukta: ϑw ˣ : HTTP, FTP, Telnet, SMTP (ζ TLS). ϑw : HTTPS, SFTP, SSH, SMTPS.

ɢw ɬ : HTTP, FTP, Telnet, SMTP (३ TLS). ɢw : HTTPS, SFTP, SSH, SMTPS.

Q563. Which of the following protocols is used for transferring email messages from one machine to another?

३ K Ȼ - protocol email Ȼ machine Ȧ Ȼ ɠ ?

(A) SNMP (B) FTP

(C) TELNET (D) SMTP

[Source: RPSC Programmer 2024 Paper-1 (with o cial key)] · Hard

(D) SMTP

Nukta: RPSC o cial key (d). = SMTP; = POP3/IMAP.

RPSC o cial key (d). = SMTP; = POP3/IMAP.

Q564. In one of the pairs below, both protocols can use multiple TCP connections between the same client and the same server. Which pair is it?

ε ȻȻ ३ Ȼ Ȼprotocols client server ɳTCP connections ʔ

(A) HTTP, FTP (B) HTTP, TELNET

(C) FTP, SMTP (D) HTTP, SMTP

[Source: CN Topic Test-02.pdf] · Hard

(A) HTTP, FTP

Nukta: ɴ'multi-connection' protocols ɾ— ″ ″ .

Ȼmulti-connection' protocols ʐ— ɾ ɳ.

Q565. Identify the true statements: (1) HTTP may use di erent TCP connections for di erent objects of a webpage if non-persistent connections are used (2) FTP uses two TCP connections, one for data and another for control (3) TELNET and FTP can only use TWO connections at a time

Ȼ(1) non-persistent connections Ȼ HTTP webpage - objects ɠ TCP connections (2) FTP TCP connections , data ɠ Ȧ control ɠ (3) TELNET FTP Ȼ TWO connections ʐ

(A) (1) only (B) (1) and (2)

(C) (2) and (3) (D) (1), (2) and (3)

[Source: CN Topic Test-03.pdf] · Hard

(B) (1) and (2)

Nukta: HTTP = multi-connection capable; FTP = ; 'only two' ζ ʏ ʏ.

HTTP = multi-connection capable; FTP = ; 'only two' ३ ʁ ʁ.

Q566. Which of the following is NOT an application layer protocol?

३ K Ȼ - application layer protocol ᵾ ?

(A) HTTP (B) IP

(C) TELNET (D) FTP

[Source: SSC CGL Internet & Networking (via EduRev bank)] · Hard

(B) IP

Nukta: Layer 7 : HTTP, FTP, SMTP, DNS, Telnet, SNMP, DHCP, TFTP...

Layer 7 : HTTP, FTP, SMTP, DNS, Telnet, SNMP, DHCP, TFTP...

Q567. Which is the FALSE statement?

- ?

(A) Telnet is used for remote login (B) FTP can transfer files between two hosts

(C) TFTP relies on TCP (D) SMTP is used to send mail messages

[Source: SSC CGL Internet & Networking (via EduRev bank)] · Hard

(C) TFTP relies on TCP

Nukta: TFTP = (UDP, port 69); FTP = (TCP, ports 20/21).

TFTP = r(UDP, port 69); FTP = r(TCP, ports 20/21).

Q568. Match TCP/IP layers and protocols — (P) Transport layer (Q) Network (internet) layer (R) Application layer; (1) ICMP (2) FTP (3) UDP. Which matching is correct?

TCP/IP layers protocols ३ — (P) Transport layer (Q) Network (internet) layer (R) Application layer; (1) ICMP (2) FTP (3) UDP. - ३ ?

(A) P-2, Q-1, R-3 (B) P-1, Q-2, R-3

(C) P-3, Q-1, R-2 (D) P-3, Q-2, R-1

[Source: Random Test- Rajasthan PYQ.pdf (OCR) + answer_keys.py Q23] · Hard

(C) P-3, Q-1, R-2

Nukta: Layers : Application = FTP/HTTP/SMTP/DNS; Transport = TCP/UDP; Internet = IP/ICMP/IGMP.

Layers : Application = FTP/HTTP/SMTP/DNS; Transport = TCP/UDP; Internet = IP/ICMP/IGMP.

Q569. Which one of the following is NOT a client-server application?

३ K Ȼ - client-server application ᵾ ?

(A) Internet Chat (B) Web browsing

(C) E-mail (D) Ping

[Source: RPSC Programmer 2011 Paper-1 (via pyqp.in)] · Hard

(D) Ping

Nukta: RPSC 2011 o cial key (4). Ping = reachability , application service ᵡ.

RPSC 2011 o cial key (4). Ping = reachability , application service ᵾ.

Q570. Which of the following is/are example(s) of stateful application layer protocols? I. HTTP II. FTP III. TCP IV. POP3

३ K Ȼ - stateful application layer protocol ? I. HTTP II. FTP III. TCP IV. POP3

(A) I and II only / I II (B) II only / II

(C) I only / I (D) II and III only / II III

[Source: GATE CS 2016 Set-1] · Hard

(B) II only

Nukta: TCP ko application protocol samajhna option jaal hai; HTTP stateful sirf modern keep-alive mein lagta hai.

TCP application protocol ʁ; HTTP ǵ ŋ ३ keep-alive Ȼstateful

Q571. Identify the correct sequence in which the following packets are transmitted by a host when a browser requests a web page from a remote server, assuming that the host has just been restarted:

ɢ host - restart õ browser remote server web page ३ K packets t' Ȼransmit Ȼ:

(A) HTTP GET request, DNS query, TCP SYN

(B) DNS query, HTTP GET request, TCP SYN

(C) DNS query, TCP SYN, HTTP GET request

(D) TCP SYN, DNS query, HTTP GET request

[Source: GATE CS 2016 Set-2] · Hard

(C) DNS query, TCP SYN, HTTP GET request

Nukta: GET pehle ja hi nahi sakta — connection bina handshake ke nahi banti; restart yaani DNS cache bhi khaali.

GET ʁ ; restart ȻDNS cache

Q572. Which of the following protocol pairs can be used to send and retrieve e-mails (in that order)?

ŋ ų Ĵ (t' Ȼ ǵ ३ K Ȼ ३ protocol ų ?

(A) SMTP and POP3 / SMTP POP3 (B) POP3 and SMTP / POP3 SMTP

(C) IMAP and SMTP / IMAP SMTP (D) SMTP and IMAP only / SMTP IMAP

[Source: GATE CS 2019] · Hard

(A) SMTP and POP3

Nukta: Ulti jodi (POP3 and SMTP) order ke against hai; IMAP bhi retrieve karta hai par sabse common jodi SMTP+POP3 hai.

Í ; IMAP retrieve ij SMTP+POP3

Q573. Consider a long-lived TCP session with an end-to-end bandwidth of 1 Gbps (= 10^9 bits/second). The session starts with a sequence number of 1234. The minimum time (in seconds, rounded to the closest integer) before this sequence number can be used again is _____.

TCP session ł bandwidth 1 Gbps (= 10^9 bits/second) sequence number 1234 ę sequence number ų ij (seconds Ø3 œ) ³ ?

(A) 34 (B) 35

(C) 43 (D) 32

[Source: GATE CS 2018] · Hard

(A) 34

Nukta: Bits mein hisaab karke 4 s; ya 2^{32} bits samajh kar galat number — bytes ka chakkar hai.

Bits Ø (4 s) 2 bits

Q574. In the TCP three-way handshake, host P sends a connection request to Q with SYN = 1, SEQ = X, ACK = 0. Q accepts the connection. Which of the following represents the information in the TCP segment header sent by Q to P?

TCP three-way handshake Øost P, Q connection request : SYN = 1, SEQ = X, ACK = 0 Q connection Ï Q Â P segment header Øł ?

(A) SYN = 1, SEQ = Y, ACK = X (B) SYN = 1, SEQ = X, ACK = Y

(C) SYN = 1, SEQ = Y, ACK = X+1 (D) SYN = 0, SEQ = Y, ACK = X+1

[Source: GATE CS 2021 Set-2] · Hard

(C) SYN = 1, SEQ = Y, ACK = X+1

Nukta: ACK = X galat (yeh khud ko X+1 expect karna hua); SYN=0 second segment jaisa hota.

ACK = X (X+1 ŵ); SYN=0 À

Q575. Consider data transfer using TCP over a 1 Gbps link. Assuming that the maximum segment lifetime (MSL) is set to 60 seconds, the minimum number of bits required for the sequence number field of the TCP header, to prevent the sequence number space from wrapping around during the MSL, is _____.

1 Gbps link TCP data transfer Maximum segment lifetime (MSL) 60 seconds MSL sequence number space wrap ģ TCP header sequence number field Ø ij ³ bits ε ?

(A) 32 (B) 33

(C) 34 (D) 36

[Source: GATE CS 2022] · Hard

(B) 33

Nukta: 32 bits mein space 60 s ß kam padta hai; 34/36 zyada hain (minimum pucha hai).

32 ; 34/36 ŏ (ij)

Q576. A TCP client and server complete data transfer. The client calls close; a FIN segment is sent to the server, which responds with an ACK received by the client. As per the TCP connection state diagram (RFC 793), in which state does the client-side TCP connection wait for the FIN from the server?

TCP client-server data transfer õ Client close ; server FIN ; server ACK client ३ RFC 793 ।state diagram client-side connection server FIN ।ų w ३ state Ȼ ?

(A) TIME_WAIT (B) FIN_WAIT_2

(C) CLOSE_WAIT (D) LAST_ACK

[Source: GATE CS 2017 Set-1] · Hard

(B) FIN_WAIT_2

Nukta: TIME_WAIT aakhri state hai (dono FIN ke baad); CLOSE_WAIT server ki hoti hai.

TIME_WAIT ØCLOSE_WAIT server ।

Q577. In the TCP three-way handshake, let the three packets exchanged during connection establishment be P1, P2 and P3 in order. Which of the following is TRUE about the TCP header flags set in these packets?

TCP three-way handshake Ȼonnection establishment packets P1, P2 P3 (t' Ȼ ʐ Ȼ TCP header flags Ȼ Ħ ?

(A) P1: SYN; P2: SYN, ACK; P3: ACK (B) P1: SYN, ACK; P2: ACK; P3: FIN

(C) P1: SYN; P2: ACK; P3: FIN (D) P1: ACK; P2: SYN; P3: ACK

[Source: GATE CS 2025 Set-1] · Hard

(A) P1: SYN; P2: SYN, ACK; P3: ACK

Nukta: FIN connection release ka flag hai; pehle packet par ACK kabhi nahi hota.

FIN release flag ; packet ACK ४

Q578. The World Wide Web (WWW) is:

World Wide Web (WWW) Ħ ?

(A) another name for the Internet / Internet Ȧ (B) a worldwide connection of computers / Ĵ Ȼ ३ Ò ê

(C) a worldwide collection of information / ३ Ò ê ŧ (D) a collection of linked information residing on computers connected by the Internet / Internet Ĵ Ȼ link ।ŋ Ȼ ŧ

[Source: EMRS PGT Computer Science] · Hard

(D) a collection of linked information residing on computers connected by the Internet

Nukta: Internet aur WWW ko same bolna sabse common galti hai; 'connection of computers' to network ki paribhasha hai.

Internet WWW ; 'connection' network

TOPIC 7 — Network Security

53 MCQs • Theory → Questions (Easy → Medium → Hard)

Ω Samjh — Section A — Security Goals & Attacks

CIA triad, attack-goal mapping (interception/modification/DoS/spoofing), active vs passive attacks, malware phases, security tools.

Security ka maksad: information ko threats se bachana. Core goals = CIA triad (Confidentiality, Integrity, Availability) + Authentication + Non-repudiation.

Security ṇ : information threats = CIA triad + Authentication + Non-repudiation

- Confidentiality — sirf authorized log dekh sakein (encryption se).
  Confidentiality — authorized (encryption)
- Integrity — data badla na jaaye (hash, digital signature se).
  Integrity — data (hash, digital signature)
- Availability — resource time par mile (DoS se bachao).
  Availability — (DoS)
- Authentication = sahi pehchaan; Non-repudiation = baad mein inkar na ho sake.
  Authentication = ; Non-repudiation =
- Attack goals: Interception (confidentiality par), Modification (integrity par), Interruption/DoS (availability par), Fabrication (authenticity par).
  Attack goals: Interception, Modification, Interruption/DoS, Fabrication
- Passive attack = sirf sunna/snooping (detection mushkil); Active attack = data badalna/interrupt karna (prevention).
  Passive attack = ; Active attack = /
- Common threats: Spoofing (nakli pehchaan), Sni ng (packets sunna), Phishing (nakli sites), Malware: Virus (file ke saath), Worm (apne aap spread), Trojan (nakli software), Time bomb (time trigger).
  Threats: Spoofing, Sni ng, Phishing, Virus, Worm, Trojan, Time bomb
- Worm resources khaata hai (CPU/memory) — khud spread hota hai, host file ki zaroorat nahi.
  Worm resources —

Goal	Kis par hamla	Bachav
Confidentiality	Interception/Sni ng	Encryption
Integrity	Modification	Hash, digital signature
Availability	DoS/DDoS	Firewall, redundancy
Authenticity	Spoofing/Fabrication	Authentication

Q579. In the virus life cycle, which phase actually performs the malicious activity?

Virus life cycle - phase malicious activity ?

(A) Infection Phase (B) Triggering Phase
(C) Dormant Phase (D) Execution (Payload) Phase

[Source: Repo: BCI Subject Test-9 (BCI/SCI Batch)] · Easy

(D) Execution (Payload) Phase

Nukta: Virus phases: Dormant (idle) → Propagation (copy) → Triggering (condition met) → Execution (payload runs).

Virus phases: Dormant (idle) → Propagation (copy) → Triggering (condition met) → Execution (payload runs)

Q580. Which backup medium provides Write-Once-Read-Many (WORM) functionality?

- backup medium Write-Once-Read-Many (WORM) functionality ?

(A) Hard disk (B) Magnetic tape

(C) Optical disk (CD-R / DVD-R) (D) Flash drive

[Source: Repo: BCI Subject Test-9 (BCI/SCI Batch)] · Easy

(C) Optical disk (CD-R / DVD-R)

Nukta: WORM = optical write-once media (CD-R/DVD-R/BD-R). Rewritable: HDD, tape (LTO), flash, CD-RW.

WORM = optical write-once media (CD-R/DVD-R/BD-R) Rewritable: HDD, tape (LTO), flash, CD-RW

Q581. Which attack becomes possible if the SAME key is reused many times in symmetric encryption?

ε symmetric encryption Ȼ key ŋ, - attack ?

(A) Replay attack (B) Known-plaintext attack

(C) Man-in-the-middle attack (D) Phishing attack

[Source: Repo: BCI Subject Test-9 (BCI/SCI Batch)] · Easy

(B) Known-plaintext attack

Nukta: One-time pads are safe ONLY with a fresh key per message. Reuse → pattern analysis. Exam favourite link: key reuse → known-plaintext.

One-time pads message ŋkey ɢw ƺ Reuse → pattern analysis Exam favourite link: key reuse → known-plaintext

Q582. Which of the following tools is mainly used for password cracking?

ȝ ĸ Ȼ - tool ħ password cracking ɠ ?

(A) Wireshark (B) Nmap

(C) Cain and Abel (D) Angry IP Scanner

[Source: Repo: BCI Subject Test-9 (BCI/SCI Batch)] · Easy

(C) Cain and Abel

Nukta: Security tools: Wireshark = packet analyser, Nmap = port scanner, Cain&Abel = password cracker, Angry IP Scanner = IP scanner, Snort = IDS.

Security tools: Wireshark = packet analyser, Nmap = port scanner, Cain&Abel = password cracker, Angry IP Scanner = IP scanner, Snort = IDS

Q583. Which port and IP address scanner is most famous among users?

Users ɥɠ ắ port IP address scanner - ?

(A) Cain and Abel (B) Angry IP Scanner

(C) Snort (D) Ettercap

[Source: Repo: BCI Subject Test-9 (BCI/SCI Batch)] · Easy

(B) Angry IP Scanner

Nukta: Cain&Abel = password cracking; Angry IP = IP/port scan; Snort = IDS; Ettercap = sni ng/ ARP spoofing.

Cain&Abel = password cracking; Angry IP = IP/port scan; Snort = IDS; Ettercap = sni ng/ARP spoofing

Q584. The term 'CIA triad' in information security refers to:

Information security ØCIA triad' Ķ ३ ?

(A) Cryptography, Integrity, Authentication
(B) Confidentiality, Integrity, Availability
(C) Confidentiality, Information, Access (D) Control, Information, Access

[Source: Repo: BCI_SCI Subject Test-01 (CN and Security)] · Medium

(B) Confidentiality, Integrity, Availability

Nukta: CIA = Confidentiality, Integrity, Availability. AAA (authentication, authorisation, accounting) is a separate trio.

CIA = Confidentiality, Integrity, Availability AAA (authentication, authorisation, accounting) trio

Q585. A modification attack violates which security goal?

Modification attack - security ł Í ?

(A) Confidentiality (B) Integrity
(C) Availability (D) Authentication

[Source: Repo: BCI_SCI Subject Test-01 (CN and Security)] · Medium

(B) Integrity

Nukta: Attack→goal: Eavesdropping/ Interception → Confidentiality. Modification → Integrity. DoS/Interruption → Availability. Spoofing → Authentication.

Attack→goal: Eavesdropping/Interception → Confidentiality Modification → Integrity DoS/ Interruption → Availability Spoofing → Authentication

Q586. Which of the following attacks violates confidentiality?

३ ĸ Ø - attack confidentiality Í ?

(A) Interception (B) Modification
(C) Fabrication (D) Interruption

[Source: Subject Test Solution-1 Part-2 (CN & Security) slides] · Medium

(A) Interception

Nukta: Attack map: Interception→Confidentiality, Modification→Integrity, Fabrication→Authenticity, Interruption→Availability.

Attack map: Interception→Confidentiality, Modification→Integrity, Fabrication→Authenticity, Interruption→Availability

Q587. A Denial of Service (DoS) attack violates which goal?

Denial of Service (DoS) attack - ł Í ?

(A) Confidentiality (B) Integrity
(C) Availability (D) Authentication

[Source: Repo: BCI_SCI Subject Test-01 (CN and Security)] · Medium

(C) Availability

Nukta: DoS = Availability attack. DDoS = distributed DoS. Smurf/ SYN-flood = DoS variants.

DoS = Availability attack DDoS = distributed DoS Smurf/ SYN-flood = DoS variants

Q588. A spoofing attack targets which security aspect?

Spoofing attack 3 security GW ?

(A) Authentication (B) Integrity

(C) Availability (D) Confidentiality

[Source: Repo: BCI_SCI Subject Test-01 (CN and Security)] · Medium

(A) Authentication

Nukta: Spoofing/impersonation → Authentication. Phishing → steals credentials (auth). Sni ng → Confidentiality.

Spoofing/impersonation → Authentication Phishing → credentials (auth) Sni ng → Confidentiality

Q589. Which of the following is an ACTIVE attack?

3 K Ȼ - ACTIVE attack ?

(A) Interception (B) Modification

(C) Tra c analysis (D) Eavesdropping

[Source: Repo: BCI_SCI Subject Test-01 (CN and Security)] · Medium

(B) Modification

Nukta: Passive = observe only (eavesdropping, tra c analysis). Active = change/disrupt (modification, fabrication, interruption/DoS).

Passive = (eavesdropping, tra c analysis) Active = / (modification, fabrication, interruption/DoS)

Q590. Match the attacks with their descriptions: (a) Phishing (b) DoS Attack (c) Man-in-the-Middle (d) Spoofing — (i) Flooding network tra c (ii) Stealing user credentials (iii) Intercepting communication (iv) Impersonation attack

Ȼ 3 3 : (a) Phishing (b) DoS Attack (c) Man-in-the-Middle (d) Spoofing — (i) ɳ ʖ (ii) (iii) Ȼ l' w (iv) ɬ

(A) a–ii, b–i, c–iii, d–iv (B) a–i, b–ii, c–iii, d–iv

(C) a–iii, b–i, c–ii, d–iv (D) a–ii, b–iii, c–i, d–iv

[Source: Repo: BCI Subject Test-9 (BCI/SCI Batch) — Q3 match (codes recovered)] · Medium

(A) a–ii, b–i, c–iii, d–iv

Nukta: Attack one-liners: Phishing=fake emails/sites, DoS=flood, MitM=middle interception, Spoofing=fake identity.

Attack one-liners: Phishing=fake emails/sites, DoS=flood, MitM=middle interception, Spoofing=fake identity

Q591. 'Junk email' is also known as:

'Junk email' Ħ ₹

(A) Spam (B) Spoof

(C) Sni er (D) Spool

[Source: RRB NTPC computer banks (via practicemock PYQ set)] · Medium

(A) Spam

Nukta: Spam = junk mail; Spoof = 0 ; Sni er = ˝ .

Spam = junk mail; Spoof = 0 ; Sni er = .

Q592. The goal of 'Integrity' ensures that:

'Integrity' ł ³ g Ì ³ :

(A) Data can be accessed only by authorised users (B) Data is accurate and has not been modified

(C) Data is always available when needed (D) Unauthorised users cannot enter the system

[Source: Repo: BCI_SCI Subject Test-01 (CN and Security)] · Hard

(B) Data is accurate and has not been modified

Nukta: Goal map: Confidentiality = authorised access only. Integrity = no unauthorised change. Availability = always accessible.

Goal map: Confidentiality = authorised access Integrity = ³ authorisation ʁ Availability = Ê

Q593. Which of the following is NOT a goal of network security?

³ K Ȼ - network security ł ʁ ?

(A) Confidentiality (B) Integrity

(C) Availability (D) Redundancy

[Source: Subject Test Solution-1 Part-2 (CN & Security) slides] · Hard

(D) Redundancy

Nukta: Security goals = CIA. Reliability goals = redundancy, fault tolerance, recovery.

Security goals = CIA Reliability goals = redundancy, fault tolerance, recovery

Q594. The primary di erence between active and passive attacks is:

Active passive attacks ħ Ħ ?

(A) Active attacks change data; passive attacks do not (B) Passive attacks destroy hardware

(C) Both are the same (D) Passive attacks are legal

[Source: Repo: BCI_SCI Subject Test-01 (CN and Security)] · Hard

(A) Active attacks change data; passive attacks do not

Nukta: Active: modification, fabrication, interruption. Passive: eavesdropping, tra c analysis. Exam one-liner: 'change vs observe'.

Active: modification, fabrication, interruption Passive: eavesdropping, tra c analysis Exam one-liner: 'change observe'

Q595. A _____ does not perform any destructive action directly; it only consumes system resources to bring the system down.

_____ ŋ ʁ ; g l' ³ g g ŋ system resources .

(A) Virus (B) Worm

(C) Trojan (D) Spyware

[Source: HTET 2024 (via KnowledgeGate solved PYQ bank)] · Hard

(B) Worm

Nukta: HTET 2024 o cial key (b). ķ Virus = ɴ ; Worm = ; Trojan = 0 ; Spyware = .

HTET 2024 o cial key (b). ŋ Virus = Ȼ ; Worm = ; Trojan = 0 ; Spyware = .

Ω Samjh — Section B — Cryptography

Plaintext/ciphertext/key, symmetric vs asymmetric ciphers, DES/AES/RSA, stream vs block, classical ciphers, digital signatures, steganography.

Cryptography = message ko chhupana. Symmetric (ek key) vs Asymmetric (public-private key pair) — dono ka apna jagah hai.

Cryptography = Symmetric (key) Asymmetric (public-private key)

- Plaintext = asli message; Ciphertext = encrypted message; Key = transformation ka secret.
  Plaintext = ; Ciphertext = encrypted; Key =
- Symmetric encryption — ek hi key encrypt+decrypt (DES, 3DES, AES, RC4, Blowfish); tez, par key distribute karna mushkil.
  Symmetric — key (DES, AES); , key
- Asymmetric encryption — key pair: public se encrypt, private se decrypt (RSA, ECC, Di e-Hellman); dheema, par key distribution aasan; digital signatures isi se.
  Asymmetric — key pair (RSA); , ; digital signatures
- Stream cipher (bit-by-bit, RC4) vs Block cipher (blocks, DES/AES).
  Stream cipher (bit-by-bit) vs Block cipher (blocks)
- Hash function (MD5, SHA) — message ka fixed digest; change = alag hash; integrity check ke liye.
  Hash (MD5, SHA) — fixed digest; integrity check
- Digital signature = private key se hash ko encrypt — authentication + non-repudiation.
  Digital signature = private key hash encrypt — authentication + non-repudiation
- Certificate/CA — public key ki authenticity dene wali trusted authority (PKI).
  Certificate/CA — (PKI)
- Substitution (letters badalna — Caesar) vs Transposition (order badalna); Steganography = message chhupana image mein (encrypt nahi).
  Substitution vs Transposition; Steganography = (encrypt)

Fakt	Symmetric	Asymmetric
Keys	shared key	public + private pair
Speed		
Key distribution		
Use	bulk data encryption	key exchange, signatures
Examples	DES, 3DES, AES	RSA, ECC, Di e-Hellman

Q596. The text that gets transformed using a cipher algorithm is called:

Cipher algorithm ɢ text Ħ ₹

(A) Complex Text (B) Transformed Text

(C) Plain Text (D) Scalar Text

[Source: Repo: BCI Subject Test-9 (BCI/SCI Batch) — Q9] · Easy

(C) Plain Text

Nukta: Cryptography flow: Plain text --(cipher + key)--> Ciphertext --(decipher + key)--> Plain text.

Cryptography flow: Plain text --(cipher + key)--> Ciphertext --(decipher + key)--> Plain text

Q597. In the digital signature technique, when the whole message is signed using an asymmetric key, the receiver uses:

Digital signature technique ₡ asymmetric key sign , ų ĵ ŋ :

(A) Her or his own public key (B) Her or his own private key

(C) The sender's public key (D) None of these

[Source: Repo: BCI Subject Test-9 (BCI/SCI Batch)] · Easy

(C) The sender's public key

Nukta: Signature direction: SIGN with private → VERIFY with public. Encryption direction: ENCRYPT with public → DECRYPT with private.

Signature ɛ : private SIGN → public VERIFY Encryption ɛ : public ENCRYPT → private DECRYPT

Q598. Which statement about plaintext, ciphertext and key is MOST accurate?

Plaintext, ciphertext key ₡ - ?

(A) Plaintext is generated after decryption using a public key (B) Ciphertext is the original readable message

(C) Plaintext is transformed into ciphertext using an algorithm and a key (D) Ciphertext and plaintext are always of the same length

[Source: Repo: BCI Subject Test-9 (BCI/SCI Batch) — Q10] · Medium

(C) Plaintext is transformed into ciphertext using an algorithm and a key

Nukta: Remember the 3 objects: plaintext (readable), ciphertext (scrambled), key (secret parameter). Encryption = f(plaintext, key).

3 objects ₡plaintext (Ĩ), ciphertext (scrambled), key (ĵ parameter) Encryption = f(plaintext, key)

Q599. Which of the following uses the SAME key for encryption and decryption?

ɜ ĸ ₡ - encryption decryption ɠ key ?

(A) Symmetric key cipher (B) Asymmetric key cipher

(C) Hash function (D) Digital signature

[Source: Repo: BCI_SCI Subject Test-01 (CN and Security)] · Medium

(A) Symmetric key cipher

Nukta: Symmetric = 1 key (fast). Asymmetric = key pair (public+private, slow). Hash = one-way digest. Signature = private-sign/public-verify.

Symmetric = 1 key ( ) Asymmetric = key pair (public+private,) Hash = one-way digest Signature = private-sign/public-verify

Q600. Which of the following uses a PAIR of keys (public and private)?

 ³ K Ȼ - keys ɬ (public private) ?

(A) Symmetric cipher (B) Asymmetric cipher

(C) DES (D) AES

[Source: Repo: BCI_SCI Subject Test-01 (CN and Security)] · Medium

(B) Asymmetric cipher

Nukta: Public-key examples: RSA, Di e-Hellman, ECC, DSA. Symmetric: DES, 3DES, AES, Blowfish, RC4.

Public-key examples: RSA, Di e-Hellman, ECC, DSA Symmetric: DES, 3DES, AES, Blowfish, RC4

Q601. Which is FASTER in encryption and decryption?

Encryption decryption Ȼ - ?

(A) Symmetric key cipher (B) Asymmetric key cipher

(C) Cannot decide (D) None of these

[Source: Repo: BCI_SCI Subject Test-01 (CN and Security)] · Medium

(A) Symmetric key cipher

Nukta: Speed: Symmetric >> Asymmetric. Hybrid systems (TLS/PGP) use asymmetric KEYS to exchange a symmetric SESSION key.

Speed: Symmetric >> Asymmetric Hybrid systems (TLS/PGP) asymmetric KEYS symmetric SESSION key -ų ȥ

Q602. What is the main disadvantage of symmetric key encryption?

Symmetric key encryption ɬ ħ Ħ ?

(A) It is very slow (B) Key distribution is di cult

(C) It uses two keys (D) It cannot encrypt text

[Source: Repo: BCI_SCI Subject Test-01 (CN and Security)] · Medium

(B) Key distribution is di cult

Nukta: Symmetric weakness = key distribution (n users → n(n-1)/2 keys). Asymmetric weakness = slow + key authenticity (certificates).

Symmetric = key distribution (n users → n(n-1)/2 keys) Asymmetric = + key authenticity (certificates)

Q603. In public-key cryptography, a message encrypted with the public key can only be decrypted by:

Public-key cryptography ȹublic key encrypt ³ ³ Â decrypt ?

(A) Another public key (B) The same public key

(C) The private key corresponding to that public key (D) A session key

[Source: Repo: BCI_SCI Subject Test-01 (CN and Security)] · Medium

(C) The private key corresponding to that public key

Nukta: Public/private are mathematically paired — like a lock (public) and its only key (private).

Public/private ɢ ę ⱬ (public) ɬ Ů (private) ɬ

Q604. Data Encryption Standard (DES) is a:

Data Encryption Standard (DES) ₹ ?

(A) Block cipher (B) Stream cipher

(C) Bit cipher (D) Byte cipher

[Source: Repo: BCI_SCI Subject Test-01 (CN and Security)] · Medium

(A) Block cipher

Nukta: DES = 64-bit block, 56-bit key (symmetric). AES = 128-bit block. RC4 = stream cipher.

DES = 64-bit block, 56-bit key (symmetric) AES = 128-bit block RC4 = stream cipher

Q605. Which cipher method encrypts one bit or byte at a time?

- cipher method ₵ bit byte encrypt ?

(A) Block cipher (B) Stream cipher

(C) Substitution cipher (D) Transposition cipher

[Source: Repo: BCI_SCI Subject Test-01 (CN and Security)] · Medium

(B) Stream cipher

Nukta: Stream = 1 bit/byte at a time (RC4, A5/1). Block = fixed-size blocks (DES/AES). Substitution/Transposition = classical techniques.

Stream = 1 bit/byte at a time (RC4, A5/1) Block = fixed-size blocks (DES/AES) Substitution/ Transposition = classical techniques

Q606. The Caesar Cipher is an example of which cryptographic technique?

Caesar Cipher ৩ cryptographic technique ?

(A) Transposition cipher (B) Substitution cipher

(C) Stream cipher (D) Block cipher

[Source: Repo: BCI_SCI Subject Test-01 (CN and Security)] · Medium

(B) Substitution cipher

Nukta: Classical types: Substitution (Caesar, Playfair) = replace; Transposition (Rail fence) = rearrange.

Classical types: Substitution (Caesar, Playfair) = replace; Transposition (Rail fence) = rearrange

Q607. The process of detecting hidden data inside a file is known as:

৩ file ৪ data ৰ ৸ত' ₹ ?

(A) Cryptanalysis (B) Steganalysis

(C) Data mining (D) Steganography

[Source: Repo: BCI_SCI Subject Test-01 (CN and Security)] · Medium

(B) Steganalysis

Nukta: Steganography = HIDING data (cover file). Steganalysis = DETECTING it. Cryptography = scrambling it.

Steganography = data ৪ (cover file) Steganalysis = Cryptography = scrambl

Q608. What is the output of a steganography process called?

Steganography ৸ত' output ₹ ?

(A) Ciphertext (B) Stego-object

(C) Encoded file (D) Key file

[Source: Repo: BCI_SCI Subject Test-01 (CN and Security)] · Medium

(B) Stego-object

Nukta: Stego terms: cover file (original carrier), secret message (hidden data), stego-object (result).

Stego terms: cover file (original carrier), secret message (hidden data), stego-object (result)

Q609. In steganography, the file used to hide the secret data is called:

Steganography Ø secret data ß ĝ ß file ĦŽ

(A) Secret file (B) Cover file

(C) Cipher file (D) Key file

[Source: Subject Test Solution-1 Part-2 (CN & Security) slides] · Medium

(B) Cover file

Nukta: Cover file = the disguise. Secret message = hidden data. Stego-object = cover + secret after embedding.

Cover file = ø Secret message = ß data Stego-object = embedding cover + secret

Q610. Which of the following is an example of asymmetric encryption?

ɜ ĸ Ø - asymmetric encryption ?

(A) AES (B) DES

(C) RSA (D) Blowfish

[Source: SSC CGL Tier-2 repeated Qs (via Oliveboard) · also BCI_SCI ST-01] · Medium

(C) RSA

Nukta: Asymmetric = RSA, ECC, Di e-Hellman, DSA. Symmetric = DES, 3DES, AES, Blowfish, RC4, IDEA.

Asymmetric = RSA, ECC, Di e-Hellman, DSA
Symmetric = DES, 3DES, AES, Blowfish, RC4, IDEA

Q611. The data to be encrypted at the sender's site is called _____ and the data to be decrypted at the receiver's end is called _____.

Sender ɬ encrypt ɜ data _____ receiver ɬ decrypt ɜ data _____ Ž

(A) Secret key, public key (B) Public key, secret key

(C) Plain text, cipher text (D) Cipher text, plain text

[Source: SBI PO Computer Awareness PYQ (via bankersadda)] · Medium

(C) Plain text, cipher text

Nukta: : Plain ( ) → encrypt → Cipher ( ) → decrypt → Plain.

: Plain ( ) → encrypt → Cipher ( ) → decrypt → Plain.

Q612. Public key is used to:

Public key ɜ ĝ ?

(A) Encrypt the data (B) Decrypt the data

(C) Both encrypt and decrypt (D) Generate session keys only

[Source: Repo: BCI_SCI Subject Test-01 (CN and Security)] · Hard

(A) Encrypt the data

Nukta: Encrypt with public → decrypt with private. SIGN with private → verify with public. Exam asks both directions!

Encrypt public → decrypt private SIGN private → verify public Exam Ø !

Q613. Which statement about symmetric and asymmetric cryptography is TRUE?

Symmetric asymmetric cryptography Ȼ - ?

(A) Symmetric cryptography provides authentication (B) Asymmetric cryptography is faster than symmetric

(C) Symmetric cryptography requires secure key exchange (D) Both use two di erent keys

[Source: Repo: BCI_SCI Subject Test-01 (CN and Security) — Q33] · Hard

(C) Symmetric cryptography requires secure key exchange

Nukta: True: symmetric needs secure key exchange. False: symmetric does not inherently authenticate; asymmetric is slower; symmetric uses ONE key.

: symmetric secure key exchange ε : symmetric ľ authentication ʁ ; asymmetric ; symmetric key

Q614. What is the key length of AES-256 encryption?

AES-256 encryption ɬ key length ɜ ?

(A) 128 bits (B) 192 bits

(C) 256 bits (D) 512 bits

[Source: Repo: BCI_SCI Subject Test-01 (CN and Security)] · Hard

(C) 256 bits

Nukta: AES variants: AES-128, AES-192, AES-256 (key sizes). Block size is always 128 bits.

AES variants: AES-128, AES-192, AES-256 (key sizes) Block size 128 bits

Q615. Which of the following statements is FALSE?

ɜ ĸ Ȼ - ?

(A) AES is a block cipher (B) RSA is a symmetric cipher

(C) Di e-Hellman is used for key exchange

(D) DES is symmetric

[Source: Repo: BCI_SCI Subject Test-01 (CN and Security)] · Hard

(B) RSA is a symmetric cipher

Nukta: False-statement MCQs: check each. AES=block , RSA=asymmetric , DH=key exchange , DES=symmetric .

False-statement MCQs: ȻAES=block , RSA=asymmetric , DH=key exchange , DES=symmetric

Q616. Suppose everyone in a group of N people wants to communicate secretly with the N−1 others using a symmetric-key cryptographic system. The number of keys required in the system as a whole is:

ɢ N Ȼ Ȼ ê ĥ Ħ ɬ N−1 Ȼ symmetric-key cryptographic system ĵ . ɜ keys ε Ȼ?

(A) 2N (B) N(N − 1)

(C) N(N − 1)/2 (D) (N − 1)²

[Source: GATE CS PYQ (via examprev)] · Hard

(C) N(N − 1)/2

Nukta: GATE key (c). Public-key N Ř ʏ — ß hybrid systems .

GATE key (c). Public-key Ȼ ľ ʁ — ǧ hybrid systems .

Ω Samjh — Section C — Firewalls, IPSec & Security Protocols

DMZ design, IPSec modes (transport/tunnel), SSL/TLS & HTTPS, security-protocol and mechanism layer mapping.

Firewall, IPSec, SSL/TLS aur VPN — network ko secure karne ke practical tools.

Firewall, IPSec, SSL/TLS VPN — network w ê ∂

- Firewall types: Packet-filtering (header rules), Stateful (connection state), Proxy/Application gateway (message level).

  Firewall: Packet-filtering, Stateful, Proxy/Application gateway

- DMZ — public servers wala beech ka zone; andar ka LAN sabse secure.

  DMZ — public servers wŮ; LAN ©w

- IPSec (Network layer security): AH (authentication header) + ESP (encryption + authentication).

  IPSec (Network layer): AH + ESP

- IPSec modes: Transport mode (sirf payload encrypt) vs Tunnel mode (poora packet encrypt — VPN ke liye).

  IPSec modes: Transport (payload) vs Tunnel (poora packet — VPN)

- VPN = public network par private tunnel (encryption ke saath) — IPSec/PPTP/L2TP se banta hai.

  VPN = public network private tunnel

- SSL/TLS — transport layer security: handshake (certificate exchange, session key) + record protocol (encryption); HTTPS = HTTP + SSL/TLS.

  SSL/TLS — handshake + records; HTTPS = HTTP + SSL/TLS

- SSH = secure remote login (Telnet ka secure version, port 22).

  SSH = ©w remote login (port 22)

- E-mail security: PGP (Pretty Good Privacy), S/MIME — encryption + signature.

  E-mail security: PGP, S/MIME

- Wireless security: WEP (kamzor) → WPA → WPA2 (AES) → WPA3.

  Wireless: WEP → WPA → WPA2 → WPA3

Technology	Layer	Kaam
Firewall	3–7	tra c filter
IPSec	3 (Network)	packet encryption/auth (VPN)
SSL/TLS	4–5 (Transport)	secure web (HTTPS)
SSH	Application	secure remote login
PGP/S-MIME	Application	secure e-mail

Q617. IPSec is MOST commonly used to implement:

IPSec ħ ३ ∮ ?

(A) Secure email (B) Secure web browsing

(C) Virtual Private Network (VPN) (D) Wireless security

[Source: Repo: BCI Subject Test-9 (BCI/SCI Batch)] · Easy

(C) Virtual Private Network (VPN)

Nukta: IPSec = VPN tunneling (ESP/AH). Secure email = PGP/SMIME. Secure web = SSL/TLS. Wireless = WPA2.

IPSec = VPN tunneling (ESP/AH) Secure email = PGP/SMIME Secure web = SSL/TLS Wireless = WPA2

Q618. In IPSec Tunnel mode:

IPSec Tunnel mode ∅

(A) Only the payload is encrypted (B) Only the header is encrypted

(C) The entire IP packet is encrypted (D) No encryption is used

[Source: Repo: BCI Subject Test-9 (BCI/SCI Batch)] · Easy

(C) The entire IP packet is encrypted

Nukta: Tunnel = whole packet + new header (gateway VPN). Transport = payload only (host-to-host).

Tunnel = packet + header (gateway VPN) Transport = payload (host-to-host)

Q619. The SSL/TLS handshake protocol is mainly used for:

SSL/TLS handshake protocol ħ ३ ∮ ?

(A) Data encryption (B) Data compression

(C) Key exchange and authentication (D) Packet routing

[Source: Repo: BCI Subject Test-9 (BCI/SCI Batch)] · Easy

(C) Key exchange and authentication

Nukta: TLS parts: Handshake = keys+authentication; Record = encryption+integrity. HTTPS = HTTP over TLS (port 443).

TLS parts: Handshake = keys+authentication; Record = encryption+integrity HTTPS = HTTP over TLS (port 443)

Q620. Match the security mechanisms with their OSI layers: (a) Firewall (b) Digital Signature (c) MAC Filtering (d) Port-based Filtering — (i) Network Layer (ii) Application Layer (iii) Data Link Layer (iv) Transport Layer

Security mechanisms OSI layers ३ : (a) Firewall (b) Digital Signature (c) MAC Filtering (d) Port-based Filtering — (i) Network Layer (ii) Application Layer (iii) Data Link Layer (iv) Transport Layer

(A) a–i, b–ii, c–iii, d–iv (B) a–ii, b–i, c–iii, d–iv

(C) a–iv, b–ii, c–i, d–iii (D) a–i, b–iii, c–ii, d–iv

[Source: Repo: BCI Subject Test-9 (BCI/SCI Batch) — Q4 match (codes recovered)] · Medium

(A) a–i, b–ii, c–iii, d–iv

Nukta: Mechanism-layers: Firewall L3/L4 (IP), Port filter L4, MAC filter L2, Digital signature L7. Packet-filtering firewall = L3/L4.

Mechanism-layers: Firewall L3/L4 (IP), Port filter L4, MAC filter L2, Digital signature L7 Packet-filtering firewall = L3/L4

Q621. Match the security protocols with their OSI layers: (a) IPSec (b) SSL/TLS (c) PGP (d) WPA2 — (i) Application Layer (ii) Network Layer (iii) Transport/Presentation Layer (iv) Data Link Layer (Wi-Fi)

Security protocols OSI layers : (a) IPSec (b) SSL/TLS (c) PGP (d) WPA2 — (i) Application Layer (ii) Network Layer (iii) Transport/ Presentation Layer (iv) Data Link Layer (Wi-Fi)

(A) a–ii, b–iii, c–i, d–iv (B) a–ii, b–i, c–iii, d–iv

(C) a–iii, b–ii, c–i, d–iv (D) a–i, b–ii, c–iii, d–iv

[Source: Repo: BCI Subject Test-9 (BCI/SCI Batch) — Q1 match (codes recovered)] · Medium

(A) a–ii, b–iii, c–i, d–iv

Nukta: Protocol layers: IPSec L3, SSL/TLS L4-7 (exam: transport/presentation), PGP L7, WPA2 L2, Kerberos L7.

Protocol layers: IPSec L3, SSL/TLS L4-7 (exam: transport/presentation), PGP L7, WPA2 L2, Kerberos L7

Q622. IPSec operates in which mode(s)?

IPSec mode(s) ?

(A) Transport mode only (B) Tunnel mode only

(C) Both transport mode and tunnel mode (D) Application mode

[Source: Repo: BCI Subject Test-9 (BCI/SCI Batch)] · Medium

(C) Both transport mode and tunnel mode

Nukta: IPSec modes: Transport (payload only, host-to-host) & Tunnel (full packet, gateway-to-gateway/VPN).

IPSec modes: Transport (payload, host-to-host) Tunnel (packet, gateway-to-gateway/VPN)

Q623. Which mode of IPSec provides end-to-end security between two hosts?

IPSec - mode hosts end-to-end security ?

(A) Transport mode (B) Tunnel mode

(C) Static mode (D) Gateway mode

[Source: Repo: CN Topic Test 5 (UNIT TEST-5)] · Medium

(A) Transport mode

Nukta: End-to-end (host↔host) = Transport mode. Gateway↔gateway (VPN) = Tunnel mode.

End-to-end (host↔host) = Transport mode Gateway↔gateway (VPN) = Tunnel mode

Q624. Which of the following is used in HTTPS for encryption?

HTTPS encryption ?

(A) SSL/TLS (B) FTP

(C) ARP (D) NAT

[Source: Repo: CN Topic Test 5 (UNIT TEST-5)] · Medium

(A) SSL/TLS

Nukta: HTTPS = HTTP + SSL/TLS (443). HTTP = 80 (no encryption). FTP = 21. ARP = IP→MAC. NAT = address translation.

HTTPS = HTTP + SSL/TLS (443) HTTP = 80 (encryption) FTP = 21 ARP = IP→MAC NAT = address translation

Q625. In which layer of the network architecture is the Secure Socket Layer (SSL) used?

Secure Socket Layer (SSL) network architecture ɬ3 layer Ȼ ?

(A) Physical layer (B) Session layer

(C) Application layer (D) Presentation layer

[Source: Unit Test-4 Solution (OCR), p6 (bank key)] · Medium

(B) Session layer

Nukta: Exam convention: SSL = Session layer (secure session setup + negotiation). Note: TLS = successor of SSL.

Exam convention: SSL = Session layer (secure session setup + negotiation) : TLS, SSL ģ ɟ

Q626. When a logic bomb is activated by a time-related event, it is known as a:

ɳlogic bomb ɜ - ɜť , ʐ

(A) Time-related bomb sequence (B) Virus

(C) Time bomb (D) Trojan horse

[Source: SBI PO Computer Awareness PYQ (via bankersadda)] · Medium

(C) Time bomb

Nukta: Logic bomb = code that explodes on a condition; if the condition is time, it is a Time bomb.

Logic bomb = ɳ ; ɳ Time bomb.

Q627. Verification based on a login ID and password is called:

Login ID password :

(A) Logging in (B) Configuration

(C) Authentication (D) Confidentiality

[Source: RBI/IBPS computer-awareness banks (via ixambee)] · Medium

(C) Authentication

Nukta: Authentication = ' ?'; Authorization = ' ₵ NÍ 0 ?'.

Authentication = ' ?'; Authorization = ' ĸ Ȼ ɜ ?'.

Q628. Access control based on a person's fingerprints is an example of:

ɜ êńĦ ɬ ģ Ȼ ɜ ə access control :

(A) Biometric identification (B) Characteristic identification

(C) Encryption (D) Logistics

[Source: SBI PO Computer Awareness PYQ (via bankersadda)] · Medium

(A) Biometric identification

Nukta: Physical identity (biometric) is stronger than something known (password).

x (password) ə (biometric).

Q629. Which of the following networks extends a private network across public networks?

३ K Ȼ - network public networks private network ३ ľ ?

(A) Virtual private network (VPN) (B) Local area network

(C) Enterprise private network (D) None of these

[Source: BPSC TRE 2024 (via KnowledgeGate solved PYQ bank)] · Medium

(A) Virtual private network (VPN)

Nukta: BPSC 2024 o cial key (a). VPN = a private limousine on the public road.

BPSC 2024 o cial key (a). VPN = ŋ३ ३ ǥ .

Q630. Which one of the following is a key function of a firewall?

३ K Ȼ firewall ħ - ?

(A) Monitoring (B) Copying

(C) Deleting (D) Moving

[Source: RRB NTPC previous-year computer banks (via adda247 50-PYQ PDF)] · Medium

(A) Monitoring

Nukta: Firewall = the network's gatekeeper — it watches, checks, stops.

Firewall = network ɬ — , , .

Q631. What is the advantage of setting up a DMZ (Demilitarized Zone) using a firewall?

Firewall DMZ (Demilitarized Zone) Ħ ?

(A) To create multiple zones of trust (B) Load balancing on the server

(C) Improving network performance (D) Bringing internet service into a home

[Source: RPSC Programmer 2024 Paper-1 (o cial key)] · Hard

(A) To create multiple zones of trust

Nukta: O cial RPSC key. DMZ = bu er zone between untrusted Internet and trusted LAN (3 zones: external / DMZ / internal).

O cial RPSC key DMZ = untrusted Internet trusted LAN bu er zone (3 zones: external / DMZ / internal)

Nukte Rapid-Fire — Exam Pointers (quick revision)

One-line question (left) with its answer + trick (right). Exam se pehle revise karo.

5 components of data communication? Data communication 5 components?	Message, Sender, Receiver, Medium, Protocol Message, Sender, Receiver, Medium, Protocol
3 criteria of a good network? Ğ network 3 criteria?	Performance, Reliability, Security Performance, Reliability, Security
4 characteristics of data communication? Data communication 4 characteristics?	Delivery, Accuracy, Timeliness, Jitter Delivery, Accuracy, Timeliness, Jitter
Jitter means? Jitter ฎ	Delay variation between packets Packets delay
Protocol elements? Protocol elements?	Syntax, Semantics, Timing Syntax, Semantics, Timing
Simplex example? Simplex ?	TV broadcast / keyboard→monitor (one-way) TV broadcast / keyboard→monitor (one-way)
Half-Duplex example? Half-Duplex ?	Walkie-talkie (both ways, one at a time) Walkie-talkie (₡ , ₡)
Full-Duplex example? Full-Duplex ?	Telephone (both ways, same time) Telephone (₡ ,)
Mesh cables formula? Mesh cables formula?	n(n-1)/2 cables, n-1 ports per device n(n-1)/2 cables, n-1 ports per device
Star / Ring / Bus cables? Star / Ring / Bus cables?	Star = n, Ring = n, Bus = 1 Star = n, Ring = n, Bus = 1
Most fault-tolerant topology? ð fault-tolerant topology?	Mesh (redundant paths) Mesh (redundant paths)
Easy to install & reconfigure? Install/reconfigure ₡ ?	Star topology Star topology
Multipoint topology? Multipoint topology?	Bus (shared backbone) Bus (shared backbone)
IEEE 802.2 / 802.3 / 802.5 / 802.11? IEEE 802.2 / 802.3 / 802.5 / 802.11?	LLC / Ethernet(MAC) / Token Ring / Wi-Fi LLC / Ethernet(MAC) / Token Ring / Wi-Fi
Network size order? Network size ť ?	PAN < LAN < MAN < WAN PAN < LAN < MAN < WAN
internet vs Internet? internet Internet?	Any network of networks vs THE global one ŋ network of networks 3 Ò ê

Intranet vs Extranet? Intranet Extranet?	Intranet = inside the organisation; Extranet = trusted outsiders (supplier) Intranet = I' ; Extranet = (supplier)
Client-Server vs P2P examples? Client-Server P2P ?	ATM = Client-Server; BitTorrent = P2P ATM = Client-Server; BitTorrent = P2P
ARPANET facts? ARPANET I ?	1969 (ARPA); TCP/IP adopted 1983; WWW 1991 (CERN, Tim Berners-Lee) 1969 (ARPA); TCP/IP adopted 1983; WWW 1991 (CERN, Tim Berners-Lee)
Internet switching? Internet - switching use ?	Packet switching (data travels in packets) Packet switching (data packets)
Circuit switching key? Circuit switching ?	Dedicated separate physical path per connection connection dedicated physical path
OSI layers mnemonic? OSI layers mnemonic?	Please Do Not Throw Sausage Pizza Away (L1→L7) Please Do Not Throw Sausage Pizza Away (L1→L7)
OSI kisne banaya, kab? OSI , ?	ISO, in 1984 — 7 layers ISO, 1984 — 7 layers
TCP/IP kisne banaya? TCP/IP ?	Cerf & Kahn (DoD/ARPA), 1970s — adopted 1983 Cerf & Kahn (DoD/ARPA), 1970s — adopted 1983

Layer counts? Layer counts?	OSI = 7, Internet stack = 5, TCP/IP suite = 4 OSI = 7, Internet stack = 5, TCP/IP suite = 4
TCP/IP mein kya missing? TCP/IP missing?	No separate Session & Presentation layers Session Presentation layers
OSI→TCP/IP mapping? OSI→TCP/IP mapping?	OSI 1+2 = Network Access; OSI 5+6+7 = Application OSI 1+2 = Network Access; OSI 5+6+7 = Application
PDU per layer? layer PDU?	Data / Segment / Packet-Datagram / Frame / Bits Data / Segment / Packet-Datagram / Frame / Bits
Encapsulation direction? Encapsulation ?	Sender: top→bottom (headers add); Receiver: bottom→top (remove) Sender: top→bottom; Receiver: bottom→top
4 levels of addresses? 4 levels addresses?	Physical(48-bit), IP(32/128-bit), Port(16-bit), Specific(URL) Physical(48-bit), IP(32/128-bit), Port(16-bit), Specific(URL)
Delivery scopes? Delivery scopes?	Bit-by-bit=Physical; Hop-to-hop=Data Link; Source-destination=Network; End-to-end/Process-process=Transport Bit-by-bit=Physical; Hop-to-hop=Data Link; Source-destination=Network; End-to-end=Transport
Presentation layer kaam? Presentation layer ?	Translation + Encryption + Compression Translation + Encryption + Compression

Session layer ke 3 kaam? Session layer 3 ?	Dialog control + Token management + Synchronization (checkpoint) Dialog control + Token management + Synchronization
Data Link ka trio? Data Link trio?	Framing + Error detection + Physical(MAC) addressing Framing + Error detection + MAC addressing
LLC vs MAC? LLC MAC?	LLC (802.2) = Network layer interface; MAC (802.3) = medium access + MAC address LLC = interface; MAC = medium access + address
Support layers grouping? Support layers grouping?	Network support = 1,2,3; User support = 5,6,7; Transport(4) = bridge Network support = 1,2,3; User support = 5,6,7; Transport = bridge
Device layers? Devices ↕ layers?	Repeater/Hub=L1; Bridge/Switch=L2; Router=L3; Gateway=L4+ Repeater/Hub=L1; Bridge/Switch=L2; Router=L3; Gateway=L4+
Reliable delivery? Reliable delivery?	Transport layer (TCP) — IP is best-e ort only Transport layer (TCP) — IP ∫ n best-e ort
Router visits formula? Router visits formula?	n routers: Network layer = n+2, Data Link = 2(n+1) n routers: Network layer = n+2, Data Link = 2(n+1)
Physical layer ka kaam? Physical layer ?	Signals, media, line/channel coding, modulation, baud rate, bit-by-bit Signals, media, coding, modulation, baud rate, bit-by-bit

IP/ICMP/ARP kis layer ke? IP/ICMP/ARP 3 layer ?	Network / Internet layer (L3) Network / Internet layer (L3)
MAC address size? MAC address size?	48 bits (6 bytes, 12 hex digits). 48 bits (6 bytes, 12 hex digits)
IPv4 size? IPv6 size? Port size? IPv4 size? IPv6 size? Port size?	32 / 128 / 16 bits. 32 / 128 / 16 bits
MAC address other names? MAC address ij ?	Physical address / hardware address / burned-in address. Physical address / hardware address / burned-in address
Broadcast MAC address? Broadcast MAC address?	FF:FF:FF:FF:FF:FF (all 48 bits = 1). FF:FF:FF:FF:FF:FF (48 bits = 1)
I/G bit trick for unicast/multicast? Unicast/multicast g I/G bit trick?	Last bit of first octet: 0 = unicast, 1 = multicast. octet 3 bit: 0 = unicast, 1 = multicast
Class ranges by first octet (A/B/C/D/E)? octet Class ranges (A/B/C/D/E)?	0-127 / 128-191 / 192-223 / 224-239 (multicast) / 240-255 (experimental). 0-127 / 128-191 / 192-223 / 224-239 (multicast) / 240-255 (experimental)
Default masks A/B/C? Default masks A/B/C?	255.0.0.0 /8, 255.255.0.0 /16, 255.255.255.0 /24. 255.0.0.0 /8, 255.255.0.0 /16, 255.255.255.0 /24

Hosts per class (A/B/C)? ų class hosts (A/B/C)?	2^24−2 (16,777,214) / 2^16−2 (65,534) / 2^8−2 (254). 2^24−2 (16,777,214) / 2^16−2 (65,534) / 2^8−2 (254)
Loopback address? Loopback address?	127.0.0.1 (whole 127.0.0.0/8 block). 127.0.0.1 (127.0.0.0/8 block)
APIPA range (DHCP fail)? APIPA range (DHCP fail)?	169.254.0.1 - 169.254.255.254. 169.254.0.1 - 169.254.255.254
Private IP blocks (RFC 1918)? Private IP blocks (RFC 1918)?	10.x / 172.16-31.x / 192.168.x. 10.x / 172.16-31.x / 192.168.x
0.0.0.0 means? 0.0.0.0 ฎ	'This host — no IP yet' (source) or default route (0.0.0.0/0). ' host — IP ษ' (source) default route (0.0.0.0/0)
Usable hosts formula? Usable hosts Ů?	2^host-bits − 2. /24→254, /25→126, /27→30, /30→2. 2^host-bits − 2 /24→254, /25→126, /27→30, /30→2
Subnets formula? Subnets Ů?	2^borrowed-bits. 3 bits → 8 subnets. 2^borrowed-bits 3 bits → 8 subnets
CIDR /24 means? CIDR /24 ฎ	24 network bits (mask 255.255.255.0). 24 network bits (mask 255.255.255.0)

/30 used for? /30 ?	Point-to-point links (2 usable hosts). Point-to-point links (2 hosts)
Subnetting trade-o ? Subnetting trade-o ?	More subnets ⇔ fewer hosts each. subnets ⇔ hosts
IPv6 has broadcast? IPv6 broadcast ?	NO — IPv6 uses multicast + anycast, no broadcast. — IPv6 multicast + anycast , broadcast
IPv6 transition techniques? IPv6 transition techniques?	Tunneling (IPv6-in-IPv4), Dual-stack, Translation (NAT64). Tunneling (IPv6-in-IPv4), Dual-stack, Translation (NAT64)
Well-known / registered / ephemeral port ranges? Well-known / registered / ephemeral port ranges?	0-1023 / 1024-49151 / 49152-65535. 0-1023 / 1024-49151 / 49152-65535
Port numbers: DNS, HTTP, HTTPS, FTP, SMTP, SSH? Port numbers: DNS, HTTP, HTTPS, FTP, SMTP, SSH?	53, 80, 443, 21, 25, 22. 53, 80, 443, 21, 25, 22
Socket = ? Socket = ?	IP address + Port number (identifies a process). IP address + Port number (process)
DNS records: A / AAAA / PTR / MX / CNAME? DNS records: A / AAAA / PTR / MX / CNAME?	name→IPv4 / name→IPv6 / IP→name / mail / alias. name→IPv4 / name→IPv6 / IP→name / mail / alias

ARP vs RARP? ARP vs RARP?	ARP = IP→MAC. RARP = MAC→IP (obsolete, use DHCP). ARP = IP→MAC RARP = MAC→IP (ų ℊ , DHCP ₡
NAT purpose? NAT ṅ ?	Map private IPs → public IP (save addresses + hide internals). Private IPs → public IP map (addresses + internals ℓ)
Per-hop vs end-to-end in a routed packet? Routed packet ₡er-hop vs end-to-end?	MAC changes per hop; IP addresses stay end-to-end. MAC hop ; IP addresses end-to-end ⱬ
Who allocates IP addresses globally? Global IP addresses allocate ?	ICANN/IANA (with RIRs). IETF makes protocols, IEEE makes LAN standards. ICANN/IANA (RIRs) IETF protocols , IEEE LAN standards
Device-layer ladder (1/2/3/7)? Device-layer ladder (1/2/3/7)?	Hub/Repeater/Modem = 1; Switch/Bridge/NIC = 2; Router = 3; Gateway = 7. Hub/Repeater/Modem = 1; Switch/Bridge/NIC = 2; Router = 3; Gateway = 7
Dumb devices? Dumb devices?	Hub and Repeater (no address reading). Bridge/switch/router = intelligent. Hub Repeater (address ʁ) Bridge/switch/router = intelligent
Hub vs switch — collision domain? Hub vs switch — collision domain?	Hub = 1 collision domain for all ports. Switch = per-port collision domain. Hub = ports ℊ 1 collision domain Switch = per-port collision domain
Who separates broadcast domains? Broadcast domains ?	Router (and VLANs on a switch). Switch alone does NOT. Router (switch VLANs) switch ʁ

Switch CAM table stores? Switch CAM table Ø ?	MAC addresses (learned from source MACs) — for fast frame forwarding. MAC addresses (source MACs) — frame forwarding
Unknown unicast at a switch? Switch unknown unicast?	Flood out all ports (except source). Known = single port. Broadcast = flood. ports flood (source) Known = port Broadcast = flood
Router between two switches + broadcast? switches router + broadcast?	Broadcast does NOT cross the router — the other side receives nothing. Broadcast router —
4 VLANs = ? broadcast domains 4 VLANs = ? broadcast domains	4 (each VLAN is its own broadcast domain). 4 (VLAN broadcast domain)
Access point operates at? Access point layer ?	Layer 2 — it is a wireless bridge (802.3 ↔ 802.11 frames). Layer 2 — wireless bridge (802.3 ↔ 802.11 frames)
What does a router do per hop? hop router	Decrements TTL, updates checksum, rewrites BOTH MAC addresses. IP addresses stay same. TTL , checksum , MAC addresses IP addresses same
Router functions NOT included? Router Ø	Collision detection (that's Layer-2 CSMA/CD on switches/NICs). Collision detection (Layer-2 CSMA/CD — switches/NICs)
Routers reassemble fragments? Routers fragments	NO — only the destination reassembles. Routers may fragment but never reassemble. — destination Routers fragment

RED technique? RED technique?	Random Early Detection — drops packets early (randomly) to avoid congestion collapse. Random Early Detection — congestion g Í (randomly) packets
Gateway = ? Gateway = ?	Protocol converter — links systems with DIFFERENT protocols (works at all 7 layers). Protocol converter — g È protocols systems (7 layers)
Default gateway means? Default gateway ñ	The router IP used to leave the local network (gateway of last resort). Local network g router IP (gateway of last resort)
Firewall = ? Firewall = ?	Security system between trusted private & untrusted outside networks — filters tra c. Trusted private untrusted outside networks security system — tra c filter
Packet-filtering vs stateful vs proxy firewall? Packet-filtering vs stateful vs proxy firewall?	L3/L4 IP rules (stateless) → tracks sessions → L7 intermediary (hides clients). L3/L4 IP rules (stateless) → sessions → L7 intermediary (clients 8)
Static NAT vs PAT? Static NAT vs PAT?	Static = 1 private ↔ 1 public (permanent). PAT = many private ↔ 1 public (ports). Static = 1 private ↔ 1 public (l') PAT = many private ↔ 1 public (ports)
MUX = ? MUX = ?	Multiplexer — combines many I/O signals into ONE line (FDM/TDM/WDM); demux splits again. Multiplexer — ñ/O signals line Œ (FDM/TDM/WDM); demux 3
Intelligent devices? Intelligent devices?	Switch + router (they read addresses & decide). Repeater + hub = dumb. Switch + router (address 3 ñ ¾ Repeater + hub = dumb

File server = ? File server = ?	A specialised computer that stores & serves files to clients. 3 computer clients files t&
Repeaters extend network how? Repeaters network ₹	Regenerate (amplify + reshape) signals — so data can travel farther without degradation. Signals regenerate (amplify + reshape) — data 3 Ȧ
Attenuation / distortion / noise? Attenuation / distortion / noise?	Strength loss / shape change / unwanted signals. Noise can only be MINIMISED, never removed. Strength loss / shape change / unwanted signals Noise , ȣ
Crosstalk cause? Crosstalk ?	Electromagnetic coupling between nearby pairs — minimised by tighter twists. ɻ pairs electromagnetic coupling — tighter twists
Manchester encoding? Manchester encoding?	Mid-bit transition = self-clocking + DC-free + noise-resistant (Physical layer). Mid-bit transition = self-clocking + DC-free + noise-resistant (Physical layer)
Encoding redundancy order? Encoding redundancy t ?	1B2B (100%) > 4B5B/8B10B (25%) > 5B6B (20%). Gigabit fiber = 8B/10B. 1B2B (100%) > 4B5B/8B10B (25%) > 5B6B (20%) Gigabit fiber = 8B/10B
Scrambler use? Scrambler ?	Make the line signal look random (break long runs) — helps clock recovery. Not compression. Line signal random (runs) — clock recovery Ȼ Compression ȣ
Shannon & Nyquist formulas? Shannon Nyquist Ů?	$C = B \cdot \log_2(1+SNR)$ (noisy); $C = 2B \cdot \log_2(L)$ (noiseless). $C = B \cdot \log_2(1+SNR)$ (noisy); $C = 2B \cdot \log_2(L)$ (noiseless)

Bit rate vs baud rate? Bit rate baud rate?	Bit rate = baud × bits per signal unit. Bit rate = baud × ų signal unit bits
Time to send n bits at b bps? b bps n bits ?	t = n/b seconds. t = n/b seconds
Analog signal parameters? Analog signal parameters?	Amplitude (height), Frequency (cycles/sec, Hz), Phase (start), Wavelength (distance/cycle). Amplitude ( ) Frequency (cycles/sec, Hz), Phase ( ), Wavelength (Å /cycle)
Quantization noise in? Quantization noise	PCM (Pulse Code Modulation) — rounding analog samples to levels. PCM (Pulse Code Modulation) — analog samples levels
Baseband vs broadband? Baseband broadband?	Baseband = digital signals directly. Broadband = modulated analog carriers (FDM). Baseband = digital signals Broadband = modulated analog carriers (FDM)
Bandwidth vs throughput? Bandwidth throughput?	Bandwidth = max capacity; Throughput = actual data per second (exam often mixes them). Bandwidth = capacity; Throughput = data (exam)
Fiber's 3 superpowers? Fiber 3 ?	EMI-immune, tap-proof, highest bandwidth (via total internal reflection). EMI- , tap- w , bandwidth (total internal reflection)
TIR condition? TIR	Angle of incidence > critical angle (denser to rarer medium). Angle of incidence > critical angle (denser rarer medium)

UTP / coax / fiber distance limits? UTP / coax / fiber Ȧ ?	100 m / 185-500 m / up to tens of km (single-mode). 100 m / 185-500 m / ǵ Ø&km (single-mode)
Connectors? Connectors?	RJ-45 (twisted pair), BNC (coax), SC/ST/LC (fiber), RJ-11 (telephone). RJ-45 (twisted pair), BNC (coax), SC/ST/LC (fiber), RJ-11 (telephone)
Coax reflection cause? Coax reflection ?	Wrong termination resistance (impedance mismatch). termination resistance (impedance mismatch)
Single-mode vs multi-mode fiber? Single-mode multi-mode fiber?	Single = long haul (km, backbone). Multi = short LAN distances (modal dispersion). Single = Ȧ (km, backbone) Multi = LAN Ȧ (modal dispersion)
Wi-Fi rate order? Wi-Fi rate ť ?	b (11 Mbps) < g (54) < n (600) < ac (multi-Gbps). b (11 Mbps) < g (54) < n (600) < ac (multi-Gbps)
Line-of-sight wireless? Line-of-sight wireless?	Infrared (and microwave links) need LOS. Radio (Wi-Fi/Bluetooth) passes walls. Infrared (microwave links) LOS ȼ Radio (Wi-Fi/Bluetooth) Ø
Microwave band? Microwave band?	300 MHz - 300 GHz (NOT 300 GHz-400 THz — that is infrared!). 300 MHz - 300 GHz (300 GHz-400 THz ȣ — infrared !)
Bluetooth = ? Bluetooth = ?	2.4 GHz short-range radio (IEEE 802.15.1 WPAN). 2.4 GHz short-range radio (IEEE 802.15.1 WPAN)

FDMA / TDMA / CDMA? FDMA / TDMA / CDMA?	Frequency bands / time slots / unique codes (same band & time). Frequency bands / time slots / unique codes (same band time)
DLL = ? DLL = ?	MAC + LLC. MAC = physical addressing + medium access; LLC = flow/error control + interface. MAC + LLC. MAC = physical addressing + medium access; LLC = flow/error control + interface.
Routing (layer ? Routing 3 layer ?	Network layer (L3) . DLL ß ķ hop ŋ . Network layer (L3) . DLL ∫ ŋ hop õ .
SW / GBN / SR windows? SW / GBN / SR windows?	SW = 1; GBN ≤ $2^n - 1$; SR ≤ 2^{n-1} (N/2). SR ß ķ frame . SW = 1; GBN ≤ $2^n - 1$; SR ≤ 2^{n-1} (N/2). SR ∫ ŋ frame .
Hamming distance d=3 Í ? Hamming distance d=3 Ħ ?	2 errors detect 1 error correct. $d \geq 2t+1$. 2 errors detect 1 error correct. $d \geq 2t+1$.
Parity Í ? Parity Ħ ?	Odd number of errors. Simple parity = 1 bit; 2D parity = rows+columns. Odd number of errors. Simple parity = 1 bit; 2D parity = rows+columns.
CRC vs Checksum? CRC vs Checksum?	DLL = CRC (FCS field); Network/Transport = checksum. Integrity = message digest. DLL = CRC (FCS field); Network/Transport = checksum. Integrity = message digest.
Ethernet frame limits? Ethernet frame limits?	Min 64 / Max 1518 bytes; data 46–1500 (MTU 1500). Min size = CSMA/CD collision detection. Min 64 / Max 1518 bytes; data 46–1500 (MTU 1500). Min size = CSMA/CD collision detection.

Frame field order? Frame field order?	Preamble → Dest MAC → Source MAC → Type → Data → FCS. FCS = CRC. Preamble → Dest MAC → Source MAC → Type → Data → FCS. FCS = CRC.
Bit stu ng? Bit stu ng?	Data N 1s 0 (flag confusion). HDLC bit-oriented; PPP byte-oriented. Data Ȼ 1s 0 (flag confusion). HDLC bit-oriented; PPP byte-oriented.
ALOHA formulas? ALOHA formulas?	Pure: 2T vulnerable, G·e ²G (~18.4%). Slotted: T vulnerable, G·e G (~36.8%). Pure: 2T vulnerable, G·e ²G (~18.4%). Slotted: T vulnerable, G·e G (~36.8%).
CSMA/CD vs CSMA/CA? CSMA/CD vs CSMA/CA?	CD = wired Ethernet (jam signal); CA = wireless (RTS/CTS, NAV). Jam = collision x . CD = wired Ethernet (jam signal); CA = wireless (RTS/CTS, NAV). Jam = collision ɬ .
Token Ring mode? Token Ring mode?	Round Robin (token passing). Reservation mode Nslot book ɾ. Round Robin (token passing). Reservation mode Ȼ slot book ʐ
IPv4 header key sizes? IPv4 header key sizes?	Min header 20B; Total Length 16-bit; Fragment O set 13-bit; TTL 8-bit; ToS = priority. Min header 20B; Total Length 16-bit; Fragment O set 13-bit; TTL 8-bit; ToS = priority.
Fragmentation vs Reassembly? Fragmentation vs Reassembly?	MF flag + Fragment O set . routers ɾ ß ķdestination. MF flag + Fragment O set . routers ʐ ɠ ŋdestination.
ARP / RARP / DNS / DHCP? ARP / RARP / DNS / DHCP?	ARP = IP→MAC; RARP = MAC→IP (obsolete); DNS = name→IP; DHCP = automatic IP (zeroconf). ARP = IP→MAC; RARP = MAC→IP (obsolete); DNS = name→IP; DHCP = automatic IP (zeroconf).

ICMP/IGMP (layer ? ICMP/IGMP 3 layer ?	Network layer. ICMP = ping/errors/PMTU; IGMP = multicast membership. Echo = query, error ¥. Network layer. ICMP = ping/errors/PMTU; IGMP = multicast membership. Echo = query, error ¤.
Routing protocol families? Routing protocol families?	RIP = Distance vector (Bellman-Ford, max 15 hops); OSPF/IS-IS = Link state (Dijkstra); BGP = Path vector (Inter-AS). RIP = Distance vector (Bellman-Ford, max 15 hops); OSPF/IS-IS = Link state (Dijkstra); BGP = Path vector (Inter-AS).
DV x ? DV ¬ ?	Count-to-infinity. : split horizon + poison reverse + hold-down. Count-to-infinity. : split horizon + poison reverse + hold-down.
STP MPLS? STP MPLS?	STP = L2 loops ; MPLS = 'Layer 2.5' (labels, QoS). STP = L2 loops ; MPLS = 'Layer 2.5' (labels, QoS).
TCP vs UDP? TCP vs UDP?	TCP = reliable, ordered, connection-oriented, broadcast ¥; UDP = connectionless, fast. TCP = reliable, ordered, connection-oriented, broadcast ¤; UDP = connectionless, fast.
Socket = ? Socket = ?	IP + Port. Ports 16-bit (0–65535). Connection = socket pair. IP + Port. Ports 16-bit (0–65535). Connection = socket pair.
TCP handshake flags? TCP handshake flags?	SYN → SYN-ACK → ACK (ESTABLISHED). SYN = G , FIN = , RST = ¥. SYN → SYN-ACK → ACK (ESTABLISHED). SYN = Ę , FIN = , RST = ń.
TCP flow vs congestion control? TCP flow vs congestion control?	Flow = receiver window (rwnd); Congestion = cwnd (slow start, avoidance). E ective = min(rwnd, cwnd). Flow = receiver window (rwnd); Congestion = cwnd (slow start, avoidance). E ective = min(rwnd, cwnd).

TCP attacks ? TCP attacks ?	SYN flood (handshake), session hijacking (random ISN). SYN flood (handshake), session hijacking (random ISN).
Port numbers? Port numbers?	21/20 FTP, 22 SSH, 23 Telnet, 25 SMTP, 53 DNS, 67/68 DHCP, 80 HTTP, 443 HTTPS, 110 POP3, 143 IMAP. 21/20 FTP, 22 SSH, 23 Telnet, 25 SMTP, 53 DNS, 67/68 DHCP, 80 HTTP, 443 HTTPS, 110 POP3, 143 IMAP.
Secure vs w protocols? Secure vs w protocols?	Secure: HTTPS, SSH, SFTP, FTPS. w : HTTP, FTP, Telnet. SSL/TLS = browser-server encryption. Secure: HTTPS, SSH, SFTP, FTPS. w : HTTP, FTP, Telnet. SSL/TLS = browser-server encryption.
Presentation layer ? Presentation layer ?	Compression, encryption, translation. Syntax = format; Semantics = meaning. Compression, encryption, translation. Syntax = format; Semantics = meaning.
FTP ?	TCP connections (control 21 + data 20); TFTP UDP (port 69); NFS = remote files local . TCP connections (control 21 + data 20); TFTP UDP (port 69); NFS = remote files local .
Ping / traceroute? Ping / traceroute?	Ping = reachability (ICMP Echo); traceroute = (Linux=UDP, Windows=ICMP). 403 = Forbidden. Ping = reachability (ICMP Echo); traceroute = (Linux=UDP, Windows=ICMP). 403 = Forbidden.
Multiplexing layer ? Multiplexing layer ?	Transport layer (ports processes share). Presentation = data . Transport layer (ports processes share). Presentation = data .
CIA triad? CIA triad?	Confidentiality, Integrity, Availability. Confidentiality, Integrity, Availability

Goal of each: interception / modification / DoS / spoofing? ł : interception / modification / DoS / spoofing?	Confidentiality / Integrity / Availability / Authentication. Confidentiality / Integrity / Availability / Authentication
Active vs passive attack? Active passive attack?	Active = changes/disrupts data. Passive = only observes (eavesdrop, tra c analysis). Active = data / ɟ Passive = (eavesdrop, tra c analysis)
Virus phases? Virus phases?	Dormant → Propagation → Triggering → Execution (payload runs). Dormant → Propagation → Triggering → Execution (payload)
WORM medium? WORM medium?	Optical disk (CD-R/DVD-R) — write once, read many. Optical disk (CD-R/DVD-R) — ǵ Ȼ ɳ Ȼ
Symmetric vs asymmetric keys? Symmetric asymmetric keys?	1 shared key (fast, key-distribution problem) vs key pair public+private (slow). 1 shared key (, key-distribution ľ) key pair public+private ( )
Encrypt/decrypt vs sign/verify direction? Encrypt/decrypt sign/verify ɛ ?	Encrypt public → decrypt private. Sign private → verify public. Encrypt public → decrypt private Sign private → verify public
DES / AES / RSA basics? DES / AES / RSA basics?	DES = 64-bit block/56-bit key. AES = 128-bit block (128/192/256 keys). RSA = asymmetric. DES = 64-bit block/56-bit key AES = 128-bit block (128/192/256 keys) RSA = asymmetric
Stream vs block cipher? Stream block cipher?	Stream = 1 bit/byte at a time (RC4). Block = fixed blocks (DES/AES). Stream = 1 bit/byte at a time (RC4) Block = fixed blocks (DES/AES)

Caesar cipher type? Caesar cipher ų ?	Classical substitution (replace letters by shifting). Classical substitution (w Ȼ ɣ)
Steganography terms? Steganography terms?	Cover file (carrier) + secret message → stego-object. Detection = steganalysis. Cover file (carrier) + secret message → stego-object = steganalysis
False-friend: RSA is symmetric? False-friend: RSA symmetric ?	NO — RSA is ASYMMETRIC. AES/DES/Blowfish are symmetric. ƅ — RSA ASYMMETRIC AES/DES/Blowfish symmetric ʐ
Security tools? Security tools?	Wireshark = sni , Nmap = port scan, Cain&Abel = password crack, Angry IP = IP scan, Snort = IDS. Wireshark = sni , Nmap = port scan, Cain&Abel = password crack, Angry IP = IP scan, Snort = IDS
Key reuse risk? Key reuse ɣ ?	Known-plaintext / frequency analysis attacks become possible. Known-plaintext / frequency analysis attacks ʐ
DMZ? DMZ?	Bu er zone between internet & trusted LAN — public servers face outside, internals stay safe. Internet trusted LAN bu er zone — public servers , internals ɢw
IPSec modes? IPSec modes?	Transport = payload only (host-to-host end-to-end). Tunnel = whole packet (gateway VPN). Transport = payload (host-to-host end-to-end) Tunnel = packet (gateway VPN)
IPSec / SSL / PGP / WPA2 layers? IPSec / SSL / PGP / WPA2 layers?	Network / Transport-Presentation / Application / Data Link (Wi-Fi). Network / Transport-Presentation / Application / Data Link (Wi-Fi)

Firewall / port filter / MAC filter / signature layers? Firewall / port filter / MAC filter / signature layers?	Network / Transport / Data Link / Application. Network / Transport / Data Link / Application
SSL/TLS handshake does? SSL/TLS handshake ?	Key exchange + server authentication (certificates). Record protocol does encryption. Key exchange + server authentication (certificates) Encryption Record protocol
HTTPS = ? HTTPS = ?	HTTP + SSL/TLS on port 443. HTTP + SSL/TLS port 443
DoS vs DDoS? DoS DDoS?	Single-source flooding vs distributed (many zombies/botnet) flooding. Single-source flooding distributed (zombies/ botnet) flooding
Attack examples map? Attack examples map?	Phishing=steal credentials; MitM=intercept; Spoofing=impersonate; Smurf=DoS variant. Phishing=credentials ; MitM=intercept; Spoofing=impersonate; Smurf=DoS variant